

TABLE 11-3. Examples of Acetone Butanol and Mixed-Solvent Fermentations

Products ^a	1 ^b	2 ^c	3 ^d	4 ^e	5 ^f	6 ^g
CO ₂	195.5	220.0	207.0	174.0	215.0	195.0
H ₂	233.0	165.9	111.1	229.4	137.0	54.0
Acetate	42.6	24.8	20.3	48.5	16.0	5.0
Butyrate	75.3	7.1	14.5	59.5		
Formate	trace				10.0	
Lactate				25.7		
Ethanol		4.9			122.0	95.0
Butanol		47.4	50.2			4.5
Acetone		22.3			28.0	6.0
Isopropanol			18.0			
Acetoin		5.7				
2,3-butanediol					12.0	39.0
C recovery (%)	97.0	98.0	93.5	98.0	105.0	100.0
O-R balance	1.02	1.01	1.05	0.99	0.90	1.13

^aAmounts expressed as millimoles per 100 mmol of glucose used.

^b*C. saccharobutylicum*. Donker, H. L. 1926. Ph.D. thesis, Technische Hooget school, Delft, The Netherlands.

^c*C. acetobutylicum*. Donker, 1926.

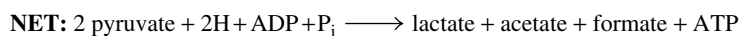
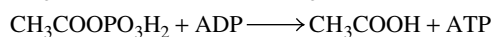
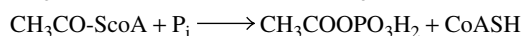
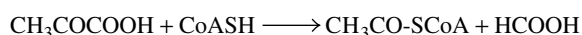
^d*C. butylicum*. Sjolander, N. W. 1937. *J. Bacteriol.* **34**:419.

^e*C. thermosaccharolyticum*. Osburn, G. L., R. W. Brown, and C. H. Werkman. 1937. *J. Biol. Chem.* **121**:685.

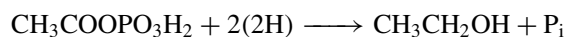
^f*B. acetobutylicum*. Osburn, et al., 1937.

^g*B. polymyxa*. Osburn, et al., 1937.

formation rather than by metabolizing hexose through an alternative pathway. Under these conditions, a dismutation of pyruvate occurs, giving rise to lactate, acetate, and formate through the following sequence:



Ethanol, when produced, is formed via reduction of acetyl phosphate:



Because of a difference in the pyruvate cleaving systems, formate, rather than carbon dioxide is the major one carbon product. Carbon dioxide is formed in small amounts by some homolactic organisms under conditions in which the fermentation is diverted from the production of lactate.

Homofermentative organisms, such as *S. faecalis*, have been shown to possess high levels of glucose-6-phosphate dehydrogenase and 6-phosphogluconate dehydrogenase.

However, a homofermentative pattern is maintained through regulatory interrelationships between the EMP and HMP pathways (see Fig. 11-3). The specific activation of lactate dehydrogenase (LDH) by FDP is characteristic of *S. bovis* and many other lactic acid-producing bacteria. The level of LDH, which is dependent on FDP for activity, decreases as fermentation becomes heterolactic. In *S. faecalis* a single NADP-linked 6PGD is observed when glucose is the primary energy source. Gluconate-adapted cells produce two 6-PGDs: one specific for NADP and the other specific for NAD. The NADP-linked enzyme is insensitive to FDP inhibition but is inhibited by ATP and other nucleotides. The G-6-P isomerase is inhibited by 6-PG.

When there is a cellular demand for ATP, these regulatory activities direct glucose carbon through the energy-producing EMP pathway. Inhibition of G-6-P isomerase by 6-PG prevents the accumulation of F-6-P. Alteration of the fermentative pattern of these organisms by change in pH or other environmental influences may result from the disruption of these regulatory functions. In *S. mutans*, triose phosphates (glyceraldehyde-3-phosphate or dihydroxyacetone phosphate) strongly inhibit pyruvate-formate lyase (PFL), the first step in the dismutation sequence. Inhibition by triose phosphates in cooperation with a reactivating effect of ferredoxin may regulate pyruvate-formate lyase activity.

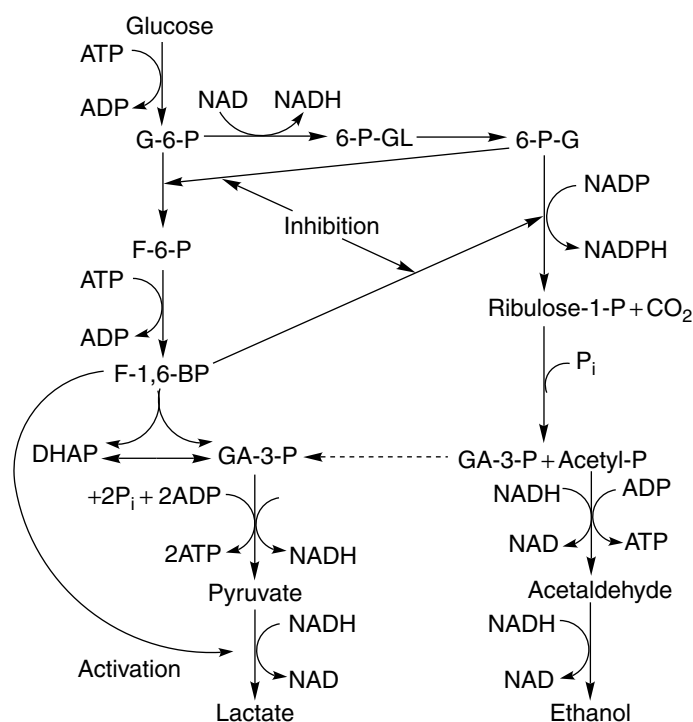


Fig. 11-3. Regulation of metabolic pathways in *Streptococcus faecalis*. 6-Phosphogluconate (6-P-G) inhibits the activity of hexose diphosphate isomerase. Fructose-1,6-bisphosphate (F-1,6-BP) inhibits the activity of 6-phosphogluconate dehydrogenase. Lactate dehydrogenase is activated by F-1,6-BP. (Source: Redrawn from Moat, A. G. 1985. Biology of lactic, acetic, and propionic acid bacteria. In *Biology of Industrial Microorganisms*, A. L. Demain and N. A. Solomon (Eds.). Benjamin-Cummings, Menlo Park, CA. pp. 143–186.)

The genus *Lactococcus* includes *L. lactis*, *L. cremoris*, and *L. diacetylactis*. These species, as well as various species of *Leuconostoc* and *Lactobacillus*, produce acetoin, diacetyl, and 2,3-butanediol. These compounds are responsible for the characteristic and desirable flavor in certain dairy products such as butter. In certain other fermentations, the presence of these compounds is undesirable, and it is of importance to be able to minimize their production. As shown in Figure 11-4, there are several routes for the production of acetoin and diacetyl. With acetate as substrate, acetyl-CoA is formed directly from acetate without the intermediary formation of pyruvate. *Lactococcus diacetylactis*, *Lactobacillus casei*, and other lactate-producing organisms that require lipoic acid for growth activate acetate to acetyl phosphate and then form acetyl-CoA via acetate kinase and phosphotransacetylase. Therefore, in media devoid of lipoic acid, these organisms cannot form acetyl-CoA from pyruvate.

In a lipoate-free medium containing glucose and acetate, acetoin is formed via both acetyl-CoA-C₂-TPP condensation and C₂-C₃ condensation routes. In *L. lactis* the gene for α -acetolactate decarboxylase (*aldB*) is clustered with the genes for the branched-chain amino acids. This enzyme appears to play a dual role in this organism by regulating valine and leucine biosynthesis as well as catalyzing the second step of the pathway to acetoin and 2,3-butanediol. Control of the shift from homolactic to mixed-acid fermentation seems to be governed by the NADH/NAD⁺ ratio related to the flux-controlling activity of glyceraldehyde-3-phosphate dehydrogenase.

During growth in milk, PFL and lactate dehydrogenase are important to sustain growth of *L. lactis* and to regenerate reduced cofactors under anaerobic conditions. The gene encoding PFL has been cloned and characterized. The enzyme is activated

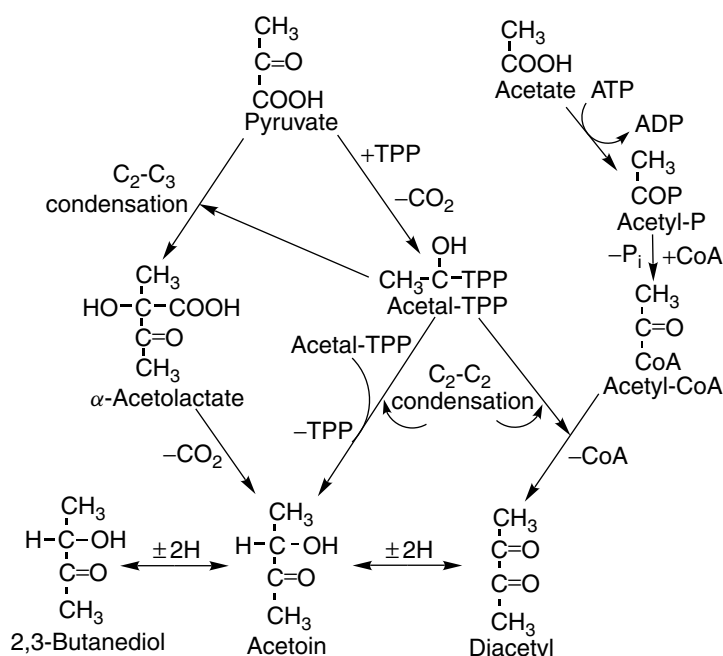


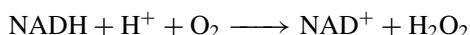
Fig. 11-4. Alternate routes to acetoin, diacetyl, and 2,3-butanediol. TPP, thiamine pyrophosphate.

via free-radical formation and is degraded in the presence of oxygen. Under anaerobic conditions, the carbon flux from pyruvate is distributed mainly between the two competing enzymes: lactate dehydrogenase (LDH) and pyruvate-formate lyase (PFL). Regulation of the shift from homofermentative to mixed-acid formation is associated primarily with the influence of allosteric effectors acting on the LDH and PFL enzymes.

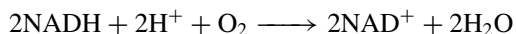
Organisms that metabolize sugars via fermentative pathways and do not use molecular oxygen as a final hydrogen acceptor or generate ATP via oxidative phosphorylation are anaerobes. However, if such organisms grow in the presence of oxygen, they may be referred to as aerotolerant. In the presence of oxygen, toxic products such as hydrogen peroxide may be generated by the autooxidation of reduced FAD, and hydroxyl radicals may be formed by the interaction of this compound with superoxide. Anaerobic organisms that tolerate the presence of oxygen are usually able to do so by virtue of the presence of enzymes that protect them from these toxic compounds (see further discussions in Chapter 18). Some fermentative organisms such as *L. plantarum* are aerotolerant simply because they do not interact with molecular oxygen. High intracellular concentrations of Mn^{2+} also aid these organisms in scavenging superoxide anion.

Streptococcus pyogenes and other lactic acid bacteria produce NADH oxidase, a unique flavoprotein that catalyzes the four-electron reduction of oxygen to water. This enzyme may actually allow certain fermentative organisms to use molecular oxygen to advantage in producing energy. Under anaerobic conditions the conversion of pyruvate to lactate or a mixture of formate, acetate, and ethanol serves to oxidize NADH back to NAD^+ . In the presence of oxygen, flavin-containing enzymes catalyze the oxidation of NADH by oxygen.

Although *S. mutans* depends on glycolysis and prefers an anaerobic environment, the presence of oxygen induces the formation of NADH oxidase and superoxide dismutase. Two types of NADH oxidase are induced in oxygen-tolerant strains of *S. mutans*. A peroxide-forming oxidase (Nox-1) catalyzes the reaction:



The second oxidase (Nox-2) catalyzes a four-electron reduction of oxygen by NADH:



The presence of free FAD enhances the activity of Nox-1 but not that of Nox-2. *S. mutans* also produces a peroxidase (AhpC) encoded by *ahpC*, a gene located upstream of the *nox-1* gene. This peroxidase enzyme complements Nox-1 by reducing the peroxide to water and emulating the action of Nox-2. However, Nox-2 appears to play an important role in aerobic energy metabolism in oxygen-tolerant *S. mutans*, whereas Nox-1 serves a negligible role despite the fact that it is the only enzyme produced when the organism is grown under anaerobic conditions. This organism also produces another antioxidant system, encoded by the *dpr* gene that is independent of the Nox-1-AhpC system and can compensate for deletions of both *nox-1* and *ahpC* in mutant strains. The product of the *dpr* gene, Dpr, is an iron-binding protein that presumably contributes to oxygen tolerance by maintaining a low level of iron.

Most lactic acid bacteria do not produce cytochromes because they are unable to synthesize heme. However, when grown in the presence of hematin, *Enterococcus*

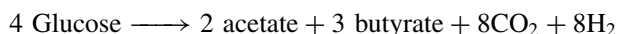
faecalis produces functional cytochromes that yield ATP via oxidative phosphorylation. An *E. faecalis* gene cluster, *cydABCD*, encodes a functional cytochrome *bd*–terminal oxidase that is produced when the cells are grown in the presence of heme. The ability to form cytochromes seems to be limited to *E. faecalis* and a few strains of *L. lactis*. Other lactic acid bacteria produce menaquinones, which are isoprenoid quinones that function in electron transport and possibly in oxidative phosphorylation.

BUTYRIC ACID – AND SOLVENT-PRODUCING FERMENTATIONS

Members of the *Clostridium*, *Butyrivibrio*, *Bacillus*, and other less-well defined flora of marsh sediments, wetwood of living trees, and anaerobic sewage digestion systems produce butyric acid, butanol, acetone, isopropanol, or 2,3-butanediol. Hydrogen, carbon dioxide, acetate, and ethanol are commonly found among the fermentation products.

Clostridium acetobutylicum utilizes the EMP pathway for glucose catabolism with the formation of ethanol, carbon dioxide, hydrogen, acetone, isopropanol, butyrate, and butanol from pyruvate via acetyl-CoA as shown in Figure 11-1. During growth, the organism first forms acetate and butyrate (acidogenic phase). As the pH drops and the culture enters the stationary phase, there is a metabolic shift to solvent production (solvetogenic phase). The pivotal reaction in these fermentations is the formation of acetyl-CoA from pyruvate. Acetyl-CoA can then undergo condensation to form acetoacetate, which may be reduced to butyrate and butanol or cleaved via decarboxylation to acetone. Acetone may be further reduced to isopropanol. Acetyl-CoA may also be reduced to acetaldehyde and ethanol. With other species such as *B. acetoethylicum* or *B. polymyxa*, lactate or 2,3-butanediol may be observed as shown in Table 11-3.

In balancing the fermentation products formed in these complex fermentations, it is not difficult to account for the carbon in the products formed. However, formulating a theoretical scheme to account for a balance of oxidized and reduced products may be more difficult. Consider the following equation:

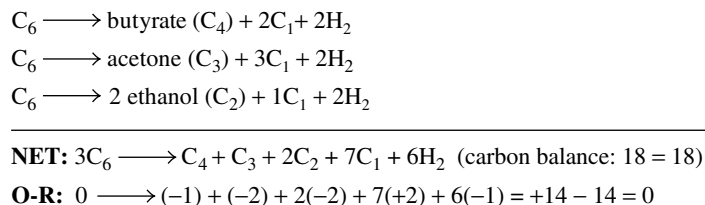


$$\text{Carbon balance : } (4 \times 6) = (2 \times 2) + (3 \times 4) + (8 \times 1) = 24$$

$$\text{O–R balance : } 0 = 0 + 3(-2) + 8(+2) + 8(-1) = +16 - 14$$

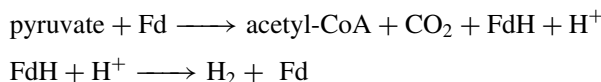
The carbon atoms balance, but there is a deficit of some reduced product. In most clostridial fermentations there is a compensatory formation of small amounts of several reduced products as shown in Table 11-3. Production of these compounds appears to be necessary because the condensation of 2 mol of acetyl-CoA to form acetoacetyl-CoA and the subsequent conversion to acetone via acetoacetate decarboxylase results in a sharp decrease in the number of hydrogen acceptors available. Acetyl-CoA can act as a hydrogen acceptor, giving rise to ethanol, and acetone can be further reduced to isopropanol. In fermentations that show the best O–R balance, ethanol, acetoin, and 2,3-butanediol are found among the final end products.

The overall equation for mixed butyrate, acetone, and ethanol fermentation can account for both the carbon and the O–R balance:



Generally, small amounts of either oxidized or reduced products will be formed to provide for a balance of the oxidized and reduced products. The oxidation values are given in Figure 11-1. Representative examples of acetone-butanol and mixed-solvent fermentations are shown in Table 11-3.

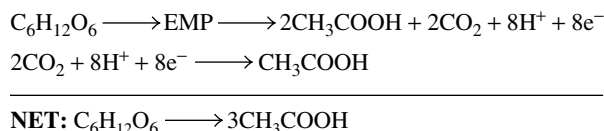
Formate is produced in relatively few of these fermentations. Although carbon dioxide and hydrogen gas are produced as in the fermentation of gram-negative organisms, in clostridia and other anaerobes hydrogen is formed through a pyruvate:ferredoxin (Fd) oxidoreductase without the intermediary production of formate. The reduced ferredoxin is converted to hydrogen by hydrogenase:



The flow of electrons from NADH to FdH to H₂ explains why these organisms produce large quantities of H₂ and do not consume H₂.

An enzyme that contains two Fe₄S₄ clusters catalyzes the oxidation of pyruvate to acetyl-CoA, carbon dioxide, and reduced ferredoxin. Ferredoxin also contains two Fe₄S₄ clusters arranged in approximately two-fold symmetry within the ferredoxin protein. Hydrogenases that result in the production of hydrogen gas contain iron-only enzymes. The *C. pasteurianum* enzyme contains five distinct iron-sulfur clusters covalently attached to the protein. One Fe₂S₂ cluster and three Fe₄S₄ clusters transfer electrons to the hydrogen center, the presumed catalytic site of the enzyme. The hydrogen center is composed of a six-iron cluster and a two-iron subcluster bound to carbon monoxide and/or cyanide. The arrangement of the iron (Fe) and sulfur (S) atoms in the four-iron cluster is shown in Figure 11-5.

Production of acetic acid from glucose by *C. thermoaceticum* is efficient in that 3 molecules of acetate are produced per molecule of glucose. This is achieved by a novel series of reactions that involves the synthesis of acetate from carbon dioxide:



It is particularly significant that the synthesis of acetate from carbon dioxide by this organism was the first demonstration of total synthesis of an organic compound from

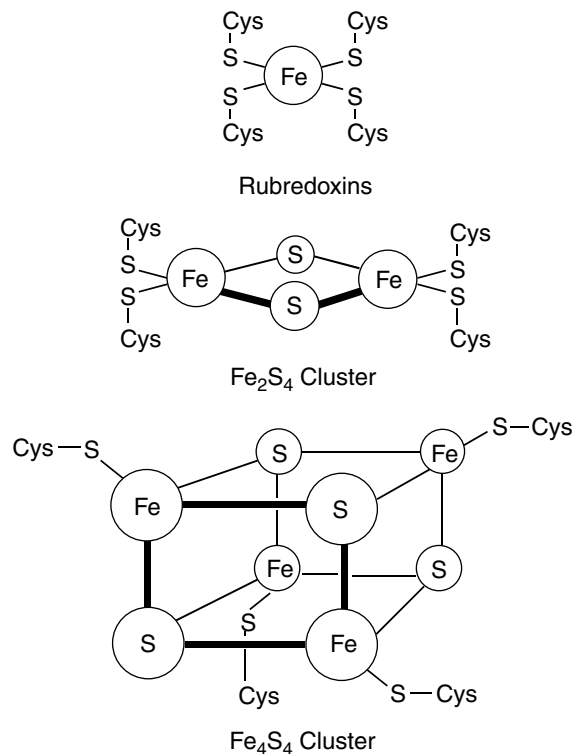


Fig. 11-5. proposed arrangements of the iron (Fe) and sulfur (S) atoms in rubredoxin, ferredoxin, and hydrogenase systems. Cys-S represents the linkage with the S in cysteine. In 8Fe/8S systems, the iron and sulfur atoms are arranged in two tetrameric Fe_4S_4 clusters.

carbon dioxide by a heterotrophic organism. This pioneering work by Harland G. Wood in the 1940s and 1950s conclusively demonstrated that all forms of life fix carbon dioxide. However, most heterotrophs can only add carbon dioxide to a preexisting compound and form a carboxyl group. Bacteria that can form an organic compound from carbon dioxide and hydrogen contain hydrogenase, an enzyme that converts hydrogen to two protons and two electrons. These electrons provide the necessary reductive capacity for the utilization of carbon dioxide. The overall reaction involves participation of ferredoxin as a reduced electron carrier and the enzymes hydrogenase, carbon monoxide dehydrogenase, and methylenetetrahydrofolate reductase.

FERMENTATIONS OF THE MIXED-ACID TYPE

Under anaerobic conditions and in the absence of alternate electron acceptors, members of the *Enterobacteriaceae* (*Escherichia*, *Enterobacter*, *Salmonella*, *Klebsiella*, and *Shigella*) ferment glucose to a mixture of acetic, formic, lactic, and succinic acids and ethanol (Table 11-4). The production of acetate and formate as major products is notable. As much as 85% of the glucose fermented by *E. coli* is metabolized via the EMP pathway. Other pathways must contribute to the products to some extent,

TABLE 11-4. Examples of Mixed-Acid Fermentations

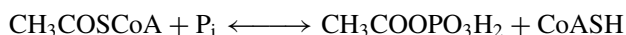
Products ^a	<i>Escherichia Coli</i>	<i>Enterobacter Aerogenes</i>	<i>Salmonella Typhi</i>
Lactate	108.8	53.4	121.7
Ethanol	41.3	59.4	25.4
Acetate	32.0	10.1	25.6
Formate	1.6	5.5	39.33
CO ₂	54.0	126.9	0
H ₂	45.2	44.2	0
Succinate	18.0	6.0	10.8
Acetoin	0	0.4	0
2,3-Butanediol	0	34.6	0
Carbon recovery (%)	100.0	99.5	93.3
O-R balance	0.99	0.99	0.97

^aAmounts are expressed as millimoles per 100 mmol glucose used.

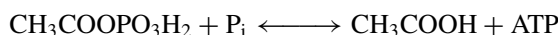
however. As opposed to clostridial fermentations in which acetyl-CoA, carbon dioxide, and dihydrogen usually arise without formate as an intermediate, formate is consistently found as a product of sugar metabolism by enteric bacteria. A CoA-dependent pyruvate-formate lyase (Pfl) encoded by the *pfl* gene initiates the sequence



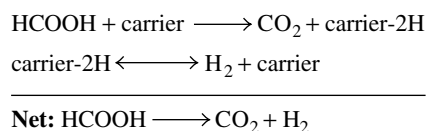
Phosphotransacetylase converts acetyl-CoA to acetyl phosphate:



Acetate kinase (encoded by the *ackA* gene in *E. coli*) generates ATP from acetyl phosphate:



This enzyme results in the production of acetate and generates a major portion of the ATP produced during anaerobic growth. The formate-hydrogen lyase (FHL) system converts formate to H₂ and CO₂. The FHL system actually consists of two enzymes: formate dehydrogenase-H (FDH-H), which yields carbon dioxide and a reduced carrier; and a hydrogenase-3, which produces H₂ from the reduced carrier:



Induction of FDH-H (encoded by *fdhF*) requires the presence of formate, molybdate, the absence of electron acceptors, such as oxygen or nitrate, and acidic pH. Expression of *fdhF* requires an alternate σ factor (NtrA) and an upstream activating sequence (UAS). Expression of *pfl* is dependent on phosphoglucosyltransferase and phosphofructokinase activities.

Lactic acid arises from pyruvate through the activity of LdhA induced under the conditions of anaerobiosis and a low pH. Production of ethanol occurs via the reduction of acetyl-CoA to ethanol through a CoA-linked acetaldehyde dehydrogenase coupled to an NAD-linked alcohol dehydrogenase.

Acetoin is formed by *Enterobacter aerogenes* but not by *E. coli*. A variety of other organisms (e.g., *Serratia*, *Erwinia*) also form acetoin or its oxidation (diacetyl) or reduction (2,3-butanediol) products, as shown in Figures 11-1 and 11-4. In gram-negative organisms, thiamine pyrophosphate is involved as a cofactor and α -acetolactate is produced as an intermediate in acetoin formation. Acetate induces the production of the acetolactate-forming enzyme, acetolactate decarboxylase, and diacetyl reductase. Diacetyl reductase also functions as an acetoin dehydrogenase in *E. aerogenes* and serves as a regulator of the balance between acetoin and 2,3-butanediol.

In *E. coli* the direct formation of diacetyl from 2-hydroxyethyl-TPP and acetyl-CoA has been demonstrated. Diacetyl is reduced to acetoin by an NADP⁺-specific diacetyl reductase that does not act on acetoin (Fig. 11-4). Acetoin is a major catabolic product of *B. subtilis* grown aerobically in glucose. Since acetoin is neutral, large amounts of glucose can be produced without excess acid production.

Two genes, *ilvBN* (coding for acetohydroxy acid synthase) and *alsS* (coding for α -acetolactate synthase), are involved in producing acetolactate from pyruvate in *B. subtilis*. Acetolactate can be converted to acetoin by spontaneous decarboxylation at low pH or via the action of acetolactate decarboxylase (encoded by *alsD*). Acetoin serves as a carbon storage compound. It is reimported from the growth medium and utilized during the stationary phase when other carbon sources have been depleted. There appears to be more than one pathway of acetoin utilization in *B. subtilis*. Three genes form an operon, *acuABC*, genes that encode acetoin utilization enzymes. Another system, the *acoABCL* operon, encoding the multicomponent acetoin dehydrogenase enzyme complex, appears to constitute the main route of acetoin catabolism in this organism. Expression of this operon is induced by acetoin and repressed by glucose. An *acoR* gene, located downstream from the *acoABCL* operon, encodes a positive regulator, which stimulates the transcription of the operon.

A survey of anaerobic fermentation balances with various substrates using a nonintrusive nuclear magnetic resonance (NMR) spectroscopy technique showed that substrates more reduced than glucose yield more reduced product as ethanol. More oxidized substrate resulted in the production of more neutral product in the form of acetate. The redox level of the substrate is important in governing the ratio of oxidized to reduced products. For example, sugar alcohols, which are more reduced than the corresponding hexoses, must yield a higher proportion of more reduced fermentation products to achieve hydrogen balance.

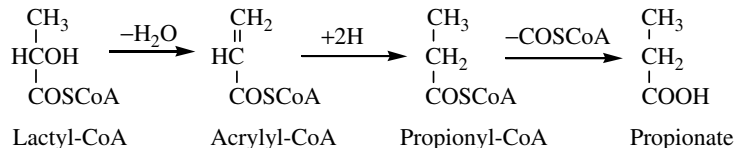
Fermentation of citrate (oxidation value: +3) yields sufficient oxidized products in the form of HCOOH (oxidation value: +1) and carbon dioxide (oxidation value: +2) to balance the oxidation/reduction equation. The fermentation of citrate by *Klebsiella pneumoniae* begins with cleavage by citrate lyase to form acetate and oxaloacetate. Oxaloacetate is decarboxylated by the oxaloacetate decarboxylase Na⁺ pump. Part of the energy from this reaction is used for citrate transport. The formed pyruvate is degraded by pyruvate-formate lyase to acetyl-CoA and formate. Phosphotransacetylase forms acetyl phosphate from acetyl-CoA and P_i. Transfer of high-energy phosphate from acetyl phosphate to ADP generates ATP, leaving acetate as a final end product. A

portion of the formate is acted upon by formate-hydrogen lyase, yielding CO_2 and H_2 . An NADP-reducing hydrogenase converts H_2 to $\text{NADPH} + \text{H}^+$, which is available for biosynthetic reactions.

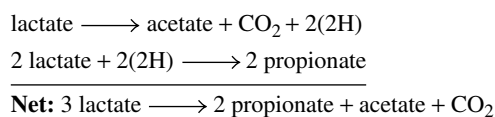
Utilization of glycerol (oxidation value: -1) involves transport into the cell via a facilitator encoded by *glpF*. Glycerol is phosphorylated by glycerol kinase (encoded by *glpK*) trapping it inside the cell as glycerol-3-phosphate (G-3-P). Under anaerobic conditions a dehydrogenase encoded by the *glpACB* operon yields dihydroxyacetone phosphate, which can then be metabolized via the triose phosphate pathway. In the presence of oxygen, an aerobic dehydrogenase encoded by the *glpD* operon accomplishes the same end. The *fnr* gene product FNR, a pleiotropic activator of genes involved in anaerobic respiration, regulates the utilization of glycerol, G-3-P, and glycerophosphate diesters. The Arc (aerobic respiratory control system) controls the expression of several genes encoding enzymes involved in aerobic respiration.

PROPIONIC ACID FERMENTATION

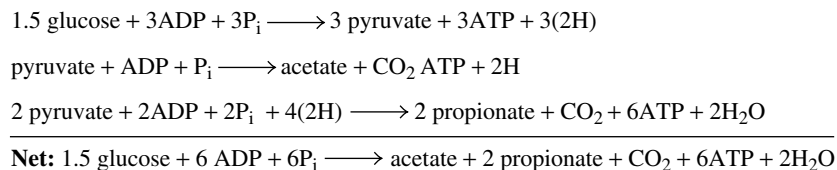
Propionate, acetate, and carbon dioxide are the major products of the fermentation of glucose, glycerol, and lactate by *Propionibacterium*, *Veillonella*, *Bacteroides*, and some species of clostridia. *Clostridium propionicum*, *Bacteroides ruminicola*, and *Peptostreptococcus* can remove water from lactate to form acrylate with subsequent reduction to propionate. Coenzyme A and lactyl CoA dehydrase mediate the reaction



The two hydrogen atoms required for the reduction of acrylyl-CoA to propionate to propionyl-CoA arise through the formation of acetate, CO_2 , and $2(2\text{H})$ from a portion of the total lactate utilized:



In *Propionibacterium* and *Veillonella*, propionate formation arises via a more complex series of reactions:



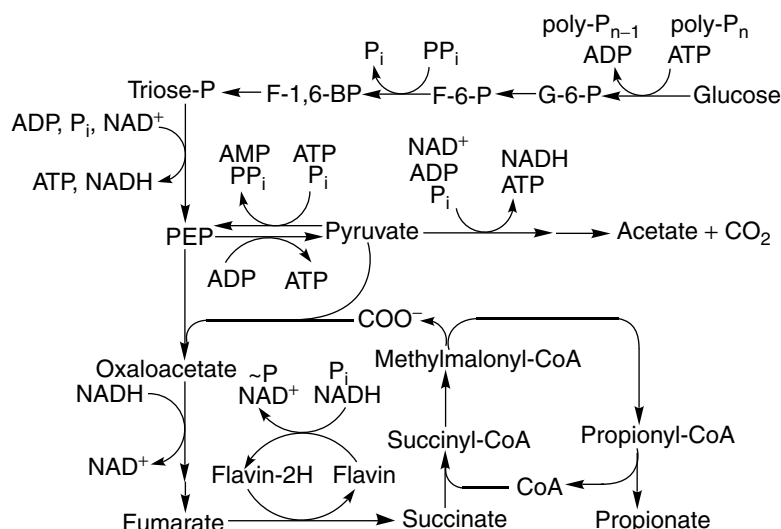
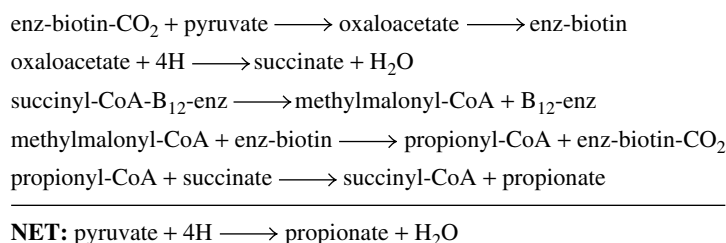


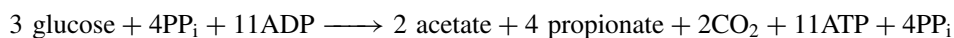
Fig. 11-6. formation of propionate, acetate, and CO₂ by *Propionibacterium*. (Source: From H. G. Wood, 1986, personal communication.)

The details of this pathway from glucose to propionate, acetate, and CO₂ in propionibacteria are shown in Figure 11-6.

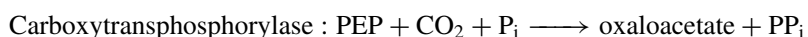
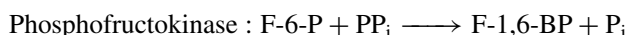
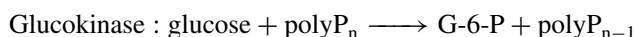
A major factor in the elucidation of this route of propionate formation was the demonstration of transcarboxylation by methylmalonyl-oxaloacetate transcarboxylase. This multimeric transcarboxylase catalyzes the reversible transfer of a carboxyl group from methylmalonyl CoA to pyruvate to form propionyl-CoA and oxaloacetate. The transferred carboxyl group is never released or exchanged with the CO₂; nor is it exchanged with the CO₂ dissolved in the medium. In this reaction, biotin plays an important catalytic role. Cobalamin (vitamin B₁₂) also serves as a cofactor in the formation of methylmalonyl-CoA from succinyl-CoA in the reaction sequence



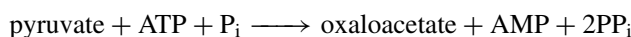
As shown in Figure 11-6, the hydrogen atoms required for the reduction of oxaloacetate to succinate are obtained from reactions in the conversion of glucose to acetate and carbon dioxide via pyruvate. Under appropriate conditions and in the presence of the requisite enzymes, yields of 11 mol of ATP per 3 mol of glucose can be achieved:



The participation of polyphosphate and PP_i also explains the high cell yields observed in the growth of propionibacteria. Demonstration of reactions in which the energy inherent in polyphosphate and PP_i may be utilized and not wasted through hydrolysis has far-reaching implications in other systems, since many biological reactions yield PP_i or polyphosphate as products. Some examples are as follows:



Net conversion of pyruvate to oxaloacetate:

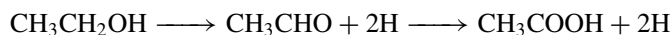


ACETIC ACID FERMENTATION

The acetic acid bacteria are divided into two genera: *Acetobacter* and *Gluconobacter*. Both species are obligate aerobes that oxidize sugars, sugar alcohols, and ethanol with the production of acetic acid as the major end product. Electrons from these oxidative reactions are transferred directly to the respiratory chain. The respiratory chain of *G. suboxydans* consists of cytochrome *c*, ubiquinone, and a terminal cytochrome-*o* ubiquinol oxidase. In *A. aceti* the composition of the respiratory chain varies according to the degree of aeration during growth.

Gluconobacter suboxydans lacks a functional TCA cycle, although all of the enzymes of this cycle except succinate dehydrogenase are present. This organism cannot ferment glucose or other carbohydrates since the EMP and Entner-Doudoroff pathways are not present. A modified pentose cycle is used for the metabolism of glucose under aerobic conditions (Fig. 11-7). *Acetobacter* species have a functional TCA cycle. The acetyl phosphate resulting from the C_2 - C_3 cleavage of pentose is oxidized to CO_2 and energy is generated via oxidative phosphorylation.

A characteristic activity of *Acetobacter* and *Gluconobacter* is the oxidation of ethanol to acetic acid. Ethanol oxidation occurs via two membrane-associated dehydrogenases: alcohol dehydrogenase and acetaldehyde dehydrogenase:



The electrons generated in ethanol oxidation are transferred directly to the respiratory chain.

Commercial production of acetic acid is carried out in two steps. Alcohol is first produced by yeast fermentation converting glucose to CO_2 and alcohol. The second stage uses *A. aceti* to convert the ethanol to acetic acid, with the reducing equivalents being transferred to oxygen via the respiratory chain as described above. The process requires carefully controlled conditions to achieve the maximum yield of acetate.

Acetate is a major product of carbohydrate metabolism in *B. subtilis*. Conversion of pyruvate to acetate occurs primarily during exponential growth. During the stationary phase, acetate is generated via butanediol. Overacidification of the medium

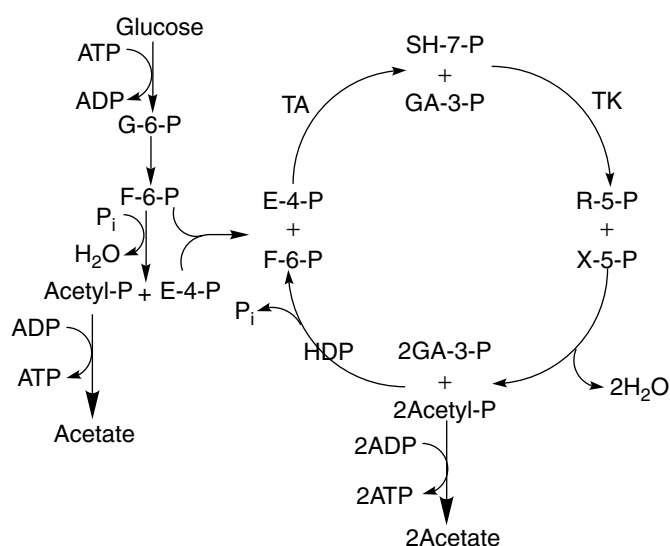


Fig. 11-7. Modified pentose cycle for *Acetobacter* and *Gluconobacter*. TA, transaldolase; TK, transketolase; E-4-P, erythrose-4-phosphate; SH-7-P, sedoheptulose-7-phosphate; R-5-P, ribose-5-phosphate; X-5-P, xylulose-5-phosphate; HDP, hexose diphosphate; GA-3-P, glyceraldehyde-3-phosphate. (Source: From A. G. Moat, 1985. *Biology of lactic, acetic, and propionic acid bacteria*. In *Biology of Industrial Microorganisms*. A. L. Demain and N. A. Solomon (Eds.). Benjamin Cummings, Menlo Park, CA, pp. 143–188.)

due to pyruvate and acetate accumulation during exponential growth is apparently alleviated by acetoin production under aerobic conditions or 2,3-butanediol under anaerobic conditions. These compounds can be reutilized in the stationary phase. The phosphotransacetylase (*pta*) gene, together with the acetate kinase (*ackA*) gene, encode the enzymes that catalyze the conversion of acetyl-CoA to acetate via acetyl~P. These two genes are organized into an operon and are involved in the maintenance of the intracellular acetyl-CoA and acetyl~P pools.

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CHAPTER 12

PHOTOSYNTHESIS AND INORGANIC METABOLISM

Organisms that use C_1 compounds (e.g., CO_2 or CH_4) as their major or sole source of carbon and energy are called **autotrophs**. **Methylo**trophs use methane (CH_4) or methanol (CH_3OH) as their sole source of carbon. Autotrophic organisms that use light as a source of energy are **photoautotrophs**. The source of utilized energy serves as a physiological distinction, as shown in Table 12-1.

CHARACTERISTICS AND METABOLISM OF AUTOTROPHS

Photosynthetic Bacteria and Cyanobacteria

Most living forms on earth are ultimately dependent on the process of photosynthesis. This process occurs in green plants, algae, cyanobacteria, and photosynthetic bacteria. A large community of marine microorganisms, generally referred to as phytoplankton, contains many species of cyanobacteria (representative examples: *Prochlorococcus*, *Synechococcus*, and *Anabaena*) that comprise the largest population of photosynthetic organisms on the planet. Many plants and microorganisms also conduct nitrogen fixation (see Chapter 14), providing a basis of continuity for all other life. Some reactions in the photosynthetic process are quite slow and inefficient. Therefore, one major aspect of the study of photosynthetic organisms is the improvement of the efficiency of the process through genetic engineering. Photosynthetic bacteria are found in the deeper waters of permanently stratified (**meromictic**) lakes where the conditions are anaerobic, but light is available.

Differentiation between the photosynthetic bacteria and the cyanobacteria (sometimes referred to in the past as blue-green algae) is based on the type of photosensitive pigments produced. Prokaryotes such as the cyanobacteria (*Anabaena*, *Synechococcus*, *Prochlorococcus*) that conduct true photosynthesis contain **chlorophyll a**, which is common to the eukaryotic algae and green plants. Water serves as the electron

TABLE 12-1. Principal Groups of Autotrophs

Energy Source	Group	Genera
H ₂	Hydrogen bacteria	<i>Ralstonia</i> (formerly <i>Alcaligenes</i>), <i>Nocardia</i> , <i>Xanthobacter</i> , <i>Pseudomonas</i> <i>derxia</i>
NH ₃	Nitrifying bacteria	<i>Nitrosolobus</i> , <i>Nitrosomonas</i> , <i>Nitrocystis</i>
NO ₂	Nitrifying bacteria	<i>Nitrobacter</i> , <i>Nitrospina</i> , <i>Nitrosococcus</i>
N ₂	Nitrogen-fixing bacteria	<i>Azotobacter</i> , <i>Anabaena</i> , <i>Prochlorococcus</i> , <i>Rhizobium</i>
H ₂ S, S, S ₂ O ₃ ²⁻	Sulfur bacteria	<i>Thiobacillus</i> , <i>Sulfolobus</i> , <i>Desulfotomaculum</i> , <i>Wolinella</i> , <i>Desulfovibrio</i> , <i>Beggiatoa</i>
Fe ²⁺	Iron bacteria	<i>Gallionella</i> , <i>Sphaerotilus</i> , <i>Thiobacillus</i> <i>ferrooxidans</i> , <i>Leptothrix</i> , <i>Shewanella</i> <i>oneidensis</i>
CH ₄ , CH ₃ OH	Methylotrophs	<i>Hyphomicrobium</i> , <i>Methylomonas</i> , <i>Methylobacterium</i> , <i>Methylosinus</i> , <i>Paracoccus</i> , <i>Pseudomonas</i>
H ₂ , CO ₂ , Formate Methylamine Trimethylamine Acetate	Methanogens	<i>Methanobacterium</i> , <i>Methanobrevibacter</i> , <i>Methanococcus</i> , <i>Methanomicrobium</i> , <i>Methogenium</i> , <i>Methanospirillum</i> , <i>Methanosarcina</i>
Light	Phototrophs	<i>Rhodobacter</i> , <i>Anabaena</i> , <i>Prochlorococcus</i> , <i>Synechococcus</i> , algae

donor and oxygen is generated by photolysis. The purple bacteria (*Thiorhodaceae*) contain **bacteriochlorophyll a** or **b**. The *Thiorhodaceae* utilize H₂S and/or inorganic compounds as electron donors, and their metabolism does not involve molecular oxygen (i.e., it is anaerobic). Green bacteria (*Chlorobacteriaceae*) contain **bacteriochlorophyll c** or **d** and small amounts of **bacteriochlorophyll a**, using H₂S and/or organic compounds as electron donors and following anaerobic metabolic pathways. The structure and biosynthesis of bacteriochlorophyll has been studied in detail and is discussed in Chapter 15.

Production of light-absorbing carotenoid pigments also represents a differentiating characteristic. All algae and green plants contain β -carotene. The purple sulfur and nonsulfur bacteria contain a variety of carotenoid pigments of both aliphatic and aryl types, whereas the green bacteria contain only aryl carotenoids (Fig. 12-1). The carotenoid pigments absorb light energy and transfer it to the chlorophyll molecules of the antenna.

Algae and the cells of higher plants contain chloroplasts. Comparable structures (**chromatophores**) are observed in the photosynthetic bacteria. The photosynthetic apparatus of *Rhodococcus sphaeroides* consists of a series of **intracytoplasmic membranes** (ICMs) that appear as vesicular invaginations originating from the cytoplasmic membrane. *R. sphaeroides* carries out anoxygenic photosynthesis but is also capable of both aerobic and anaerobic respiration as well as fermentation.

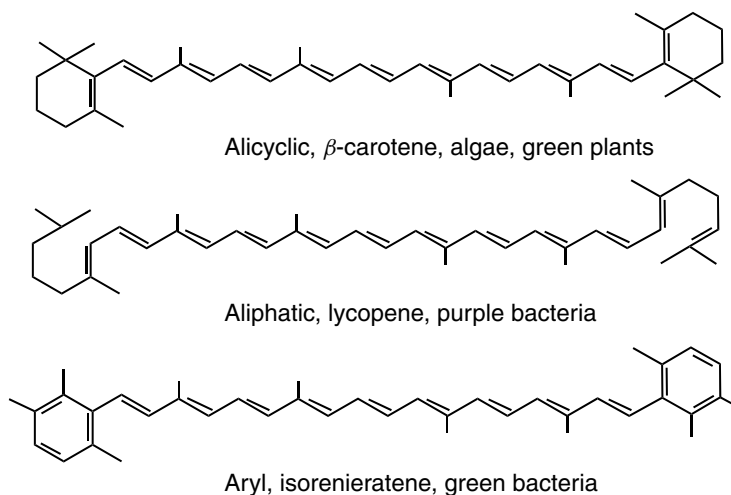
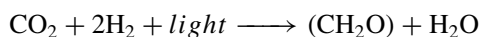
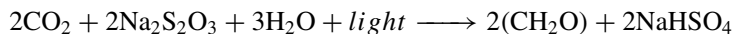
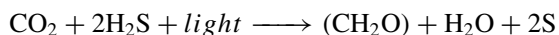
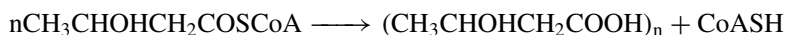
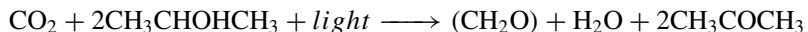


Fig. 12-1. Examples of carotenoid pigments produced by plants, algae, and photosynthetic bacteria. Although there are many variations, all carotenoids are of one of these three basic types.

The *Chlorobacteriaceae* (green bacteria) contain vesicles enclosed within a thin nonunit membrane that is not directly associated with the cell membrane. Metabolically, the green bacteria are strict anaerobic organisms that are obligately photosynthetic. They utilize H_2S , thiosulfate, or H_2 as an electron donor and CO_2 as the carbon source:



The purple bacteria contain two groups: the purple sulfur bacteria (*Thiorhodaceae*) that use H_2S as an electron donor and the purple nonsulfur bacteria (*Athiorhodaceae*) that depend on organic compounds such as short-chain fatty acids for photosynthetic metabolism. Poly- β -hydroxybutyrate is the end product:



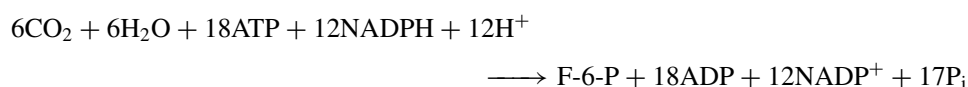
Poly- β -hydroxybutyrate serves as a major storage reserve material in these organisms. It is also an important reserve energy source in many other organisms. The cyanobacteria are considered to be very early evolutionary forms because of their lack of dependence on oxygen and on the basis of molecular evidence derived from 16S rRNA sequencing. Phylogenetic analysis of *c*-type cytochromes and rRNA sequences

has established a relationship between cyanobacteria and the chloroplasts of green algae and higher plants. These lines of evidence provide support for the concept of prokaryotic origins of chloroplasts along similar lines of development attributed to mitochondria.

Autotrophic CO₂ Fixation and Mechanisms of Photosynthesis

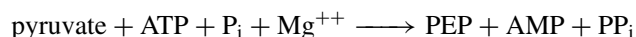
Photoautotrophs and chemoautotrophs, in which CO₂ serves as the sole or principal source of cellular carbohydrate, fix CO₂ via either the reductive pentose phosphate pathway (Calvin) cycle or the reductive C₄-dicarboxylic acid pathway. These systems were first discovered in green plants. Originally, all green plants were thought to assimilate atmospheric CO₂ via the reductive pentose pathway (Fig. 12-2) in which phosphoglyceric acid (PGA) is the first stable product (hence the designation C₃ plants). Subsequently, an alternative pathway of CO₂ fixation was discovered in which C₄ dicarboxylic acids (oxaloacetate and malate) were found as the primary products of photosynthesis. Within a taxonomic category, plants with C₃ photosynthesis are considered to be ancestral to those with C₄ primary photosynthetic products.

In photosynthetic and autotrophic bacteria, CO₂ fixation occurs primarily via the reductive pentose phosphate pathway (Fig. 12-2). In this system reduction of 1 mol of CO₂ to the oxidation level of carbohydrate involves the oxidation of 2 mol of NADPH and the hydrolysis of 3 mol of ATP. Only two of the reactions, phosphoribulokinase and ribulose biphosphate carboxylase (RuBisCO), are specific to photosynthetic or chemoautotrophic organisms. The other reactions are held in common with the carbohydrate metabolism of nonphotosynthetic organisms. The reductive pentose cycle constitutes the **dark reaction** of photosynthesis. Six turns of the cycle result in the synthesis of 1 mol of hexose (F-6-P):



The remainder is recycled through the reductive pathway as shown in Figure 12-2.

The reductive C₄-dicarboxylic acid pathway (Fig. 12-3) is present in a number of photosynthetic bacteria. In some organisms, such as the *Chlorobium*, it is the only cyclic pathway for CO₂ assimilation. Organisms that use the C₄ pathway possess the enzyme pyruvate-orthophosphate dikinase, which synthesizes phosphoenolpyruvate (PEP):



This enzyme differs from the PEP synthase of *E. coli* and other bacteria that can utilize C₄ acids in that it produces orthophosphate rather than monophosphate. *Chlorobium thiosulfatophilum*, a member of the green sulfur bacteria, requires P_i in addition to Mg⁺⁺ and ATP for the formation of PEP from pyruvate, supporting the fact that in photosynthetic bacteria, such as C₄ plants, pyruvate-orthophosphate dikinase rather than PEP synthase is used to form PEP in the photosynthetic assimilation of CO₂. The reductive carboxylic acid cycle is essentially a reverse of the TCA cycle in which pyruvate oxidase and α-ketoglutarate oxidase systems are replaced by ferredoxin-dependent pyruvate synthetase and α-ketoglutarate synthetase. This system is also of major importance in the metabolism of anaerobic bacteria.

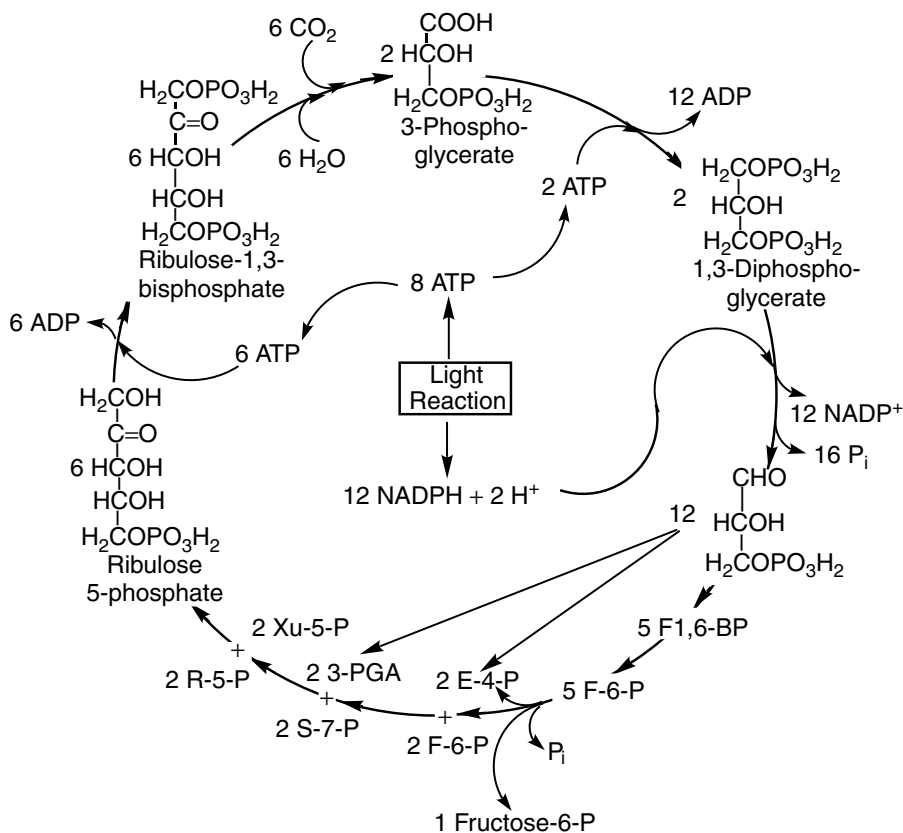
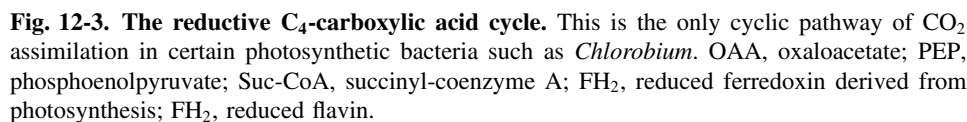


Fig. 12-2. The reductive pentose phosphate pathway. Since 3-phosphoglycerate is the first stable product of atmospheric CO₂ fixation, this pathway is sometimes referred to as the C₃ pathway. This cycle of reactions constitutes the dark reaction of photosynthesis because the energy required in the form of ATP has already been generated during photophosphorylation.

Photosynthesis, whether in green plants, algae, cyanobacteria, or photosynthetic bacteria, begins with the absorption of light by a pigment molecule and the delivery of the absorbed light energy to electron carriers that can transduce the energy into chemical form. The function of the light-harvesting pigments or **antennae** is to capture photons. Energy in the excited pigments is channeled into a complex called the **reaction center**. The reaction center functions as a battery that transfers electrons across the photosynthetic membrane and provides the energy for the fixation of carbon dioxide. The components of the bacterial photosynthetic reaction center have been studied in considerable detail. A somewhat simplistic diagram of the reaction center is shown in Figure 12-4.

Within the reaction center there are four bacteriochlorophyll molecules. Two of these are referred to as a **special pair** because they absorb light and transfer it to an electron. The two additional bacteriochlorophyll molecules appear to be inactive and are referred to as “voyeur chlorophyll.” Once the energy of the photon has been transferred to an



The terminal steps resulting in the phosphorylation of ADP represent an additional series of electron transfer reactions involving ferredoxin, NADP^+ , and cytochromes. The generation of ATP in photosynthesis is a process comparable to that utilized in coupling phosphorylation to electron transport during respiration (i.e., chemiosmotic coupling to an ATPase). In noncyclic photophosphorylation (Fig. 12-5) electrons are transferred from chlorophyll to ferredoxin, flavoprotein, and then to NADP^+ . An electron donor (water in plants and algae; H_2 , H_2S , or various organic compounds in photosynthetic bacteria) transfers electrons to cytochrome, producing the chemical energy needed to phosphorylate ADP. In cyclic photophosphorylation, ATP is generated from ADP and P_i with no other net chemical change (Fig. 12-5). Since cyclic

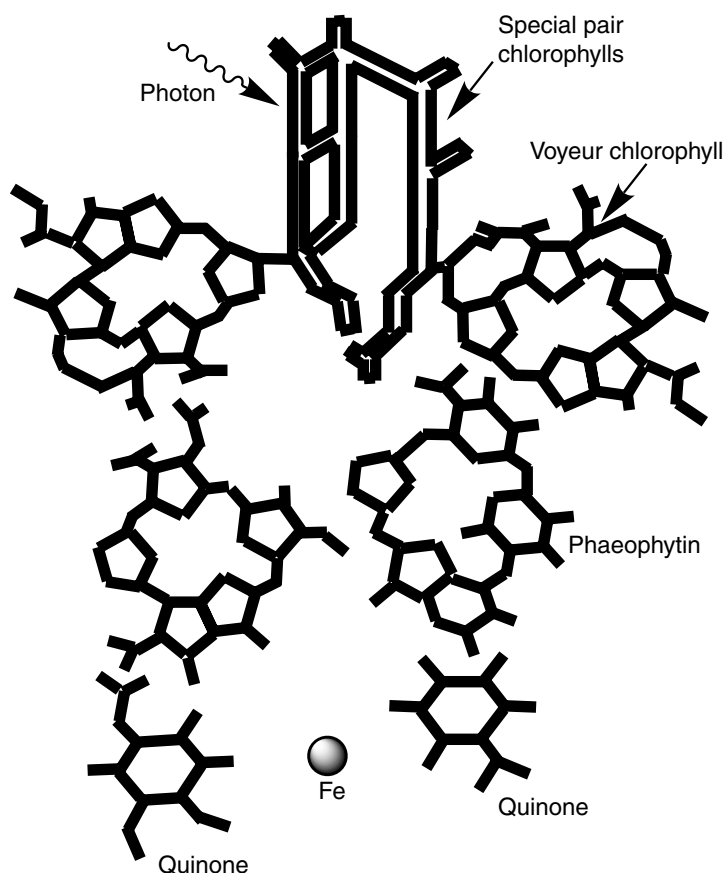


Fig. 12-4. Conceptual drawing of the bacterial photosynthetic reaction center.

photophosphorylation does not generate $\text{NADH} + \text{H}^+$, compounds such as H_2 , H_2S , or other available compounds provide reducing power.

In cyanobacteria and the eukaryotic red algae, phycobiliproteins are the most prominent light-harvesting polypeptides of the cell. These polypeptides are highly pigmented, water-soluble proteins that make up a major portion of the soluble cell protein. The major phycobiliproteins are phycoerythrin, phycocyanin, and allophycocyanin. Phycobilisome complexes appear as rows of closely spaced granules at the outer surface of the photosynthetic (thylakoid) membranes of red algae and cyanobacteria. The composition of the phycobilisome complex of the filamentous cyanobacterium *Fremyella diplosiphon* is altered by growth under red light as compared to green light. The differences in composition of the phycobilisome structure are the result of altered expression of the genes coding for phycobiliproteins.

Hydrogen Bacteria

Members of this group include the examples given in Table 12-1. These organisms utilize H_2 to provide energy and reducing power for growth and CO_2 fixation. Most of these bacteria are facultatively autotrophic and grow readily on organic substrates.

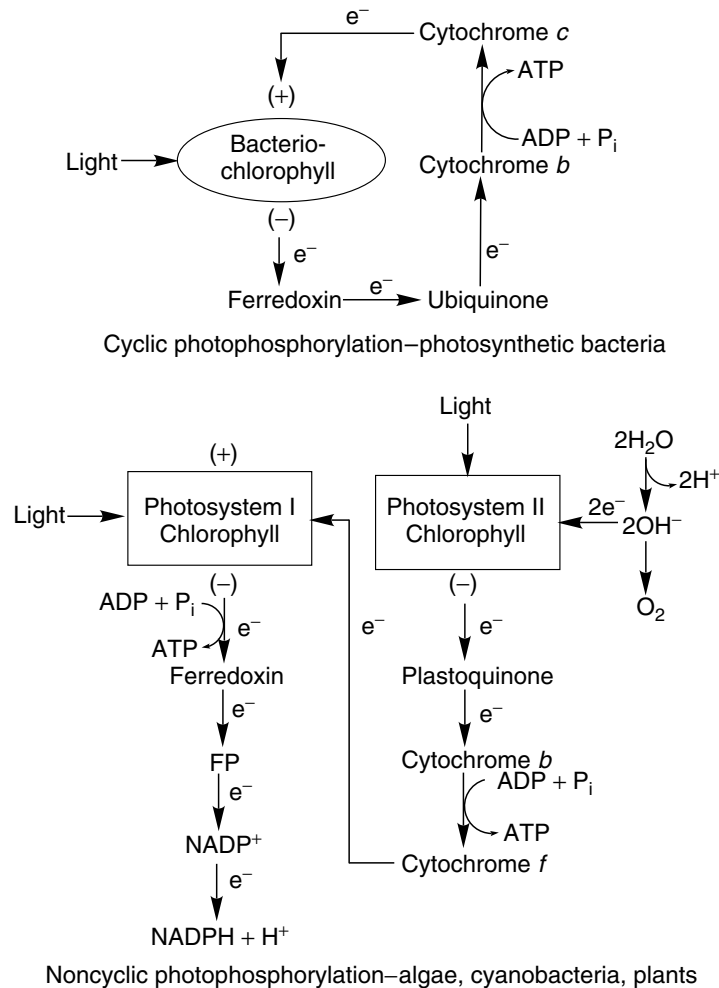
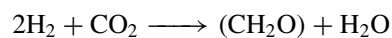


Fig. 12-5. Comparison of cyclic and noncyclic photophosphorylation. In cyclic photophosphorylation ATP is produced, but no reducing equivalents are generated. In the noncyclic pathway, two molecules of ATP are produced, reduced NADP is generated, and oxygen is produced by photolysis of water.

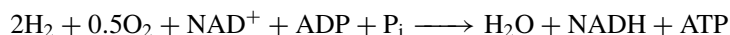
Many heterotrophic bacteria are capable of using H_2 to provide reducing power and energy for metabolic purposes but cannot support CO_2 fixation.

The facultative chemolithoautotrophic proteobacterium, *Ralstonia eutropha* (formerly *Alcaligenes eutrophus*), can utilize H_2 as a sole source of energy. Two energy-generating (NiFe) hydrogenases are present. One, a membrane-bound hydrogenase, is primarily involved in electron transport-coupled phosphorylation, whereas the other is a cytoplasmic enzyme that reduces NAD to provide reducing equivalents.

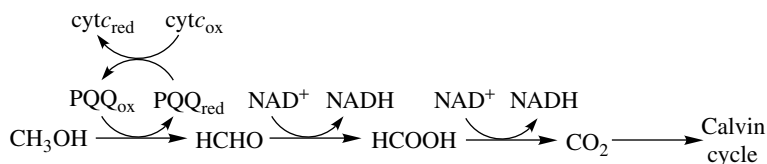
Reduction of CO_2 by H_2 can be shown as



Utilizable energy in the form of ADP is generated from the oxidation of H_2 by hydrogenase:



Carbon dioxide is assimilated autotrophically through the essential reactions of phosphoribulokinase and ribulose-1,5-bisphosphate carboxylase and the Calvin cycle (Fig. 12-3). In *Ralstonia eutropha* phosphoribulokinase is partially inactivated when an autotrophic culture is shifted to heterotrophic growth with pyruvate as the source of carbon and energy. Reactivation of phosphoribulokinase occurs after exhaustion of pyruvate from the medium. The hydrogen autotroph, *Xanthobacter*, can grow autotrophically with either hydrogen or methanol as an energy source. Hydrogen is oxidized by a membrane-bound hydrogenase. Methanol is oxidized to formaldehyde, formate, and then to CO_2 by the sequential action of methanol dehydrogenase, formaldehyde dehydrogenase, and formate dehydrogenase:

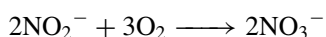


Nitrifying Bacteria

These organisms are important participants in the nitrogen cycle; hence their activities and relationship to nitrogen metabolism are discussed in greater detail in Chapter 14. *Nitrosomonas* is the most common organism found in the soil that oxidizes ammonia to nitrite:



Nitrobacter is the most common organism in soil that oxidizes nitrite to nitrate:

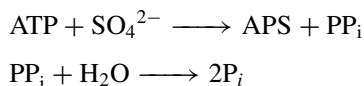


Both *Nitrosomonas* and *Nitrobacter* have been shown to possess a specialized mechanism for ATP production, and reduced NAD is required for the assimilation of carbon dioxide. A portion of the ATP derived from oxidative phosphorylation at the cytochrome level is used to reduce pyridine nucleotides.

Sulfur Bacteria

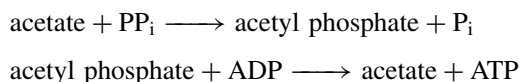
All members of this group, *Thiorhodaceae*, are capable of growth on elemental sulfur. Many can utilize thiosulfate ($S_2O_3^{2-}$) as well. The biochemistry of two of the sulfate-reducing bacteria, *Desulfovibrio* and *Desulfotomaculum*, are fundamentally different. In the case of *Desulfovibrio*, the PP_i produced during the formation of adenylyl sulfate

(APS) from ATP and sulfate in the first step of sulfate reduction is hydrolyzed to P_i by inorganic pyrophosphatase:

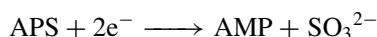


By this process, the energy in PP_i is dissipated by hydrolysis, and to obtain a net yield of ATP during growth on lactate plus sulfate, *Desulfovibrio* must carry out electron transfer-coupled phosphorylation.

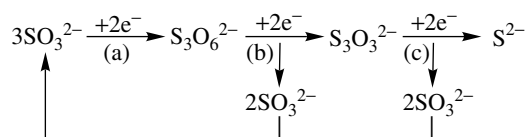
By comparison, *Desulfotomaculum* conserves the bond energy in PP_i by means of the enzyme acetate: PP_i phosphotransferase and the subsequent formation of ATP by acetate kinase:



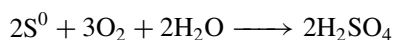
These reactions allow *Desulfotomaculum* to use PP_i as a source of energy for growth with acetate and sulfate. The conversion of APS (adenosine phosphosulfate) to sulfite by APS reductase requires the addition of two electrons:



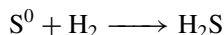
The further reduction of sulfite to sulfide requires the action of sulfite reductase (a), trithionate reductase (b), and thiosulfate reductase (c), and the recycling of sulfite:



The sulfur-dependent archaea found in the vicinity of hot springs are able to grow chemoautotrophically using CO_2 as the sole carbon source and the oxidation of elemental sulfur with oxygen yielding sulfuric acid:

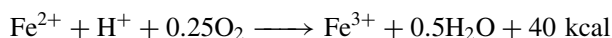


However, *Sulfolobus ambivalens* is able to live by an anaerobic mode of chemoautotrophy using CO_2 as the sole carbon source but using H_2 for the reduction of sulfur to H_2S :



Iron Bacteria

Thiobacillus ferrooxidans, *Gallionella*, *Leptothrix*, *Sulfolobus*, *Sphaerotilus*, and *Shewanella oneidensis* are capable of oxidizing ferrous iron to ferric iron as a means of generating biologically useful energy:



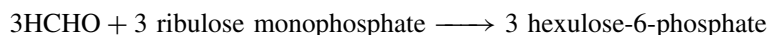
Thiobacillus ferrooxidans is an obligate autotroph. While it can be grown heterotrophically in the absence of an oxidizable iron source, continued cultivation on an organic substrate renders it incapable of growth with ferrous iron as the sole energy source. *T. ferrooxidans* differs from other autotrophic organisms in that it cannot revert to an autotrophic mode of life after prolonged cultivation on organic substrates. The transition of *T. ferrooxidans* to obligate organotrophy is governed by a number of factors including the pH of the medium, the incubation temperature, the availability of oxygen, the age of the cells at the time of transition, and the type of energy and carbon source available. Conversion to organotrophy results in a gradual loss of the ability to oxidize Fe^{2+} and cessation of CO_2 fixation. *Gallionella*, *Sphaerotilus*, and other iron-oxidizing organisms appear to be facultative and can be readily grown as heterotrophs and then returned to growth on iron.

Shewanella oneidensis, a metal-reducing bacterium found in soils, can use ferric iron as a terminal electron acceptor. This organism is able to reductively dissolve Fe^{3+} -containing minerals such as goethite ($\alpha\text{-FeOOH}$) or hematite ($\alpha\text{-Fe}_2\text{O}_3$). Under anaerobic conditions *S. oneidensis* can apparently generate two energized membranes using a system of proteins that shuttle electrons from an energy source in the cytoplasm, across the plasma membrane and periplasmic space, to the outer membrane. Once in the outer membrane, iron reductases appear to transfer electrons directly to Fe^{3+} in the crystal structure of minerals, causing a weakening of the iron-oxygen bond and reductive dissolution of the mineral. Using atomic force microscopy it has been possible to show that the affinity between *S. oneidensis* and goethite increases by two to five times under anaerobic conditions. An iron reductase within the outer membrane is apparently mobilized and specifically interacts with the goethite surface to facilitate the electron transfer process.

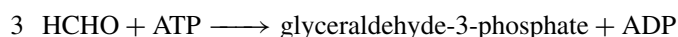
METHYLOTROPHS

Methylotrophic bacteria are able to utilize methane, methanol, methylamine, or formate as the sole source of carbon and energy. The term **methanotroph** designates methylotrophs that can use methane for carbon and energy. There are also several species of yeasts and molds that can use methane or methanol. Most methylotrophs are obligate in that they can only use C_1 compounds. The general pathway of oxidative reactions is shown in Figure 12-6.

Two types of methylotrophic bacteria have been identified on the basis of the mode of assimilation of formaldehyde. **Type I methylotrophs** use the ribulose monophosphate pathway for formaldehyde assimilation:



The hexulose-6-phosphate is metabolized via the central pathway to form glyceraldehyde-3-phosphate. The overall reaction is



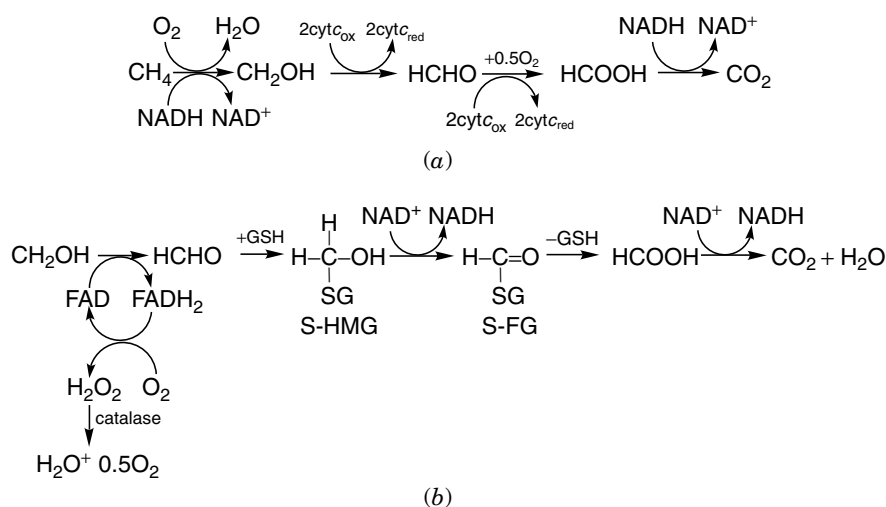
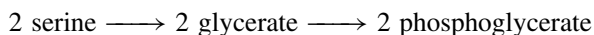
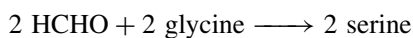
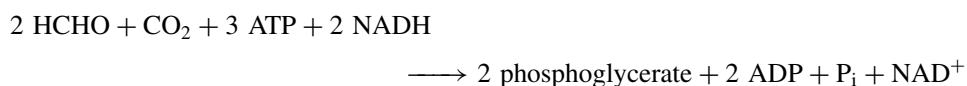


Fig. 12-6. (a) General pathway of oxidative reactions in methylotrophs. (b) Pathway for the conversion of methanol to CO_2 and H_2 in a methylotrophic yeast. GSH, glutathione; S-HMG, S-hydroxymethyl glutathione; S-FG, S-formylglutathione.

Type II methylotrophs use the serine pathway for formaldehyde assimilation:



The overall reaction is



Bacterial methylotrophs include *Paracoccus denitrificans* and several species of *Pseudomonas*, *Hyphomicrobium*, and *Xanthobacter*. However, most of the obligate methylotrophs belong to the genera *Methylophilus*, *Methylobacterium*, *Methylococcus*, *Methylosinus*, or *Methylomonas*.

Methylotrophic yeast includes *Hansenula*, *Candida*, *Torulopsis*, and *Pichia*. The metabolic pathway for the conversion of methanol to CO_2 and H_2O appears to be similar for several of these yeasts. The pathway of *Pichia pastoris* involves alcohol oxidase, catalase, formaldehyde dehydrogenase, S-formylglutathione, and formate dehydrogenase in the sequence of reactions shown in Figure 12-6. In yeasts, the alcohol dehydrogenase and catalase reactions take place in peroxisomes, membranous organelles containing flavin-linked oxidases that regenerate oxidized flavin by reaction with O_2 . Synthesis of some of these enzymes is tightly regulated and several of the genes involved in methanol utilization appear to be controlled at one level by a

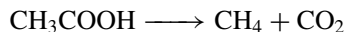
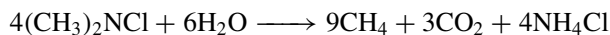
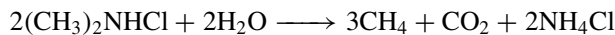
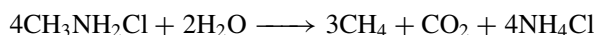
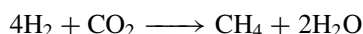
glucose catabolite repression–depression mechanism. The structural genes for alcohol dehydrogenase and two other enzymes in the sequence are regulated by methanol at the level of transcription.

Methane is produced in anaerobic environments such as natural wetlands but is also a major agricultural and industrial by-product. As the most abundant organic gas in the atmosphere it absorbs terrestrial radiation (infrared radiation) more effectively than does CO₂. As a result, methane contributes more heavily to global warming. Methanotrophic bacteria are distributed widely and play a significant role in moderating global warming by oxidizing most of the methane before it reaches the atmosphere. Methanol-oxidizing organisms are useful for the production of single-cell protein; microbial cells are used as animal feed supplements. Growing such organisms on materials that would otherwise be disposed of as waste provides an important means of recycling these materials into useful products. Some of the methylotrophs display a wide range of biotransformations of potential commercial importance.

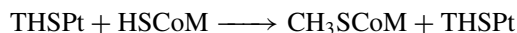
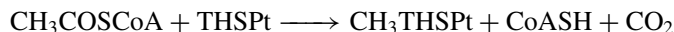
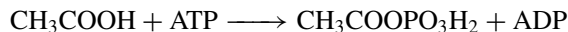
Methanogens

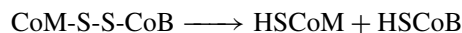
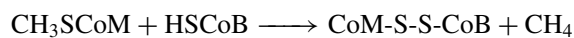
Methanogenic organisms gain energy by using H₂ to reduce CO₂ to CH₄. These organisms can also decarboxylate acetate to CO₂ and CH₄. Methane formation represents the terminal portion of a complex series of anaerobic reactions that occur in nature and involve a number of organisms that degrade biopolymers such as cellulose, starch, or proteins to acetate, H₂, and CO₂. Conversion of complex organic material to these simple products requires the action of both primary and secondary fermenters from the clostridia and other anaerobic organisms. Primary fermenters can yield acetate, H₂, and CO₂. Other products require additional degradation by the secondary fermenters. Methanogenic microorganisms conduct the last portion of the conversion sequence to yield CH₄ as a final product.

Methanogens belong to the archaea (archaeobacteria). The major genera are listed in Table 12-1. Most methanogens can produce CH₄ from H₂ and CO₂ as shown in the first equation below. Only the Methanosarcinales (e.g., *Methanosarcina barkeri*) can reduce other substrates to CH₄ according to the following equations:



Conversion of acetate to CH₄ and CO₂ involves the following intermediary steps:





Reduction of CO_2 to CH_4 follows the pathway shown in Figure 12-7 and involves the function of several unique coenzymes: methanofuran (MFR); tetrahydromethanopterin (H_4MPT); deazaflavin F_{420} as an electron donor; coenzyme M ($\text{HS-CH}_2\text{CH}_2\text{SO}_3^-$, or HSCoM); and coenzyme B (HSCoB , 7-mercaptoheptanoylthreonine phosphate). Several of these coenzymes were once thought to be present only in methanogenic archaea. However, it has now been shown that the C_1 transfer enzymes and their cofactors, CoM and CoB, function in methylotrophic bacteria as well. The chemical structures of these cofactors are shown in Figure 12-8. All methanogens use the major energy-yielding step associated with the reduction of a methyl group to CH_4 , although different species may obtain electrons for the reductive step from the oxidation of a variety of substrates.

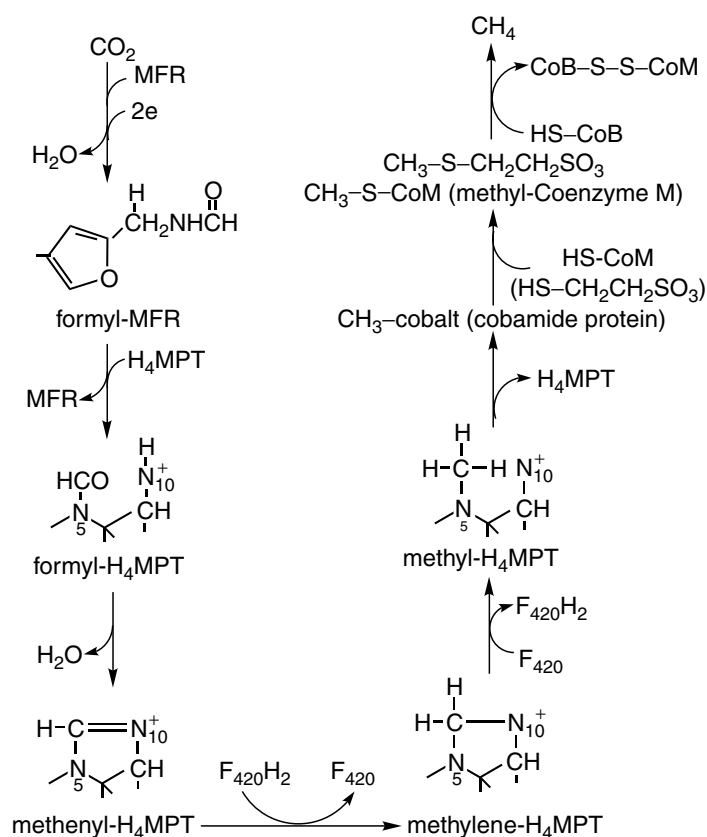


Fig. 12-7. Pathway for the reduction of CO_2 to CH_4 by methanogens. MFR, methanofuran; H_4MPT , tetrahydromethanopterin; F_{420}H_2 , deazaflavin F_{420} , electron donor; HS-CoB , coenzyme B (7-mercaptoheptanoylthreonine phosphate), electron donor. Complete structures of the unusual cofactors that participate in methanogenesis are shown in Figure 12-8. (Source: From R. S. Wolfe, *ASM News* 62:529–534, 1996.)

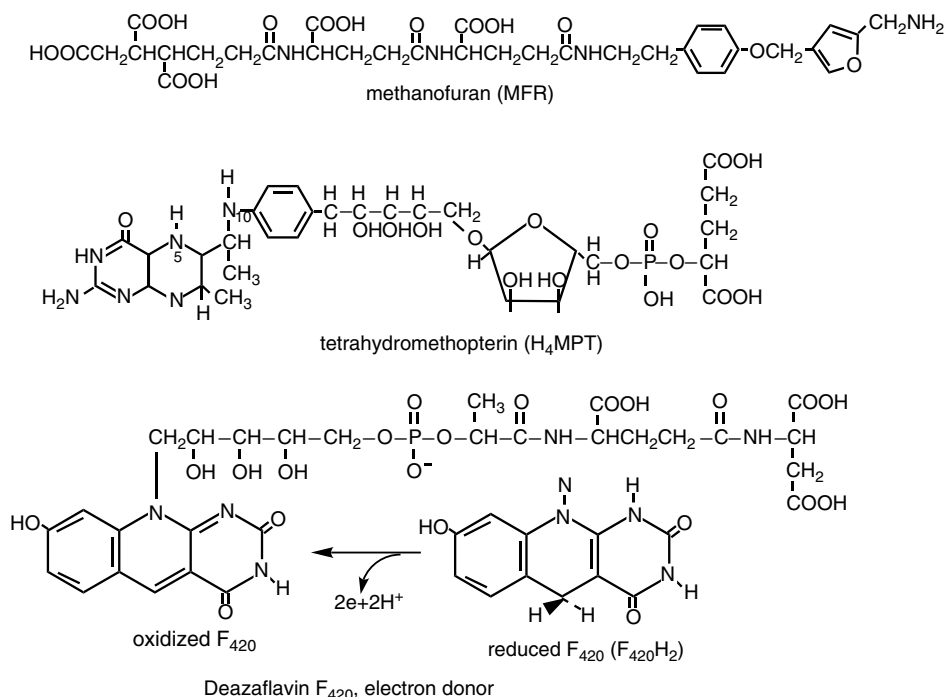


Fig. 12-8. Structures of the unusual coenzymes that participate in methanogenesis. The pathway of reduction of CO₂ to CH₄ is shown in Fig. 12-7. (Source: From R. S. Wolfe, *ASM News* **62**:529–534, 1996.)

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CHAPTER 13

LIPIDS AND STEROLS

The major lipid-containing components of microbial cells are membranous structures such as the cytoplasmic membrane, the outer membrane of gram-negative bacteria, the intracytoplasmic membranes of photosynthetic bacteria, and the nuclear and mitochondrial membranes of eukaryotic cells. Other lipids play important roles as electron carriers, enzyme cofactors, transport agents in cell wall synthesis, and light-absorbing pigments.

A common feature of all lipids is their insolubility in water. Membrane lipids are **amphipathic**—that is, they have **hydrophilic** (water-soluble) and **hydrophobic** (water-insoluble) regions that cause them to orient into bilayers. This characteristic provides a solubility barrier to polar solutes in an aqueous environment. Phospholipids comprise the bimolecular leaflet structure of the cytoplasmic membrane, which is both flexible and self-sealing. Monounsaturated and ring-containing fatty acids contribute to the flexibility and fluidity that enables membranes to undergo changes in shape that accompany cell growth or movement.

In the fungi, sterols are associated with the cytoplasmic membrane and are involved with permeability functions as indicated by the leakage of metabolites following the binding of polyene antibiotics, such as nystatin or amphotericin B, to the sterol groups. Certain lipids, for example, glycosyldiglycerides and lipoteichoic acids, are found in the cell surface of gram-positive but not gram-negative bacteria. Conversely, gram-negative bacteria display an outer membrane whose outer face is composed largely of lipopolysaccharides (LPS) while gram-positive bacteria are devoid of these structures. Glycolipids are present in non-sterol-requiring *Mycoplasma* but not in the sterol-requiring members of this group, suggesting replacement of sterols by glycolipids in the membrane structures of the first group. In fungi there is an association of lipids and sterols with the development of respiratory activity.

Although there is still a great deal to be determined with regard to the precise role of lipids in various cellular activities, our knowledge of the biosynthesis and metabolism of lipids and the regulation of these activities is well developed.

LIPID COMPOSITION OF MICROORGANISMS

Representative examples of the major types of naturally occurring fatty acids are shown in Table 13-1. Of the examples shown, not all are found in microbial cells. In contrast to eukaryotic organisms, bacterial cells contain few fatty acids with greater than 19 carbon atoms, the shorter chain-length acids being primarily of importance as metabolic intermediates in fatty acid biosynthesis. The predominant saturated fatty acid in microorganisms is palmitic acid (C₁₈). Lesser quantities of stearic (C₁₈), myristic (C₁₄), and lauric (C₁₂) acids are observed. The principal unsaturated fatty acids are monoenoic acids (one unsaturated position). Di-, tri-, or polyenoic acids are not found in bacteria. Many unusual fatty acids are present in smaller quantities. Branched, hydroxylated, or methylated fatty acids and cyclopropane ring-containing fatty acids are present in many microorganisms. Corynic acid (a branched, dihydroxy fatty acid with 52 carbon atoms) and mycolic acid (a branched, dihydroxy fatty acid with 87 to 88 carbon atoms) are examples of fatty acids unique to the *Corynebacterium* and *Mycobacterium*, suggesting a close taxonomic relationship between these two genera.

Yeasts and molds have a fatty acid composition more closely related to that of plants and their seeds. The common brewer's and baker's yeasts contain about 5 to 8% lipids (dry weight), with palmitoleic acid being a major constituent. However, when baker's yeast is grown under anaerobic conditions, palmitic acid synthesis increases at the expense of palmitoleic acid. Many yeasts and molds accumulate lipids, usually as triglycerides containing a high percentage of palmitic acid, particularly near the end of the growth cycle.

Archaea produce a number of unusual lipid structures that contain ether linkages. These organisms are divided into three major groups: halophiles, methanogens, and thermophiles. Thermophiles thrive in extreme environments that are uninhabitable by any other living forms because their unusual structural and biochemical factors aid in their survival and functioning under these harsh conditions.

Sterols are commonly present in fungi but are rarely observed in bacteria. Exceptions are found: *Mycoplasma* incorporates sterols from the growth medium into cell membranes; *Streptococcus pneumoniae* incorporates cholesterol into cytoplasmic membranes; and methanotropic bacteria synthesize sterols in significant quantities.

By means of high-resolution gas chromatography, an accurate system of species identification has been developed on the basis of fatty acid composition.

Straight-Chain Fatty Acids

Straight-chain fatty acids are the major constituents of membrane phospholipids. Palmitic, stearic, hexadecanoic, octadecenoic, cyclopropanic, 10-methylhexadecanoic, 2- or 3-hydroxyl fatty acids, and a few others are commonly observed in bacterial cells. Saturated and unsaturated straight-chain fatty acids with less than 10 or 12 carbon atoms are usually intermediates of degradative pathways or biosynthesis of longer-chain fatty acids.

The conditions under which microbial cells are grown can markedly influence their fatty acid constituents. Composition of the growth medium, availability of oxygen, temperature, pH, and the age of the culture can each affect the distribution of fatty acids. Young cultures of *E. coli* contain large quantities of C₁₆- and C₁₈-monoenoic acids and only small amounts of C₁₇- and C₁₉- cyclopropane

TABLE 13-1. Representative Examples of Naturally Occurring Fatty Acids

Number of C atoms	Generic Name	Common Name	Composition
<i>Saturated Fatty Acids</i> (—C—C—C—)			
1	Methanoic	Formic	HCOOH
2	Ethanoic	Acetic	CH ₃ COOH
3	Propanoic	Propionic	C ₂ H ₅ COOH
4	Butanoic	Butyric	C ₃ H ₇ COOH
6	Hexanoic	Caproic	C ₅ H ₁₁ COOH
16	Hexadecanoic	Palmitic	C ₁₅ H ₃₁ COOH
18	Octadecanoic	Stearic	C ₁₇ H ₃₅ COOH
<i>Unsaturated Fatty Acids (Monoenoic)</i> (—C—C=C—C—)			
4	<i>trans</i> -2-Butenoic	Crotonic	C ₄ H ₆ O ₂
16	<i>cis</i> -9-Hexadecenoic	Palmitoleic	C ₁₆ H ₃₀ O ₂
18	<i>cis</i> -9-Octadecenoic	Oleic	C ₁₈ H ₃₄ O ₂
18	<i>trans</i> -11-Octadecenoic	Vaccenic	C ₁₈ H ₃₄ O ₂
<i>Unsaturated Fatty Acids (Dienoic)</i> (—C—C=C—C=C—)			
6	2,4-Hexadienoic	Sorbic	C ₆ H ₈ O ₂
18	<i>cis</i> -9, <i>cis</i> -12-Octadecadienoic	α -Linoleic	C ₁₈ H ₃₂
<i>Unsaturated Fatty Acids (Trienoic)</i> (—C=C—C=C—C=C—)			
18	<i>cis</i> -6, <i>cis</i> -9, <i>cis</i> -12-Octadecatrienoic	γ -Linoleic	C ₁₈ H ₃₀ O ₂
18	<i>cis</i> -9, <i>cis</i> -12, <i>cis</i> -15-Octadecatrienoic	α -Linoleic	C ₁₈ H ₃₀ O ₂
<i>Unsaturated Fatty Acids (Tetraenoic)</i> (—C=C—C=C—C=C—C=C—)			
20	5,8,11,14-Eicosatetraenoic	Arachidonic	C ₂₀ H ₃₂ O ₂
<i>Hydroxyalkanoic Acids</i> (—CH ₂ —CHOH—COOH)			
12	2-Hydroxydodecanoic	2-Hydroxylauric	C ₁₂ H ₂₄ O ₂
18	2-Hydroxyoctadecanoic	2-Hydroxystearic	C ₁₈ H ₃₆ O ₂
<i>Keto, Epoxy, and Cyclo Fatty Acids</i>			
5	4-ketopentanoic (—C—CO—C—)	Levulinic	C ₅ H ₈ O ₂
19	ω -(2- <i>n</i> -Octylcycloprop-1-enyl)-octanoic (9,10-methyleneoctadec-9-enoic)	Sterculic	C ₁₉ H ₃₄ O ₂
	$\text{CH}_3-(\text{CH}_2)_7-\underset{\text{CH}_2}{\text{C}=\text{C}}-(\text{CH}_2)_7-\text{COOH}$		

TABLE 13-1. (continued)

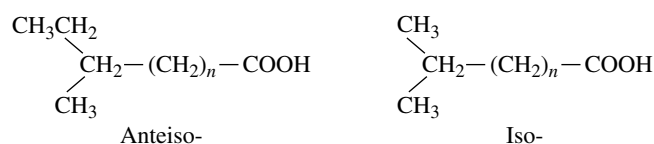
<i>Hydroxy Unsaturated Fatty Acids</i>			
18	D-12-Hydroxy- <i>cis</i> -9-octadecenoic	Ricinoleic	C ₁₉ H ₃₄ O ₂
<i>Branched-Chain Fatty Acids</i>			
5	3-Methylbutanoic	Isovaleric	C ₅ H ₁₀ O ₂
16	14-Methylpentadecanoic	<i>iso</i> -Palmitic	C ₁₈ H ₃₂ O ₂
19	1-D-10-Methyloctadecanoic	Tuberculostearic	C ₁₉ H ₃₈ O ₂
20	18-Methylnonadecanoic	<i>iso</i> -Arachidic	C ₂₀ H ₄₀ O ₂

fatty acids. In the late exponential phase of growth, the cyclopropane fatty acids are greatly increased at the expense of the unsaturated fatty acids. In *S. aureus* about 5% of the total fatty acids are unsaturated when the organism is grown on a minimal defined medium. When human serum is added to the medium, the proportion of unsaturated fatty acid increases to 27%. Under aeration *S. aureus* displays a decrease in the percentage of saturated fatty acids. In anaerobically grown staphylococci, C₁₈ and C₂₀ acids comprise 59% of the total fatty acids. Unsaturated fatty acids are not present in *B. acidocaldarius*, an organism characterized by its tolerance of high temperature (50–70 °C) and acidity (pH 2–5). This organism notably produces a high proportion of unusual lipids. The ability of the organism to grow under these extreme conditions is attributed, in part, to the unusual fatty acid composition.

Fungi contain the same major classes of lipids and fatty acids as bacteria, but the relative amounts of each class may vary. In *Phycomyces*, yeasts, *Euascomycetes*, and *Basidiomycetes* the C₁₆ and C₁₈ fatty acids predominate. The fatty acids of *S. cerevisiae* are also in the C₁₆ and C₁₈ class, with only 1 to 2% of the total fatty acids having chain lengths greater than 18 carbon atoms. These range from C₁₉ to C₃₄, with C₂₆-unsaturated acids constituting 18% of the fatty acids with greater than 18 carbon atoms at the midexponential phase of growth and 64% at the late exponential phase.

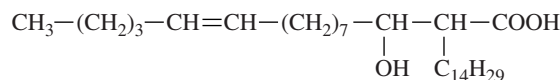
Branched-Chain Fatty Acids

Among bacterial genera, relatively high concentrations of branched-chain fatty acids have been found in *Bacillus*, *Staphylococcus*, *Corynebacterium*, *Mycobacterium*, *Pseudomonas*, and *Spirochaeta*. The most common types of branched-chain fatty acids observed are of the *iso*- and *anteiso*-configuration:

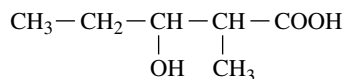


iso- and *anteiso*-fatty acids are characteristic constituents of lipids in many different organisms.

Bacillus subtilis was one of the first bacterial species in which branched-chain fatty acids were identified. At least six different branched-chain fatty acids representing 60% of the total fatty acids have been found. Corynemycolenic acid, produced by *Corynebacterium*, has the following structure:



Three branched-chain hydroxy acids, 2-hydroxy-9-methyl-decanoic acid, 3-hydroxy-9-methyl-decanoic acid, and 3-hydroxy-11-methyl-dodecanoic acid, have been identified in *Pseudomonas maltophilia*. Several species of *Mycobacterium* contain 2-methyl-3-hydroxypentanoic acid:

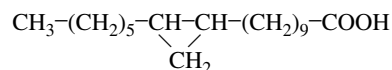


Many other 3-hydroxy fatty acids have been found in microorganisms. Members of the genus *Bacteroides* contain hydroxy fatty acids as major constituents. The LPS from *B. fragilis* contains 13-methyltetradecanoic, D-3-hydroxypentadecanoic, D-3-hydroxyhexadecanoic, D-3-hydroxy-15-methylhexadecanoic, and D-3-hydroxyheptadecanoic acids. The D-3-hydroxyheptadecanoic acids predominate. Lesser amounts of the iso-branched C₁₅, straight-chain C₁₆, and anteiso-branched C₁₇ acids are present.

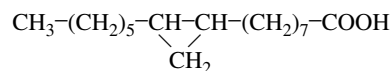
The relatively high level of hydroxy acid is too great to be accounted for simply as cell-envelope-bound LPS fatty acid. Many of these hydroxy acids may be associated with the sphingolipids present in a high level in *Bacteroides*. Poly-β-hydroxybutyrate is present in prominent intracellular granules and serves as an energy reserve polymer in bacteria, cyanobacteria, and eukaryotic organisms. 3-Hydroxy fatty acids can also serve as precursors to the monoenoic fatty acids. See further discussion on biosynthesis of the branched chain fatty acids in the "Biosynthesis of Fatty Acids" section.

Ring-Containing Fatty Acids

In 1962, Hoffman discovered the first ring-containing fatty acid, lactobacillic acid, in *L. arabinosus*. It was shown to be *cis*-11,12-methyleneoctadecanoic acid:



Greater than 15% of the fatty acids of most lactobacilli are ring-containing fatty acids, whereas less than 5% of the fatty acids of *Bifidobacterium* are ring-containing fatty acids. Along with lactobacillic acid, two other cyclopropane fatty acids are commonly found in bacteria. A C₁₇-cyclopropane fatty acid, *cis*-9,10-methylenehexadecanoic acid, is present in *E. coli* and other gram-negative organisms and has the following structure:



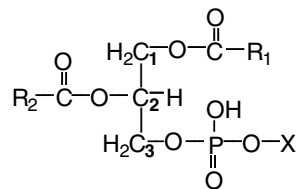
The marked inhibition of cyclopropane fatty acid synthesis, which accompanies the induction of filamentous forms of *E. coli*, suggests that ring-containing fatty acids play an important structural role in the cell envelope. Another commonly occurring cyclopropane fatty acid, 9,10-methyleneoctadec-9-enoic acid, is shown in Table 13-1.

Cyclopropane fatty acids are formed from unsaturated fatty acids present in phospholipids. They are formed only in organisms capable of making the corresponding unsaturated fatty acids (palmitoleic, *cis*-vaccenic, and oleic acids). Once formed, the unsaturated position is converted to a cyclopropane ring by the transfer of a methyl group from *S*-adenosylmethione catalyzed by the cyclopropane fatty acid synthase. This enzyme binds to bilayers of unsaturated fatty acid-containing phospholipids and forms cyclopropane rings on both faces of the membrane bilayer. Modification of the unsaturated positions in the fatty acids of the membrane occurs at the onset of the stationary phase of growth. Protection of the fatty acid from oxidation or from chemical degradation is offered as one explanation for this change in structure. Cyclopropane fatty acid synthase is induced by rapid limitation of oxygen, by initiation of respiration, or by the addition of nitrate or thiosulfate to the culture medium. Although the precise role of cyclopropane fatty acids remains elusive, there does seem to be considerable evidence to suggest that they play a role in the pathogenicity of *M. tuberculosis* and *Helicobacter pylori*.

Alk-1-enyl Ethers (Plasmalogens)

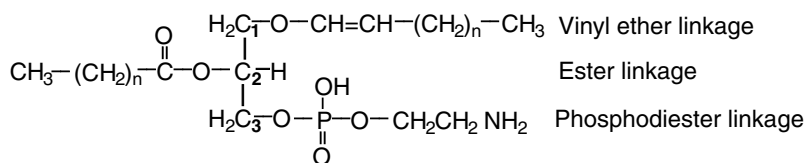
Phosphoglycerides that yield aldehydes upon hydrolysis contain long-chain fatty acids in ether linkage with the C-1 of glycerol and are called **plasmalogens**. These long-chain aldehydes are otherwise similar in composition to long-chain fatty acids. For example, the long-chain aldehydes derived from plasmalogens of *C. butyricum* are comparable to the long-chain fatty acids of this organism. The other carbon atoms of the glycerol backbone are linked to a fatty acid at C-2 and a phosphate ester at C-3 (Fig. 13-1). Although plasmalogens are known to occur in higher organisms, surveys of a wide variety of microorganisms have failed to reveal their presence in aerobic, facultative, or microaerophilic bacteria. In anaerobic genera, the plasmalogens occur in addition to, rather than in place of, the usual phospholipids. Strict anaerobes from the genera *Bacteroides*, *Clostridium*, *Desulfovibrio*, *Peptostreptococcus*, *Propionibacterium*, *Ruminococcus*, *Selenomonas*, *Sphaerophorus*, *Treponema*, and *Veillonella* all contain plasmalogens, the molar ratio of aldehydes/phosphorus varying from 0.004 to 1.04.

The major components of the plasmalogens of *C. butyricum* are 16:0 (saturated C₁₆ fatty acid), 16:1 (monounsaturated C₁₆ fatty acid), 17:cyc (C₁₇ cyclopropane fatty acid), 18:0 (saturated C₁₈ fatty acid), 18:1 (monounsaturated C₁₈ fatty acid), and 19:cyc (C₁₉ cyclopropane fatty acid). The fatty aldehydes obtained from the plasmalogens of *Selenomonas ruminantium* consist of normal saturated and monounsaturated fatty aldehydes, with 12 to 18 carbon-chain lengths predominating. A marine bacterium contains phosphatidic acid mixed with its plasmalogen analogs. Phosphatidylserine and its plasmalogen analog are major phosphoglycerides of the strictly anaerobic rumen bacterium *Megasphaera elsdenii*. Phosphoglycerides of the anaerobic lactate-fermenting *Veillonella parvula* and *Anaerovibrio lipolytica* also contain a high proportion of heptadecenoic acyl and alk-1-enyl ether moieties.

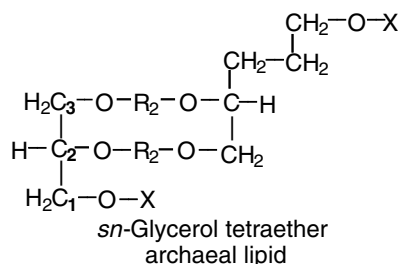
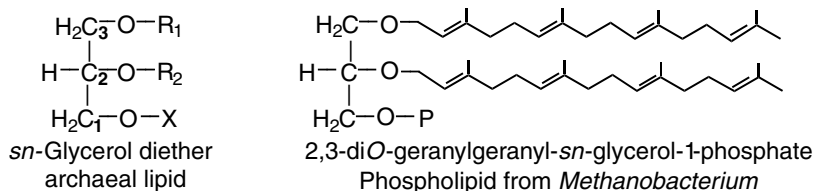


R₁,R₂= Fatty acids

General phosphoglyceride structure



Plasmalogen structure



R₁ = Phytanyl (C₂₀ polyisoprenol alcohol in ether linkage)

R_2 = Biphytanyl (C_{40} polyisoprenoid alcohol in ether linkage)

X = H; saccharide; or phosphate derivative

Fig. 13-1. Structure of phospholipids, plasmalogens, and archaeal membrane lipids. A fundamental feature of the Archaea is that the glycerophosphate backbone is *sn*-glycerol-1-phosphate, the enantiomer of that found in bacteria and eukaryotes.

Alkyl Ethers

Long-chain alcohols bound to glycerol in alkyl ether linkage occur in the Archaea, a group of bacteria that exist primarily in extremely harsh environments. Their ability to thrive in these extreme conditions is considered to be due, at least in part, to these unusual lipid structures. The Archaea contain predominantly di-*O*-alkyl analogs of phosphatidylglycerophosphate. The lipids of these organisms are formed by condensation of glycerol or more complex polyols with isoprenoid alcohols containing

20, 25, or 40 carbon atoms. A cardinal feature of all archaeal phospholipids is the presence of 2,3-di-*O*-*sn*-glycerol as the backbone structure as opposed to the *sn*-1,2-glycerol found in most other naturally occurring glycerophosphates or diacylglycerols (see Fig. 13-1). To date, no exceptions have been found to this fundamental difference in stereoconfiguration.

Phospholipids (Phosphoglycerides)

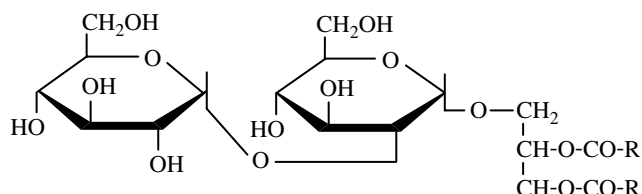
Mono-, di-, and triglycerides are the major lipid components of plants and animals. In contrast, bacteria contain relatively low concentrations of glycerides, suggesting that they play a minor role in microbial structure or function. In mammals, a high percentage of stored lipids are in the form of glycerides, while in microorganisms the phospholipids (phosphoglycerides) are the predominant type (Fig. 13-1). Mammalian cells contain phosphatidylcholine as the most common phospholipid. Bacteria usually contain phosphatidylethanolamine, phosphatidylserine, phosphatidylglycerol, and phosphatidic acid. Lysophosphatidyl compounds, which lack one of the three fatty acid chains, are sometimes found in microorganisms. Phosphatidylinositol is low in content in most bacteria but is found in higher concentration in mycobacteria, fungi, and protozoa.

Gram-negative bacteria contain large amounts of phospholipids. Together with LPS, they constitute up to 20 to 40% of the dry weight of the cell envelope. Some of these lipids, especially C₅₅-isoprenols, serve as lipid carriers in the transferase enzymes involved in peptidoglycan synthesis (see Chapter 7). The phospholipids of the cell membrane are considered to exist as bimolecular leaflets. Many marine and estuarine bacteria contain significant quantities of lysophosphatidylethanolamine. Cardiolipin (diphosphatidylglycerol) and cardiolipin synthetase are absent in many nonfermentative isolates of marine bacteria. There are two metabolic pools of phosphatidylglycerol and cardiolipin in *B. stearothermophilus*, *E. coli*, *S. aureus*, and *H. parainfluenzae*. Cardiolipin is an important constituent of the membrane of *H. parainfluenzae*.

As mentioned previously, the archaea produce a variety of unusual lipid structures that contain ether linkages. Although quite different in their chemical structure, most of these lipids are comparable to bacterial phospholipids in that they contain single or bipolar head groups with hydrophobic alkyl side chains. They form membrane bilayers with the polar head group toward the aqueous phase that are indistinguishable from those of bacteria. Neutrophilic halophiles contain C₂₀ phytanyl substituents (Fig. 13-1). However, in alkaliphilic halophiles, C₂₅ sesterpanylic compounds may predominate. The membranes of methanogens contain diphytanyl-glycerol diether and dibiphytanyldiglycerol tetraether. The presence of tetraether lipids in methanogens correlates with the ease of demonstrating freeze-fracture planes in these organisms. Above the 45 to 50% level of tetraether core lipids, the frequency of membrane fracture is markedly reduced. The membrane structures of thermophiles show considerable variation, but C₂₀ phytanyl and C₄₀ biphytanyl chains are found in *Desulfurococcus*, *Thermoproteus*, *Thermofilum*, and *Pyrodictum*. *Sulfolobus* contains a variety of tetraethers of two classes. The first class is comprised of glycerol-dialkylglycerol-tetraethers that contain two *sn*-2,3-glycerol components bridged by two isoprenoid C₄₀ diols through ether linkages. In the second class, referred to as glycerol-dialkylnonitol tetraethers, a branched nonitol replaces one of the glycerols.

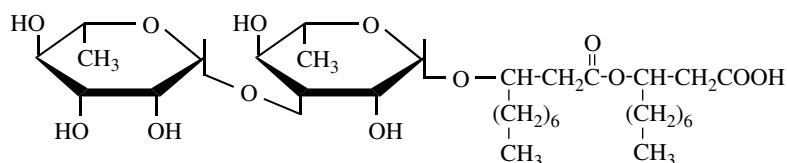
Glycolipids

Glycosyldiglycerides may contain glucose, galactose, or mannose. The structure of the glycolipid α -D-glucopyranosyl-(1,2)- α -D-glucopyranosyl-(2,3)-diglyceride is

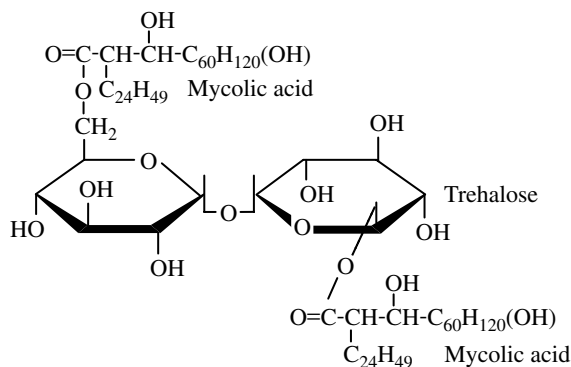


Monoglycosyl derivatives apparently represent biosynthetic precursors of the diglycosyl derivatives. Glycosyldiglycerides are widely distributed in gram-positive bacilli, but thus far have not been found in gram-negative bacilli. They have either α,α or β,β configurations and do not seem to occur in the α,β form. The nonsterol-requiring *Mycoplasma* contains glycolipids, suggesting that glycolipids replace sterols in either the structure and/or the function of the membrane.

In most glycolipids the carbohydrate moiety is linked to the glycerol component. However, glycolipid structures have been observed in which the carbohydrate portion is attached directly to the fatty acid. A rhamnolipid found in *P. aeruginosa* contains two molecules of rhamnose and two molecules of decanoic acid:

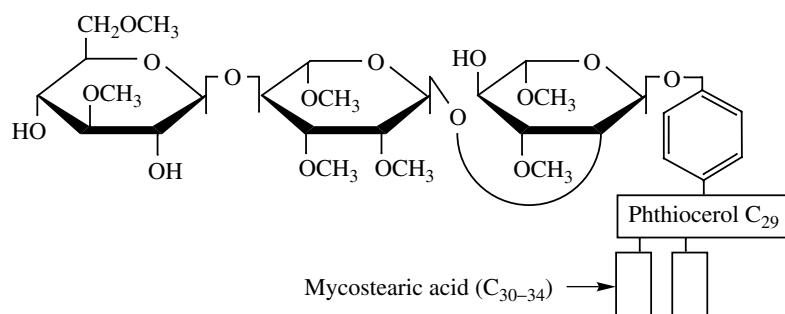


A glycolipid of major importance in *M. tuberculosis*, the **cord factor**, is associated with the virulence of this organism as well as with the characteristic serpentine growth pattern. Cord factor has been shown to contain two molecules of mycolic acid attached to trehalose. In human and bovine strains of *M. tuberculosis*, the substituent side chain is $C_{24}H_{49}$, as shown in the following structure:



The side chain in the cord factor in avian and saprophytic mycobacteria is $C_{22}H_{45}$. Other variations in the chemical structure of mycolic acids have been found as a larger number of microorganisms have been examined.

A unique phenolic glycolipid in *M. leprae* contains 3,6-di-*O*-methyl-glucose, 2,3-di-*O*-methylrhamnose, and 3-*O*-methylrhamnose linked to phenol-dimycoseroyl phthiocerol:



This unusual antigen of *M. leprae* suppresses the *in vitro* mitogenic response of lymphocytes from lepromatous patients. The suppressor T cells of these patients recognize the specific terminal trisaccharide moiety that triggers the suppression. Lymphocytes from patients with tuberculoid leprosy, lepromin-positive contacts, or normal donors show no comparable suppression of lymphocyte proliferation. Removal of the mycoserotic acid side chain has no effect on *in vitro* suppression. However, absence of the 3' terminal methyl group or removal of the terminal sugar abolishes or significantly reduces the suppressive effect. Comparable glycolipids from *M. kansasii* and *M. bovis* have no effect on the mitogenic response.

BIOSYNTHESIS OF FATTY ACIDS

Fatty acid biosynthesis has been studied extensively in bacteria and yeast. Acetyl-CoA is the ultimate precursor of fatty acid carbons. Acetyl-CoA carboxylase, a biotin-containing enzyme that catalyzes the ATP-dependent fixation of CO_2 into acetyl-, propionyl-, and butyryl-CoA, catalyzes the first committed reaction in the pathway (Fig. 13-2). The reaction occurs in two stages: the carboxylation of biotin with bicarbonate is catalyzed by biotin carboxylase, and transfer of the carboxyl group from carboxybiotin to acetyl-CoA to form malonyl-CoA is mediated by carboxyltransferase (Fig. 13-3). The first stage is followed by a series of reactions that ultimately lead to the synthesis of long-chain fatty acids at the C_{16} (palmitic) and C_{18} (stearic) chain length (Fig. 13-2). Initially, it was puzzling to find that CO_2 , an essential reactant in the system, was not incorporated into the fatty acids. This occurs because the CO_2 used to form malonyl-CoA is subsequently removed in the condensation step. It is actually the liberation of CO_2 that shifts the equilibrium in the direction of synthesis.

Because of its crucial role as the first committed step in fatty acid biosynthesis, the acetyl-CoA carboxylase system has been considered as one of the most likely sites of regulation of fatty acid biosynthesis. In mammalian systems, several mechanisms of regulation of acetyl-CoA carboxylase have been proposed. Allosteric regulation by

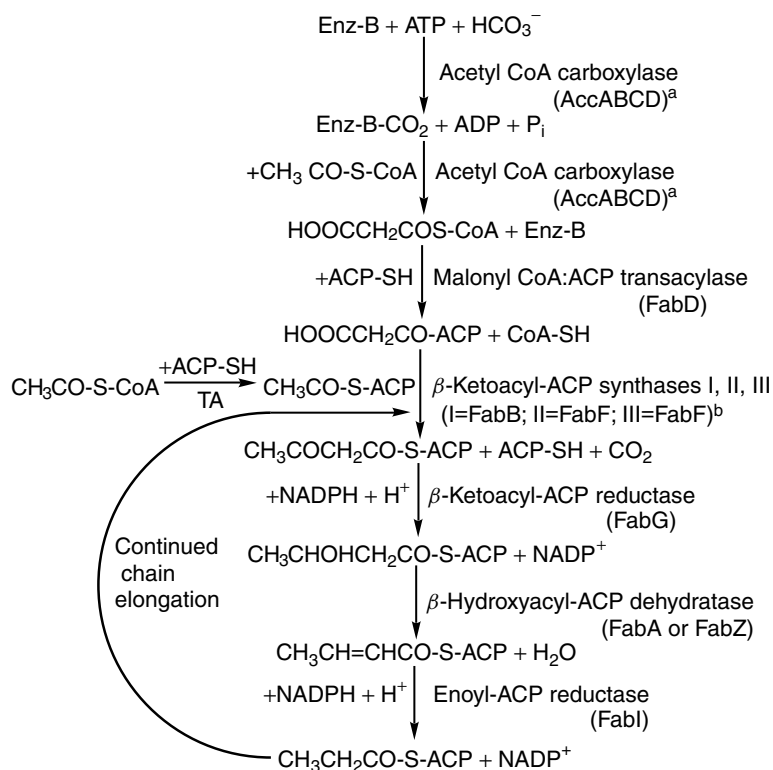


Fig. 13-2. Reactions in fatty acid biosynthesis. Enz-B, enzyme-biotin; CoA, coenzyme A; ACP, acyl carrier protein; TA, acetyl-CoA-ACP transacylase. The three-letter designations are for the structural proteins. ^aSee Figure 13-3 for details of acetyl-CoA carboxylase system involving enzymes AccABCD. ^bThe FabH enzyme, β -ketoacyl-ACP synthase III, catalyzes the condensation of acetyl-CoA and malonyl-ACP, whereas FabB and FabF, β -ketoacyl-ACP synthases I and II, are active with C_2 - C_{14} -ACP but inactive with C_{16} -ACP, and are inactive with CoA derivatives.

citrate and isocitrate, feedback inhibition by the end product (a long-chain acyl-CoA), covalent modification by phosphorylation and dephosphorylation, and regulation at the level of gene expression have all been implicated as possible mechanisms. In *E. coli* the four components of the acetyl-CoA carboxylase system function as an enzyme complex. The genes that encode the biotin carboxyl carrier protein (BCCP), *accB*, and biotin carboxylase, *accC*, are cotranscribed (*accBC*) and map at 72 minutes. The genes encoding the carboxyltransferase α (*accA*) and β (*accD*) subunits map at different regions of the chromosome. The rates of transcription of the genes encoding all four subunits of acetyl-CoA carboxylase are directly related to the rate of cell growth. The promoter sequences of these genes and certain features of their respective promoter regions indicate a role in regulation of the complex.

Fatty acid biosynthesis in bacteria is conducted by a soluble, disassociated system designated as **type II**. At least eight individual enzyme components are readily separated and purified from *E. coli* and *P. shermanii*. The system requires acetyl-CoA, malonyl-CoA, NADH, NADPH, and ACP for fatty acid synthesis. In *E. coli* and other

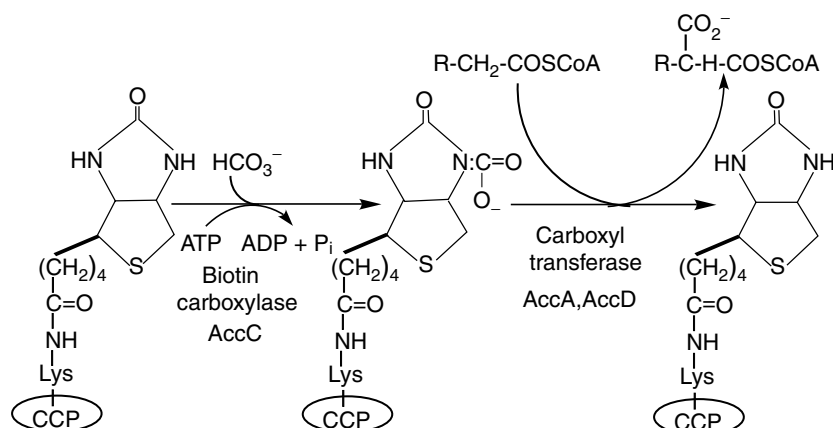
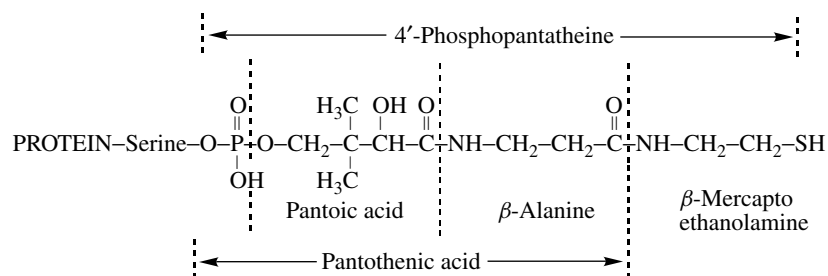


Fig. 13-3. The role of biotin in the acetyl-CoA carboxylase system. This is the first committed step in fatty acid biosynthesis. The *E. coli* complex consists of four subunits. The three-letter designations are the products of the corresponding structural genes in *E. coli*.

bacteria, 12 or more genes code for the enzymes depicted in Figure 13-2. The genes *acpP*, *fabD*, *fabF*, *fabG*, and *fabH* are encoded in a *fab* gene cluster that has been termed the *fab* operon. The enzyme β -ketoacyl-ACP synthase I (FabB) is required for the elongation of unsaturated fatty acids. Mutants lacking FabB activity are unable to synthesize either palmitoleic or *cis*-vaccenic acids and require fatty acids for growth.

β -Ketoacyl-ACP synthase II (FabF) is responsible for the temperature-dependent regulation of fatty acid composition. Mutants lacking FabF are deficient in *cis*-vaccenic acid but grow normally. β -Ketoacyl-ACP synthase III (FabH) catalyzes the formation of acetoacyl-ACP. The *fabA* gene encodes 3-dehydrodecanoyl-ACP dehydratase. Malonyl-CoA-ACP transacylase (FabD) catalyzes the transacylation of ACP with malonate. Enoyl-ACP reductase (FabI) conducts the final step in the elongation process. The product is recycled through the last four steps to yield long-chain fatty acids. In *E. coli*, FabI is the target of triclosan, a broad-spectrum antibiotic active against many bacteria and fungi. Mutations in the *fabI* gene confer triclosan resistance. Triclosan is widely used as an ingredient in antiseptic soaps, toothpastes, and other materials. The development of triclosan resistance may select for multiple resistance to other agents, including many clinically useful antibiotics. Triclosan (also known as irgasan) provides for the selective growth of *P. aeruginosa* on pseudomonas isolation agar.

The ACP that functions in fatty acid synthesis is very stable to heat and acid treatment. The structure of the prosthetic group is



The composition of this prosthetic group is very similar to that of coenzyme A. The ACP is specific to the fatty acid synthase system and cannot replace coenzyme A.

In *E. coli* unsaturated fatty acids are synthesized simultaneously with saturated fatty acids. The divergence occurs at the C_{10} stage (β -hydroxy-decanoate). At this step there is apparently a competition between β -hydroxyacyl-ACP dehydrase, which forms α,β,γ -*cis* double bonds, as shown in Figure 13-4. Continued elongation leads to the formation of unsaturated fatty acids. As a result of this mode of unsaturated fatty acid synthesis, only monounsaturated fatty acids are formed, and *cis*-vaccenic acid ($18:1^{\Delta^{11}}$) is the major unsaturated fatty acid in organisms that use the anaerobic pathway (see Fig. 13-4 and Table 13-2).

In yeasts, fungi, and other eukaryotes, a multienzyme complex called **type I** carries out fatty acid synthesis. Mycobacteria also contain a **type I** fatty acid synthase system. The **type I** fatty acid synthase in yeast consists of two multifunctional peptide chains each having six subunits ($\alpha_6\beta_6$) that carry out the same succession of reactions shown in Figure 13-2. An acyl carrier protein component serves as the carrier to move the growing fatty acid chain to the active sites. In yeast the elongation is terminated at

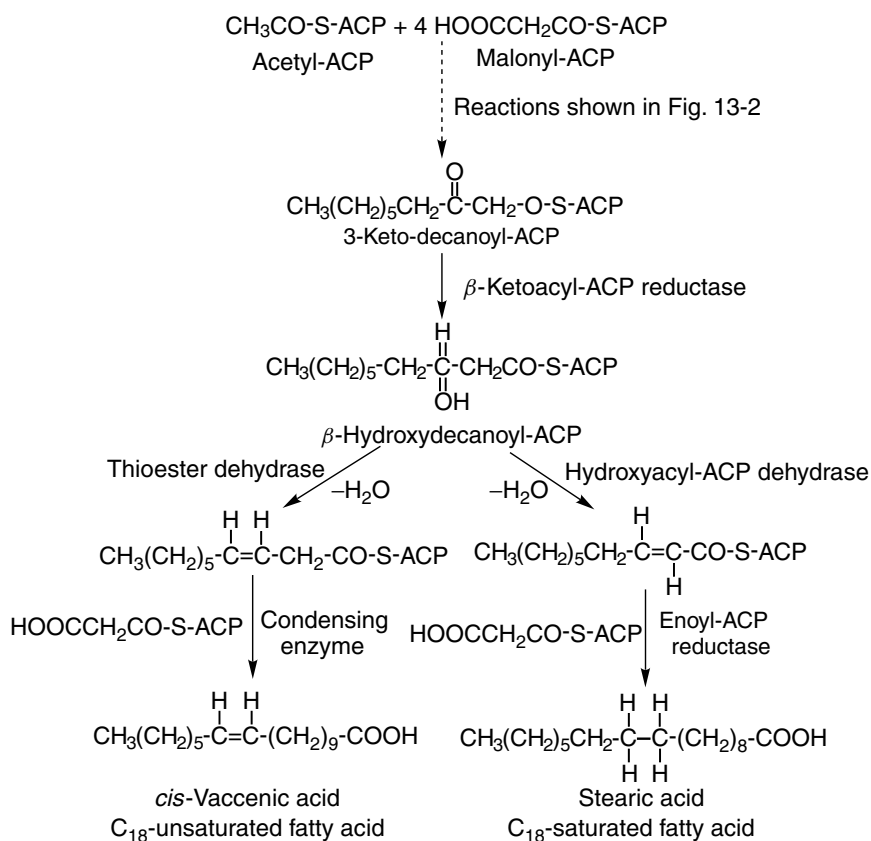
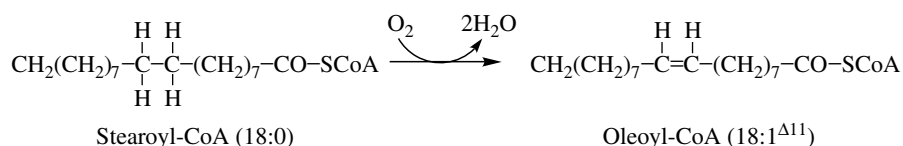


Fig. 13-4. Anaerobic fatty acid biosynthesis showing divergence of the saturated and unsaturated pathways. The pathway to saturated fatty acids shown in Figure 13-2 is followed until the C_{10} level (β -hydroxydecanoyl-ACP), at which point there is a competition between the enzymes involved in forming saturated and unsaturated fatty acids.

TABLE 13-2. Distribution of Pathways for Biosynthesis of Long-Chain Fatty Acids

Anaerobic Pathway	Aerobic Pathway
<i>Escherichia coli</i>	<i>Alcaligenes faecalis</i>
<i>Salmonella enterica</i>	<i>Corynebacterium diphtheriae</i>
<i>Serratia marcescens</i>	<i>Mycobacterium phlei</i>
<i>Azotobacter agilis</i>	<i>Bacillus</i> (several species)
<i>Agrobacterium tumefaciens</i>	<i>Micrococcus luteus</i>
<i>Lactobacillus plantarum</i>	<i>Beggiatoa</i> spp.
<i>Staphylococcus haemolyticus</i>	<i>Myxobacter</i> spp.
<i>Clostridium pasteurianum</i>	<i>Leptospira canicola</i>
<i>Clostridium butyricum</i>	<i>Saccharomyces cerevisiae</i>
<i>Caulobacter crescentus</i>	<i>Candida lipolytica</i>
<i>Propionibacterium</i>	<i>Neurospora crassa</i>
<i>Cloroflexus auranticus</i>	<i>Stigmatella auranticus</i>
<i>Chlorobium limicola</i>	

the C₁₆ (palmitic acid) to C₁₈ (stearic) level, and a transferase forms a coenzyme A derivative. Mammals, yeast, algae, protozoa, and some highly aerobic bacteria (*Alcaligenes*, *Mycobacterium*, *Corynebacterium*, *Bacillus*, and others as shown in Table 13-2) use the **aerobic pathway** to synthesize long-chain unsaturated fatty acids. In *S. cerevisiae* oxidative desaturation by a specific desaturase, Ole1p, introduces a double bond between positions 9 and 10 after the long-chain saturated fatty acid has been synthesized:



Approximately 70 to 80% of the total fatty acids in membrane lipids of the yeast *Saccharomyces cerevisiae* consist of the unsaturated fatty acids palmitoleic acid (16:1) and oleic acid (18:1). The remaining fatty acids are saturated, consisting primarily of palmitic acid (16:0) and lesser amounts of stearic acid (18:0) and myristic acid (14:0).

Most other fungi contain the di- and trienoic acids linoleic acid (18:2) and α -linolenic acid (18:3). Yeast utilizes fatty acid desaturases for the formation of unsaturated fatty acids. The Δ -9 desaturase catalyzes the formation of the double bond between the ninth and tenth carbon atoms of both palmitoyl (16:0) and stearoyl (18:0) CoA substrates to form 16:1 and 18:1 unsaturated fatty acids. The activity of the Δ -9 desaturase is markedly reduced by the addition of unsaturated fatty acids, presumably because the level of the desaturase mRNA is repressed under these conditions. Longer-chain fatty acids of 24 to 26 carbon atoms are synthesized by a membrane-bound elongase containing two components: Elo2p and Elo3p. These very long-chain fatty acids comprise only 1% of the cellular fatty acid pool, but they are essential for the ceramide found in sphingolipids and the lipid portion of glycosylphosphatidylinositol-anchored membrane proteins.

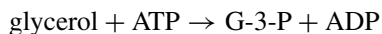
Under anaerobic conditions yeast exhibit a requirement for unsaturated fatty acids that stems from their requirement for coupling oxidative phosphorylation to ATP synthesis in the yeast mitochondrion. This effect is obvious since yeast use the aerobic pathway (desaturase enzyme) for the formation of unsaturated fatty acids. However, studies with mutant strains show that *S. cerevisiae* also requires unsaturated fatty acids for aerobic growth. The required amount is 4-fold higher than that required for anaerobic growth.

BIOSYNTHESIS OF PHOSPHOLIPIDS

Phospholipids (mainly phosphatidylcholine, phosphatidylethanolamine, phosphatidylinositol, and phosphatidylserine) are the major structural components of cellular membranes. Biosynthesis of these important compounds, more correctly referred to as phosphoglycerides, begins with the formation of an *sn*-glycerol-3-phosphate (G-3-P). The G-3-P may be formed from dihydroxyacetone phosphate (DOHAP) derived from hexose cleavage by the action of the biosynthetic G-3-P dehydrogenase encoded by *gpsA*:



sn-Glycerol-3-phosphate may also be produced from glycerol via the action of glycerol kinase encoded by *glpK*:



The first committed steps in phosphoglyceride biosynthesis involve coupling two molecules of fatty acid to G-3-P (Fig. 13-5). This reaction is catalyzed by two acyl transferases. Glycerol-3-phosphate acyltransferase (PlsB) adds one unsaturated fatty acid to form 1-acyl-glycerol-3-phosphate. An unsaturated fatty acid is added to the C-2 position by 1-acylglycerophosphate acyltransferase (PlsK) to form phosphatidic acid. In bacteria, a fatty acid-ACP is the donor rather than a fatty acid-CoA. The next step involves formation of cytidine diphosphoglyceride (CDP-diglyceride) by phosphatidate cytidyltransferase, as shown in Figure 13-5.

In mammalian cells, the seeds of plants, and in lower eukaryotes, triglycerides are formed in considerable quantities and serve as storage compounds. In microorganisms, particularly bacteria, phospholipids are more commonly observed. *E. coli* normally contains approximately 70% phosphatidylethanolamine, 25% phosphatidylglycerol, and 5% cardiolipin.

At this point the pathway branches. Addition of L-serine by phosphatidylserine synthase (encoded by *pssA*) yields phosphatidyl serine. Decarboxylation of phosphatidylserine by phosphatidylserine decarboxylase (encoded by *psd*) yields phosphatidylethanolamine, one of the major phospholipids found in bacteria. In the other branch of the pathway, addition of a second molecule of G-3-P to the CDP-diglyceride by phosphatidylglycerophosphate synthetase (encoded by *pgsA*) results in the release of CMP and the formation of phosphatidylglycerolphosphate. Removal of P_i by phosphatidylglycerolphosphate phosphatase (encoded by *pgsA*, *pgsB*) yields phosphatidyl glycerol. In bacteria, two molecules of phosphatidylglycerol are condensed by

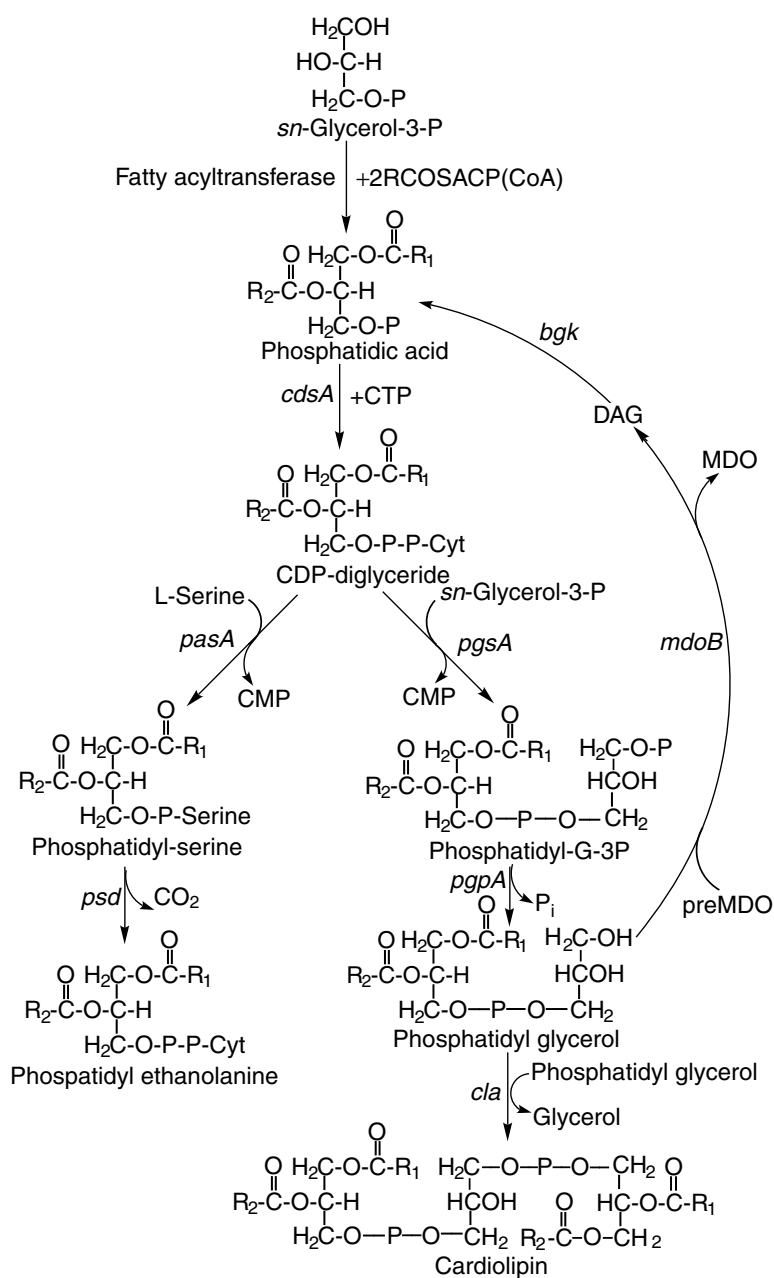
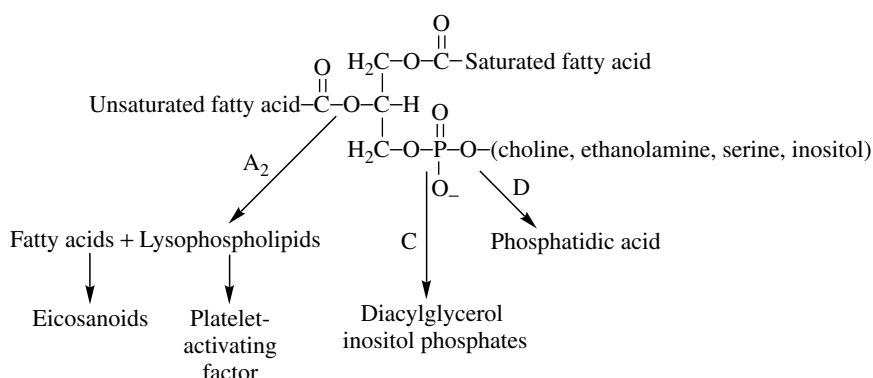


Fig. 13-5. Biosynthesis of phosphoglycerides (phospholipids). The structural genes for the *E. coli* enzymes are: *plsB*, *sn*-glycerol-3-phosphate acyltransferase; *plsK*, 1-acylglycerol-3-phosphate acyltransferase; *cdsA*, phosphatidate cytidyltransferase; *pgsA*, phosphatidylserine synthase; *psd*, phosphatidylserine decarboxylase; *pgpA*, phosphatidylglycerophosphate synthase; *pgpA*, phosphatidylglycerophosphate phosphatase; *cls*, cardiolipin (diphosphatidylglycerol) synthase.

cardiolipin synthase (encoded by *cls*), releasing glycerol to form cardiolipin (diphosphatidylglycerol). In mammalian systems, the formation of cardiolipin involves the formation of a CDP diglyceride as an intermediate.

In addition to their role as structural components of the cell membrane, another significant aspect of phospholipids is the production of derivatives that serve as messengers in cellular regulation. Phospholipids are acted upon by several classes of phospholipases (A₂, C, and D) to produce a variety of intracellular and intercellular messengers:



Many of these regulatory compounds are of major significance in mammalian systems. However, the formation of phosphatidylinositol and its phosphorylated derivatives in yeast is an important aspect of yeast signal transduction. Phospholipid breakdown by various classes of phospholipases gives rise to compounds that serve as messengers in the utilization of signal transduction cascades in the regulation of the cell cycle, response to environmental stress, and the process of sporulation in yeast.

DEGRADATION OF FATTY ACIDS

Most bacteria do not readily degrade fatty acids. However, once induced, they may become very active fatty acid oxidizers. For example, *E. coli* will grow on long-chain fatty acids only after a prolonged lag phase. Cells induced to grow on a fatty acid, such as palmitate, will grow immediately on fatty acids without any appreciable lag. The activities of the enzymes of fatty acid oxidation shown in Figure 13-6 increase or decrease coordinately, depending on the conditions under which the cells were previously grown. Reversal of the strong glucose inhibition of fatty acid utilization by cAMP indicates that catabolite repression is the mechanism involved.

The structural genes (*fad* genes) encoding enzymes of fatty acid degradation are observed at four distinct loci on the *E. coli* gene map and encode at least five enzyme activities involved in the transport, acylation, and β -oxidation of medium-chain (C₆-C₁₀) and long-chain (C₁₂-C₁₈) fatty acids. Long-chain fatty acids can induce the *fad* genes, whereas medium-chain fatty acids cannot. Therefore, the wild-type *E. coli* can utilize long-chain fatty acids, such as oleate (C₁₈ : 1), but not medium-chain length, such as decanoate (C₁₀ phenotype), as the sole carbon and energy source. The *fadR* gene is a multifunctional regulatory gene mapping at 25.5 minutes that exerts negative

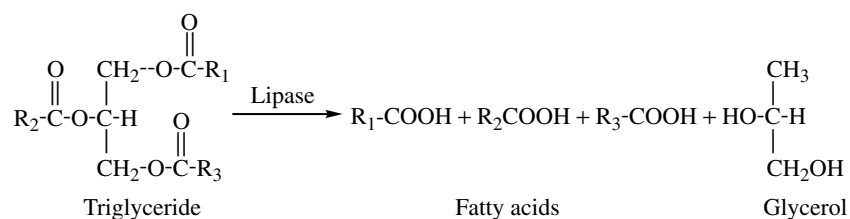
control over the *fad* regulon. The product of the *fadR* gene (FadR) is a diffusible protein that exerts control over fatty acid degradation by decreasing the transcription of the *fad* structural genes.

In addition to the *fad* enzymes, expression of the glyoxylate shunt enzymes is also required for the growth of *E. coli* on acetate or fatty acids as a sole carbon source. In wild-type *E. coli*, repression of the acetate (*ace*) operon is under the control of *fadR* and *iclR* regulatory genes for the *aceBA* operon). Both the *fadR* and *iclR* genes regulate the glyoxylate shunt at the level of transcription. The *fadR* gene seems to play a role in the regulation of unsaturated fatty acid biosynthesis as evidenced by the fact that *fadR* mutants synthesize significantly less unsaturated fatty acids than wild-type cells. A functional *fadR* gene is required for optimal synthesis of unsaturated fatty acids.

The *fadAB* and *fadR* genes have been cloned and characterized with regard to their specific functions. In *E. coli* a multifunctional enzyme complex is encoded by the *fadAB* operon. Two protein subunits (42,000 and 78,000 Da) in this enzyme complex exhibit 3-ketoacyl-CoA thiolase, 3-hydroxyacyl-CoA epimerase, and enoyl-CoA hydratase. Two identical small subunits are encoded within the *fadA* gene, which contains thiolase activity. The other four activities are associated with the large subunit encoded by *fadB*. The intact complex is required for function. Both *fadA* and *fadB* are transcribed as a single transcriptional unit with the direction of transcription being from *fadA* to *fadB*. Plasmids carrying the *fadR* gene suppress the expression of the *fadR* operon. A 19,000 Da protein, FadR, exerts control over the *fad*, *ace*, and *fab* structural genes.

Neurospora crassa also possesses an inducible β -oxidation system. Activities of the enzymes in this system are enhanced after a shift from a sucrose to an acetate growth medium. The induction is even more pronounced after growth in the presence of oleate as the sole carbon and energy source. The enzymes of this pathway are localized in the glyoxysome microbodies, which are distinct from the peroxisomes (subcellular organelles whose main function is to process substances for elimination). The β -oxidation system in *N. crassa* relies on acyl-CoA dehydrogenase rather than acyl-CoA oxidase in the first step. The system does not seem to be under the same type of coordinate regulation as the fatty acid oxidation system in *E. coli*. By comparison, β -oxidation in yeasts takes place in peroxisomes. Peroxisomal β -oxidation involves catalase in the reoxidation of FADH₂. These organelles have a novel long-chain acyl-CoA synthetase whose product is exclusively used for β -oxidation and not for lipid synthesis. Lipolytic yeasts, such as *Candida rugosa*, secrete lipases that initiate the process by degrading triacylglycerol to fatty acids and glycerol as described previously.

Some of the earliest investigations into the metabolism of fatty acids and triglycerides dealt primarily with the degradation of triglycerides by lipases and further oxidation of the liberated fatty acids. Lipases attack triglycerides in the following manner:



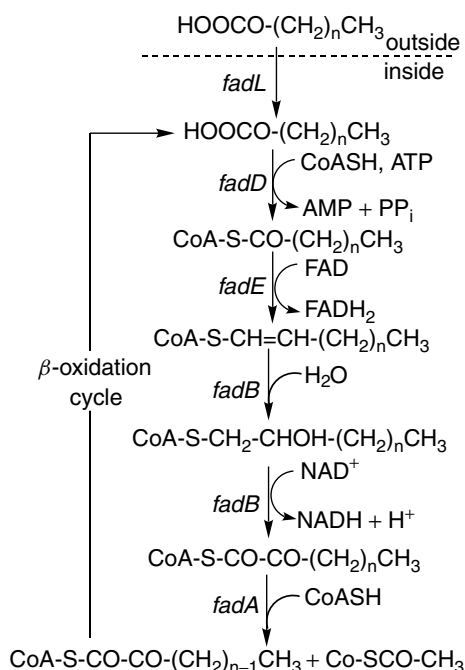
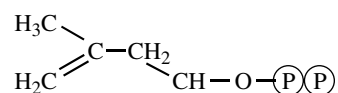


Fig. 13-6. Degradation of fatty acids (β -oxidation cycle). The *E. coli* structural genes and enzymes are: *fadL*, fatty acid transport protein; *fadD*, acyl-CoA synthetase; *fadE*, acyl-CoA dehydrogenase; *fadB*, 3-hydroxyacyl-CoA dehydrogenase; *fadA*, β -ketoacyl-CoA thiolase.

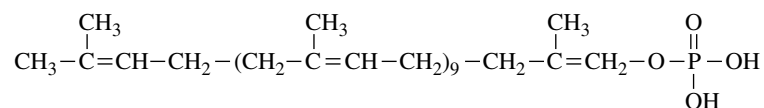
The liberated glycerol is oxidized to glycerol-3-phosphate, which is then metabolized via the triose phosphate portion of the EMP pathway. The free fatty acids are converted to acyl-CoA derivatives by acyl-CoA synthetase and subsequently oxidized via the sequence of reactions shown in Figure 13-6. These reactions are repeated until the fatty acids have been reduced to acetyl-CoA. The two-carbon fragments are then metabolized by the glyoxylate cycle. In organisms that do not have a functional glyoxylate cycle, acetoacetate, β -hydroxybutyrate, and acetone may be produced.

BIOSYNTHESIS OF ISOPRENOIDS

Isoprenoids are compounds composed of repeating five carbon isopentenyl diphosphate (IPP) units. In bacteria, polyprenyl phosphates such as undecaprenol are involved in cell wall biosynthesis (Chapter 7). The electron transport agents menaquinone and ubiquinone (Chapters 9 and 15) contain isoprenols. Carotenoids (Chapter 12) are composed of isoprenoid units. The sterols, chlorophylls, and plastoquinones found in eukaryotic organisms also contain isoprenoids. All isoprenoids are synthesized by consecutive condensations of the five-carbon monomer isopentenyl diphosphate (IPP):



In bacteria, the C₅₅-isoprenol, bactoprenol, which participates in cell wall biosynthesis, is composed of repeating isoprenyl subunits and has the following structure:



The central isoprenoid precursor, isopentenyl diphosphate (IPP), can be synthesized via the mevalonate pathway as shown in Figure 13-7. In staphylococci, streptococci, and enterococci, genes are present that encode all of the enzymes of the mevalonate pathway, but none of the genes coding for enzymes of the GAP-pyruvate pathway (see below) are found. The mevalonate pathway has been demonstrated in *Myxococcus*, *Lactobacillus*, *Methanococcus*, *Methanobacterium*, *Pyrococcus*, *Archeoglobus*,

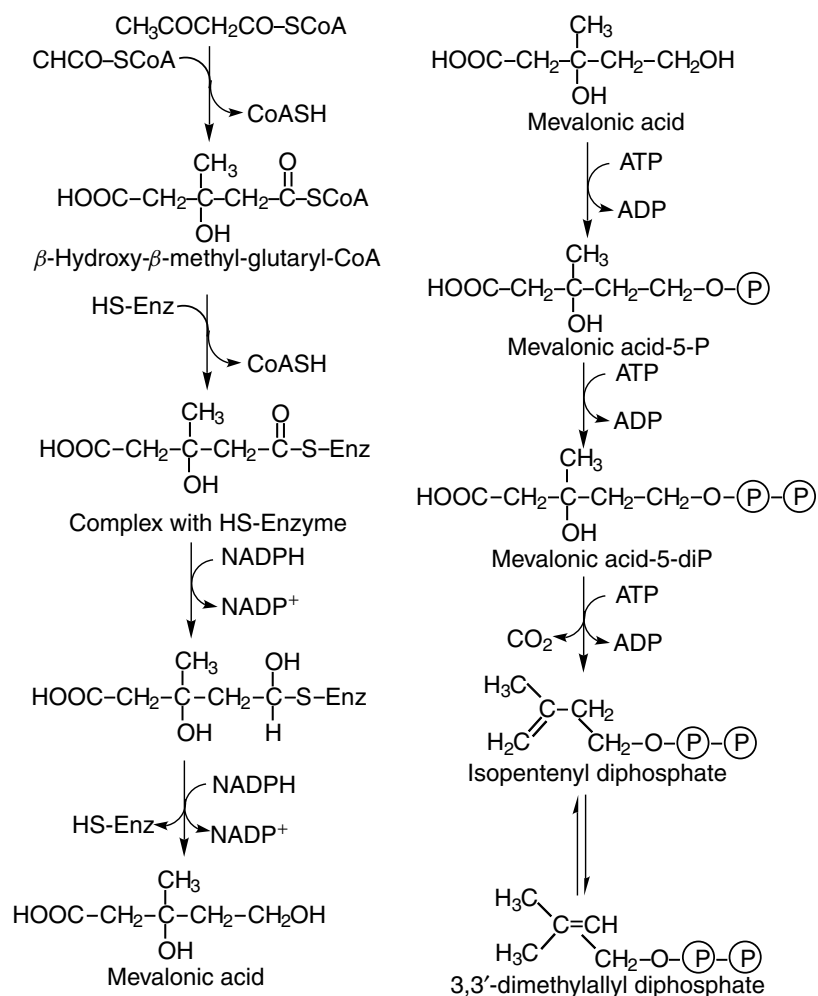
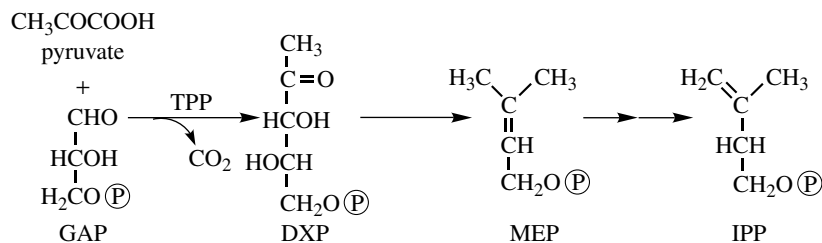


Fig. 13-7. Biosynthesis of mevalonic acid and its conversion to isoprenoid structural form.

and *Streptomyces* species. In all species using the mevalonate pathway, hydroxymethylglutaryl coenzyme A (HMG-CoA) synthetase and HMG-CoA reductase genes are closely linked. Genes encoding mevalonate kinase, mevalonate decarboxylase, and phosphomevalonate kinase apparently constitute an operon. These enzymes are essential for the growth of *S. pneumoniae* and presumably other gram-positive cocci. Phylogenetic analysis provides evidence to suggest that gram-positive cocci obtained the genes for the mevalonate pathway through one or more horizontal gene transfer events involving primitive eukaryotes.

Some organisms use an alternative pathway to IPP, the pyruvate-GAP pathway. This pathway involves condensation of pyruvate and glyceraldehyde-3-phosphate (GAP) to form 1-deoxy-D-xylulose 5-phosphate (DXP). Subsequent conversion of DXP to 2-C-methyl-D-erythritol 4-phosphate (MEP) leads to the formation of isopentenyl diphosphate:



Alternate pathway to isopentenylphosphate (IPP)

Recent studies reveal that almost all of the eubacteria except for *S. aureus* and other gram-positive cocci and *B. burgdorferi* utilize only the nonmevalonate pathway for isoprenoid biosynthesis. Most gram-negative bacteria and *B. subtilis* possess only components of the GAP-pyruvate pathway. The nonmevalonate pathway has also been demonstrated in *Chlamydia trachomatis*, *Aquifex aeolicus*, *Synechocystis*, *Scenedesmus*, and *M. tuberculosis*.

Yeasts and molds contain sterols in their cytoplasmic membranes. A few prokaryotic organisms such as *Mycoplasma* have sterols in their membranes. Sterols function in membrane permeability. In fungi, a branch of the pathway of fatty acid synthesis leads to the formation of mevalonic acid, the first intermediate in sterol biosynthesis. Through a complex series of reactions, mevalonic acid is converted to farnesyl pyrophosphate and then to squalene. Squalene then undergoes several cyclization reactions to ultimately give rise to lanosterol, the precursor of many sterol derivatives. The pathways leading to the formation of sterols are shown in Figures 13-7 and 13-8.

The pathway to mevalonic acid (Fig. 13-7) has been demonstrated in yeast as well as in mammalian liver. The condensing enzyme responsible for the coupling of acetoacetyl-CoA and acetyl-CoA has been isolated and purified from yeasts. Hydroxymethylglutaryl-CoA can also arise from malonyl-CoA via reactions that are identical to those responsible for the initiation of fatty acid biosynthesis. Free hydroxymethylglutarate is not readily utilized as a precursor of steroids by yeast or mammalian liver, presumably because of the absence of an activating enzyme. On the other hand, *Phycomyces blakesleeanus* can readily incorporate this compound since it has the necessary activating enzyme.

The *Mycoplasmataceae* are separated into two groups on the basis of their nutritional requirements for lipid components. The genus *Mycoplasma* requires sterols for growth,

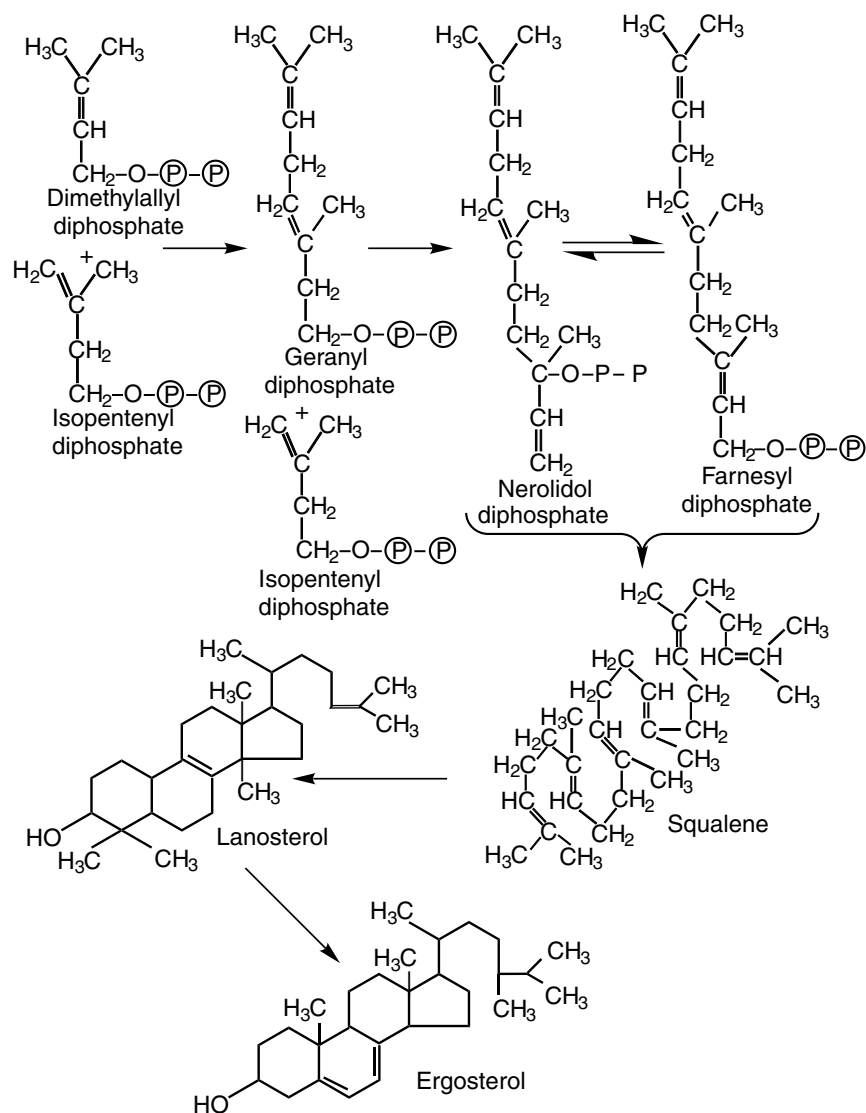


Fig. 13-8. Biosynthesis of nerolidol and farnesyl pyrophosphates and their conversion to squalene and ergosterol. It should be noted that there are several intermediate steps between squalene and lanosterol and between lanosterol and ergosterol.

whereas *Acholeplasma* synthesize carotenoid compounds that assume the functional capacity of sterols. Sterol and carotenol can apparently substitute for one another. Cholesterol added to the culture medium is used in place of carotenoids whose synthesis is repressed under these growth conditions (for examples of carotenoid structures, see Chapter 12).

Acholeplasma laidlawii can synthesize carotenoids from acetate. The organism contains a specific acetokinase and phosphotransacetylase, both of which are necessary for the synthesis of acetyl-CoA, and the β -ketothiolase and CoA transferase

enzymes required to make acetoacetyl-CoA. The presence of β -hydroxymethylglutaryl-CoA–condensing enzyme and reductase activity in *Acholeplasma*, together with the ability to incorporate ^{14}C -labeled acetate into mevalonate, indicates that these organisms use the same pathway as yeast for the synthesis of mevalonate. The absence of these enzyme activities in *Mycoplasma hominis* explains its growth requirement for sterol. In other *Mycoplasma* species (e.g., *M. gallisepticum*), the metabolic block occurs subsequent to the mevalonate step in the biosynthetic pathway to terpenoids.

In order to participate in the biosynthesis of sterols, mevalonic acid must first be converted into the isoprenoid structural form, as shown in Figure 13-7. This sequence of reactions is present in yeast. There is some evidence that 5-diphospho-3-phosphomevalonic acid, a very unstable intermediate, is formed. This would account for the arrangement of the double bonds in the isopentenyl structure. Isopentenyl diphosphate and dimethylallyl diphosphate are condensed with the elimination of pyrophosphate to form geranyl pyrophosphate. This compound condenses with another molecule of isopentenyl diphosphate, yielding nerolidol diphosphate and, ultimately, *trans, trans*-farnesyl diphosphate (Fig. 13-8). The reaction sequence leading to the formation of nerolidol and farnesyl diphosphate has been demonstrated in yeast.

Squalene is formed by the condensation of one molecule of farnesyl diphosphate and one molecule of nerolidol diphosphate (Fig. 13-8). The conversion of squalene to ergosterol in yeast or to cholesterol in mammals involves a number of intermediary steps. In yeast, an epoxide intermediate precedes the formation of lanosterol. At least eight additional steps are required to convert lanosterol to ergosterol. *S*-adenosylmethionine donates the additional carbon (C_{28}) at the lanosterol level. In mammalian systems, the conversion of squalene to cholesterol requires at least two sterol carrier proteins (SCP_1 and SCP_2) to bind the substrate and make its reaction to the sterol-synthesizing enzymes present in the microsomes. These sterol carrier proteins differ in several respects from the acyl carrier protein involved in fatty acid biosynthesis. However, both of these compounds serve the analogous function of maintaining the solubility and reactivity of the substrate.

Yeasts, including *S. cerevisiae*, require sterols for growth. Under aerobic conditions, most yeasts are able to synthesize the required level of ergosterol. Under anaerobic conditions, yeast cannot synthesize sterols or unsaturated fatty acids and require their addition to the growth medium. Yeast mutants auxotrophic for sterols and unsaturated fatty acids are altered in the fatty acid composition of their mitochondrial phospholipids. Incorporation of fatty acids into phospholipids varies with the sterol and unsaturated fatty acids supplied. Ergosterol, in the presence of linoleic or linolenic acids or a mixture of palmitoleic and oleic acids, permits excellent growth. Substitution of other sterols, such as cholesterol, or addition of oleic acid as the sole fatty acid, results in poor growth.

The genetic basis for limiting sterol synthesis in *S. cerevisiae* resides in the regulation of hydroxymethylglutaryl-CoA reductase activity. A decrease in the specific activity of this enzyme correlates with accumulation of squalene during supplementation with ergosterol or mevalonolactone. The addition of ergosterol results in feedback inhibition of hydroxymethylglutaryl-CoA reductase.

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CHAPTER 14

NITROGEN METABOLISM

Microorganisms play a major role in the nitrogen cycle. A unique group of bacteria fix atmospheric nitrogen (dinitrogen) into ammonia and assimilate the ammonia into amino acids. Certain nitrogen-fixing bacteria have established a symbiotic relationship with plants where they fix atmospheric nitrogen into forms that can be utilized by the host plants. Degradation of nitrogenous compounds by microorganisms is also an important aspect of the nitrogen cycle. Without this degradation and subsequent return of nitrogen from a wide variety of complex natural and artificial compounds to the nitrogen cycle, higher forms of life could not exist. An in-depth knowledge of these processes may aid materially in overcoming the imbalances created in the nitrogen cycle by overloading it with waste and excretory materials and improper use of agricultural lands by overabundant applications of fertilizers, herbicides, and pesticides.

In this chapter, we consider the contributions of microorganisms to the nitrogen cycle and the underlying mechanisms of the processes of nitrogen fixation, metabolism of inorganic nitrogen compounds, and assimilation of inorganic nitrogen into amino acids. While urea is not an inorganic compound, it is widely used as a fertilizer because it is readily broken down into carbon dioxide and ammonia. For this reason, urea metabolism is also described in this chapter. Reactions involved in the interconversion of the amino acids, especially amino group transfer (transamination), and other important reactions of amino acids are discussed. The biosynthesis of amino acids, purines, pyrimidines, and other nitrogen-containing compounds is covered in Chapters 15 and 16.

BIOLOGICAL NITROGEN FIXATION

Dutch farmers associated the establishment of rich stands of clover with the improvement of subsequent crops grown in the same fields. Although they were

unaware of the precise nature of the relationship, they noted the development of nodules on the root system of clover. These observations provided the foundations for systematic crop rotation implemented by Townsend in England during the agricultural revolution of the eighteenth century. The photograph in Figure 14-1 shows extensive nodule development on the root system of a soybean plant, which is a legume. A legume is a plant that bears seed pods, such as pea or soybean. These plants often form a symbiotic association with bacteria that form prominent nodules on the roots. These nodules in the root system harbor millions of bacteria that convert atmospheric nitrogen (dinitrogen) to ammonia (NH_3).

The works of Atwater and of Hellriegel and Wilfarth, published in the 1880s, were among the first to provide scientific data supporting the importance of the root nodules and the bacteria within them in the process of nitrogen fixation by clover and other leguminous plants. Despite criticisms by scientists of the time, their reports initiated a concerted effort to determine the nature of the symbiotic organisms associated with the

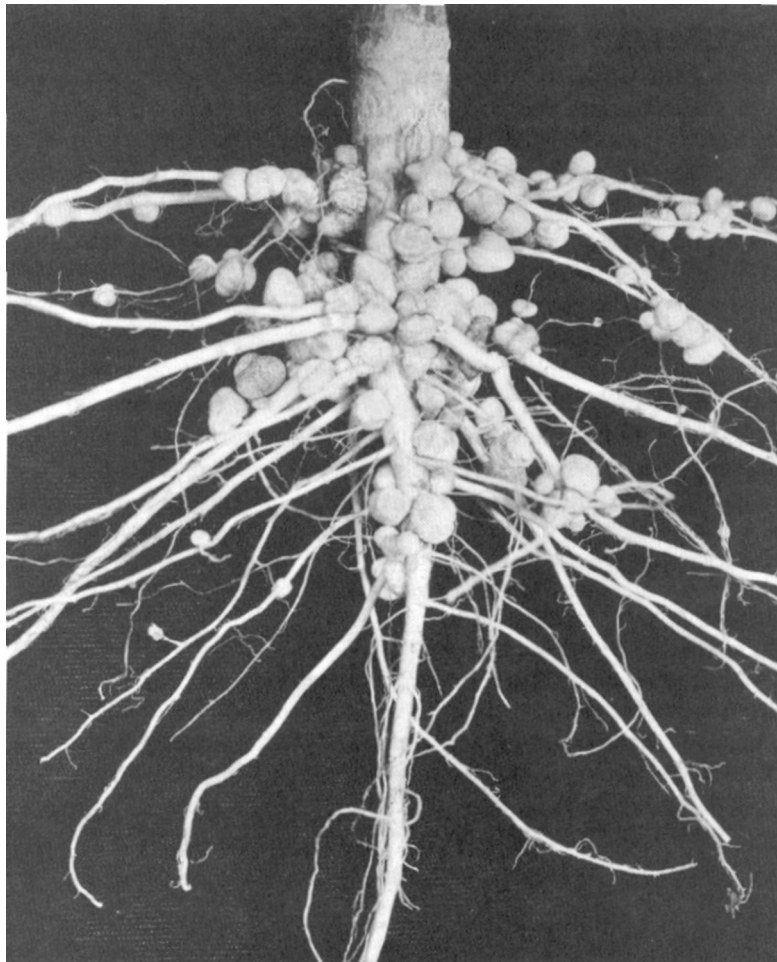


Fig. 14-1. Extensive root nodule development in the root system of soybean (*Glycine max*), a leguminous plant. (Photo courtesy of R. S. Smith, Milwaukee, WI.)

root nodules of legumes (and other plants), as well as characterization of the numerous free-living microorganisms capable of nitrogen fixation.

For many years, it has been considered that the ammonia produced by the bacteria in the nodule was transferred to the plant where it was then assimilated into amino acids and other nitrogenous compounds. More recently, it has been determined that under certain conditions the bacteria assimilate the ammonia into the amino acid alanine, which is then transported into the plant system. A number of examples of nitrogen-fixing organisms and their associated plant partners are listed in Table 14-1.

Symbiotic nitrogen-fixing organisms and the photosynthetic nitrogen fixers appear to account for most of the atmospheric nitrogen assimilated into organic forms in nature. Under conditions in which fixed nitrogen is low, root-nodulated angiosperms and gymnosperms and the cyanobacteria may be especially valuable in improving soil fertility. The estimated amount of nitrogen fixed by symbiotic bacterial-inoculated legumes in the United States approximately equals the amount of nitrogen supplied by farmers as nitrogen fertilizer. In the past, the contribution of free-living forms was probably underestimated. With the introduction of the acetylene reduction technique* for assessing the nitrogen-fixing potential of organisms in natural environments, it has been possible to show the presence of nitrogen-fixing organisms in a number of settings, including the intestinal tract of nonruminant mammals. Although free-living organisms, in general, appear less efficient in their ability to fix nitrogen, their number, variety, and ubiquitous distribution suggest that they are of major ecological importance. The cyanobacteria have a distinct advantage in that the energy (as ATP) and the reducing power (as NADH) required for nitrogen fixation can be supplied by photosynthesis, a process that makes it possible for them to become established in environments unfavorable for the development of other nitrogen-fixing organisms.

Rapidly rising energy and labor costs have made it less economical to increase plant growth by the use of ammonia fertilizer. Commercial production of ammonia by the Haber-Bosch process (catalyst-mediated reduction of hydrogen with nitrogen under high pressure and temperature) is expensive. Furthermore, continued application of chemical fertilizers at a high rate has threatened water supplies and the ecological balance of rivers and streams. Thus, attention has been focused on the improvement of plant yields through the development of new associations between nitrogen-fixing bacteria and plants. In the pursuit of this goal, intensive efforts have been directed toward studies of the genetics, biochemistry, and ecology of both free-living and symbiotic nitrogen-fixing organisms. It has already become apparent that the photosynthetic capacity of plants is one limiting factor in nitrogen fixation. Work on the selection and breeding of plant strains with greater photosynthetic efficiency has also been intensified. Another long-range goal has been to attempt development of an association between nitrogen-fixing bacteria and the root system of highly efficient photosynthetic plants such as corn.

* Originally, mass spectrometry was used to measure the amount of ^{15}N -labeled nitrogen gas reduced to ammonia by nitrogenase. The discovery that nitrogenase can reduce acetylene to ethylene proved very useful, since this reaction can be measured by the somewhat simpler technique of gas chromatography.

TABLE 14-1. Examples of Nitrogen-Fixing Organisms*Symbiotic Association of Various Genera with Leguminous Plants*

Azorhizobium caulinodans, tropical legume (*Sesbania rostrata*)
Allorhizobium undicola, lotus (*Lotus albus*)
Bradyrhizobium japonicum, soybean (*Glycine max*)
Mesorhizobium amorphae, false indigo (*Amorpha fruticosa*)
Rhizobium trifolii, clover (*Trifolium*, *Crotolaria*)
Sinorhizobium meliloti, alfalfa (*Medicago sativa*)

Symbiotic Association of Actinomycetes with Angiosperms

Frankia sp., alder (*Alnus*)
Frankia sp., bog myrtle or sweet gale (*Myrica*)
Frankia sp., oleasters (*Shepherdia*, *Eleagnus*, *Hippophae*)
Frankia sp., New Jersey tea (*Ceanothus*)

Symbiotic Association with Leaf-Nodulating Plants

Klebsiella aerogenes

Symbiotic Association of Marine Bacteria with Bivalves

Aerobic, chemoheterotrophic sp., bivalves (*Teredinidae*)

Symbiotic Association with Marine Diatoms

Richelia intracellularis (cyanobacterium), *Rhizosolenia*

Associative Interaction with Grasses

Azospirillum brasiliense, tropical grasses
Azospirillum lipoferum, tropical grasses, maize
Azotobacter paspali, tropical grass (*Paspalum notatum*)

Free-Living Bacteria and Cyanobacteria

Aerobic, heterotrophic

Azotobacter, *Derxia*, *Azomonas*, *Biejerinkia*, *Nocardia*, *Pseudomonas*

Aerobic, phototrophic

Anabaena, *Calothrix*, *Nostoc*, *Gleotheca*, *Cylindrospermum*,
Aphanocapsa

Facultative, heterotrophic

Enterobacter cloacae, *Klebsiella pneumoniae*, *Bacillus polymyxa*,
Desulfovibrio desulfuricans, *D. gigas*, *Achromobacter*

Anaerobic, heterotrophic

Clostridium pasteurianum, *C. butyricum*, *Propionispira arboris*

Anaerobic, phototrophic

Chromatium vinosum, *Rhodospirillum rubrum*, *Rhodopseudomonas*
sphaeroides, *R. capsulata*, *Rhodomicrobium vernielli*, *Rhodocyclus*,
Chlorobium limicola

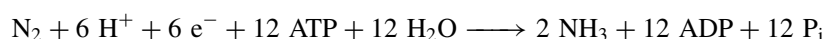
Nonphotosynthetic, autotrophic

Methanobacterium, *Methylococcus*, *Methylosinus*, *Methanococcus*,
Methanococcus, *Methanosarcina*

THE NITROGEN FIXATION PROCESS

Fixation of atmospheric dinitrogen (N_2 or $\text{N}\equiv\text{N}$) is accomplished by a variety of bacteria and cyanobacteria utilizing a multicomponent **nitrogenase system**. Despite the variety of organisms capable of fixing nitrogen, the nitrogenase complex appears to be remarkably similar in most organisms (Fig. 14-2). Nitrogenase consists of two oxygen-sensitive proteins. **Component I (dinitrogenase)** is a molybdenum-iron protein containing two subunits. **Component II (dinitrogenase reductase)** is an iron-sulfur protein that transfers electrons to dinitrogenase. These proteins, together with ATP, Mg^{2+} , and a source of electrons, are essential for nitrogen-fixing activity.

The overall process of nitrogen fixation is accomplished at considerable expense of energy, requiring from 12 to 16 molecules of ATP and 6 to 8 electrons, depending on the manner in which the equation is viewed. In one form, the equation may be written



This form of the equation does not take into account the fact that dihydrogen (H_2) is an obligate product of the nitrogenase reaction. If H_2 is considered, then the equation

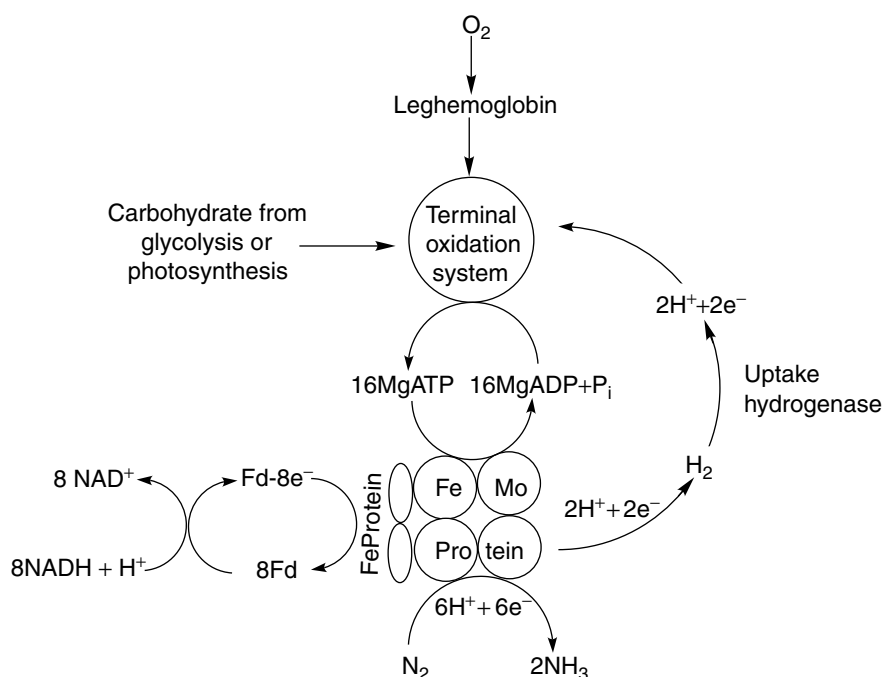
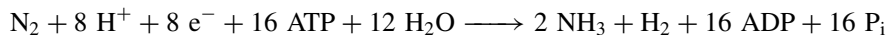
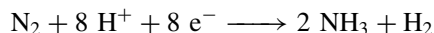


Fig. 14-2. The nitrogenase complex and the associated activities required for nitrogen fixation. The FeMo protein, dinitrogenase, is also referred to as component I. The Fe protein, dinitrogenase reductase (component II), contains a $4\text{Fe}-4\text{S}$ cluster that is not shown in the diagram. Fd, ferredoxin. The overall reaction requires $8 \text{NADH} + \text{H}^+$. Six of these are used to reduce N_2 to NH_3 and two are used to form H_2 . The uptake hydrogenase returns H to the system, thus conserving energy.

can be shown as



Thus, the theoretical stoichiometric relationship between N_2 fixation and H_2 production is related by the equation



The production of hydrogen during nitrogen fixation is an energy-expensive process. In actuality, most aerobic nitrogen fixers rarely evolve H_2 because a membrane-bound uptake hydrogenase recycles the H_2 and produces ATP through respiration to help support the ATP requirements of the system.

An anaerobic environment is essential for nitrogenase activity because of the oxygen lability of both proteins in the complex. An uptake hydrogenase coupled to a dioxygen-consuming pathway helps to maintain an anaerobic environment. Anaerobiosis is essential because oxygen represses the hydrogen uptake system. In the cyanobacteria, the problem of conducting oxygen-evolving photosynthesis and nitrogen fixation simultaneously is circumvented by sequestration of the nitrogenase system in specialized cells called heterocysts. Otherwise, they can only fix N_2 in the dark under anoxic conditions.

Components of the Nitrogenase System

The nitrogen fixation (*nif*) genes in *K. pneumoniae* are listed in Table 14-2 along with their known product or function. The nitrogenase complex of this organism contains two separable proteins. Component I (dinitrogenase) is a $\alpha_2\beta_2$ tetramer of 240 kDa encoded by *nifK* and *nifD*. The iron-molybdenum cofactor (FeMo-co) of the dinitrogenase is synthesized under the direction of six *nif* genes (*nifQ*, *B*, *V*, *N*, *E*, *H*). Mutants of *nifB* and *nifNE* accumulate an inactive apo-component I (ApoI). ApoI is an oligomer containing an additional protein, the product of *nifY*, which disassociates from the complex upon activation by the addition of FeMo-co with restoration of the ability to

TABLE 14-2. Nitrogen Fixation (*nif*) Genes and Their Products in *Klebsiella pneumoniae*^a

Gene	Product	Gene	Product
Q	FeMoCo, Mo uptake	U	Protein associated with FeS center
B	FeMoCo synthesis	X	Regulatory protein
A	Positive regulator	N	FeMoCo
L	Negative regulator	E	FeMoCo synthesis
F	Flavodoxin, electron acceptor	Y	Unknown function
M	Fe protein activation	T	Unknown function
Z	Insertion of FeMoCo protein	K	MoFe protein β -subunits
W	Activation of dinitrogenase	D	MoFe protein α -subunits
V	Homocitrate synthesis	H	Dinitrogenase reductase
S	Protein associated with FeS center	J	Pyruvate oxidoreductase

^aThe genes are listed in the order in which they appear on the detailed chromosome map of *Klebsiella pneumoniae*.

fix N_2 . Component II (dinitrogenase reductase, encoded by *nifH*) is an α_2 protein (ca. 60 kDa) containing a single four-iron four-sulfur (Fe_4S_4) center. This protein binds and hydrolyzes Mg ATP when an electron is transferred from reduced ferredoxin to dinitrogenase. Ferredoxin and/or flavodoxin can serve as electron donors.

Regulation of the 17 *nif* genes in *K. pneumoniae* is under the direction of the *nifLA* operon. The NtrC protein activates the transcription of the *nifLA* operon under the conditions of nitrogen limitation. The NifA protein is a positive regulatory factor required for *nif* gene transcription (Table 14-2). The *nifL* gene product interacts with the NifA protein to prevent NifA activation in the presence of fixed nitrogen (ammonia or amino acids) or oxygen. The functions of *nifZ*, *W*, *U*, *S*, *X*, and *T* are still not well defined. However, there is evidence that the products of the *nifW* and *nifZ* genes may be involved in processing one of the structural components of nitrogenase and that the product of the *nifX* genes is a positive regulator of the *nif* regulon in response to ammonia and oxygen (Fig. 14-3).

The nitrogen-fixing systems of *Azotobacter* species have been well characterized. Three different nitrogenases found in *A. vinelandii* and *A. chroococcum* are regulated by the Mo or V content of the culture medium. Nitrogenase 1 is produced by both organisms in the presence of Mo. Dinitrogenase reductase is composed of two identical subunits encoded by *nifH*. Dinitrogenase 1 is a tetramer of two pairs of nonidentical subunits encoded by *nifD* and *nifK*. The three structural genes appear in the order *nifHDK* and comprise an operon. Nitrogenase 2 is produced by both *A. vinelandii* and *A. chroococcum* grown in a nitrogen-free medium in the presence of V. Dinitrogenase reductase 2 is a dimer encoded by *vnfH*. Dinitrogenase 2, encoded by *vnfD* and

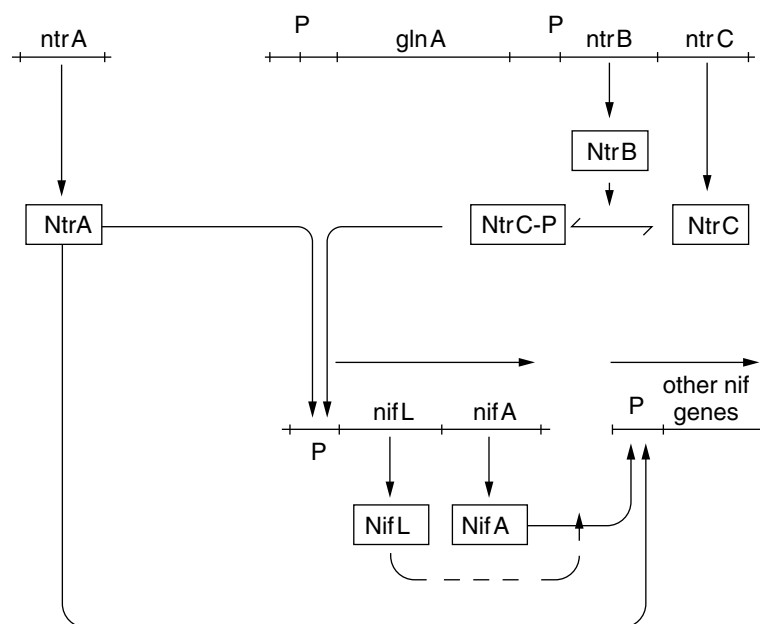


Fig. 14-3. Model for *nif* regulation in *Klebsiella pneumoniae*. The thin arrows indicate regulatory functions and the thick horizontal arrows represent transcripts. (Source: From Dixon, R. A., *J. Gen. Microbiol.* **130**:2745–2755, 1984.)

vnfK, is a tetramer composed of two pairs of subunits containing Fe and V. A third nitrogenase, nitrogenase 3, is encoded by alternate nitrogen fixation (*anfHDK*) genes, which are expressed in *A. vinelandii* only in the absence of Mo and V. Dinitrogenase reductase 3 contains two identical subunits and dinitrogenase 3 is present in two active configurations, $\alpha_2\beta_2$ and $\alpha_1\beta_2$. The regulatory genes *nifA*, *vnfA*, and *anfA* are required for the expression of nitrogenases 1, 2, and 3, respectively. An additional regulatory gene, *nfrX*, is required for growth on N_2 (diazotrophic growth) in the presence or absence of Mo.

The nitrogenase system from *Clostridium pasteurianum* is similar to that found in *K. pneumoniae* and *A. vinelandii*. Two proteins are required for nitrogen fixation and ATP-dependent H_2 evolution. One protein, component I, is an FeMo protein having dinitrogenase activity. A second protein, the Fe protein, or component II, has dinitrogenase reductase activity. Although there are structural similarities between the nitrogenase complexes of various organisms, the primary structure of those from *C. pasteurianum* is quite different from those of *K. pneumoniae* and *A. vinelandii*. The nitrogenase components from a wide variety of bacteria can interact successfully with one another. The FeMo and Fe proteins from different organisms show a remarkably high degree of successful formation of active hybrid nitrogenases. However, the components of *C. pasteurianum* nitrogenase are much less effective in forming heterologous complexes than are mixtures of the components from gram-negative organisms.

A number of phototrophic bacteria including strict anaerobes such as *Chromatium*, *Chlorobium*, and *Rhodospirillum rubrum* can fix nitrogen anaerobically in the light because their photosynthetic systems do not evolve O_2 . However, the most important phototrophs from an ecological viewpoint are the cyanobacteria. They are capable of performing oxygenic photosynthesis and fixing CO_2 through the reductive pentose phosphate pathway (see Chapter 12). Cyanobacteria lack α -ketoglutarate dehydrogenase but utilize this intermediate as a substrate for nitrogen assimilation. Nitrate, ammonia, urea, and N_2 are used as nitrogen sources. In the absence of fixed nitrogen, most cyanobacteria are able to fix N_2 using a nitrogenase system similar to that described in other diazotrophic organisms. However, they face a dilemma in that nitrogenase is extremely sensitive to oxygen. *Anabaena* and *Nostoc* form heterocysts, which are specialized cells that contain most of the nitrogenase activity. In heterocysts, the photosynthetic system is effectively inactivated and any residual O_2 is eliminated by highly active respiration.

Some cyanobacteria such as *Anabaena variabilis* produce two Mo-dependent nitrogenases: Nif1 functions in heterocysts while Nif2 functions under anoxic conditions in vegetative cells. A number of cyanobacteria that do not form heterocysts are also known. They apparently operate under microaerophilic conditions when fixing nitrogen. Among the free-living nitrogen-fixing bacteria, the cyanobacteria contribute much more significantly to the soil nitrogen because they obtain their energy from photosynthesis and are therefore capable of colonizing soil or other habitats where conditions are sparse. They also function symbiotically with primitive plants such as lichens, liverworts, and water ferns.

Of the many genes required for heterocyst function, only *ntcA*, *hanA*, and *hetR* appear to be required for the initiation of heterocyst development. NtcA functions as a nitrogen-dependent global transcriptional regulator in all cyanobacteria. In *Anabaena*, *ntcA* mutants fail to grow on nitrate, do not form heterocysts, or undergo nitrogenase

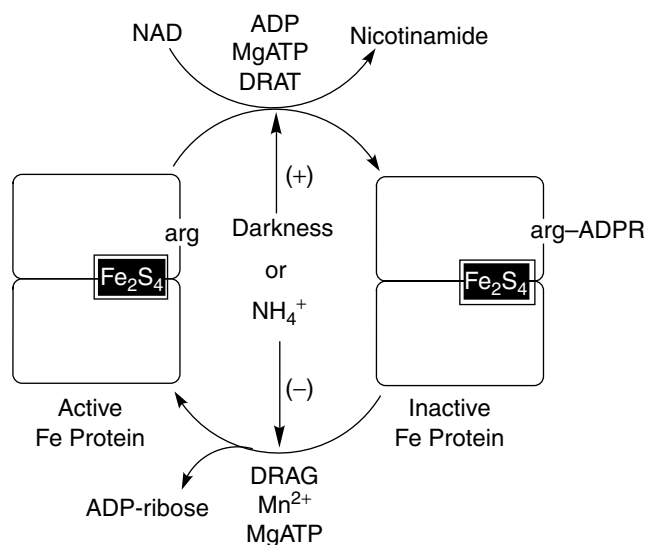


Fig. 14-4. Diagrammatic view of regulation of nitrogenase by ADP ribosylation of the Fe protein in *Rhodospirillum rubrum*. DRAG, dinitrogenase reductase-activating glycohydrolase; DRAT, dinitrogenase reductase ADP ribosyltransferase; Fe₂S₄, the iron-sulfur in the Fe protein; arg, arginine 101 position in the Fe protein. (Simplified version of diagram in C. M., Halbleib et al., *J. Biol. Chem.* **275**:3493–3500, 2000.)

synthesis. The *hetF* gene product is essential for heterocyst development in the filamentous cyanobacterium *Nostoc punctiforme*. HetF appears to cooperate with HetR in a positive regulatory pathway in differentiating heterocysts.

Nitrogenase synthesis in a variety of organisms is generally subject to the close regulatory controls at the level of *nif* gene transcription. However, many of these organisms also display a posttranslational regulation of nitrogenase activity effected by small extracellular concentrations of ammonia. In some organisms this rapid and reversible inhibition of nitrogenase, termed **ammonia switch-off**, is the result of a covalent modification of the Fe protein (dinitrogenase reductase) in response to the addition of ammonia. Formation of an inactive form of nitrogenase in the photosynthetic bacterium *R. rubrum* results from ADP ribosylation of an arginine residue in the Fe-protein component (Fig. 14-4). This activity is catalyzed by dinitrogenase reductase ADP ribosyltransferase (DRAT). Activation of the modified protein by removal of the ADP-ribose moiety is catalyzed by dinitrogenase reductase activating glycohydrolase (DRAG). Other systems in which covalent modification of the Fe protein occurs include *A. brasiliense*, *A. lipoferrum*, and *A. vinelandii*. In *A. amazonense* a different, noncovalent inhibitory mechanism results in only a partial inhibition of nitrogenase activity by ammonia. In *Rhodospirillum rubrum* nitrogen fixation is also tightly regulated by transcriptional regulation of *nif* gene expression.

SYMBIOTIC NITROGEN FIXATION

The successful establishment of a symbiotic relationship between a bacterium such as *Sinorhizobium meliloti* and its host plant (*Medicago sativa*, alfalfa), culminating in the

formation of nodules that conduct active nitrogen fixation, is a complex process that occurs in several stages: attachment of the bacteria to root hairs, root hair curling, formation of a “shepherd’s crook,” development of infection threads within root hairs, growth of the threads toward the inner cortex of the root, and formation of a nodule meristem in the inner root cortex (Fig. 14-5). These cytological changes in the root hair provide a means of passage of the organism into the internal root system where it infects a root cell. This infection causes the cell to swell and divide, forming a thick mass of cells called the root nodule. In the root nodule a differentiated form of the bacterium, termed a **bacteroid**, is capable of nitrogen fixation.

A portion of the genomes of both the plant and the bacteria are expressed only in the symbiotic state. Considering the complexity of the interaction between the bacterium and the host plant in the development of effective symbiosis, it is not surprising that attempts to develop strains that are more efficient in their nitrogen-fixing capability have been hampered by the fact that indigenous strains in the soil are more competitive than laboratory-developed strains in their ability to initiate nodule development. Nevertheless, continued efforts to engineer more effective strains of nitrogen-fixing bacteria are important because of the increase in the cost of nitrogen fertilizer and the undesirable contamination of rivers and streams that occurs through the heavy application of it.

The nodulation process requires the exchange of a series of signals between the plant and bacteria. Initially, plant genes coding for the production of flavonoids are expressed, and these compounds are excreted from the roots. Flavonoids activate NodD1 and start

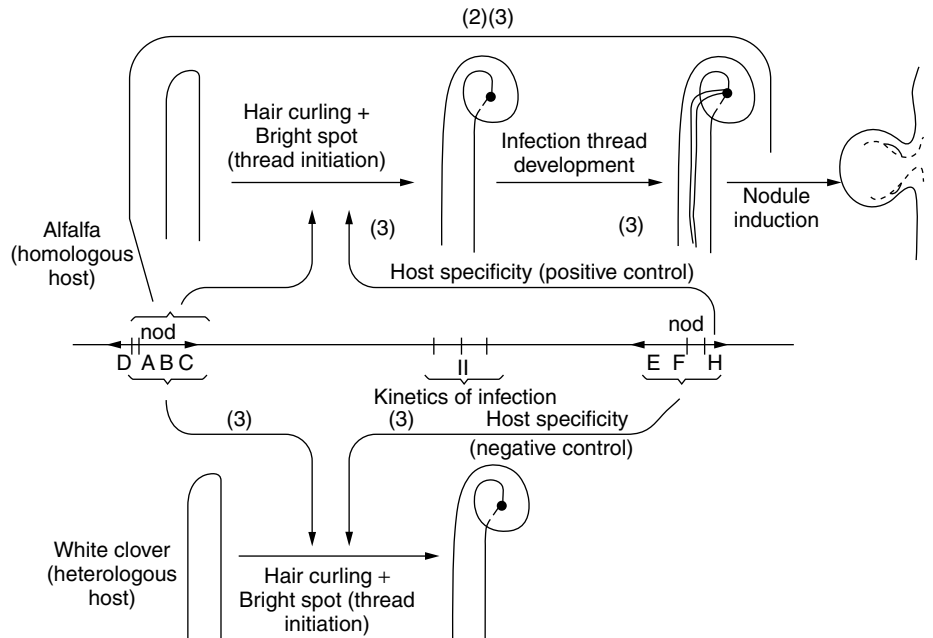


Fig. 14-5. Root infection and modulation by *Sinorhizobium meliloti* in *Medicago sativa* (alfalfa). Assignment of development step control in common and specific *nod* loci of *S. meliloti*. Thick arrows represent development steps. Thin arrows represent genetic control. (Source: From Debelle, F., et al., *J. Bacteriol* **168**:1075–1086, 1986.)

the sequence of Nod (nodulation factor) synthesis. Nod factors induce root hair curling and initiate nodule development (Fig. 14-6). Invasion of the root nodule by rhizobia is dependent on the production of at least one of several complex polysaccharides: succinoglycan, exopolysaccharide II, or a capsular polysaccharide (CPS or KPS) containing 3-deoxy-D-manno-2-octulosonic acid (see Fig. 14-7).

The early events in root nodule formation by rhizobia in leguminous plants are under the control of genes located on a large plasmid called **symbiosis** or **Sym plasmid**. Plant

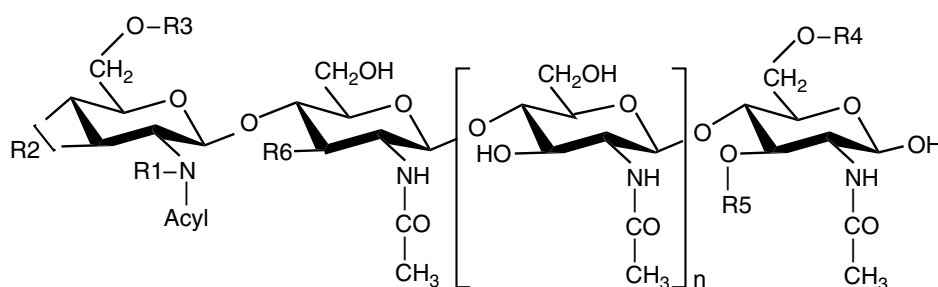


Fig. 14-6. General structure of the Nod factors produced by Rhizobia. Acyl substituents are most often C₁₆, C₁₈, or C₂₀ saturated or unsaturated fatty acids; R1 is usually a methyl group or H; R2 is frequently a carbamoyl group or OH; R3 may be H, a carbamoyl group, or an acetyl group; R4 may be fucose, methylated fucose, acetylated fucose, sulfate, or H; R5 is usually H, but may occasionally be an arabinosyl substituent; R6 is usually OH, but may occasionally be fucose or an acetyl group; n usually varies between 1 and 2, but may occasionally be 3 or 0.

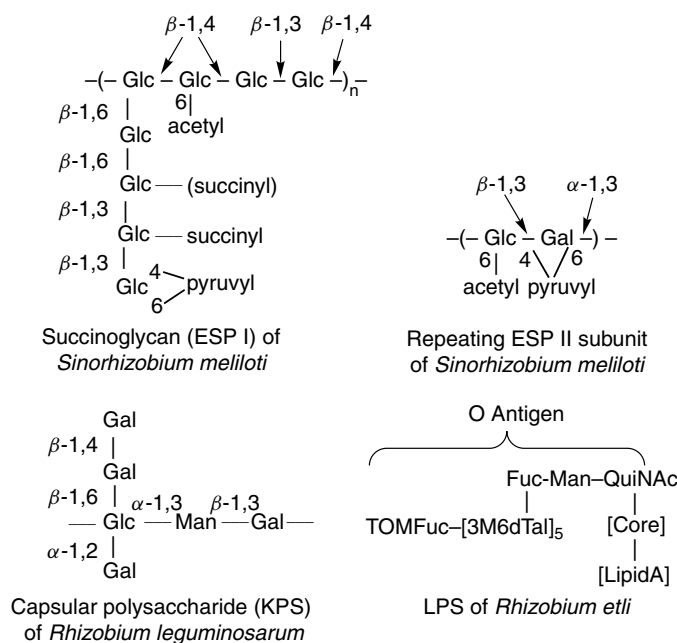


Fig. 14-7. Polysaccharides involved in nodule development by rhizobia. Glc, glucose; Gal, galactose; Man, mannose; TOMFuc, tri-O-methylated fucose; 3M6dTal, 3-O-methyl-6-deoxytalose; Fuc, fucose; QuiNAc, *N*-acetyl quinovosamine (2-amino-2,6-dideoxyglucose.)

TABLE 14-3. Genes Involved in Nodulation and Their Known Function or Activity

Designation	Function or Activity
<i>nodD</i>	Positive transcription activator; binds to the <i>nod</i> box upstream of all inducible operons
<i>nod</i> box	Highly conserved DNA sequence; <i>cis</i> -acting regulatory element of <i>nod</i> gene expression
<i>nodA</i>	<i>N</i> -Acyltransferase
<i>nodB</i>	Oligochitin deacetylase
<i>nodC</i>	Oligochitin synthase
<i>nodFEL</i>	Operon involved in addition of acyl substituents to core lipooligosaccharide
<i>nodH</i>	Addition of substituents to core lipooligosaccharide
<i>nodI</i>	Involved in efficiency of nodulation and development of infectious threads
<i>nodJ</i>	Involved in efficiency of nodulation and development of infectious threads
<i>nodM</i>	Fructose-6-phosphate:glutamine amidotransferase
<i>nodP</i>	Sulfate + APS (adenylyl sulfate) → PAPS (3'-phosphoadenylyl sulfate)
<i>nodQ</i>	Addition of substituents to core lipooligosaccharide
<i>nodS</i>	<i>N</i> -methyltransferase
<i>nodU</i>	6-O-Carbamoyltransferase
<i>nodX</i>	Addition of acetyl substituent to core lipooligosaccharide
<i>nodZ</i>	6-O-Fucosyltransferase
<i>noeA</i>	Sulfuryltransferase
<i>noeC</i>	Addition of substituent to core lipooligosaccharide
<i>noeI</i>	2-O-Methyltransferase
<i>noeJ</i>	Mannose-1-phosphate guanylttransferase
<i>noeL</i>	GDP-Mannose-4,6-dehydratase
<i>noeK</i>	Phosphomannose mutase
<i>nolL</i>	O-Acetyltransferase
<i>nolO</i>	Carbamoyltransferase
<i>nolK</i>	Fucose synthase

flavonoids, in concert with NodD, induce the expression of *nod* genes. The NodD protein serves as a positive transcription activator. It binds to the *nod* box, located upstream of all inducible *nod* genes and operons. The *nod* box, a highly conserved DNA sequence, appears to function as a *cis*-acting regulatory element of *nod* gene expression. The *nodABC* genes common to all rhizobia are required for the synthesis of a lipooligosaccharide that triggers root nodule formation. Addition of various substituents to the core compound imparts host specificity to the lipooligosaccharide. Addition of these components occurs under the influence of *nodH* and *nodQ* and the *nodFEL* operon (see Table 14-3). The NodI and NodJ proteins are involved in efficiency of nodulation and play a role in the normal development of infection threads. Once bacteroid development is complete, the inducible *nod* genes are no longer transcribed. This transcription switch-off prior to the release of the bacteria from the infection thread is a general phenomenon observed in all rhizobia and is the result of a negative regulatory control mechanism.

Under the conditions of symbiosis, the host plant provides the bacteroid with reduced carbon in the form of C₄-dicarboxylic acids (succinate, malate, and fumarate). These compounds serve as energy sources for the fixation of nitrogen to ammonia. Dicarboxylic acids are present in high concentration in the nodule and are the most effective substrates for respiration and subsequent ATP-utilizing nitrogen fixation in the bacteroid. The dicarboxylic acid transport genes (*dct*) are located on a megaplasmid. Mutants defective in dicarboxylate transport form ineffective nodules. The products of *dctB* and *dctD* regulate the expression of *dctA*, which encodes a transport protein. DctB is a sensor protein that activates DctD by phosphorylation. The DctD protein activates transcription at the σ^{54} -dependent *dctA* promoter. In addition to activating *dctA* transcription, DctD can repress the expression of *dctA*. In uninfected cells, inactive DctD binds to the *dctA* promoter and prevents its activation by NtrC.

For many years, the paradigm of symbiotic nitrogen fixation has been the release of ammonia directly from the bacteroid to the plant. Active ammonia transport systems have been described in some symbiotic microorganisms. However, recent studies reveal that alanine, rather than ammonia, is the form of nitrogen transported to the plant. It has been shown that rhizobia can assimilate ammonia into pyruvate, forming alanine via alanine dehydrogenase. The equilibrium of the alanine dehydrogenase reaction favors ammonia assimilation.

Labeling experiments with ¹⁵N₂ using highly purified nodule bacteroids of *B. japonicum* show that recently formed ammonia from nitrogenase was incorporated into alanine and then transported to the plant from the bacteroid. Secretion of labeled alanine was dependent on nitrogenase activity as bacteroids exposed to inactivating levels of oxygen or to the energy uncoupling agent CCCP (carbonyl cyanide *m*-chlorophenylhydrazone) did not excrete alanine. The peribacteroid membrane is inverted with respect to the bacteroid, so alanine secretion by the bacteroid to the plant cytoplasm can occur quickly, but it cannot be readily reabsorbed. It is considered that alanine serves primarily as a transport mechanism for fixed nitrogen, since most organisms use glutamate as the central compound in nitrogen metabolism (see "General Reactions of Amino Acids".)

INORGANIC NITROGEN METABOLISM

The assimilation of inorganic nitrogen ends with the incorporation of ammonia into organic compounds. Since ammonia is the only form of inorganic nitrogen that can be directly assimilated into amino acids, the ability of an organism to utilize other forms of inorganic nitrogen depends on the presence of enzymes or enzyme systems that are able to convert these compounds to ammonia. This process is referred to as **denitrification**. The reverse process, converting ammonia to nitrate and nitrite, is called **nitrification** (Fig. 14-8). Table 14-4 provides examples of organisms known to carry out reactions of inorganic nitrogen. Although it is not an inorganic compound, the metabolism of urea is discussed in this section because it is so readily degraded to ammonia and carbon dioxide by ureases found in many genera of bacteria and fungi.

Many of the organisms that conduct the reactions shown in Table 14-4 do not actually assimilate the nitrogen. In some organisms, nitrate may be used as a terminal electron acceptor in place of oxygen. The end product is nitrite or N₂. Nitrate respiration yields biologically useful energy under anaerobic conditions. Nitrate assimilation occurs by sequential reduction to nitrite, hydroxylamine, and, finally, ammonia. Organisms that

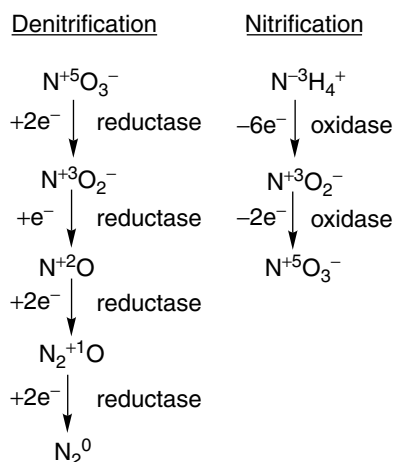
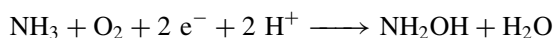


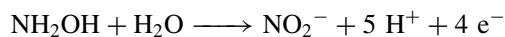
Fig. 14-8. Pathways of inorganic nitrogen metabolism. Denitrification and nitrification. The superscript numbers on the nitrogen indicate the valence state or oxidation level of the nitrogen in the compounds.

conduct both nitrate respiration and nitrate assimilation contain two nitrate reductases. In *E. coli* and *Neurospora* the assimilatory enzyme is a soluble cytoplasmic enzyme, whereas the respiratory enzyme is membrane-bound and sensitive to oxygen. Both enzymes contain flavin as the prosthetic group. In most organisms the assimilatory enzyme is repressed by ammonia and induced in the presence of either nitrate or nitrite. Algae readily utilize nitrate as a source of inorganic nitrogen, but the route of assimilation is less well characterized than in bacteria and fungi. Ammonia is considered to be the final product of nitrate reduction by algae.

Denitrification by members of the obligately chemolithotrophic *Nitrobacteriaceae* is considered to be the major source of assimilable nitrogen (ammonia) in soils. The ammonia oxidizer, *Nitrosomas europaea*, and the nitrite oxidizer, *Nitrobacter winogradskyi*, have been widely studied. *Nitrosomonas europaea* obtains all of its energy for growth from the oxidation of ammonia to nitrite. The oxidation of ammonia to hydroxylamine is an oxygen-dependent reaction catalyzed by ammonia monooxygenases:



Hydroxylamine is further oxidized to nitrite by hydroxylamine oxidoreductase:



Two of the four electrons generated from hydroxylamine oxidation are used to support the oxidation of additional ammonia molecules, while the other two electrons enter the electron transfer chain and are used to support ATP synthesis via oxidative phosphorylation and CO_2 reduction. The oxidative chain from NO_2^- to O_2 in *N. winogradskyi* consists of membrane-bound cytochrome *c* oxidase. *Nitrobacter* is obligately dependent on nitrite as the sole source of nitrogen and energy and requires carbon dioxide for growth. *Nitrobacter* is, therefore, dependent on *Nitrosomonas* or other organisms in the natural environment for its nitrogen supply.

TABLE 14-4. Known Biological Reactions of Inorganic Nitrogen

Reaction	Electrons Required	Representative Organisms Performing Reaction	Requirements (comments)
$\text{NO}_3^- \rightarrow \text{NO}_2^-$	2	<i>Escherichia coli</i> , <i>Micrococcus denitrificans</i> , <i>Bacillus subtilis</i> , <i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i> , <i>Neurospora crassa</i> , <i>Achromobacter fischeri</i>	Varies among organisms NAD, NADP, FAD, FMN, Mo, cytochromes <i>c</i> , <i>b</i>
$\text{NO}_2^- \rightarrow \text{NH}_4^+$	6	<i>Bacillus pumilus</i> , <i>Neurospora crassa</i> , <i>Bacillus pumilis</i> , <i>Clostridium pasteurianum</i> , <i>Desulfovibrio desulfuricans</i>	NADPH, NADH, FAD, Cu, Fe
NO_3^- or $\text{NO}_2^- \rightarrow \text{NO}$, N_2O , and N_2 Denitrification	Varies from 1 to 10	<i>Micrococcus denitrificans</i> , <i>Denitrobacillus</i> , <i>Pseudomonas</i> , <i>Spirillum itersonii</i> , <i>Bacillus licheniformis</i> , <i>Achromobacter</i> , <i>T. thioparus</i> <i>Thiobacillus denitrificans</i>	See requirements for individual reactions
$\text{NO}_2^- \rightarrow \text{NO}$ Nitrite reduction	1	<i>Pseudomonas stutzeri</i> , <i>E. coli</i> , <i>Bacillus subtilis</i> , <i>P. aeruginosa</i>	NAD, NADP, FMN, FAD, Fe^{2+}
$\text{NO} \rightarrow \text{N}_2$ or N_2O Nitric oxide reductase	2 or 4	<i>Pseudomonas stutzeri</i> , <i>E. coli</i> , <i>Bacillus subtilis</i> , <i>Pseudomonas aeruginosa</i> , <i>Clostridium pasteurianum</i> , <i>Pseudomonas aeruginosa</i>	NAD, NADP, FAD, FMN, Fe^{2+}
$\text{N}_2\text{O} \rightarrow \text{N}_2$ Nitrous oxide reductase	2	<i>Pseudomonas stutzeri</i>	Cu
$\text{H}_2\text{N}_2\text{O}_2 \rightarrow 2\text{NH}_4^+$ Hyponitrite assimilation	8	<i>Neurospora crassa</i> , <i>E. coli</i>	NADPH, Fe^{2+}

(continued overleaf)

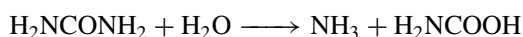
TABLE 14-4. (continued)

Reaction	Electrons required	Representative Organisms Performing Reaction	Requirements (comments)
$N_2 \rightarrow 2NH_4^+$ Nitrogen fixation	6	See Table 14-1 for representative organisms	Mo, Fe, V
$NH_2OH \rightarrow NH_4^+$ Hydroxylamine reductase	2	<i>Neurospora crassa</i> , <i>Azotobacter vinelandii</i> , <i>Desulfovibrio desulfuricans</i> , <i>Bacillus pumilis</i> , <i>Clostridium pasteurianum</i> , <i>Pseudomonas</i> <i>aeruginosa</i>	Varies—NADH, FAD, Mn^{2+}
$NH_2NH_2 \rightarrow 2NH_4^+$ Hydrazine reductase	2	<i>Micrococcus lactilyticus</i>	Measured hydrogen uptake with hydrazine added
$NH_4^+ \rightarrow$ organic compounds	In some cases, 2	Large group	Varies
$NH_4^+ \rightarrow NO_2^-$ or $NH_2OH \rightarrow NO_2^-$ Nitrification	–6 or –4	<i>Nitrosomonas</i> spp.	Acceptor—cytochrome <i>c</i> or phenazine methosulfate
$NO_2^- \rightarrow NO_3^-$ Nitrite reductase	–2	<i>Nitrobacter</i> spp.	Cytochrome, Fe^{3+}

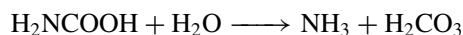
Nitrite formation by heterotrophic soil organisms appears to be rather limited. Nitrate formation, on the other hand, is accomplished by a significant number and variety of heterotrophic species present in the soil microflora. *Aspergillus flavus* is particularly active and must be considered as a major source of nitrate in soils.

Denitrifying bacteria, such as *Pseudomonas stutzeri*, use the reduction of nitrous oxide (N_2O) to dinitrogen (N_2) for the generation of energy. Reduction of N_2O is usually the terminal step of bacterial denitrification, proceeding from nitrate to nitrite to nitric oxide to nitrous oxide and the final reduction of nitrous oxide to N_2 by N_2O reductase. A gene cluster containing the genes necessary for the reduction of nitrite (*nir*), nitric oxide (*nor*), and nitrous oxide (*nos*) has been identified in *P. stutzeri*. There are at least 15 genes in this cluster arranged in the order *nos-nir-nor*. The structural gene *nosZ* codes for the copper-containing enzyme N_2O reductase. Three other genes, *nosDFY*, are involved in the formation of the copper chromophore of the enzyme. A regulatory gene, *nosR*, is present within the *nos* region of the denitrification gene cluster of *P. stutzeri*. It is essential for the expression of the *nosZ* gene, and there is indirect evidence that the transcriptional regulator Fnr may also be involved in *nosZ* expression (see Chapter 5 for a discussion of regulatory mechanisms).

Urea is a simple organic compound with the structure H_2NCONH_2 . It is widely used as an inexpensive nitrogen fertilizer because it is readily degraded to ammonia and carbamate by the enzyme urease:



Carbamate spontaneously decomposes to yield ammonia and carbonic acid:



Complete urease gene clusters have been found in *Bacillus* spp, *Helicobacter pylori*, *Klebsiella aerogenes*, *Proteus mirabilis*, and *Yersinia enterocolitica*. In addition, a number of other bacteria, including soil bacteria such as *Sinorhizobium meliloti*, have been studied in some detail with regard to the genetics of urease production and its regulation. The enzyme has been identified as a virulence factor in *H. pylori* and other organisms that cause infections in humans and animals. Urease activity can serve as a diagnostic test for infection by *H. pylori*. Urease is notable because it was the first enzyme to be crystallized—a feat accomplished by Dr. James B. Sumner of Cornell University in 1926 and for which he later received the Nobel Prize.

In enteric bacteria urease expression is positively regulated and transcription is activated either in the absence of an assimilable nitrogen source or in the presence of urea. On the other hand, in *Streptococcus salivarius*, urease expression is derepressed at low pH and is enhanced in the presence of excess carbohydrate. Ureolysis by *S. salivarius* appears to provide protection of the organisms against acid damage and may also enable the organisms to acquire nitrogen when carbohydrates are present in excess. Wild-type strains of *S. salivarius* are protected against acid killing through physiologically relevant concentrations of urea, whereas a urease-deficient mutant is not. This organism can use urea as a source of nitrogen for growth exclusively through a urease-dependent pathway.

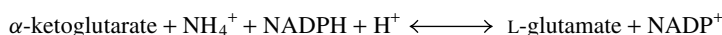
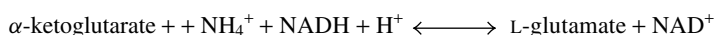
The bacterial urease from *K. pneumoniae* is composed of three basic subunits, UreA, UreB, and UreC, which appear as a trimer in the intact enzyme. Each UreABC

unit contains two coordinated nickel ions, separated from one another by 3.6 Å and bridged by a carbamylated lysine residue. Within this center are three metal-coordinated water molecules, four histidine ligands, one aspartate ligand, and two noncoordinated histidines that function in the conversion of urea into ammonia and carbon dioxide.

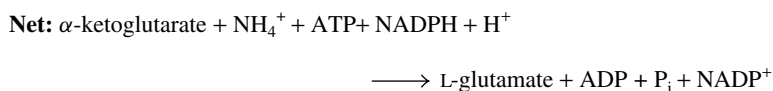
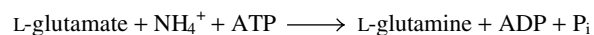
ASSIMILATION OF INORGANIC NITROGEN

Many microorganisms synthesize all of their amino acids and other nitrogenous compounds using ammonia and carbon chains derived from carbohydrate metabolism. The available pathways for ammonia assimilation are quite limited, however. Of all the amino acids found in proteins and other cellular constituents, ammonia can be directly assimilated into only a few. These amino acids then serve as donors of their amino nitrogen via transamination to keto acid precursors to form all of the other amino acids. The potential routes of ammonia assimilation are through the synthesis of glutamate, alanine, or aspartate. The major enzymes involved in ammonia assimilation are the glutamate dehydrogenases (GDH) and two enzymes that operate in tandem, glutamine synthetase (GS) and glutamate synthase, as follows:

Glutamate dehydrogenases (GDH)



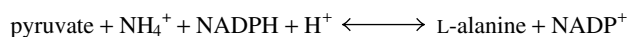
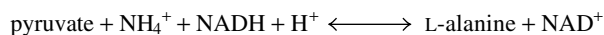
Glutamine synthase (GS)–glutamate synthase (GOGAT)



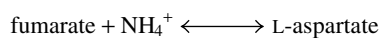
The abbreviation GOGAT frequently used for this enzyme system is derived from its previous trivial name, glutamine amide-2-oxoglutarate amino transferase.

Other enzymes that may play a role in ammonia assimilation in some organisms are alanine dehydrogenases and aspartase:

Alanine dehydrogenases



Aspartase

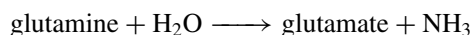


The route(s) of ammonia assimilation varies from one organism to another depending on the ammonia assimilation enzymes present. In the majority of organisms that have been studied, glutamate is the most widely used route for ammonia assimilation. In organisms capable of synthesizing both the NADP-linked glutamate dehydrogenase and the GS–GOGAT system, the NADP–GDH is functional at high concentrations of ammonia while the GS–GOGAT system is most active at concentrations of ammonia below 1 mM. These pathways are highly regulated as the concentration of ammonia, glutamate, and glutamine are key sensors that relate the nitrogen status of the organism. The yeast, *Saccharomyces cerevisiae*, assimilates at least two-thirds of its amino nitrogen requirement via glutamate, and, when the ammonia concentration in the culture medium is high, utilizes the NADP–GDH for glutamate synthesis. At low concentrations of ammonia, the GS–GOGAT system is operative. In nitrogen-fixing species of *Bacillus* (*B. polymyxa*, *B. macerans*), NADP–GDH activity is several-fold higher than that of GS and is the predominant pathway for ammonia assimilation.

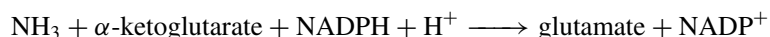
A few organisms appear to be incapable of forming the GS–GOGAT system, and, therefore utilize one or more of the alternative routes for ammonia assimilation. *Rhodospirillum rubrum*, which does not fix N₂, uses exogenously supplied ammonia via the NADP-linked GDH. The only ammonia assimilation pathway in *Streptococcus sanguis*, *S. bovis*, *S. mutans*, and *S. salivarius* is NADP–GDH regardless of the external ammonia concentration. In some *Bacillus* species, the NADP–ADH and NADP–GDH enzymes are highest in activity when ammonia is in high concentration. Under N₂-fixing conditions rhizobia appear to form alanine via ADH and export alanine rather than ammonia to the plant host.

Aspartase is sufficiently active in *Klebsiella aerogenes* to function as an important assimilatory enzyme, particularly after growth on C₄-dicarboxylic acids. Under most conditions of growth, however, aspartase appears to serve in a dissimilatory (ammonia-releasing) role.

As described in Chapter 5, the GS–GOGAT system and related enzymes are highly regulated by the Ntr (nitrogen regulation) system, particularly in *E. coli* and *K. aerogenes*. On the other hand, in *Bacillus* spp. there is no evidence for a global regulatory system analogous to the Ntr system. Activity of the GS–GOGAT system in *B. subtilis* and *B. licheniformis* is regulated by the available nitrogen source (feedback inhibition). The GS regulatory protein GlnR negatively regulates transcription of the *B. subtilis* GS structural gene, while GltC stimulates the expression of GS. In *B. licheniformis*, the GOGAT enzyme (GltS) consists of two unequal subunits. The larger subunit catalyzes the glutaminase reaction:



An ammonia transfer reaction is catalyzed by the small subunit:



In *B. subtilis*, GS (GlnA) and a regulatory protein, GlnR, are encoded in an operon. Regulation of the *glnRA* operon involves the action of both GlnR and GS. Here GlnR is a repressor that interferes with transcription under conditions of nitrogen excess.

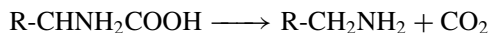
In the nitrogen-fixing anaerobe *Clostridium kluyveri*, the NADP–GDH pathway plays an important role in ammonia assimilation in ammonia-grown cells but plays only a minor role to that of the GS–GOGAT pathway in nitrogen-fixing cells — conditions in which the intracellular ammonia concentration is low. In *C. butyricum* the GS–GOGAT system is the predominant pathway for ammonia assimilation with either ammonia or N₂ as the source of nitrogen. In *C. acetobutylicum* there is no evidence for a global Ntr system and the GS enzyme is not regulated by adenylation. Instead, a promoter, P₃, which controls the transcription of an antisense RNA, is present in the downstream region of *glnA* DNA.

In phototrophic bacteria, such as the nonsulfur purple bacterium *Rhodobacter capsulatum*, the NADP–ADH aminating activity can function as an alternative route for ammonia assimilation when GS is inactive. The ADH is induced in cells grown on pyruvate plus nitrate, pyruvate plus ammonia, or L-alanine under both light-anaerobic and dark-heterotrophic conditions. Aminating activity is strictly NADPH dependent, whereas deaminating activity is strictly NAD dependent.

GENERAL REACTIONS OF AMINO ACIDS

Amino Acid Decarboxylases

Microorganisms exhibit decarboxylase activity for many amino acids including aspartate, glutamate, ornithine, lysine, arginine, tyrosine, phenylalanine, cysteic acid, diaminopimelic acid, hydroxyphenyl serine, histidine, tryptophan, 5-hydroxytryptophan, and possibly others. The general reaction for all of these enzymes is



As far as is known, pyridoxal phosphate is the coenzyme of all the amino acid decarboxylases, with the exception of histidine decarboxylase, for which the cofactor is pyruvate. All of the amino acid decarboxylases are essentially irreversible and therefore are not of importance in the biosynthesis of most amino acids. However, in the case of diaminopimelic acid (DAP) decarboxylase, lysine is the final product. Amino acid decarboxylases are produced at low pH and their range of optimal activity is pH 3–5. Thus, excess acidity resulting from the production of acid end products may be regulated by amino acid decarboxylase activity, particularly in anaerobic proteolytic organisms such as the clostridia.

The diamine putrescine, the decarboxylation product of ornithine, is an essential growth factor for several organisms and is a biosynthetic precursor of both spermidine and spermine, polyamines found in a wide variety of microorganisms (see Chapter 15). Spermine is present in eukaryotic organisms but is found in only a few bacteria (e.g., *Pseudomonas aeruginosa* and *Bacillus stearothermophilus*). Spermidine is more widely distributed, being found in bacteria and fungi as well as in higher organisms.

Germinating conidia of *N. crassa* produce an active glutamate decarboxylase. This is the first step in a pathway that leads to a rapid increase in aspartate in the amino acid pool of this organism.

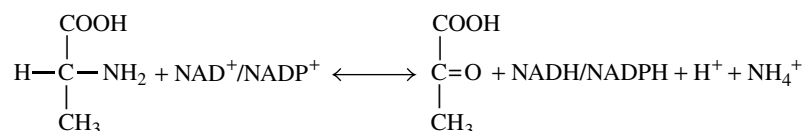
Amino Acid Deaminases

Deamination of amino acids occurs via several quite different reactions:

1. Oxidative deamination
 - a. NAD^+ - or NADP^+ -linked deamination
 - b. FAD- or FMN-linked deamination
2. Nonoxidative deamination

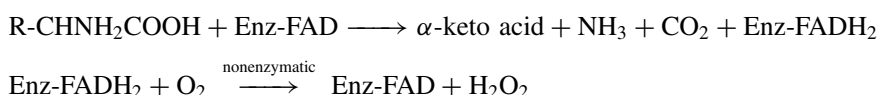
The oxidative deamination of glutamic acid by the reversible glutamate dehydrogenases has been discussed earlier in relation to ammonia assimilation. In most organisms the NAD -specific enzyme operates catabolically, whereas the NADP -linked enzyme functions in glutamate synthesis.

Alanine dehydrogenase (ADH) is present in members of the genus *Bacillus*, the photosynthetic nonpurple sulfur bacteria *Rhodobacter*, and others. The reaction proceeds via an α -imino intermediate:



The ubiquitous distribution and high activity of ADH in the aerobic bacilli has been considered as evidence that the reverse reaction catalyzed by this enzyme may serve as a major route of ammonia assimilation in these organisms. This conclusion has also been drawn in the case of the symbiotic rhizobia.

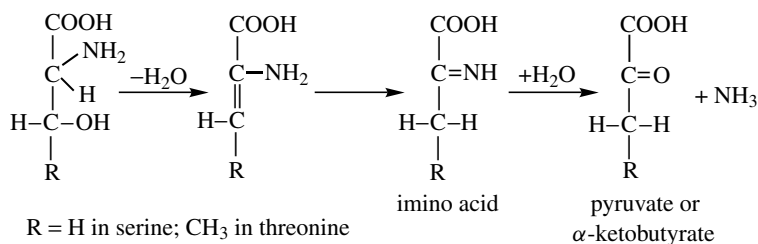
Amino acid oxidases, sometimes called aerobic or oxidative deaminases, involve reactions catalyzed by enzymes containing FAD (flavin adenine dinucleotide) or FMN (flavin mononucleotide). The reaction proceeds in two stages:



Reduced FAD (or reduced FMN) may react nonenzymatically with molecular oxygen as shown above or it may transfer the hydrogen to other hydrogen acceptors. The amino acid oxidases are nonspecific in that a single enzyme may catalyze the oxidation of a variety of amino acids, for example, methionine, phenylalanine, tyrosine, leucine, isoleucine, valine, norvaline, alanine, tryptophan, and cysteine in decreasing order of activity. Another may oxidize proline, hydroxyproline, citrulline, histidine, and arginine. The rate of oxidation may differ for each amino acid, and the order of activity may differ with the source of the enzyme. Both D- and L-amino acid oxidases are known, but a single enzyme is usually specific for one configuration. These reactions yield no useful energy if the reduced FAD is oxidized nonenzymatically by molecular oxygen. It is highly unlikely that these enzymes play any significant role in the assimilation of ammonia.

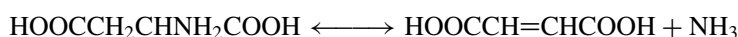
Nonoxidative deaminases, such as aspartic acid deaminase (aspartase), serine, and threonine deaminases (dehydratases), and cysteine desulfhydrase, are specific in their substrate requirements.

Serine and threonine deaminases (dehydratases) catalyze the following type of reaction:



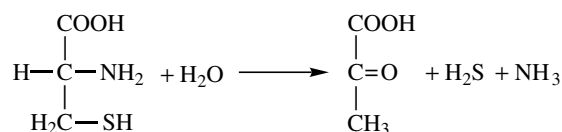
An intramolecular transfer of hydrogen atoms occurs and water is removed (dehydration) to produce an imino acid via a β -elimination reaction. In the second stage of the reaction, water reacts nonenzymatically with the imino acid to release ammonia. Pyridoxal-5'-phosphate is the cofactor in this reaction. The two deaminating enzymes have been shown to be distinct. The level of L-serine deaminase varies as a function of nitrogen nutrition, carbon source, and the supply of glycine and leucine. Glycine and leucine induce the formation of serine deaminase. The enzyme seems to play a role in a number of pathways in which serine is generated and further metabolized as part of the main carbon pathway. Both *E. coli* and *S. enterica* produce biodegradative as well as biosynthetic threonine deaminases. The latter provides α -ketobutyrate as an intermediate in the formation of isoleucine. The biodegradative enzyme is induced under anaerobic conditions in amino acid-rich medium, requires cAMP for its synthesis, and is sensitive to catabolite repression by glucose.

Aspartase is present in a number of organisms. It converts aspartate to fumarate and ammonia:



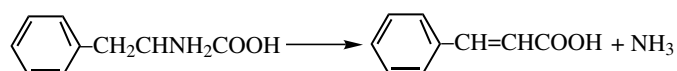
There is no hydrogen exchange in this reaction. Instead, an intramolecular transfer of hydrogen occurs, and pyridoxal-5-phosphate is not involved as a cofactor in the reaction. The reaction is reversible, providing a potential route for ammonia assimilation. Although the equilibrium of the reaction is such that it could serve as an ammonia assimilation pathway, it is considered to function primarily in a catabolic capacity.

Cysteine desulfhydrase (dehydratase) has a reaction mechanism similar to that of serine and threonine deaminases except that hydrogen sulfide, rather than water, is removed via a β -elimination reaction:



Dehydratases and desulfhydrases are essentially irreversible reactions and are not considered reactions that could readily participate in ammonia assimilation.

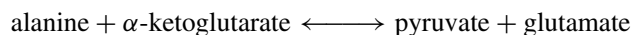
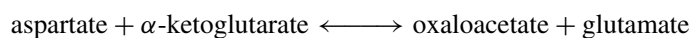
Phenylalanine deaminase (ammonia lyase) is produced by yeasts, molds, and bacteria and catalyzes the nonoxidative deamination of L-phenylalanine to *trans*-cinnamic acid:



A dehydroalanine group serves as a cofactor in the reaction. The enzyme is of interest because of its potential for use in the treatment and diagnosis of phenylketonuria and has industrial applications in the synthesis of L-phenylalanine from *trans*-cinnamic acid. In *Rhodotorula glutinis* this enzyme serves as the initial step in a metabolic pathway, leading to the formation of benzoate and 4-hydroxybenzoate.

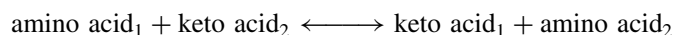
Amino Acid Transaminases (Aminotransferases)

In 1945, Lichstein and Cohen first demonstrated transaminase activity in bacteria. Transfer of the α -amino nitrogen between glutamate, aspartate, alanine, and their corresponding α -keto acids was shown to be similar to reactions observed in mammalian tissues. The two major transaminases demonstrated were aspartate amino transferase and alanine aminotransferase:



Pyridoxal phosphate is the coenzyme for all known transaminase reactions.

Transamination was first thought to be a relatively limited activity confined to these three amino acids and their keto analogs. However, in the early 1950s, Feldman and Gunsalus, and Rudman and Meister, showed that a number of amino acids would undergo transamination with α -ketoglutarate to form the corresponding amino acids. Transaminases are fully reversible reactions, so glutamate is a key donor for the synthesis of most other amino acids. This result led to a more generalized view of transamination in which any amino acid and any keto acid could exchange the amino group:



Transaminase activity is ubiquitous as evidenced by the fact that keto analogs can replace many of the amino acids for the growth of amino acid-requiring mutants. The reaction is not totally universal, however. Four major transaminases have been identified in *E. coli* as shown in Table 14-5. By these criteria, strains lacking transaminase C (*avtA* mutants) should have no nutritional requirement, whereas strains lacking transaminase B (*ilvE* mutants) should require isoleucine. In practice, most *ilvE* mutants require only isoleucine, but some show additional requirements for valine or leucine as a result of reduced expression of either *avtA* or other genes distal to *ilvE*. These transaminases have not, in general, been well identified in other organisms.

TABLE 14-5. Major Transaminases in *Escherichia coli*

Transaminase	Interacts with	Gene
Aromatic amino acids	Tyrosine Phenylalanine Glutamate Leucine Aspartate Methionine	<i>tyrB</i>
Aspartate (transaminase A)	Aspartate Glutamate Tyrosine Phenylalanine	<i>aspC</i>
Transaminase B (branched-chain aminotransferase)	Glutamate Leucine Isoleucine Valine Phenylalanine Methionine	<i>ilvE</i>
Transaminase C (alanine-valine aminotransferase)	Alanine Valine α -Aminobutyrate	<i>avtA</i>

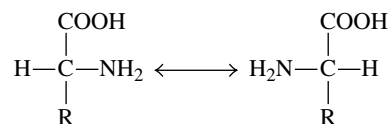
However, most of these activities appear to be present in organisms that display a general ability to synthesize the common amino acids.

The aspartate aminotransferase (L-aspartate:2-oxoglutarate aminotransferase) is ubiquitous. However, the enzyme has been studied in detail only in *E. coli*, *P. putida*, *S. cerevisiae*, and the archeon *Sulfolobus solfataricus*. A thermophilic *Bacillus* species produces an aspartate aminotransferase that shows some sequence similarity in the N-terminal region between the eubacterial and archaeal enzymes.

Other transaminases are known. For example, *S. enterica* displays a glutamine amidotransferase encoded within the *trpD* gene. It serves a dual role in transferring the amino group of glutamine to chorismic acid in the synthesis of anthranilic acid. Glutamine amidotransferase activity is also present in a tryptophan gene (*trypE*) in *Bacillus pumilis*.

Amino Acid Racemases

A number of microorganisms contain enzymes that catalyze the conversion of D-amino acids to L-amino acids via the general reaction



Most biochemical compounds are asymmetric and there is a tendency for one form to predominate over the other. Amino acids in naturally occurring proteins are usually in

L-configuration. However, the cell walls and polypeptide capsules of many organisms contain D-amino acids. Since most biosynthetic reactions lead to the synthesis of the L-amino acids, racemases are necessary for the conversion of certain amino acids to the D configuration for the synthesis of these specialized cell structures. As an example, D-alanine is a structural component of the cell wall of *Enterococcus faecalis* as well as several other gram-positive organisms. When grown in a medium lacking pyridoxal, D-alanine becomes a specific growth requirement for *E. faecalis* because alanine racemase is inactive under the conditions of pyridoxal deficiency. D-Amino acids are found in the polypeptide capsules of members of the genus *Bacillus* and in peptide antibiotics, providing other indications of the importance of racemases in microbial metabolism.

Role of Pyridoxal-5'-Phosphate in Enzymatic Reactions with Amino Acids

In the foregoing discussion of the various reactions in which amino acids participate, it was mentioned that pyridoxal-5'-phosphate (PLP), the coenzyme form of vitamin B₆, functions in many of these reactions. PLP is the most versatile of all enzyme cofactors in that it can catalyze several distinct chemical reactions. Such versatility seems to be due, in part, to the PLP cofactor acting as an electron sink. For example, a proton may be removed from the α -carbon of the amino acid substrate, with the resultant stabilization of the carbanion at the C $_{\alpha}$ or C'-4 position; or the electrons may flow into the ring, neutralizing a positively charged pyridine nitrogen (i.e., a quinoid structure may be one of the intermediates). It is assumed that appropriate groups on the apoenzyme hold the PLP in the precise alignment necessary for rupture of the —C—X bond (Fig. 14-9).

Depending on the type of electron shifts that take place, nine main types of reactions are recognized: transamination, β decarboxylation, α decarboxylation, aldol cleavage,

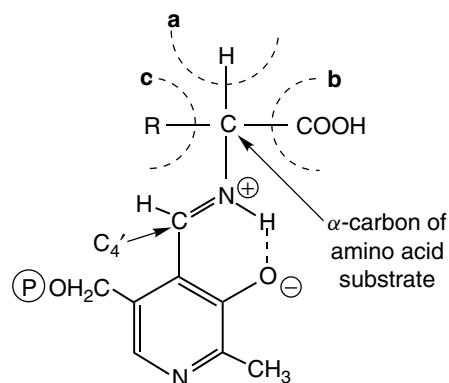


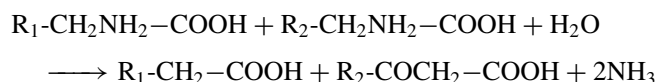
Fig. 14-9. Functions of pyridoxal-5'-phosphate (vitamin B₆) in reactions of amino acids.

The product of the enzyme-catalyzed reaction will depend on which of the four bonds projecting from the α -carbon atom is split. Reaction (a) occurs in racemization, transamination, β elimination, γ elimination, β replacement, γ replacement, and β decarboxylation. Reaction (b) occurs in α -decarboxylation. Reaction (c) occurs in aldol cleavage. Although all three of these types of reactions are well known, almost invariably each enzyme is quite specific as to which bond it will break. This remarkable specificity occurs because the enzyme can "hold" the substrate in such a way that only the required bonds can be broken. This alignment is probably due to the strategic positioning of certain groups within the active site of the enzyme.

cleavage, γ elimination, β elimination, γ displacement, or β displacement may occur. In addition, various PLP enzymes carry out unique reactions that do not fall into any of these categories—for example, dialkyl amino acid transaminase, tryptophan synthase, threonine synthase, and δ -amino-levulinate synthase. Three enzymes are necessary for the reduction of cytidine diphosphate (CDP) sugars to 1,3-dideoxy sugars. One of these enzymes contains pyridoxamine. Pyridoxamine, not pyridoxal, is a required growth factor for certain organisms. These are the only known cases where the cofactor requirement is for the amine rather than the pyridoxal form. The 1,3-dideoxy sugars are important components of certain bacterial cell walls.

THE STICKLAND REACTION

Some members of the genus *Clostridium* can utilize amino acids as a source of energy by means of coupled oxidation–reduction reactions involving certain amino acids as hydrogen donors and others as receptors:



A specific example of such a reaction involving glycine and alanine takes place according to the following reaction scheme:

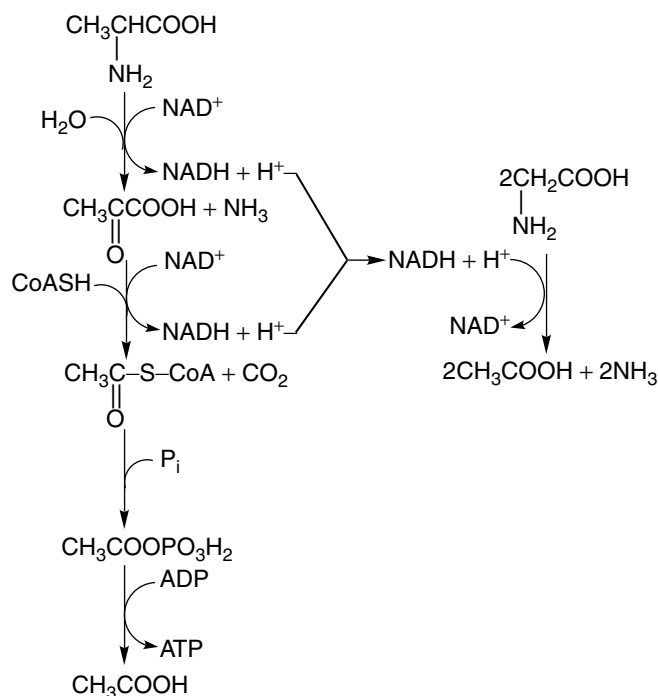
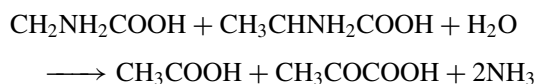


Fig. 14-10. Mechanism of the Stickland reaction.

This type of reaction leads to the formation of short-chain fatty acids and keto acids. The Stickland reaction involves several steps in which NAD first accepts a hydrogen atom from the amino acid donor and transfers it to the acceptor amino acids. The latter reaction is catalyzed by an amino acid reductase (Fig. 14-10). Certain amino acids serve preferentially as hydrogen donors and others as hydrogen acceptors. The rate of the reaction is quite rapid, and this system is used by proteolytic clostridia to gain energy via substrate-level phosphorylation.

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CHAPTER 15

BIOSYNTHESIS AND METABOLISM OF AMINO ACIDS

Amino acid biosynthesis is discussed most conveniently in the context of families of amino acids that originate from a common precursor:

1. **Glutamate or α -Ketoglutarate Family.** Glutamate, glutamine, glutathione, proline, arginine, putrescine, spermine, spermidine, and in yeasts and molds, lysine. A tetrapyrrole (heme) precursor, δ -aminolevulinate, arises from glutamate in some organisms.
2. **Aspartate Family.** Aspartate, asparagine, threonine, methionine, isoleucine, and, in bacteria, lysine.
3. **Pyruvate Family.** Alanine, valine, leucine, and isoleucine.
4. **Serine-Glycine or Triose Family.** Serine, glycine, cysteine, and cystine. In yeasts and molds, mammals, and some bacteria, δ -aminolevulinate is formed by the condensation of glycine and succinate.
5. **Aromatic Amino Acid Family.** Phenylalanine, tyrosine, and tryptophan. Additional compounds that can originate from the common aromatic pathway include enterochelin, *p*-aminobenzoate, ubiquinone, menaquinone, and NAD.
6. **Histidine.** This amino acid originates as an offshoot of the purine pathway in that a portion of the histidine molecule is derived from the intact purine ring.

In some instances, it is convenient to discuss the catabolic pathways of the amino acids in conjunction with the discussion of their biosynthesis.

THE GLUTAMATE OR α -KETOGLUTARATE FAMILY

Glutamine and Glutathione Synthesis

The importance of glutamate as one of the primary amino acids involved in the assimilation of ammonia is discussed in Chapter 14. Glutamate and glutamine play a

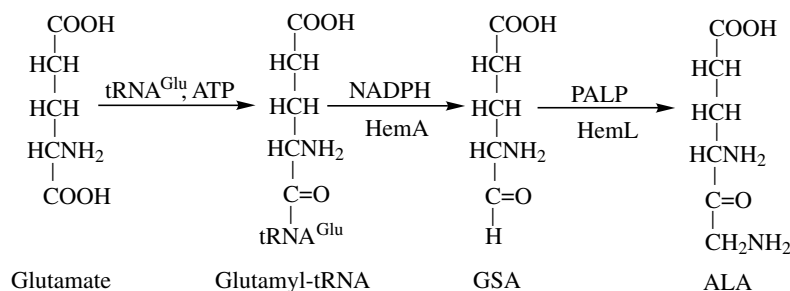
central role in amino acid biosynthesis by the ready transfer of amino or amide groups, respectively, in the synthesis of other amino acids by transamination or transamidation reactions. Glutamine is synthesized from glutamate with the participation of ammonia and ATP. Glutathione, a disulfide-containing amino acid whose functions have only recently begun to be understood in detail, is synthesized in two steps. The coupling of L-glutamate and L-cysteine in the presence of ATP to form γ -glutamylcysteine is catalyzed by a specific synthase. In the presence of glycine and ATP, glutathione synthase forms glutathione (Fig. 15-1).

The Proline Pathway

The pathway to proline (Fig. 15-1) involves formation of γ -glutamylphosphate from L-glutamate and ATP by γ -glutamyl kinase (ProB). In the presence of NADPH, γ -glutamylphosphate is reduced to glutamate γ -semialdehyde by ProA. Glutamate γ -semialdehyde cyclizes spontaneously to form 1-pyrroline-5-carboxylate. Pyrroline-5-carboxylate is converted to proline by a specific reductase (ProC).

Aminolevulinate Synthesis

In *S. enterica* and *E. coli*, δ -aminolevulinic acid (ALA), the first committed precursor to tetrapyrroles, arises from glutamate. This C₅ pathway, which was originally thought to occur primarily in plants and algae, is now firmly established as a major route to ALA in several bacterial species. Prior to this discovery, the condensation of glycine and succinyl-CoA by the enzyme ALA synthase (the C₄ pathway) was considered to be the only route of ALA formation. It is still the major route of ALA formation in mammals, fungi, and certain bacteria, such as *Rhodopseudomonas sphaeroides*, *R. capsulatus*, and *Bradyrhizobium japonicum*. The C₅ pathway to ALA involves conversion of glutamate to glutamyl-tRNA^{Glu} by glutamyl-tRNA synthetase, reduction to glutamate γ -semialdehyde (GSA) by an NADPH-dependent glutamyl-tRNA reductase (HemA), and transamination by glutamate γ -semialdehyde aminomutase (HemL) to form ALA:



The details of the pathway of tetrapyrrole (heme) biosynthesis are considered in the section on tetrapyrrole pathways.

The Arginine Pathway

Bacteria and fungi synthesize ornithine via a series of *N*-acetyl derivatives (Fig. 15-2). The function of the *N*-acetyl group is to prevent the premature cyclization of 1-pyrroline-5-carboxylate to proline. There is a divergence in the pathway in different

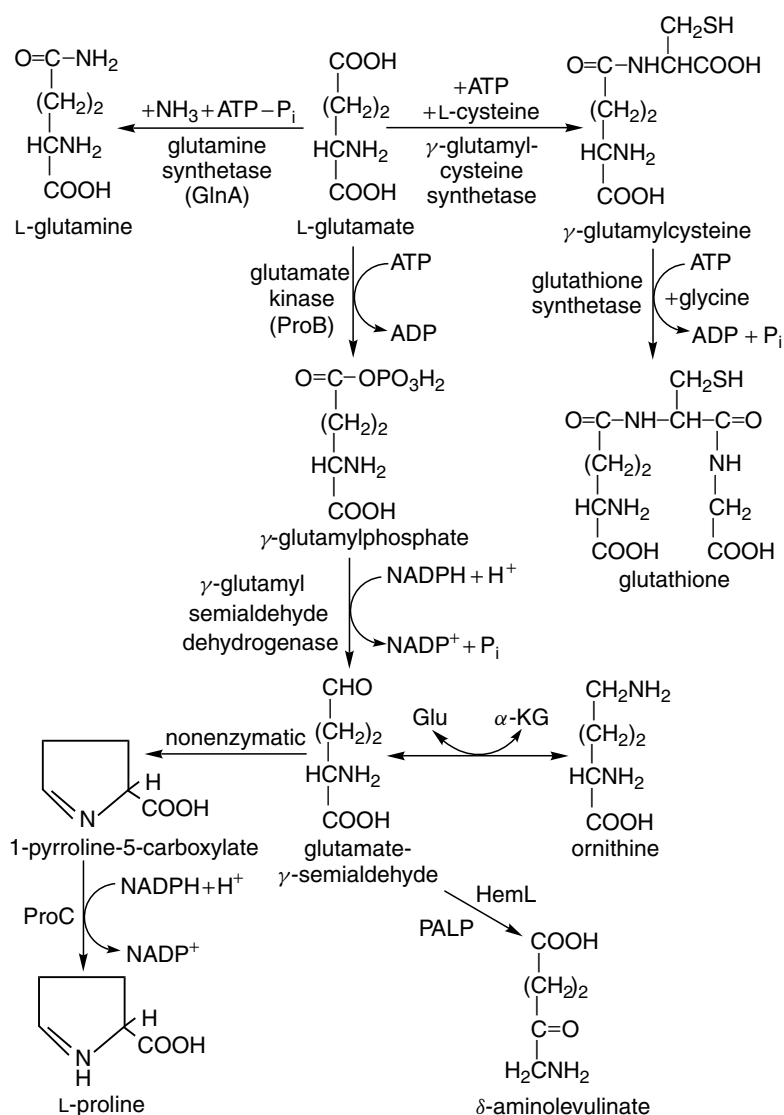


Fig. 15-1. Pathways from glutamate to glutamine, glutathione, proline, ornithine, and ALA. The transaminase that converts glutamate γ -semialdehyde to ornithine is reversible. However, it is generally considered that this reaction is primarily involved in ornithine degradation. Ornithine is normally synthesized via the pathway shown in Figure 10-2. ProC, 1-pyrroline-5-carboxylate reductase; PALP, pyridoxal phosphate; HemL, glutamate γ -semialdehyde aminomutase.

organisms depending on the manner in which the acetyl group is removed. In *Enterobacteriaceae* and *Bacillaceae*, *N*-acetylornithine is deacylated via acetylornithine deacetylase (ArgE). In *N. gonorrhoeae*, *Pseudomonadaceae*, cyanobacteria, photosynthetic bacteria, and yeasts and molds, the acetyl group of *N*-acetylornithine is recycled by ornithine acetyltransferase (ArgJ).

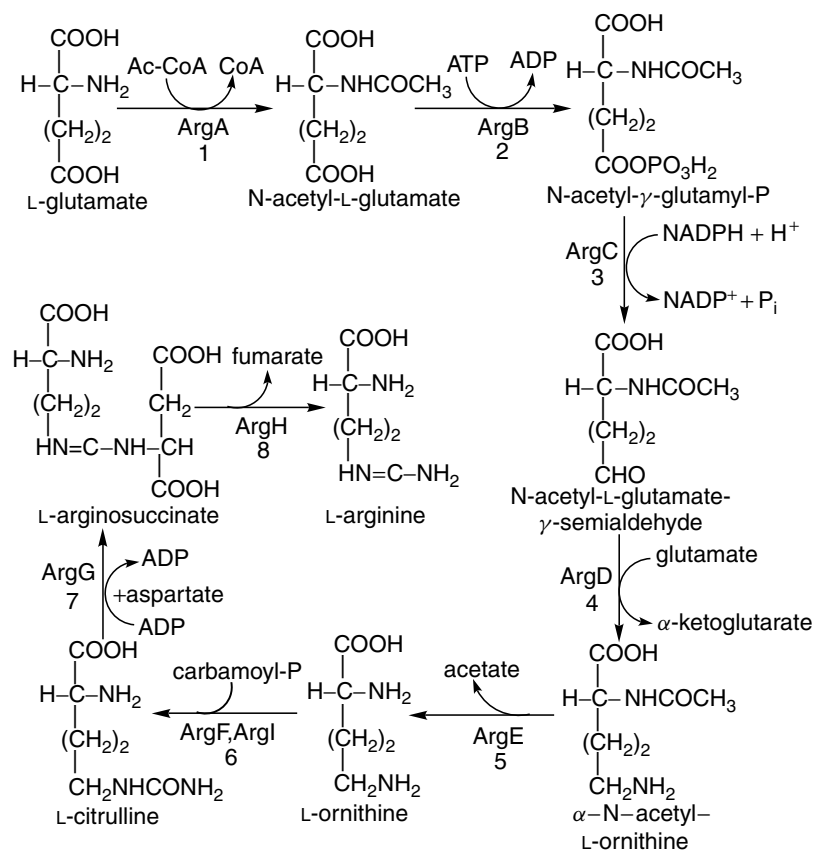
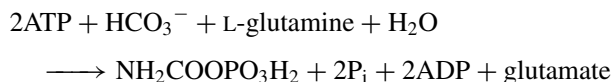


Fig. 15-2. Pathway of arginine biosynthesis. The enzyme designations for *Escherichia coli* are: ArgA, acetylglutamate synthetase; ArgB, acetylglutamate kinase; ArgC, acetylglutamate γ-semialdehyde dehydrogenase; ArgD, acetylornithine-δ-transaminase; ArgF, ArgI, ornithine transcarbamylase; ArgG, arginosuccinate synthase; ArgH, arginosuccinase. In yeasts, molds, *E. coli*, *P. mirabilis*, *S. marcescens*, and some other enterobacteria, step 5 occurs as shown. In *Pseudomonas fluorescens*, *Micrococcus glutamicus*, *Anabaena variabilis*, and several other bacteria, step 5 involves transacylation between glutamate and acetylornithine.

The eight enzymes involved in the arginine pathway (Fig. 15-2) are found in *E. coli*, *S. enterica*, *Proteus*, *Pseudomonas*, *B. licheniformis*, *B. subtilis*, *B. sphaericus*, *S. bovis*, *N. gonorrhoeae*, *N. crassa*, *A. niger*, *S. cerevisiae*, and *C. albicans*.

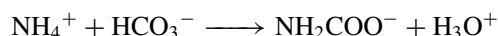
Carbamoyl phosphate is a common precursor in the biosynthesis of arginine and pyrimidines. In *E. coli* and *S. enterica* a single carbamoyl phosphate synthetase catalyzes the reaction



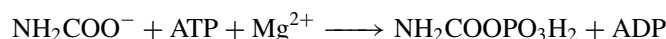
Ammonia can replace glutamine as a nitrogen donor in vitro, but glutamine is the physiologically preferred substrate, and K⁺ and Mg²⁺ are required participants. The

enzyme is composed of two nonidentical subunits. The smaller subunit, encoded by *carA*, acts as a glutamine amidotransferase. The larger subunit, encoded by *carB*, carries out the remaining functions. Since carbamoyl phosphate is involved in two major biosynthetic pathways, expression of the *carAB* operon is regulated by both arginine and pyrimidines. The enzyme is also subject to allosteric control by intermediates in both pathways. Ornithine stimulates enzyme activity, whereas UMP is inhibitory. The enzyme is activated by IMP and PRPP, coordinating its activity with purine biosynthesis as well. In some earlier studies, certain organisms appeared to depend on the nonenzymatic formation of carboxylamine and its conversion by carbamoyl phosphokinase:

Nonenzymatic:



Carbamoyl phosphokinase:



The relative importance of this reaction in ammonia assimilation as compared to the GDH or GS–GOGAT systems has never been adequately assessed.

In *S. cerevisiae* and *N. crassa* there are two carbamoyl phosphate synthetases. One enzyme, carbamoyl phosphate synthetase A, is linked to the arginine pathway and is subject to repression by arginine. The other enzyme, carbamoyl phosphate synthetase P, is linked to the pyrimidine biosynthesis pathway and is subject to both repression and feedback inhibition by pyrimidines. Localization of carbamoyl phosphate synthetase A within mitochondria seems to play a major role in the channeling of this precursor. In yeast there seems to be little channeling of carbamoyl phosphate, since both carbamoyl phosphate synthetases are in the cytoplasm and contribute to a common pool of carbamoyl phosphate.

Some microorganisms are able to generate usable energy as ATP by the anaerobic degradation of arginine using the arginine deiminase (ADI) pathway (Fig. 15-3). In *Bacillus licheniformis* a complex of three enzymes (ArcA, ArcB, and ArcC) and an arginine-ornithine antiporter (ArcD) are involved in arginine degradation. The ArcD protein mediates proton motive force–driven uptake of arginine and ornithine and the stoichiometric exchange between arginine and ornithine. The *arc* genes of *B. licheniformis* appear in the same order (*arcABDC*) as in *C. perfringens*. In *P. aeruginosa* these genes appear in the order *arcDABC*. *Lactobacillus sake* contains genes for the arginine deiminase pathway that include *arcD*, an arginine-ornithine antiporter, and a transaminase-encoding gene, *arcT*. The ADI pathway is active in a number of other aerobic spore-forming bacilli as well as in *Mycoplasma*, *Streptococcus*, *Lactococcus*, *Lactobacillus*, *Clostridium*, *Pseudomonas*, and the cyanobacterium *Synechocystis*.

Arginine degradation in *E. coli*, *K. pneumoniae*, and *P. aeruginosa* may follow yet another catabolic pathway: the arginine succinyltransferase (AST) pathway. In *E. coli* the enzymes in this pathway are encoded by genes in the *astCADBE* operon representing the structural genes for arginine succinyltransferase, succinylarginine dihydrolyase, succinylornithine transaminase, succinylglutamic semialdehyde dehydrogenase,

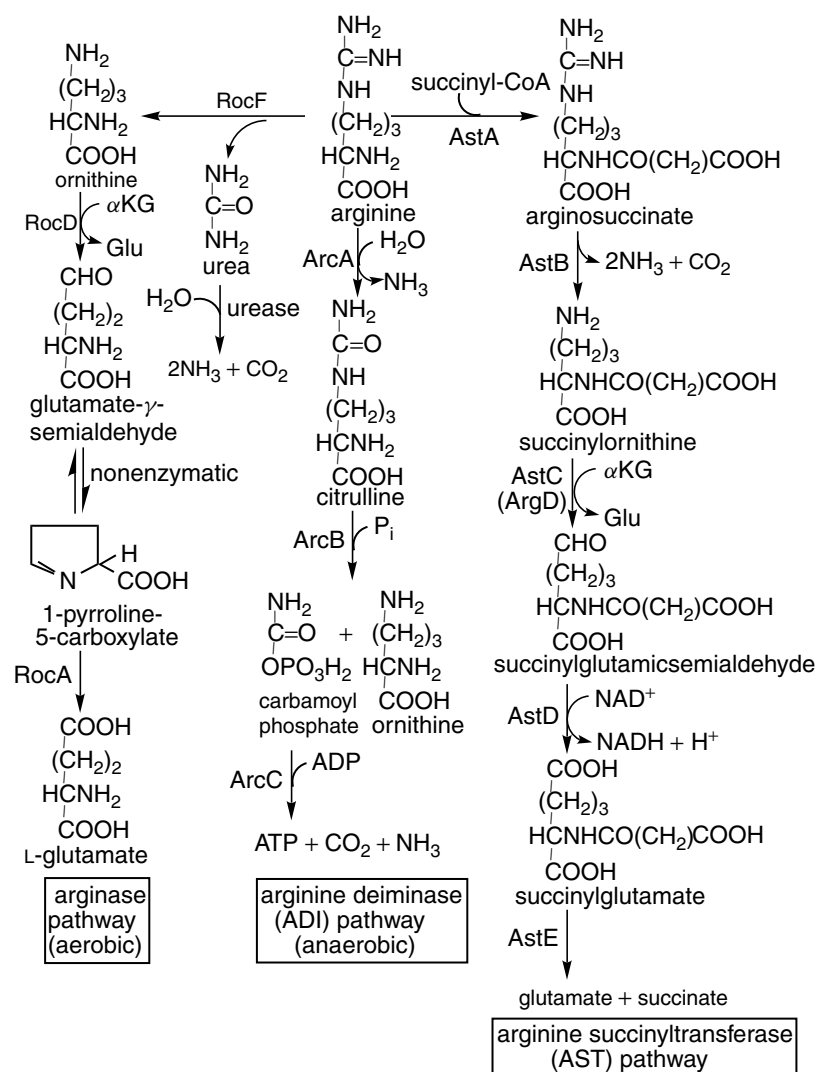
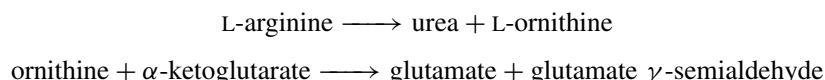


Fig. 15.3. Arginine catabolic pathways. Enzyme designations are: RocF, arginase; RocD, ornithine aminotransferase; RocA, pyrroline-5-carboxylate dehydrogenase; ArcA, arginine deiminase; ArcB, catabolic ornithine carbamoyltransferase; ArcC, carbamate kinase; AstA, arginine succinyltransferase; AstB, succinylarginine dihydrolase; AstC, succinylornithine transaminase; AstD, succinylglutamate semialdehyde dehydrogenase; AstE, succinylglutamate desuccinylase. Note that in *P. aeruginosa* the enzymes of the arginine succinyltransferase pathway are designated Aru (for arginine utilization) and are represented by the designations AruA, AruB, AruC, AruD, and AruE, respectively.

and succinylglutamate desuccinylase (Fig. 15-3). The final products are glutamate and succinate. Disruption of any of the genes in this pathway prevents arginine catabolism and impairs ornithine utilization. In *P. aeruginosa* these same genes are designated *aru* (for arginine utilization). Arginine metabolism is of considerable significance in *P. aeruginosa* as evidenced by the strong chemotactic activity for this amino acid and

the fact that there are four different catabolic pathways for arginine utilization: arginine deiminase, arginine succinyltransferase (AST), arginine dehydrogenase, and arginine decarboxylase.

In *S. cerevisiae*, *N. crassa*, and in mammalian cells, arginine is degraded via the arginase pathway. In fungi, the arginine catabolic enzymes arginase (encoded by *CAR1*) and ornithine transaminase (encoded by *CAR1*) work in tandem to degrade arginine:



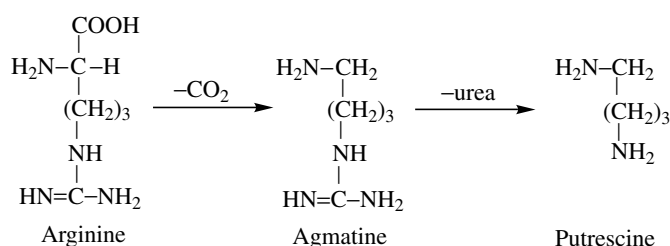
Arginase activity in mammals is so rapid that insufficient amounts of arginine are available for protein synthesis and arginine is required as one of the essential amino acid nutrients. In yeasts and molds, the presence of an alternative route of arginine synthesis and regulatory controls over arginase activity prevent the occurrence of arginine deprivation.

In *N. crassa*, arginine metabolism is compartmentalized between the cytoplasm, the mitochondrion, and vacuoles. In addition, arginase activity is inhibited both in vitro and in vivo by citrulline and ornithine, eliminating a potentially wasteful catabolism of endogenous arginine. In *S. cerevisiae*, which degrades arginine and proline via the common intermediate 1-pyrroline-5-carboxylate, there is also a compartmentalized system for the metabolism of these amino acids. The proline biosynthetic enzyme 1-pyrroline-5-carboxylate reductase is located in the cytoplasm, whereas the proline-degrading enzyme 1-pyrroline-5-carboxylate dehydrogenase is in a particulate fraction of the cell, presumably in the mitochondrion. Arginase is encoded by *CAR1*, which is inducible by arginine. Regulated expression of the *CAR1* gene requires three upstream activation sequences and an upstream repression sequence.

Polyamine Biosynthesis

Polyamines are widely distributed in bacteria, yeasts, and molds as well as in higher forms. A number of growth processes are affected by polyamines. However, their precise role in governing cell growth and differentiation has not been established. The major pathway to polyamines in yeast is via the decarboxylation of ornithine to putrescine (Fig. 15-4). Ornithine decarboxylase is the rate-limiting step and appears to be a common intermediate in most organisms.

In bacteria and plants an alternate pathway to putrescine proceeds via decarboxylation of arginine to agmatine by a biosynthetic arginine decarboxylase (encoded by *speA* in *E. coli*). Agmatine ureohydrolase removes urea from agmatine to yield putrescine:



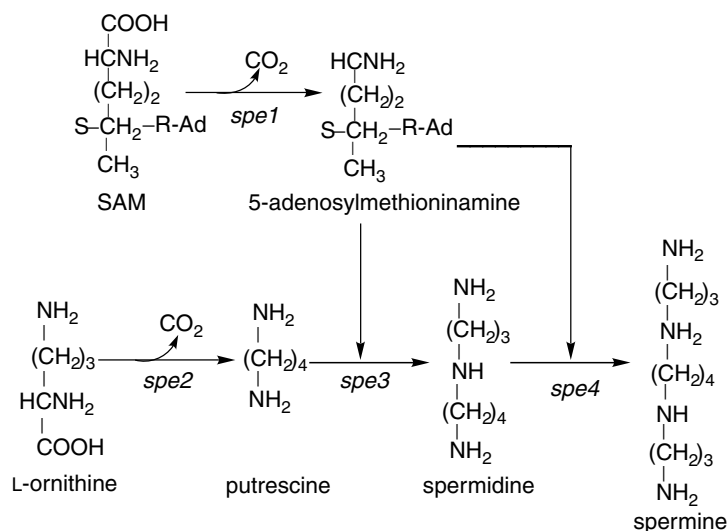


Fig. 15.4. Polyamine biosynthesis in *Saccharomyces cerevisiae*. Genes coding for the enzymes in the pathway are: *spe1*, ornithine decarboxylase; *spe2*, S-adenosylmethionine (SAM) decarboxylase; *spe3*, aminopropyl transferase (spermidine synthase); *spe4*, spermine synthase. (Source: Redrawn from Tabor, C. W., et al., *Fed. Proc.* **41**:3084–3088, 1982.)

Exogenous arginine acts as a signal for the selective utilization of this pathway in *E. coli*. This organism lacks arginase and cannot convert arginine to ornithine. When arginine is added to the growth medium, ornithine levels decline due to inhibition of arginine biosynthesis.

The α -Ketoadipate Pathway to Lysine

In yeasts and molds, and in a rare thermophilic species of bacteria, the biosynthetic pathway to lysine emanates from α -ketoglutarate (Fig. 15-5). The bacterial pathway to lysine, which is initiated by the condensation of pyruvate and aspartate- β -semialdehyde, is part of the aspartate family of amino acid biosynthetic routes and is shown in Figure 15-6. These completely divergent routes of lysine biosynthesis represent a major phylogenetic difference between bacteria and fungi. Only recently has any exception to this difference been demonstrated. So far, the only bacterium that has been shown to utilize the aminoadipate pathway to lysine is an extreme thermophile, *Thermus thermophilus*. This represents the first reported incidence where lysine is not synthesized via diaminopimelate in bacteria.

The series of reactions from homocitrate to α -ketoadipate are analogous to the reactions involved in the conversion of citrate to α -ketoglutarate in the citric acid cycle. The formation of homocitrate from acetyl-CoA and α -ketoglutarate is inhibited by lysine, indicating a feedback control mechanism for the pathway. The initial step in the pathway is also subject to repression by lysine.

Biosynthesis of the β -lactam antibiotics (penicillins and cephalosporins and their derivatives) occurs by a branch from the lysine pathway in *Penicillium chrysogenum*, *Cephalosporium acremonium*, *Streptomyces clavuligerus*, and related organisms. In virtually all organisms the pathway is initiated by the condensation of

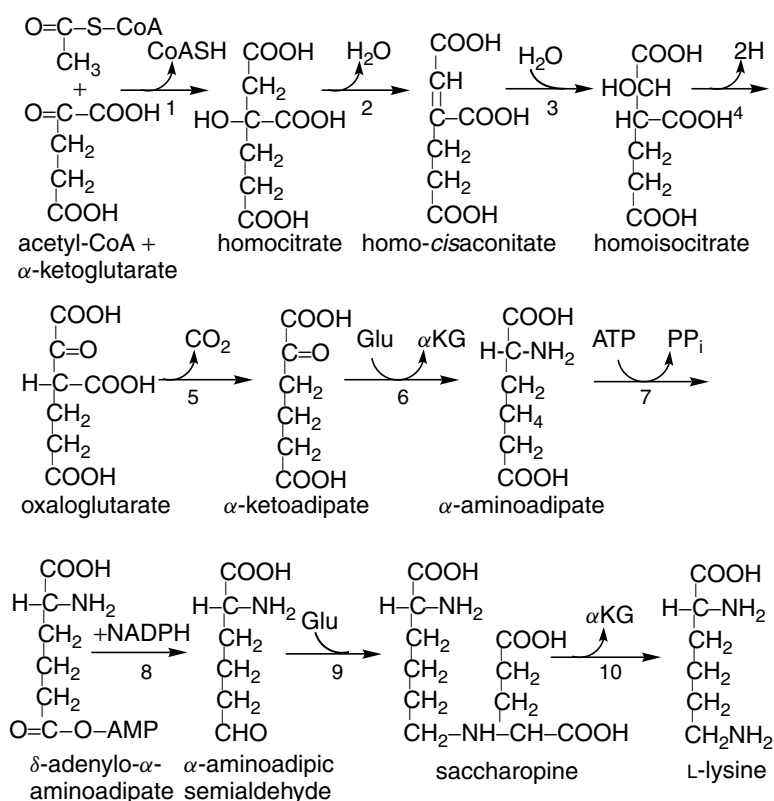
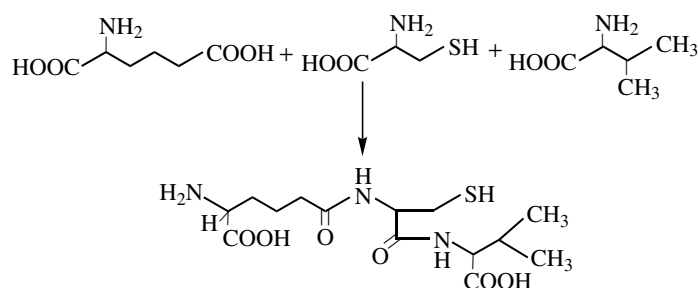


Fig. 15-5. Biosynthesis of lysine in yeasts and molds. The enzymes involved are: 1, homocitrate synthase; 2, homoaconitate dehydratase (*LYS7*); 3, homoaconitate hydratase (*LYS4*); 4 and 5, NAD^+ -mediated homoisocitrate dehydrogenase; 6, α -aminoadipate transaminase; 7, α -aminoadipate semialdehyde dehydrogenase; 8, NADPH -mediated reduction of δ -adenylo-aminodipate to form aminoadipate semialdehyde; 9, saccharopine dehydrogenase (NADP^+ , L-glutamate forming); 10, saccharopine dehydrogenase (NAD^+ , lysine forming). The gene designations are for *S. cerevisiae*. To date only one bacterium, an extremely thermophilic organism, *Thermus thermophilus*, has been shown to follow this pathway.

L- α -aminoadipate, L-cysteine, and L-valine to form the tripeptide δ -(L- α -aminoadipyl)-L-cysteinyl-L-valine (ACV) mediated by the ATP-dependent ACV synthetase:



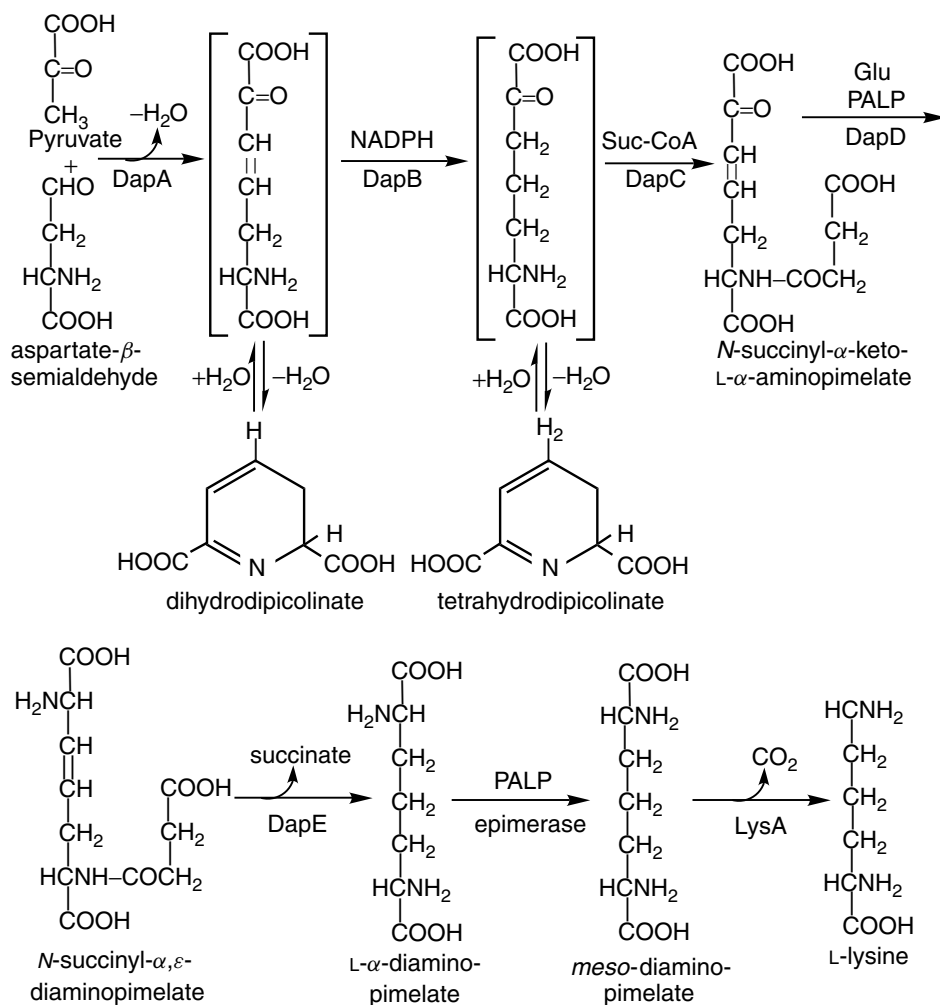
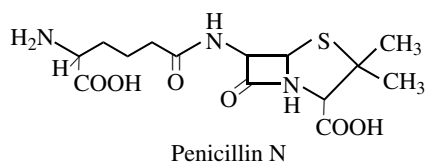


Fig. 15-6. Biosynthesis of lysine in bacteria. The enzyme designations are for *Escherichia coli*. DapA, dihydrodipicolinate synthase; DapB, dihydrodipicolinate reductase; DapC, tetrahydrodipicolinate succinylase; DapD, succinyl diaminopimelate aminotransferase; DapE, succinyl diaminopimelate desuccinylase; LysA, diaminopimelate decarboxylase; PALP, pyridoxal phosphate.

Subsequent ring closures form the β-lactam and thiazolidine ring systems to yield the basic penicillin double-ring structure:



Expansion of the ring to add an additional carbon atom gives rise to the cephalosporin series of derivatives.

THE ASPARTATE AND PYRUVATE FAMILIES

The aspartate and pyruvate families of amino acids are discussed together since there is a distinct overlap in the enzymes involved in the terminal steps of the biosynthesis of the branched-chain amino acids as shown in Figure 15-7. Valine and leucine carbon chains are derived from pyruvate. In bacteria, threonine, isoleucine, methionine, and lysine all emanate from aspartate. Fungi utilize similar pathways with the exception of lysine, which is synthesized via a completely different pathway as discussed previously.

Asparagine Synthesis

Asparagine is synthesized from aspartate, ammonia (or glutamine), and ATP by the enzyme asparagine synthetase. Two types of asparagine synthetase are known. One is

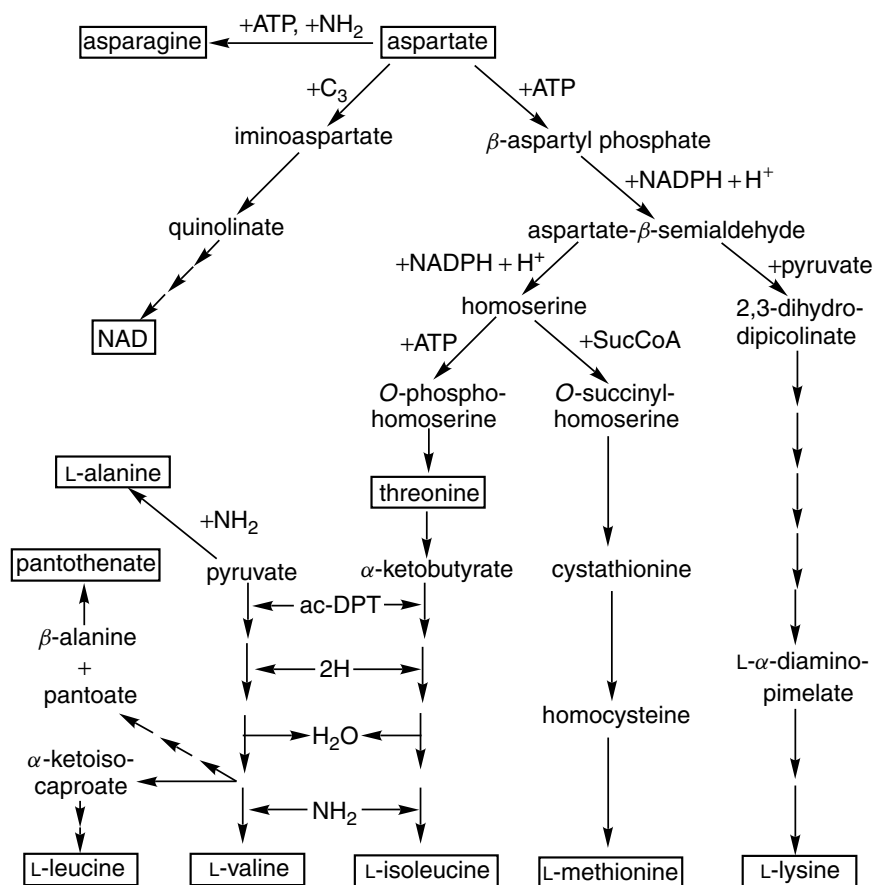
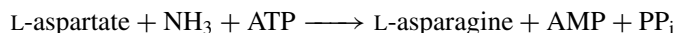
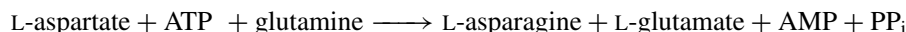


Fig. 15-7. Outline of the pyruvate and aspartate families of amino acids. Compounds enclosed in boxes represent final end products. SucCoA, succinyl-coenzyme A; ac-DPT, acetal diphosphothiamine.

AsnA, which can use only ammonia as the amino donor:



The other is AsnB, which can use either glutamine or ammonia as the nitrogen donor, although glutamine is the preferred substrate:



Both AsnA and AsnB are produced by *E. coli* and *K. aerogenes*, and activity of either enzyme is sufficient to provide an adequate supply of asparagine. A deficiency of both enzymes is required for asparagine auxotrophy. Eukaryotes apparently produce only the glutamine-dependent type of asparagine synthetase. *Bacillus subtilis* produces three genes coding for the AsnB-type asparagine synthetase (*asnB*, *asnH*, and *asnO* or *yisO*), but no gene coding for the ammonia-dependent AsnA-type enzyme has been found. The three genes are expressed differently during growth. In a medium favoring sporulation, expression of *asnB* was found only during exponential growth, that of *asnH* was elevated at the transition between exponential growth and stationary phase, and that of *asnO* occurred only during sporulation. Strains lacking *asnO* failed to sporulate, indicating a specific involvement of this gene in the sporulation process.

The Aspartate Pathway

In both bacteria and fungi, the enzymes aspartokinase and aspartate semialdehyde dehydrogenase initiate the aspartate pathway of amino acid biosynthesis (Fig. 15-8).

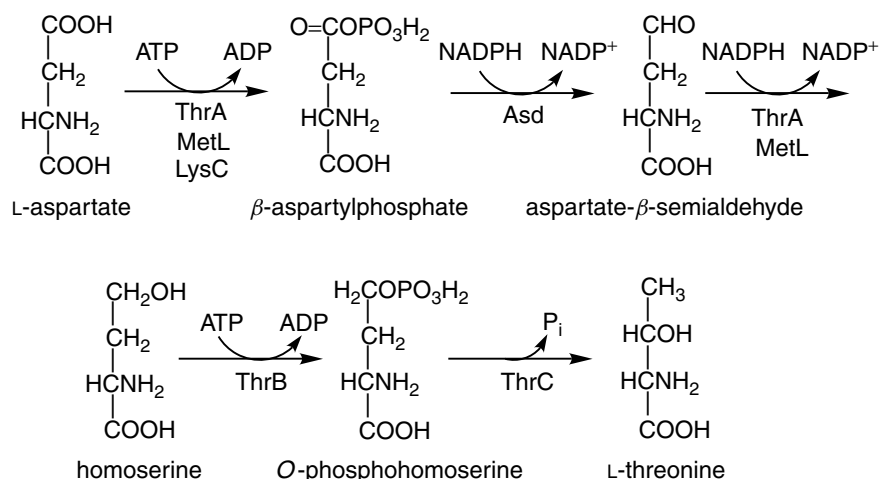


Fig. 15-8. Initial steps in the aspartate pathway of amino acid biosynthesis. The branch leading to threonine is shown here. Enzyme designations are for *E. coli*. ThrA, threonine-specific aspartokinase I–homoserine dehydrogenase I; MetL, methionine-specific aspartokinase II–homoserine dehydrogenase II; LysC, lysine-specific aspartokinase III; Asd, aspartate semialdehyde dehydrogenase; ThrB, homoserine kinase; ThrC, threonine synthetase.

In *E. coli* there are three aspartokinases, each of which is specific to the end product of the pathway involved in its synthesis. Aspartokinase I (ThrA) is specific for the threonine branch, aspartokinase II (MetL) is specific for the methionine branch, and aspartokinase III (LysC) is specific for the lysine branch. Aspartate semialdehyde dehydrogenase (Asd) forms aspartate- β -semialdehyde by removal of phosphate from aspartyl phosphate and reduction using NADPH. Aspartate β -semialdehyde stands at the first branch point in the multibranched pathway. Reduction of aspartate- β -semialdehyde to homoserine is catalyzed by homoserine dehydrogenase I (ThrA), which is specific for the threonine branch of the pathway, or homoserine dehydrogenase II (MetL), which is specific for the methionine branch. O-phosphohomoserine is converted to threonine by threonine synthetase (ThrC).

The Bacterial Pathway to Lysine

Condensation of aspartate β -semialdehyde with pyruvate yields dihydrodipicolinate, the first intermediate in the bacterial pathway to lysine (Fig. 15-6). This pathway is of special interest because dipicolinic acid is produced during sporulation in *Bacillus* species and either diaminopimelic acid or lysine is present in the peptidoglycan structures of all prokaryotes that produce a rigid cell wall. This pathway to lysine has been shown to occur in *E. coli*, *S. enterica*, *S. aureus*, *E. faecalis*, *B. subtilis*, and the cyanobacteria, and, with the exception of one highly thermophilic species mentioned earlier, seems to occur only in bacteria.

The biosynthetic pathway depicted in Figure 15-6 involves the formation of succinylated derivatives of α -keto-L- α -diaminopimelate and diaminopimelate as intermediates. *E. coli*, *S. enterica*, *S. aureus*, and a number of other bacteria utilize this pathway. Certain *Bacillus* species form acetylated derivatives exclusively. A few organisms (e.g., *B. sphaericus*, *C. glutamicum*) convert tetrahydrodipicolinate (piperidine-2,6-dicarboxylate) to D,L-diaminopimelate in a single step via diaminopimelate dehydrogenase. However, in *C. glutamicum* the pathway using succinylated intermediates appears to function along with the direct dehydrogenase pathway. This organism is of interest because it is used for the commercial production of lysine. The dehydrogenase pathway is apparently a prerequisite for handling increased flow of metabolites to D,L-diaminopimelate and lysine.

Because of the insolubility of iron at physiological pH, microorganisms have evolved a variety of systems (siderophores) to facilitate the acquisition of iron (see Chapter 7). *E. coli*, *Shigella*, and *Salmonella* synthesize an iron-chelating siderophore, aerobactin, from lysine. The iron-regulated aerobactin operon is found on a ColV-K30 plasmid. This operon consists of at least five genes for synthesis (*iuc*, iron uptake chelate) and transport (*iut*, iron uptake transport) of aerobactin. The biosynthetic pathway starts with oxidation of lysine to N^ϵ -hydroxylysine by N^ϵ -lysine monooxygenase (LucD). As shown in Figure 15-9, the hydroxyl derivative is acetylated by LucB (*N*-acetyltransferase) to form N^ϵ -acetyl- N^ϵ -hydroxylysine. Citrate addition to form aerobactin is catalyzed by aerobactin synthetase (LucC). Similar siderophores called rhizobactins are produced by rhizobia. In *Sinorhizobium meliloti* the process begins with the conversion of L-aspartic- β -semialdehyde to α -ketoglutarate, conversion of L-glutamate to L-2,4-diaminobutyric acid, and subsequent condensation with 1,3-diaminopropane and citrate to yield rhizobactin 1021, a structure comparable to aerobactin.

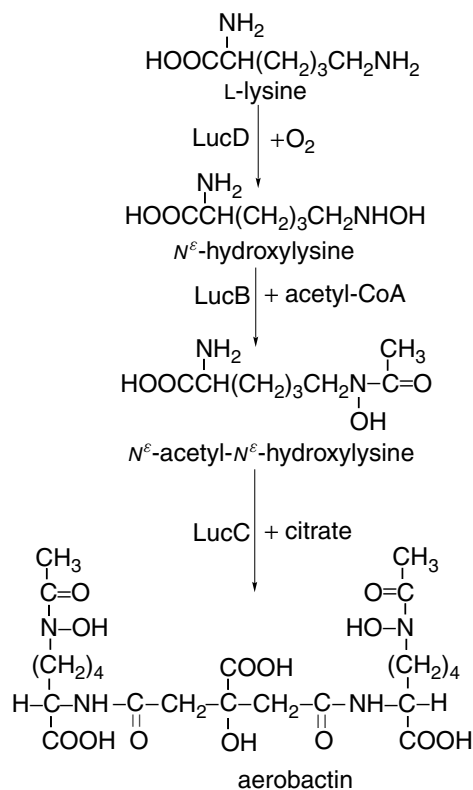


Fig. 15-9. Aerobactin synthesis from lysine. The enzyme designations for *E. coli* are: LucD, N^ϵ -lysine monooxygenase; LucB, N^ϵ -acetyltransferase; LucC, aerobactin synthase.

Threonine, Isoleucine, and Methionine Formation

Another branch point in the aspartate pathway occurs at homoserine. One branch leads to the formation of threonine (Fig. 15-8) and ultimately isoleucine. Homoserine can also be converted to methionine via several alternate routes. In *E. coli*, *S. enterica*, and other enteric organisms, conversion of homoserine to homocysteine involves either of two alternative pathways (Fig. 15-10). The enzyme homoserine succinyltransferase (MetA) condenses succinyl-CoA and homoserine to yield *O*-succinylhomoserine. Cystathionine γ -synthase is apparently capable of catalyzing the direct reaction of hydrogen sulfide (H_2S) or methylsulfide with *O*-succinylhomoserine to form homocysteine.

The methylation of homocysteine to form methionine may occur via either of two enzymes. One enzyme (MetH) is vitamin B_{12} dependent and requires NADH, FAD, SAM, and either 5-methyltetrahydrofolate or its triglutamyl derivative. A vitamin B_{12} -independent enzyme (MetE) requires only Mg^{2+} and 5-methyltetrahydropteroyltriglutamate. The 5-methyltetrahydrofolate is formed from serine and tetrahydrofolate (THF) through the action of serine hydroxymethyltransferase (GlyA), which converts THF to 5,10-methylene-THF. The enzyme 5,10-methylenetetrahydrofolate reductase (MetF) uses reduced FAD (FADH_2) to convert 5,10-methylene-THF to 5-methyl-THF.

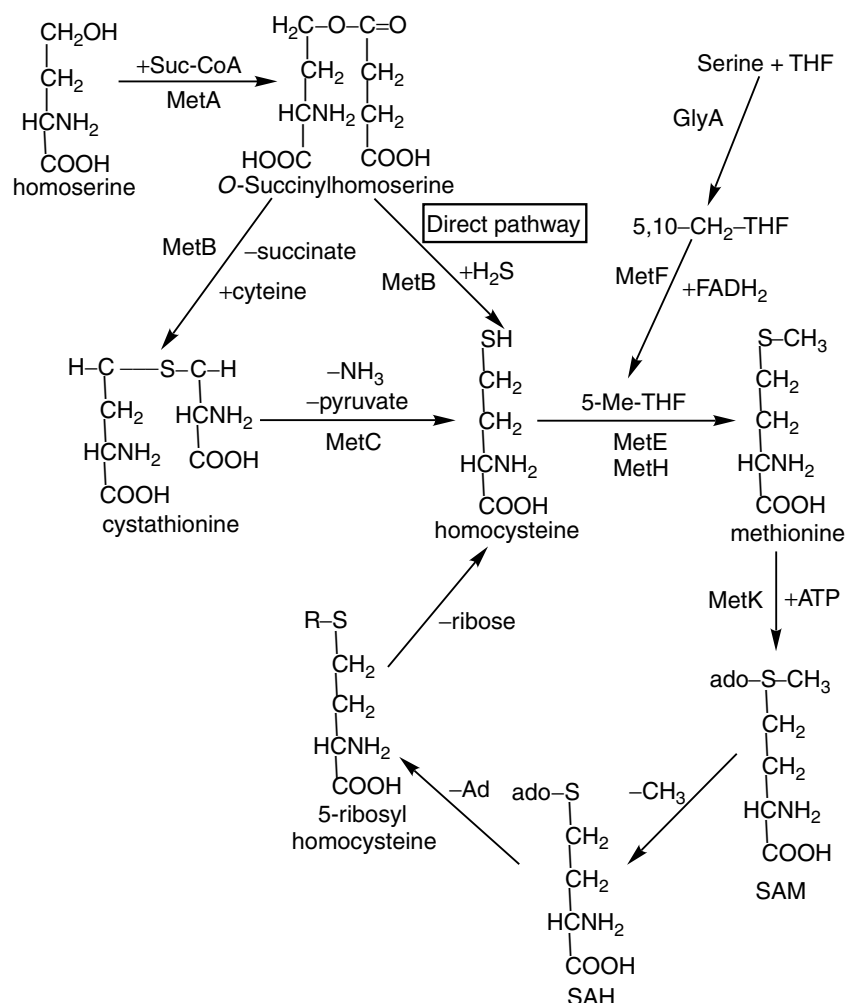


Fig. 15-10. Alternate pathways from homoserine to methionine. The enzyme designations are for *E. coli*. MetA, homoserine acyltransferase; MetB, cystathionine- γ -synthase; MetC, cystathionase; 5-Me-THF, 5-methyltetrahydrofolate; 5,10-CH₂-THF, 5,10-methylenetetrahydrofolate; MetF, 5,10-methylenetetrahydrofolate reductase; GlyA, serine hydroxymethyltransferase; MetE, tetrahydropteroyltriglutamate methyltransferase; MetH, vitamin B₁₂-dependent homocysteine-*N*⁵-methylenetetrahydrofolate transmethylase (requires catalytic amounts of SAM and FADH); MetK, s-adenosylmethionine synthetase.

In *Neurospora*, the conversion of homoserine to homocysteine occurs via an analogous sequence of reactions in which homoserine and acetyl-CoA are condensed to form *O*-acetylhomoserine. The reaction is catalyzed by homoserine acetyltransferase. In this sequence, cystathionine γ -synthase exchanges cysteine for the acetyl group in *O*-acetylhomoserine to release acetate and form cystathionine. This pathway is also followed in *Aspergillus nidulans*.

An alternate route of homocysteine formation involving the direct sulfhydrylation of *O*-acetylhomoserine, *O*-succinylhomoserine, or *O*-phosphohomoserine is mediated

by homocysteine synthase:



One or both of these activities has been reported in *N. crassa*, *S. cerevisiae*, and *E. coli*. The first of these reactions appears to be the main pathway in yeast; however, alternative routes for methionine biosynthesis are also available.

Isoleucine, Valine, and Leucine Biosynthesis

The isoleucine carbon skeleton is derived, in part, from aspartate via the deamination of threonine. It is convenient to discuss the biosynthesis of all three branched-chain amino acids at the same time because of the close interrelationship of the pathways. In the isoleucine-valine pathway, the same enzymes catalyze four of the steps in both sequences as shown in Figure 15-11. The immediate precursor of valine, α -ketoisovalerate, represents another branch point. Condensation with acetyl-CoA initiates the series of reactions leading to leucine synthesis, as shown in Figure 15-11. Via another series of reactions, α -ketoisovalerate is converted to pantoic acid, a precursor of pantothenic acid. The basic series of reactions leading to the formation of the branched-chain amino acids appears to be quite similar in most microorganisms. For example, the isopropylmalate pathway to leucine is widespread among diverse organisms capable of leucine biosynthesis.

Occasionally, organisms are found with novel routes of biosynthesis. In *Leptospira*, there is evidence that isoleucine carbon arises via a pathway involving citramalate as an intermediate. The origins of the carbon skeletons of the other amino acids are the same as in other prokaryotes.

Regulation of the Aspartate Family

Regulation of the biosynthesis of the amino acids of the aspartate family is complex because of the multiple branches and, to some extent, the interrelationships between the aspartate and pyruvate families. In *E. coli* K-12, primary regulation is exerted at two points. The major one is at the aspartokinase reaction, which regulates the flow of carbon to all of the amino acids formed (Fig. 15-12). The second site is the conversion of aspartate- β -semialdehyde to homoserine, a reaction catalyzed by homoserine dehydrogenases I and II. Aspartokinase I and homoserine dehydrogenase I activities are associated with a single multifunctional enzyme, ThrA, which is subject to allosteric inhibition by threonine. The synthesis of ThrA is repressed by a combination of threonine and isoleucine.

The structural genes for aspartokinase I-homoserine dehydrogenase I (*thrA*), homoserine kinase (*thrB*), and threonine synthase (*thrC*) lie in close proximity to one another on the *E. coli* map and constitute an operon. Aspartokinase II and homoserine dehydrogenase II activities are also catalyzed by a bifunctional protein (MetL).

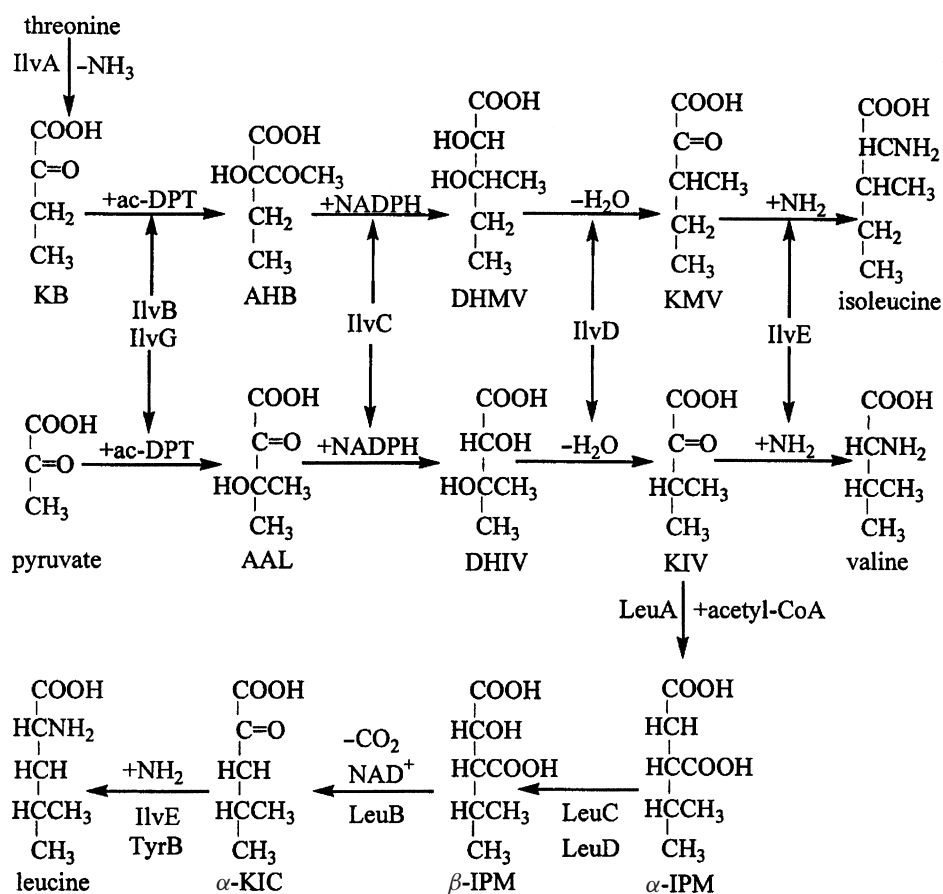


Fig. 15-11. Biosynthesis of the branched-chain amino acids, isoleucine, valine, and leucine. Enzyme designations are for *E. coli*. IlvA, threonine deaminase; IlvB, valine-sensitive acetohydroxy acid synthase I; IlvG, acetohydroxy acid synthase II; IlvC, acetohydroxy acid isomero-reductase; LeuA, α -isopropylmalate synthase; LeuC, LeuD, isopropylmalate dehydratase; LeuB, β -isopropylmalate dehydrogenase; TyrB, aromatic amino acid aminotransferase. KB, α -keto-butyrate; AHB, α -aceto- α -hydroxybutyrate; DHMV, α,β -dihydroxy β -methylisovalerate; KMV, α -keto- β -methylvalerate; α -IPM, α -isopropylmalate; β -IPM, β -isopropylmalate; α -KIC, α -keto-isocaproate.

Transcription of *metL* is regulated by methionine. However, neither of the two activities is feedback inhibited by methionine, threonine, SAM, or by combinations of these compounds. Aspartokinase III (LysC) is regulated by lysine through both feedback and repression mechanisms. Comparable regulatory systems for the aspartate pathway are present in other members of the *Enterobacteriaceae*, particularly *Salmonella*, *Enterobacter*, *Edwardsiella*, *Serratia*, and *Proteus*.

In *E. coli*, feedback inhibition plays a prominent role in regulation of the lysine pathway. A constitutive mutant for the synthesis of the lysine-sensitive aspartokinase

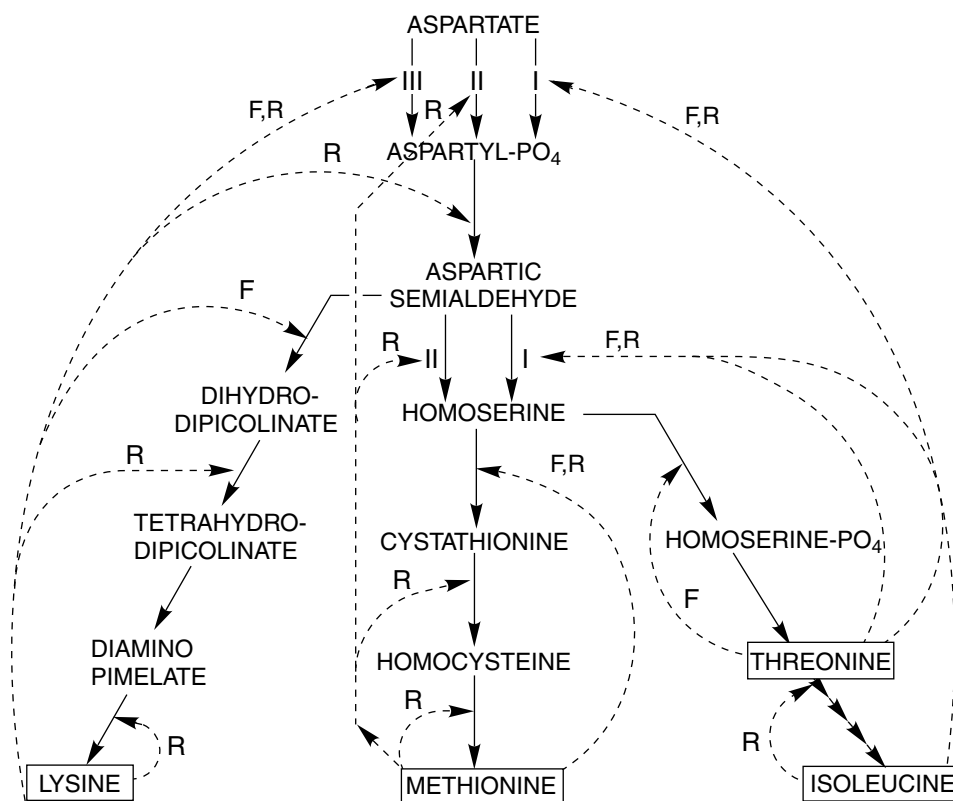


Fig. 15-12. End-product regulation of the aspartate family of amino acids in *E. coli*. F, feedback inhibition; R, repression of enzyme synthesis. Aspartokinase I–homoserine dehydrogenase I is a bifunctional enzyme encoded by *thrA*. Aspartokinase II–homoserine dehydrogenase II is a bifunctional enzyme encoded by *metL*. Aspartokinase III is encoded by *lysC*.

III (*lysC*) has been isolated. The aspartokinase III protein may serve as an inducer of diaminopimelate decarboxylase synthesis and thus regulate lysine synthesis in this manner. In some organisms, **concerted feedback** or **multivalent inhibition** is a more common type of regulation of the enzymes of the aspartate family. In this type of inhibition, all of the products of the pathway act together to effect the inhibition. Single aspartokinase enzymes subject to inhibition by lysine and threonine have been reported.

THE SERINE-GLYCINE FAMILY

In bacterial, fungal, and mammalian systems, metabolic relationships between glycine and serine may be described by the general scheme shown in Figure 15-13. Serine is a precursor of L-cysteine (Fig. 15-14). Glycine contributes carbon and nitrogen in the biosynthesis of purines, porphyrins, and other metabolites. As shown in Figure 15-13, serine hydroxymethyltransferase (SHMT), the *glyA* gene product, converts serine to

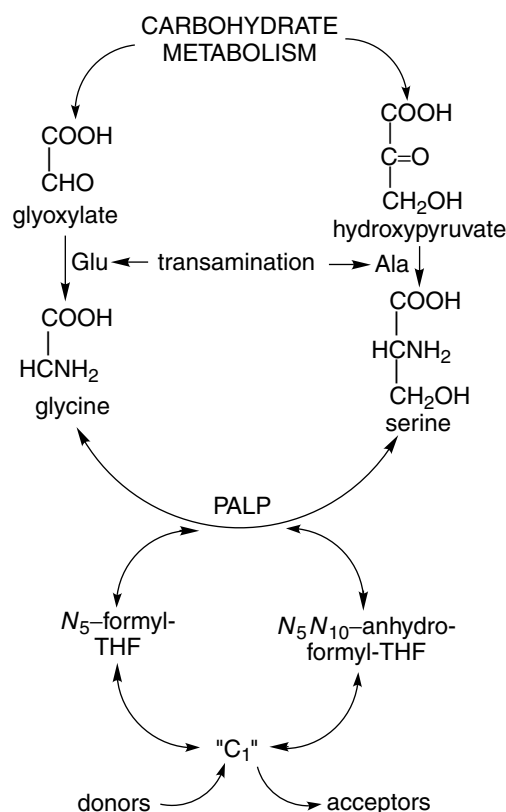
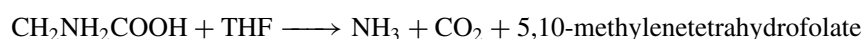


Fig. 15-13. Glycine-serine interrelationships. THF, tetrahydrofolate; PALP, pyridoxal phosphate.

glycine and 5,10 methylenetetrahydrofolate, a major contributor of one-carbon units, to the formation of methionine, purines, and thymine. Oxidative cleavage of glycine by the glycine cleavage (GCV) enzyme system provides a second source of one-carbon units:



The GCV system has been described in several bacterial species including *E. coli*, *Peptococcus glycinophilus*, and *Arthrobacter globiformis*. It consists of four proteins: the P protein, a pyridoxal phosphate enzyme; the H protein, a lipoate-containing hydrogen carrier; the T protein that transfers the methylene carbon to THF; and the L-protein, a lipoamide dehydrogenase encoded by *lpd* that is common to the pyruvate and α -ketoglutarate dehydrogenase complexes. The genes encoding these enzymes form an operon that maps at 65.2 minutes on the *E. coli* chromosome. The regulatory proteins Lrp, GcvA, and PurR are involved in the regulation of the glycine-inducible GCV system.

Serine, glycine, and cysteine are synthesized via a pathway emanating from 3-phosphoglycerate and proceeding through a series of phosphorylated intermediates (Fig. 15-14). Serine may be derived from glycine formed from glyoxylate by

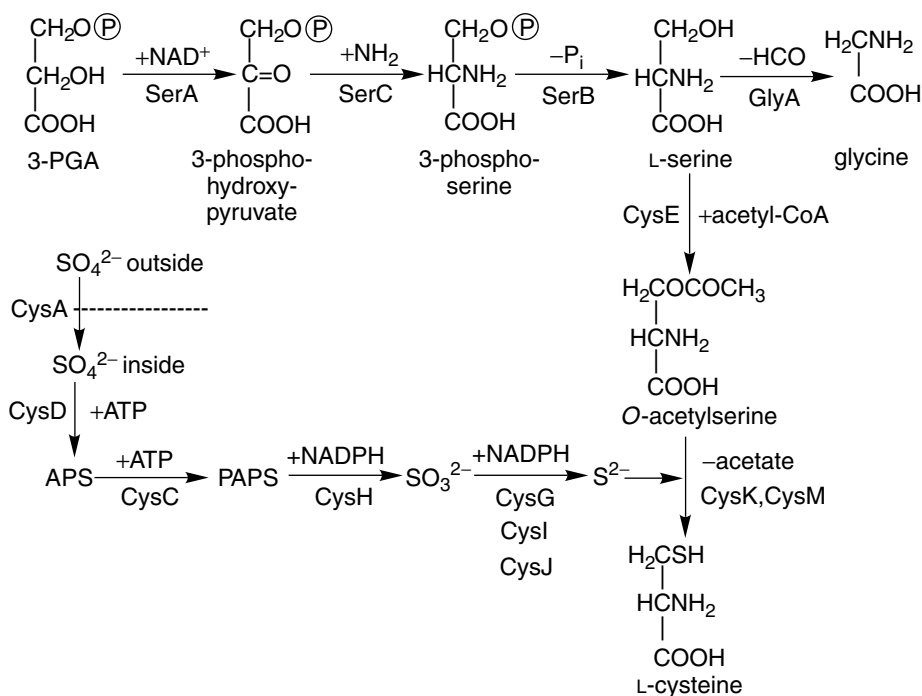


Fig. 15-14. Biosynthesis of serine, glycine, and cysteine. The enzyme designations for *E. coli* are: SerA, 3-phosphoglycerate (3-PGA) dehydrogenase; SerC, phosphoserine aminotransferase; SerB, phosphoserine phosphatase; GlyA, serine hydroxymethyltransferase; CysA, sulfate permease; CysD, sulfate adenylyltransferase; CysC, adenylylsulfate (APS) kinase; CysH, 3'-phosphoadenylyl sulfate (PAPS) reductase; CysG, CysI, CysJ, sulfite reductase; CysE, serine acetyltransferase; CysK, acetylserine sulphydrylase A; CysM, acetylserine sulphydrylase B. The *cys* operon is regulated by *cysB*.

transamination. Deamination of threonine to α -ketobutyrate with cleavage to acetyl-CoA and glycine can also give rise to serine.

Bakers yeast possesses the phosphorylated pathway to serine from 3-phosphoglycerate as well as the glyoxylate pathway. Regulation of the phosphorylated pathway by serine feedback inhibition of 3-phosphoglycerate dehydrogenase and regulation of glyoxylate transaminase has been demonstrated in *S. cerevisiae*. In bacteria the phosphorylated pathway appears to be the main route of serine formation. The bacterial pathway is also regulated by feedback inhibition of 3-phosphoglycerate dehydrogenase.

Sulfur is a constituent of several important biological components including cysteine, methionine, thiamine, biotin, lipoic acid, and coenzyme A. Sulfur usually occurs as sulfate and is transported into the cell via a sulfate permease. In most bacteria, sulfate is incorporated into organic compounds via the formation of adenylylsulfate (APS) and phosphoadenylyl sulfate (PAPS) with reduction to sulfide and incorporation into L-cysteine. Cysteine provides the sulfur atom for other sulfur-containing compounds, including methionine. The converging pathways of sulfate reduction and the formation of O-acetylserine shown in Figure 15-14 appears to be the most common route for the synthesis of cysteine. However, cysteine may also arise from serine and hydrogen

sulfide by the action of serine sulfhydrase (cysteine synthase). Cysteine is also formed by transsulfuration between homocysteine and serine (Fig. 15-10). In *K. aerogenes* the reduced sulfur of methionine can be recycled into cysteine. Under conditions where methionine serves as the sole source of sulfur, a pathway involving SAM forms homocysteine (Fig. 15-10). The yeast, *S. cerevisiae*, and some bacteria assimilate sulfide into homoserine to form homocysteine (Fig. 15-10). The homocysteine can then be converted to methionine.

Cysteine biosynthesis is regulated by the genes that code for the enzymes in the pathways of sulfate reduction and the formation of *O*-acetylserine from serine (Fig. 15-14). *O*-Acetylserine induces sulfate permease activity and also induces the rest of the enzymes in the sulfate reduction sequence. In *S. enterica* each of the enzymes in the sequence is repressed by cysteine. Sulfate permease (CysA) and serine acetyltransferase (CysE) are also inhibited by cysteine via a feedback mechanism. Although the biosyntheses of cysteine and methionine are obviously interrelated, it is the sulfate transport system and the pathway involved in the activation of sulfur that govern the synthesis of cysteine through positive control mechanisms, whereas methionine biosynthesis is under negative control. However, it has been proposed that the *O*-acetyl derivatives play a parallel role in the regulation of the synthesis of methionine and cysteine by inducing the initial steps in the pathway. Three of the genes controlling the pathway of sulfate activation (*cysIJH*) are arranged in a cluster forming an operon. Other regulatory interrelationships also exist between the cysteine and methionine pathways.

Aminolevulinate and the Pathway to Tetrapyrroles

As discussed earlier, aminolevulinate (ALA) can be synthesized by condensation of glycine and succinyl-CoA (C_4 pathway) or from glutamate (C_5 pathway). In either case, ALA is the first committed step in the formation of tetrapyrroles, as shown in Figure 15-15. The pathway diverges at uroporphyrinogen III (UroIII), one branch leading to the synthesis of siroheme and vitamin B_{12} (cobalamin) and the other to the formation of protoporphyrin IX. At this point, the pathway branches again, one branch leading to heme and the other to bacteriochlorophyll.

THE AROMATIC AMINO ACID PATHWAY

Phenylalanine, Tyrosine, and Tryptophan

Demonstration of shikimate, anthranilate, and indole as precursors of tryptophan and other aromatic amino acids in fungi and bacteria provided the earliest clues as to the intermediates in the aromatic amino acid pathway. Inhibition by analogs of the aromatic amino acids, isolation of mutants with multiple requirements for aromatic compounds, and other nutritional and biochemical findings indicated that these compounds shared a common origin. Shikimate substituted for the aromatic amino acid requirements of many auxotrophs, indicating that it was the common precursor of at least five different aromatic compounds. As a result of these findings, shikimate was considered to be the major branch point compound in the aromatic pathway. However, in both bacteria and fungi it was ultimately shown that chorismate (from the Greek meaning “to branch”) serves in this capacity (Fig. 15-16).

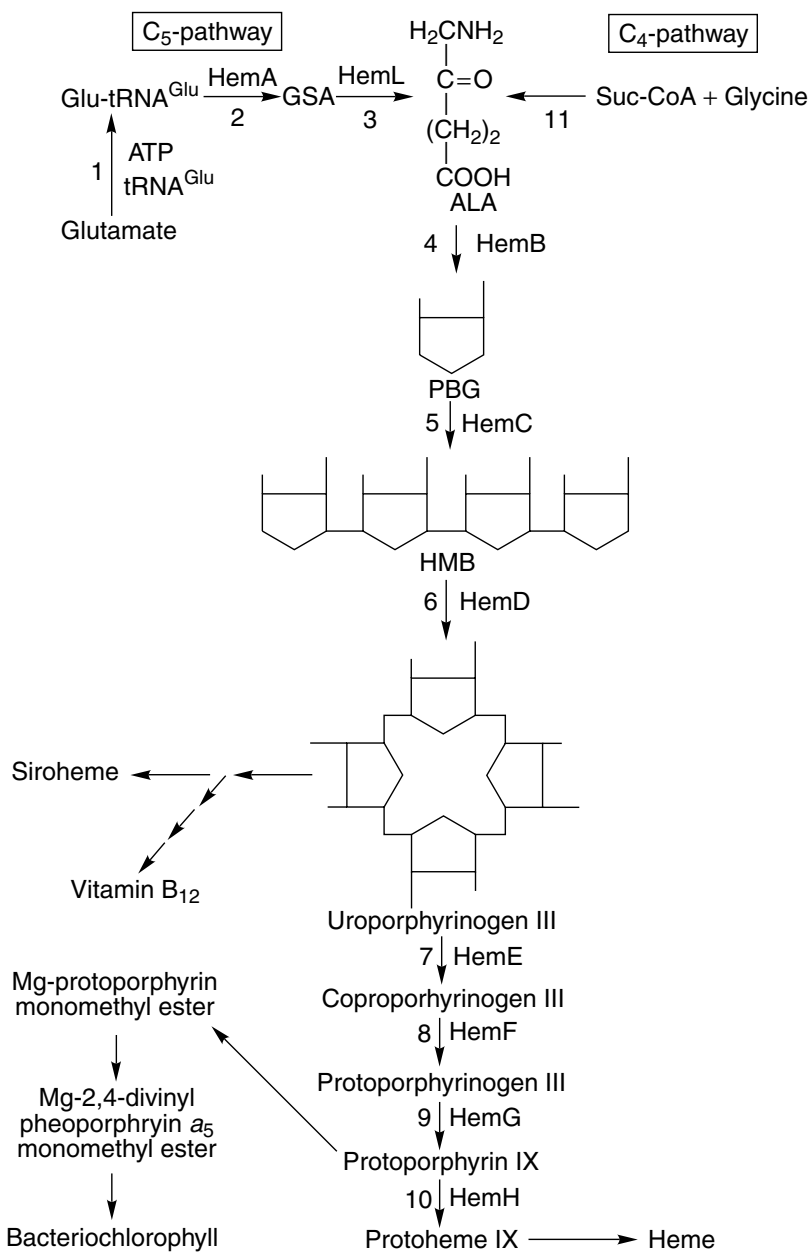


Fig. 15-15. Tetrapyrrole biosynthesis. The enzymes in the pathway are (1) glutamyl-tRNA synthase, (2) NAD(P)H glutamyl-tRNA reductase, (3) GSA (glutamate-γ-semialdehyde) 2,1-aminotransferase, (4) PBG (porphobilinogen) synthase, (5) HMB (hydroxymethylbilane) synthase, (6) UroIII (uroporphyrinogen) synthase, (7) uroporphyrinogen III decarboxylase, (8) coproporphyrinogen III oxidase, (9) protoporphyrinogen IX oxidase, (10) ferrochelatase, (11) ALA synthase.

The Common Aromatic Amino Acid Pathway

The aromatic amino acids phenylalanine, tyrosine, tryptophan, and several other related aromatic compounds are produced via a common pathway that begins with the condensation of erythrose-4-phosphate (E-4-P) and phosphoenolpyruvate (PEP) to form 3-deoxy-D-arabino-heptulosonate 7-phosphate (DAHP). The enzymes participating in the common aromatic pathway and its branches to tyrosine and phenylalanine are shown in Figure 15-16. DAHP is converted to shikimate and then to chorismate.

Chorismate stands at a branch point leading to the formation of several other amino acids and related metabolites. The formation of many additional aromatic compounds via branches of this pathway emphasizes the need for a complex regulatory system. Furthermore, a mutant blocked in the common pathway prior to a branch point may exhibit multiple nutritional requirements. Although there are many similarities in the aromatic amino acid pathway in all bacteria and fungi, there are several differences in detail regarding the manner in which tryptophan, tyrosine, and phenylalanine are formed. Even more apparent are the number of variations in the ways in which other related compounds are formed through branches or extensions from the common pathway.

In *E. coli* and *S. enterica*, regulation of the common aromatic pathway is modulated through three unlinked genes, *aroF*, *aroG*, and *aroH*, which encode isozymes of the first enzyme, DAHP synthetase, sensitive to tyrosine, phenylalanine, and tryptophan, respectively. Although all three DAHP synthetases are regulated transcriptionally, feedback inhibition is quantitatively the major control mechanism in vivo. In *S. enterica* the structural gene for the phenylalanine-inhibitable DAHP synthetase (*aroH*) is linked to *gal*. The structural gene for the tryptophan-associated isozyme (*aroF*) is linked to *aroE*, whereas that for the tyrosine-inhibitable isozyme (*aroG*) is linked to *pheA* and *tyrA*. The *pheA* and *tyrA* gene products are the respective branch-point enzymes leading to the phenylalanine and tyrosine pathways. Only the tyrosine-related isozyme (*aroG*) is completely repressed, as well as feedback inhibited, by low levels of tyrosine.

In *E. coli* there is a similar pattern of isozymic control of DAHP synthetase and control of the genes in the aromatic amino acid pathway. The Tyr repressor, the *tyrR* gene product, mediated by tyrosine and phenylalanine, respectively, represses expression of *aroF*. The TyrR protein also regulates the expression of several other genes concerned with aromatic amino acid synthesis or transport. These include *aroL*, *aroP*, *tyrB*, *tyrP*, and *mtr* (resistance to the tryptophan analog 5-methyltryptophan). Expression of *aroH* is controlled by the Trp repressor (encoded by *trpR*), which also regulates the expression of the *trp* operon, the *mir* gene, as well as the expression of *trpR* itself.

In a wide variety of fungi, the genes coding for enzymes 2 to 6 in the prechorismate pathway (Fig. 15-16) are arranged in a cluster, designated the *arom* gene cluster. In most fungi these enzymes sediment in a complex aggregate on centrifugation in sucrose density gradients. The sedimentation coefficients for them are very similar in representative examples of Basidiomycetes (*Coprinus lagopus*, *Ustilago maydis*), Ascomycetes (*N. crassa*, *S. cerevisiae*, *A. nidulans*), and Phycomycetes (*Rhizopus stolonifer*, *Phycomyces nitens*, *Absidia glauca*). By comparison, the five enzymes catalyzing the conversion of DAHP to chorismate in bacteria are physically separable in *E. coli*, *S. enterica*, *E. aerogenes*, *Anabaena variabilis*, and *Chlamydomonas reinhardtii*. In *Euglena gracilis* an enzyme aggregate with five activities has been found. This aggregate can be disassociated into smaller components.

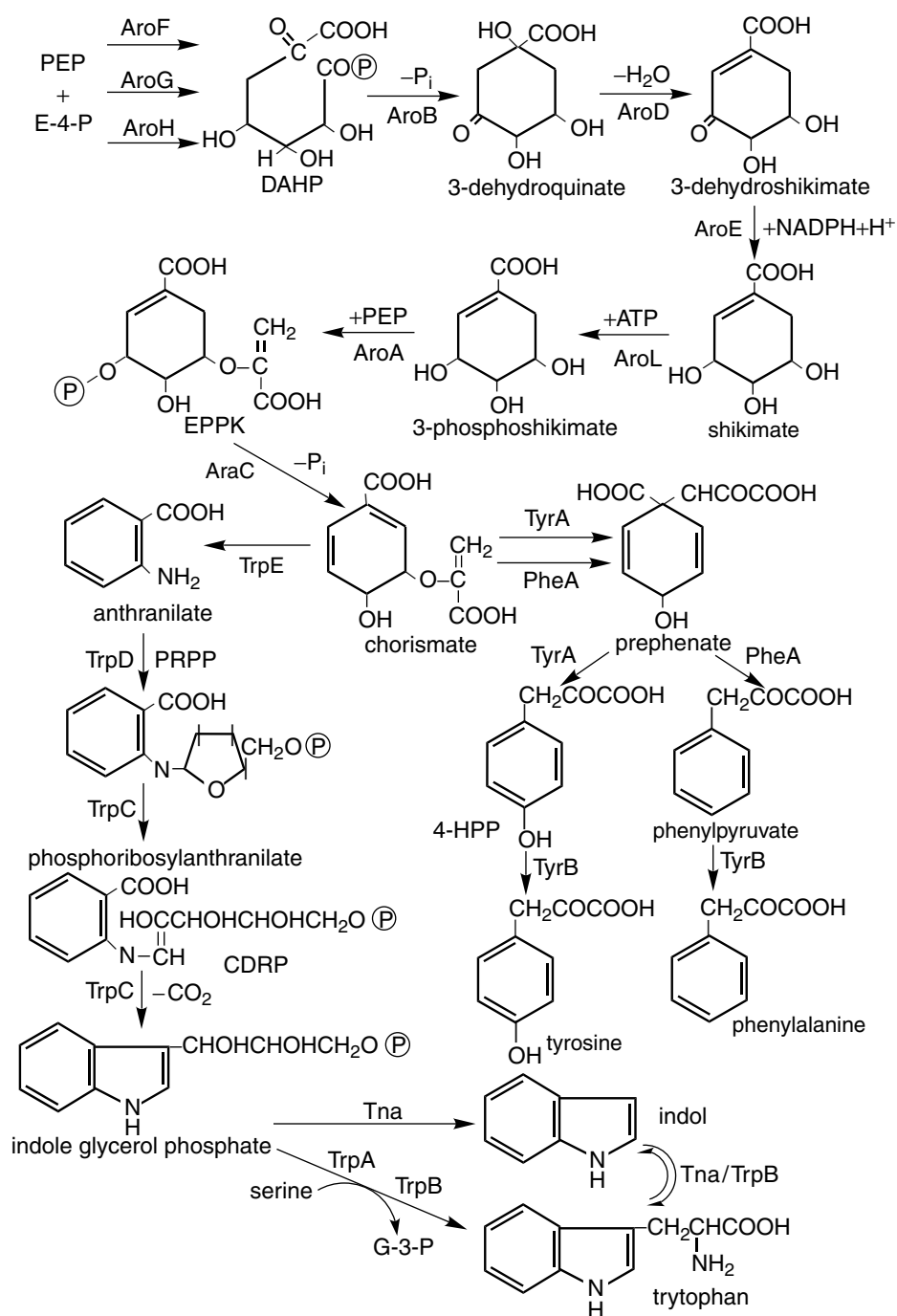
In some bacteria, the genes coding for enzymes of the chorismate pathway appear to be scattered on the chromosome. However, in *B. subtilis*, a cluster of contiguous genes for tryptophan biosynthesis represents a polycistronic operon. The *trpEDCFBA* genes form an operon similar to operons found in enteric bacteria. However, in *B. subtilis*, the *trp* genes are part of a **supraoperon** containing *trpEDCFBA-hisH-tyrA-aroE*, as shown in Figure 15-17a. The *aroFBH* genes may also be present at the 5' end of this supraoperon. All *aroF* mutants (chorismate synthetase) also lack dehydroquinate synthase (AroB) activity. The gene that specifies AroB is closely linked to *aroF*. Both genes are part of the *aro* gene cluster. Mutants lacking chorismate mutase activity also lack DAHP synthetase and shikimate kinase activity, presumably as a result of their aggregation in a multienzyme complex. As an indication of the complexity of the supraoperon, the *mtrAB* operon of *B. subtilis* encodes GTP cyclohydrolase I (MtrA), an enzyme involved in folate biosynthesis. The *mtrB* gene is a *trans*-acting RNA-binding regulatory protein activated by tryptophan. Transcription termination at the attenuator preceding the *trp* gene cluster presumably occurs as a consequence of binding of the activated MtrB protein to the nascent transcript (see later discussion on interpathway regulation in Chapter 16).

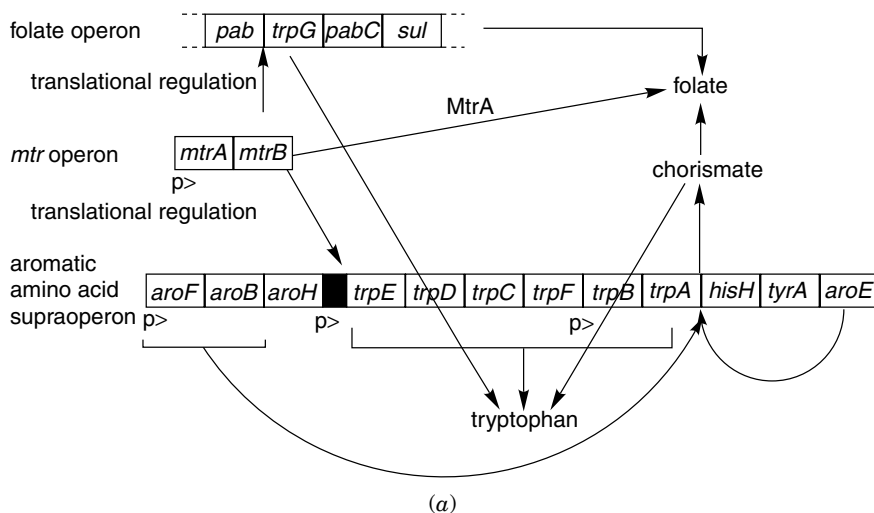
The genes controlling tryptophan synthesis in enteric bacteria are also arranged in an operon. The *E. coli trp* operon and the enzymes under its control are shown in Figure 15-17b. There are five structural genes for the enzymes in the operon and they are induced coordinately in response to the availability of tryptophan. In *E. coli* the activities of the first two enzymes are catalyzed by an aggregate formed from the products of the first two genes in the operon. A single protein catalyzes steps 3 and 4. Tryptophan synthase converts indoleglycerol phosphate to indol and then couples indol to serine to form tryptophan. The enzyme is a complex formed from the products of the last two genes (*trpA* and *tryB*). A regulatory site preceding the structural genes regulates the synthesis of mRNA^{Trp} and the enzymes of the tryptophan pathway, thus reducing operon expression. This **attenuator** function apparently occurs at the level of transcription by providing a region in which transcription is terminated (see Chapter 5 for additional details concerning regulation of gene expression).

Pathways to Tyrosine and Phenylalanine

The biosynthesis of L-tyrosine and L-phenylalanine from chorismate can occur via alternate routes in many organisms (Fig. 15-18). In *E. coli*, TyrA (chorismate mutase

Fig. 15-16. Aromatic amino acid pathways. The enzyme designations are for *E. coli*. AroF, AroG, AroH, 3-hydroxy-L-arabino-heptulosonate 7-phosphate (DHAP) synthetases; AroB, 3-dehydroquinate synthase; AroD, 3-dehydroquinate dehydratase; AroE, dehydroshikimate reductase; AroL, shikimate kinase II; AroA, 5-enolpyruvylshikimate 5-phosphate synthase; AroC, chorismate synthase; PheA, bifunctional enzyme, chorismate mutase P-prephenate dehydrogenase; TyrA, bifunctional enzyme, chorismate mutase T-prephenate dehydrogenase; TyrB, tyrosine aminotransferase; TrpE, anthranilate synthase; TrpD, anthranilate phosphoribosyl transferase; trpC, bifunctional enzyme, *N*-(5-phosphoribosyl) anthranilate isomerase-indole-3-glycerophosphate synthetase; TrpA, TrpB, tryptophan synthase; Tna, tryptophanase. PEP, phosphoenolpyruvate; E-4-P, erythrose-4-phosphate; EPPK, 3-enoyl-pyruvyl-3-phosphoshikimate; CDRP, 1-(*O*-carboxy-phenylamino)-1-deoxyribulose-5-phosphate; 4-HPP, 4-hydroxyphenylpyruvate.



Bacillus subtilis***Escherichia coli***

chorismate $\xrightarrow{1}$ AA $\xrightarrow{2}$ CP-PRA $\xrightarrow{3}$ CDRP $\xrightarrow{4}$ inGP $\xrightarrow{5}$ tryptophan

p1,O	a	trpE	trpG/D	p2	trpC/F	trpB	trpA
------	---	------	--------	----	--------	------	------

	1	2	3	4	5a	5b
p1 = promoter	AA	PR	PR-AA	inGP	tryptophan	synthase
O = operator	synthase	transferase	isomerase	synthase	β	α
a = attenuator						
p2 = internal promoter						

Responds to:

AA	indole	indole	indole	indole	indole	
tryptophan	tryptophan	tryptophan	tryptophan	tryptophan	tryptophan	tryptophan

Accumulates:

none	AA	AA	CDR	indoleG	indole	
------	----	----	-----	---------	--------	--

(b)

Fig. 15-17. Organization of the genes for tryptophan biosynthesis. (A) The aromatic amino acid supraoperon of *B. subtilis* showing interrelationships with the genes from the folate operon and the *mtr* operon. The MtrA protein (GTP cyclohydrolase I) catalyzes formation of the initial pteridine ring (dihydroneopterin triphosphates) of folate. The proteins AroF, AroB, and AroE are required for chorismate synthesis. The proteins Pab, TrpG, and PabC are involved in *p*-aminobenzoate PAB synthesis from chorismate. Sul catalyzes condensation of PAB and the pteridine ring. MtrB is considered to bind to a target sequence that overlaps the *trpG* ribosome-binding site, resulting in transcriptional regulation of the *trp* gene cluster. The *trpEDCFBA* and *trpG* gene products are necessary for tryptophan biosynthesis from chorismate in some strains of *Bacillus*. p> indicates positions of known promoters; ■ denotes position of the *trp* attenuator; = = = indicates open reading frames upstream of *pab* and downstream of *sul*. (Source: Redrawn from Babitzke, P. et al., *J. Bacteriol.* **174**:2059-2064, 1992). (B). Organization of the genes of the tryptophan operon of *E. coli*. AA, anthranilate; CP-PRA, *N*-(*o*-carboxyphenyl)-phosphoribosylamine; CDRP, 1-(*o*-carboxyphenylamino)-1-deoxyribose 5-phosphate; inGP, indoleglycerolphosphate.

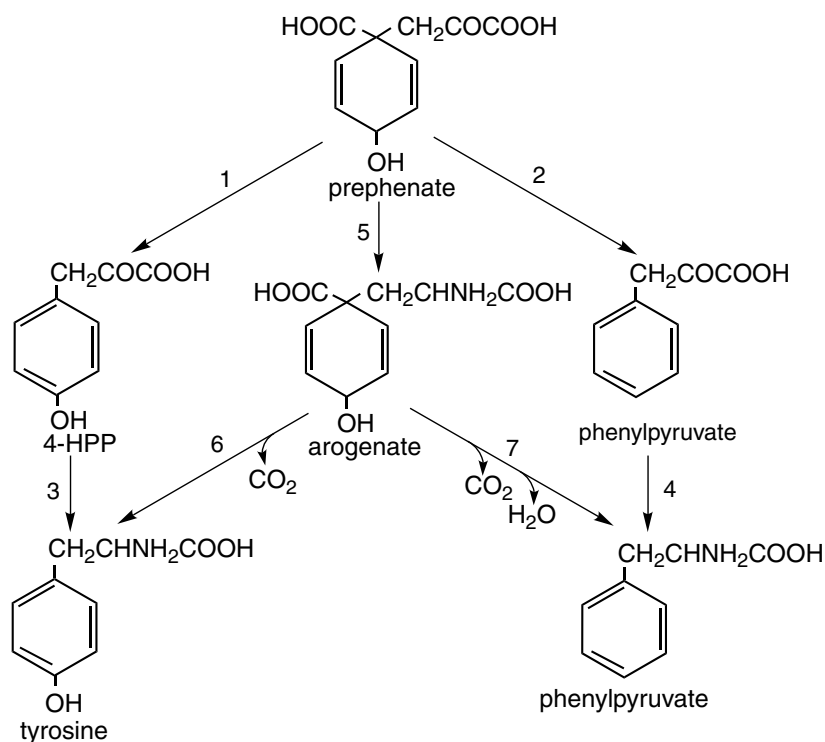


Fig. 15-18. Alternate pathways to phenylalanine and tyrosine. The pathways shown are present in *Pseudomonas aeruginosa*. Various organisms may express any combination of these pathways. The dehydrogenases may be either NAD^+ or NADP^+ linked. The enzymes are (1) prephenate dehydrogenase, (2) prephenate dehydratase, (3) 4-hydroxyphenylpyruvate (4-HPP) aminotransferase, (4) phenylpyruvate aminotransferase, (5) prephenate aminotransferase, (6) arogenate dehydratase, and (7) arogenate dehydrogenase.

T-prephenate dehydrogenase) is a bifunctional enzyme that can convert chorismate to prephenate and then to phenylpyruvate. Alternatively, prephenate may be converted to arogenate by transamination, with subsequent conversion of arogenate to either tyrosine (by arogenate dehydrogenase) or phenylalanine (by arogenate dehydratase). Utilization of these alternate pathways varies greatly from one organism to another. In *E. coli*, *S. enterica*, *B. subtilis*, and other common bacteria, the prephenate pathway to phenylpyruvate and phenylalanine and to 4-hydroxyphenylpyruvate and tyrosine is the predominant route as shown in Figure 15-15.

By comparison, *Euglena gracilis* uses arogenate as the sole precursor of both tyrosine and phenylalanine via a pathway in which arogenate rather than prephenate is the branch point. The arogenate pathway is common in the cyanobacteria. However, the degree to which it is used and the manner in which the alternative pathways are regulated differ widely from one species to another. In *P. aeruginosa* and other pseudomonads, these alternate pathways coexist. Their presence accounts for the unusual resistance of these organisms to analogs of the aromatic amino acids. Under certain circumstances, the arogenate pathway appears to function as an unregulated overflow route for tyrosine production. The arogenate pathway is a

obligatory route to tyrosine in *Corynebacterium glutamicum*, *Brevibacterium flavum*, and *B. ammoniogenes*. By comparison, arogenate appears to be a deadend metabolite in *N. crassa*, the prephenate pathway being used as the major route to tyrosine and phenylalanine.

Salmonella enterica and *E. coli* synthesize tyrosine and phenylalanine by virtually separate pathways in that there are separate DAHP synthetases and chorismate mutases coded for by genes specific to either the tyrosine or phenylalanine pathways. The chorismate mutase T and prephenate dehydrogenase coded by *tyrA* are specific for tyrosine formation while chorismate mutase P and prephenate dehydratase coded by *pheA* are specific for phenylalanine biosynthesis. In *S. enterica* the genes *aroF* and *tyrA*, which specify the tyrosine-repressible DAHP synthase (AroF) and chorismate mutase T-prephenate dehydrogenase, comprise an operon. The genes *aroG* and *pheA*, which specify phenylalanine-repressible DAHP synthase (AroG) and chorismate mutase P-prephenate dehydratase (PheA), regulate the synthesis of phenylalanine as shown in Figure 15-19. These bifunctional enzymes are found in *S. enterica*, *E. coli*, and *E. aerogenes*. They are not found as bifunctional enzymes in *N. crassa*, *S. cerevisiae*, or *B. subtilis*.

Within the genera *Pseudomonas*, *Xanthomonas*, and *Alcaligenes*, variation in the enzymes of tyrosine biosynthesis comprises five groups that compare favorably with rRNA–DNA hybridization groups. The rRNA homology groups I, IV, and V all lack activity for arogenate/NADP dehydrogenase. Group II species possess arogenate dehydrogenase (with lack of specificity for NAD or NADP) and are sensitive to feedback inhibition by tyrosine. The arogenate dehydrogenase of species in group III is insensitive to feedback inhibition. Group IV displays prephenate/NADP dehydrogenase activity, whereas groups I and V lack this activity.

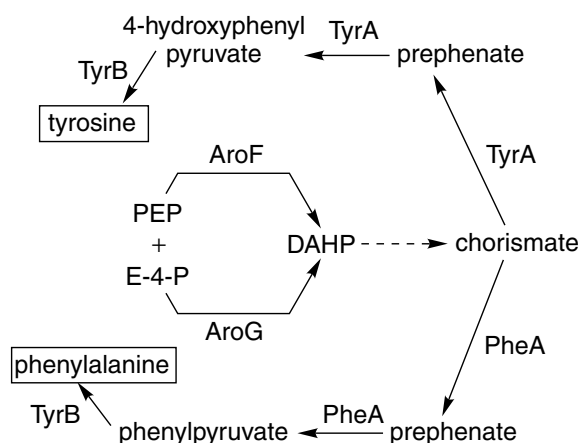
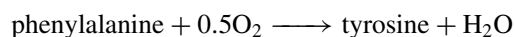


Fig. 15-19. Genes and enzymes of phenylalanine and tyrosine biosynthesis in *Escherichia coli* and *Salmonella enterica*. PEP, phosphoenolpyruvate; E-4-P, erythrose-4-phosphate; DAHP, 3-deoxy-D-arabinoheptulonate 7-phosphate. Enzyme designations are: AroF, tyrosine-repressible DAHP synthase; AroG, phenylalanine-repressible DAHP synthase; TyrA, chorismate mutase T-prephenate dehydrogenase; PheA, chorismate mutase P-prephenate dehydratase; TyrB, tyrosine aminotransferase.

Mammalian species appear to have the ability to convert phenylalanine to tyrosine through the action of phenylalanine hydroxylase:



In microorganisms this activity is uncommon. Dual requirements for tyrosine and phenylalanine generally accompany mutational loss of enzyme activity at a point prior to the branch point in the synthesis of these amino acids. *Pseudomonas* forms an inducible phenylalanine hydroxylase. Since hydroxylation reactions often represent the initiating step in the degradation of aromatic compounds, as discussed in Chapter 10, it seems plausible that a highly degradative organism, such as *Pseudomonas*, would possess such activity for degradative purposes and only secondarily to provide tyrosine from phenylalanine.

***p*-Aminobenzoate and Folate Biosynthesis**

Many bacteria and fungi, including *E. coli*, *Streptomyces griseus*, and *Pseudomonas acidovorans*, convert chorismate to *p*-aminobenzoate (PAB) via a two-step process (Fig. 15-20). In the first step, chorismate is converted to 4-amino-4-deoxychorismate (ADC) by ADC synthase (PabA, PabB) in the presence of glutamine. In the absence of PabA and glutamine, PabB can convert chorismate to ADC if an excess of ammonia is provided. ADC is converted to PAB by removal of the pyruvyl group and concomitant aromatization of the ring by a PALP-containing enzyme ADC lyase (PabC).

PAB is then condensed with 6-hydroxymethyl-7,8-pterin pyrophosphate (DHPS) by dihydropteroate synthase (SulA). The resultant dihydropteroate is then converted to 7,8-dihydrofolate by the addition of glutamate by dihydrofolate synthetase (FolC). After 7,8-dihydrofolate is reduced to tetrahydrofolate by dihydrofolate reductase, the synthase enzyme then serves to add additional glutamate units to form folylpolyglutamate.

The pteridine component of folate is derived from guanosine triphosphate (GTP). GTP cyclohydrolase I (FolE; in *B. subtilis* this enzyme (mtrA) is the product of the *mtrA* gene) catalyzes the first committed step in the biosynthesis of the pterin ring of folate. In this reaction the C-8 carbon of the GTP is removed as formate and the remaining enzyme-bound intermediate is converted to 7,8-dihydro-D-erythro-neopterin (H₂-neopterin) 3'-phosphate. Removal of the phosphate residues yields 7,8-dihydroneopterin. 7,8-Dihydroneopterin is then converted to 6-hydroxymethyl-7,8-dihydropterin by the action of dihydrohydroxymethylpterin pyrophosphokinase (FolK in *E. coli*; SulD in *Streptococcus pneumoniae*). In *S. pneumoniae*, SulD is a bifunctional enzyme with both 6-hydroxymethyl-7,8-dihydropterin kinase and 7,8-dihydroneopterin aldolase activities.

In the archaea *Methanococcus thermophila* and *Methanobacterium thermoautotrophicum*, ΔH GTP cyclohydrolase I is absent and GTP is transformed via a multistep process into H-neopterin 2':3'-cyclic phosphate. This compound is then hydrolyzed to H-pterin via a phosphorylated intermediate. Formation of the pterin ring is catalyzed by two or more enzymes rather than the GTP cyclohydrolase I enzyme present in eukaryotes and bacteria.

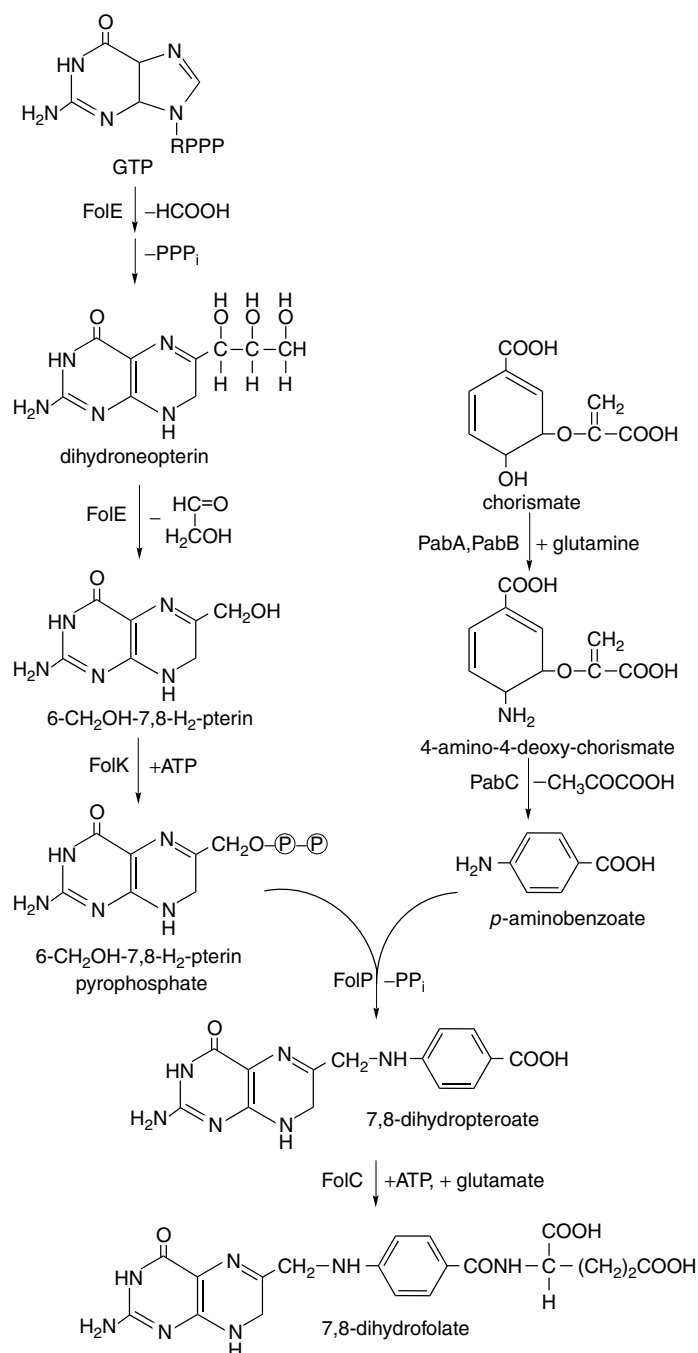


Fig. 15-20. PABA and folate biosynthesis. The enzyme designations for *Escherichia coli* are: PabA/PabB, 4-amino-4-deoxychorismate (ADC) synthase; PabC, ADC lyase; FolE, GTP cyclohydrolase; FolK, dihydro-hydroxymethylpterin pyrophosphokinase; FolP, dihydropterotate synthetase; FolC, dihydrofolate:folylpolyglutamate synthetase. [Pterin is the trivial name for 2-amino-4-hydroxypteridine; neopterin is the trivial name for 6(D-erythro-1',2',3'-trihydroxypropyl) pterin.]

Enterobactin Biosynthesis

The catechol siderophore enterobactin (enterochelin) is an iron-chelating compound that facilitates the transport of iron (see Chapter 7). It is a cyclic trimer of *N*-2,3-dihydroxybenzoylserine, which is synthesized from chorismate. The first stage of this pathway is the formation of 2,3-dihydroxybenzoate via the intermediates isochorismate and 2,3-dihydro-2,3-dihydroxybenzoate as shown in Figure 15-21. The

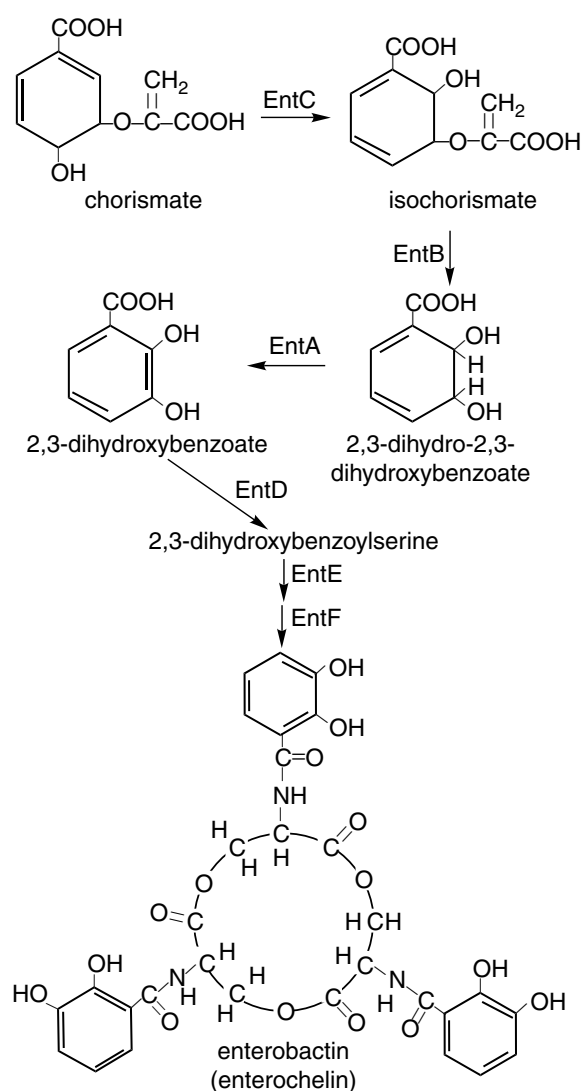


Fig. 15-21. Biosynthesis of enterobactin (enterochelin). The enzyme designations for *E. coli* are: EntC, isochorismate synthase; EntB, 2,3-dihydroxybenzoate synthetase; EntA, 2,3-dihydro-2,3-dihydroxybenzoate dehydrogenase; EntD, 2,3-dihydroxybenzoylserine synthetase; EntE, EntF serve to condense three molecules of 2,3-dihydroxybenzoylserine into the cyclic enterobactin structure.

enzymes catalyzing these steps are isochorismate synthetase (EntC), 2,3-dihydro-2,3-dihydroxybenzoate synthetase (EntB), and 2,3-dihydro-2,3-dihydroxybenzoate dehydrogenase (EntA). In *E. coli* eight genes govern the biosynthesis of enterobactin and its function. These genes occur in a cluster at 13 minutes on the *E. coli* linkage map. The second stage of enterobactin synthesis is catalyzed by a sequence of enzymes (EntD, EntE, EntF, and EntG) present in a multienzyme complex. EntD couples 2,3-dihydroxybenzoate and serine to form 2,3-dihydroxybenzoylserine. The remaining enzymes catalyze the coupling of three molecules of 2,3-dihydroxybenzoylserine to form the cyclical enterobactin structure.

The Pathway to Ubiquinone

Ubiquinone (coenzyme Q), the only nonprotein component of the electron transport chain, functions in terminal respiration (Chapter 9). Ubiquinone is synthesized from chorismate via the pathway shown in Figure 15-22. The first intermediate specific to the ubiquinone pathway, 4-hydroxybenzoate, is formed by the action of chorismate lyase (UbiC). In some organisms tyrosine can serve as a source of 4-hydroxybenzoate through the intermediary formation of 4-hydroxyphenylpyruvate. An octaprenyl group (R_8) is added by the action of UbiA (4-hydroxybenzoate octaprenyltransferase). Via decarboxylation, 3-octaprenyl-4-hydroxybenzoate is converted to 2-octaprenylphenol. UbiB, UbiH, and UbiF catalyze the three monooxygenase steps in the pathway. UbiE, an S-adenosylmethionine:2-demethylquinol methyltransferase, catalyzes the methylation steps in both the ubiquinone and menaquinone pathways.

Menaquinone (Vitamin K) Biosynthesis

Menaquinone (vitamin K₂) is a methylnaphthoquinone with the structure shown in Figure 15-23. Biosynthesis of menaquinone originates with the addition of α -ketoglutarate to the ring of chorismate and concomitant removal of pyruvate and CO₂. The details of the pathway are shown in Figure 15-23. All naturally occurring menaquinones have *trans*-configurations in the double bonds of the prenyl side chain. Menaquinone is essential for electron transfer to fumarate during anaerobic growth of *E. coli* (see Chapter 9).

Biosynthesis of Nicotinamide Adenine Dinucleotide (NAD)

Both NAD and NADP, the functional forms of nicotinic acid, are synthesized from tryptophan in mammals, *N. crassa*, and *S. cerevisiae*, as shown in Figure 15-24. The bacterium *Xanthomonas pruni* also synthesizes NAD from tryptophan. However, most bacteria use an entirely different pathway to NAD that involves the condensation of aspartate and dihydroxyacetone phosphate (DOHAP) and the subsequent conversion of the condensation product (iminoaspartate) to QA as shown in Figure 15-24. The pathway from QA to NAD appears to be identical in all organisms regardless of the mode of QA formation. The enteric bacteria (*E. coli* and *S. enterica*) and most other common bacteria (including *M. tuberculosis*) use the aspartate-DOHAP route to QA. Under anaerobic conditions, *S. cerevisiae* uses the aspartate-DOHAP pathway. This change most likely occurs because several oxidative steps in the pathway from tryptophan to quinolinate require molecular oxygen. For this reason, the bacterial

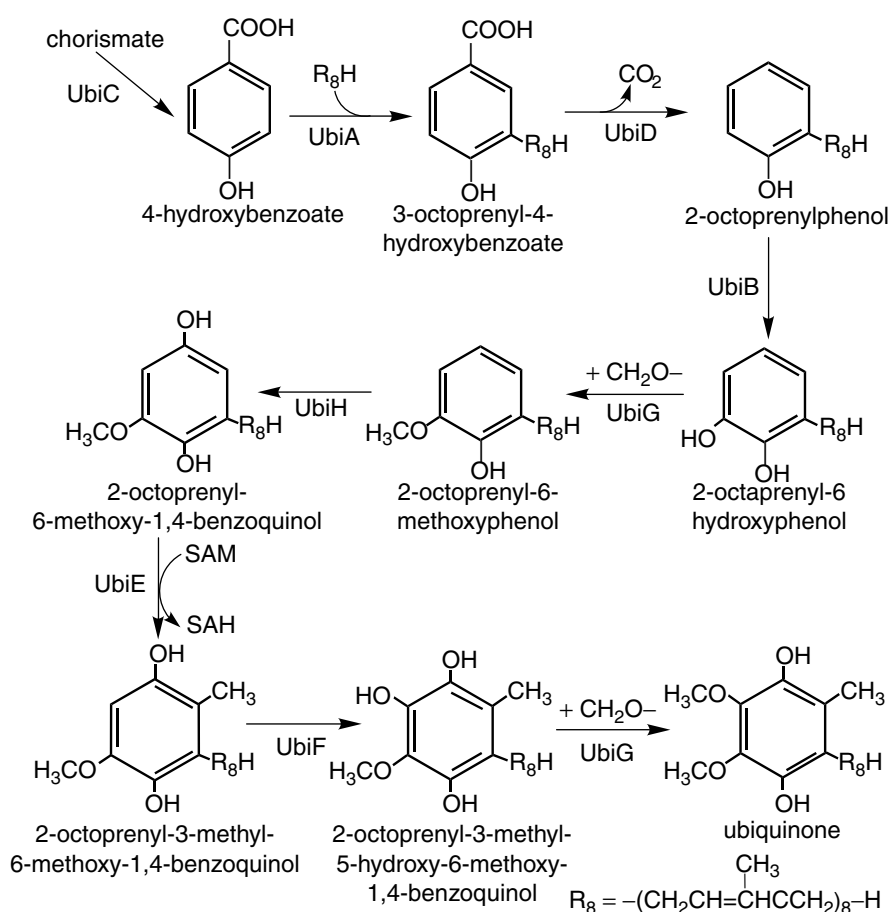


Fig. 15-22. Biosynthesis of ubiquinone (coenzyme Q). The enzyme designations are for *E. coli*. UbiC, chorismate lyase; UbiA, 4-hydroxybenzoate polyprenyl (R) transferase; UbiD, decarboxylation of 3-octaprenyl-4-hydroxybenzoate to 2-octaprenylphenol; UbiB, conversion of 2-octaprenylphenol to 2-octaprenyl-6-hydroxyphenol (flavin reductase); UbiG, o-methylation of 2-octaprenyl-6-hydroxyphenol to 2-octaprenyl-6-methoxyphenol; UbiH, conversion of 2-octaprenyl-6-methoxyphenol to 2-octaprenyl-6-methoxy-1,4-benzoquinol; UbiE, S-adenosyl-L-methionine (SAM)-dependent methyltransferase converts 2-octaprenyl-6-methoxy-1,4-benzoquinol to 2-octaprenyl-3-methyl-6-methoxy-1,4-benzoquinol; UbiF, converts 2-octaprenyl-3-methyl-6-methoxy-1,4-benzoquinol to 2-octaprenyl-3-methyl-5-hydroxy-6-methoxy-1,4-benzoquinol; UbiG, o-methylation of 2-octaprenyl-3-methyl-5-hydroxy-6-methoxy-1,4-benzoquinol to ubiquinone 8.

pathway is referred to as anaerobic. The anaerobe, *C. butylicum*, follows a unique pathway via condensation of aspartate and formate to yield the intermediate *N*-formylaspartate. The addition of acetate and subsequent ring closure yields quinolinate (Figure 15-24).

Organisms that demonstrate a specific nutritional requirement for nicotinamide utilize a unique route of NAD biosynthesis that involves conversion of nicotinamide

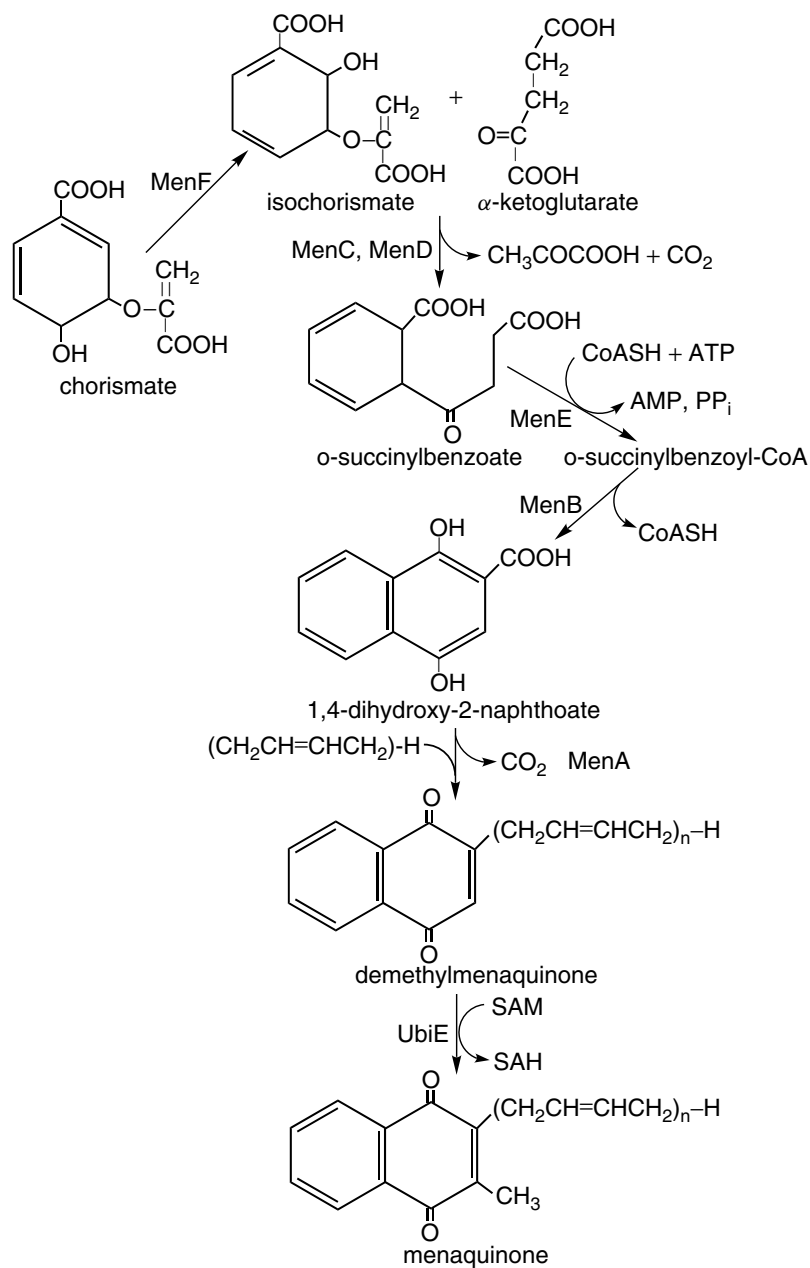


Fig. 15-23. Menaquinone biosynthesis. The enzyme designations are for *Escherichia coli*. MenF, menaquinone-specific isochorismate synthase; MenC, o-succinylbenzoate synthase II; MenD, o-succinylbenzoate synthase I [o-succinylbenzoate is the trivial name for 4-(2-carboxy-phenyl)-4-oxybutyrate]; MenE, o-succinylbenzoate-CoA synthase; MenB, 1,4-dihydroxy-2-naphthoate synthase; MenA, 1,4-dihydroxy-L-naphthoate octaprenyltransferase; UbiE, S-adenosylmethionine (SAM):2-demethylmenaquinone methyltransferase. SAH, S-adenosylhomocysteine.

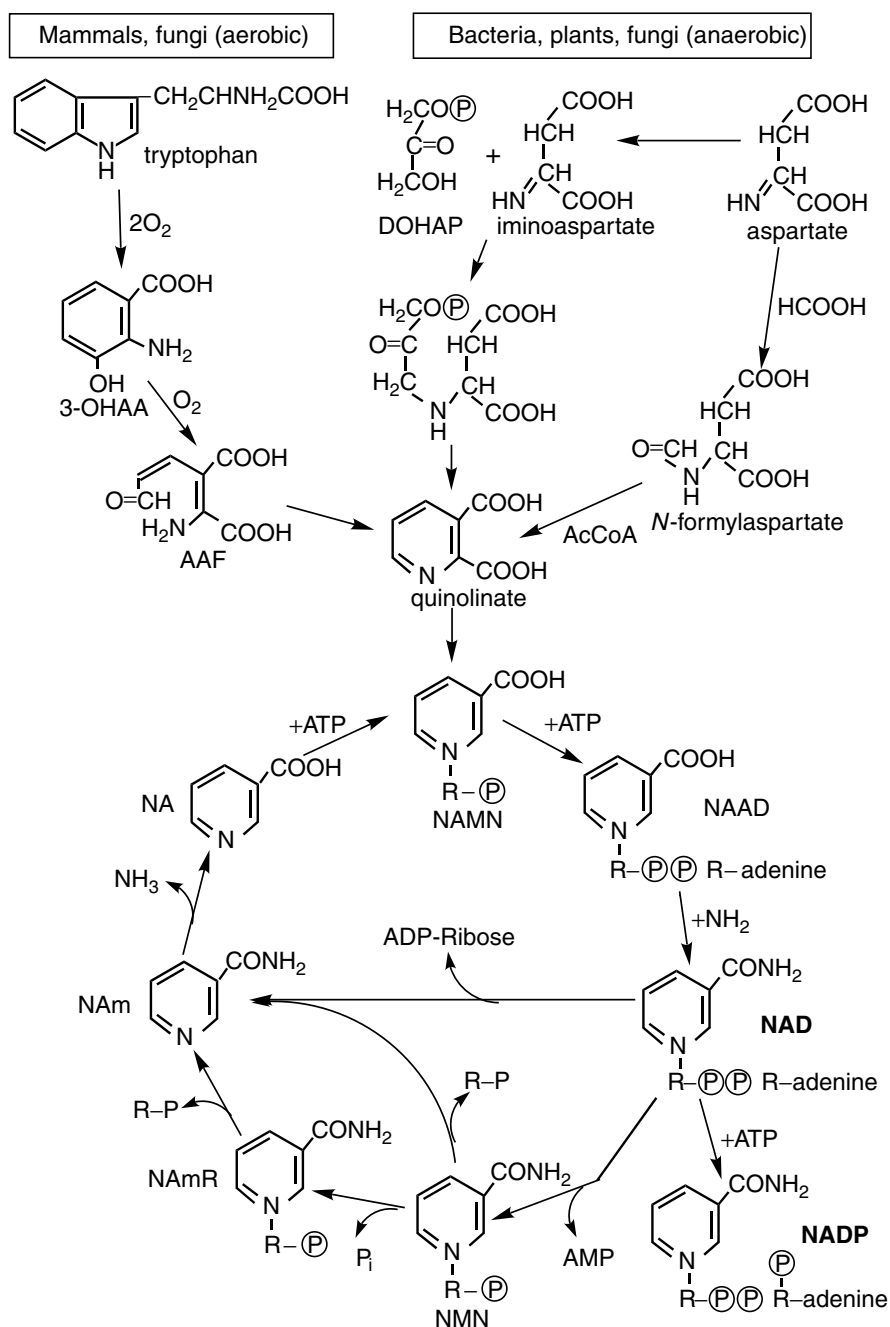
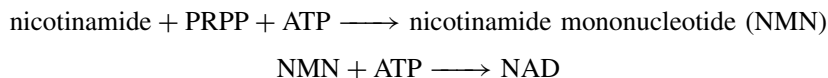


Fig. 15-24. Alternate pathways for NAD biosynthesis and the pyridine nucleotide cycles. 3-OHAA, 3-hydroxyanthranilate; AAF, 2-acrolein-3-aminofumarate; DOHAP, dihydroxyacetone phosphate; AcCoA, acetyl coenzyme A; NAMN, nicotinic acid mononucleotide; NAAD, nicotinic acid adenine dinucleotide; NAD, nicotinamide adenine dinucleotide; NADP, nicotinamide adenine dinucleotide phosphate; NMN, nicotinamide mononucleotide; NamR, nicotinamide riboside; NA, nicotinic acid; Nam, nicotinamide.

to NAD without prior deamidation:



Lactobacillus fructosus and *Haemophilus influenzae* use this route for NAD biosynthesis. It is used for the same purpose in the nucleus of mammalian cells. The second reaction, catalyzed by NAD pyrophosphorylase (ATP:NMN adenylyltransferase), is considered to be located exclusively in the nucleus of mammalian cells and can be used as an identifying marker for testing the purity of isolated nuclei.

Bacteria use NAD as a substrate in several important reactions including DNA repair. DNA ligases catalyze the formation of phosphodiester bonds at single-strand breaks between adjacent 3'-OH and 5'-phosphate termini in double-stranded DNA. DNA ligases that require ATP for adenylation are found in eukaryotic cells and in bacteriophages. All known bacterial DNA ligases require NAD rather than ATP. The overall reaction for the NAD⁺-dependent DNA ligase is



A comparison of the amino acid composition and DNA sequences indicate that the NAD-dependent ligases are phylogenetically unrelated to the ATP-dependent DNA ligases. The first step of DNA ligation in bacteria requires adenylation of the ε-amino group of lysine 112 at the reactive site of the enzyme by NAD creating an adenylated enzyme intermediate with AMP covalently bound to the enzyme and releasing NMN. In the second step of the reaction, the adenylate group is transferred from Lys-112 to the terminal 5'-phosphate at the DNA nick, resulting in the formation of a phosphodiester bond and sealing the DNA strand. The fact that NAD-dependent ligases are unique to bacteria suggests that this enzyme may be a selective target for new antibacterial agents.

Enzymes referred to as ADP ribosylases or NAD glycohydrolases catalyze reactions in which the ADP-ribosyl moiety of NAD is transferred to an acceptor to yield mono- or poly-ADP-ribosylated derivatives:



Cholera toxin is an ADP ribosylase. In Chapter 14 the role of NAD glycohydrolase in regulating nitrogen fixation is discussed. In the light organ of the Hawaiian squid, the symbiont *Vibrio fischeri* uses an NAD glycohydrolase to regulate its activity. Streptococci produce NADase, a characteristic used as an identifying marker.

Utilization of NAD as a substrate in a number of reactions emphasizes the need for rapid recycling of NAD by means of salvage pathways or pyridine nucleotide cycles (PNCs). A number of alternative PNCs have been shown to operate in microorganisms as shown in Figure 15-24.

Some organisms that degrade NAD to nicotinamide and nicotinic acid cannot recycle these products and accumulate them in the culture medium. The human strains, but

not bovine or avian strains of *M. tuberculosis*, can be identified on the basis of this characteristic. The anaerobe *Clostridium butylicum* also accumulates nicotinic acid because it is unable to recycle it. Although these organisms can synthesize NAD, they lack the requisite enzymes to form nucleotide derivatives of nicotinic acid.

HISTIDINE BIOSYNTHESIS

The biosynthesis of histidine occurs via a unique pathway that is more closely linked to the metabolism of pentoses and purines than to any of the other amino acid families. The pathway of histidine biosynthesis, shown in Figure 15-25, is initiated by the coupling of phosphoribosyl pyrophosphate (PRPP) with ATP at N-1 of the purine ring. This is followed by opening of the ring. The N-1 and N-2 of the imidazole ring of histidine are thus derived from the N-1 and N-2 of the adenine ring. Aminoimidazolecarboxamide ribonucleotide (AICRP) derivatives are normally bound to the enzymes involved in their formation and rearrangement. Cleavage of the open-ring structures gives rise to imidazole-glycerol phosphate (IGP) and AICRP, which is recycled via the purine biosynthetic pathway to reform the purine nucleotide. There do not appear to be any branches from this pathway to other end products. The chemical intermediates and the enzymes involved in histidine biosynthesis are virtually identical in a variety of bacteria and fungi, but the arrangement of the genes controlling the synthesis and operation of the pathway differs markedly from one organism to another.

In *S. enterica*, 8 genes arranged in a single operon control 10 steps in the synthesis of histidine (Fig. 15-26). The genes in *Staphylococcus aureus* are similarly arranged. In yeast, the *his4* region specifies three of the enzyme activities of the histidine pathway, namely, the second, third, and tenth steps. The enzymes responsible for these actions are tightly associated in a multifunctional complex. There is no evidence for aggregation of the remainder of the enzymes, nor are the genes specifying the other enzymes contiguous with these genes or with each other. This contrasts with the close linkage of the 10 *his* genes in *S. enterica* but is comparable to many other gene–enzyme relationships in fungi. The equivalent genes coding for the second, third, and tenth steps in *Neurospora* are also arranged in a cluster. They code for a complex composed of nonidentical isomers. Purified preparations of the complex contain all of these enzyme activities.

The arrangement of the genes in the *his* operon of *S. cerevisiae* is shown in Figure 15-26. The genes in the *his* operon do not correspond to the order of the reactions in the pathway. Mechanisms for regulation of histidine biosynthesis include (1) operon repression mediated at an attenuated site in response to the availability of histidyl-tRNA^{his}, (2) transcriptional control responsive to the general availability of amino acids involving the general signal molecule guanosine tetraphosphate, and (3) feedback inhibition of the first enzyme of the pathway (ATP–phosphoribosyltransferase, ATP–PRT). Details of the various mechanisms of regulation are presented in Chapter 5.

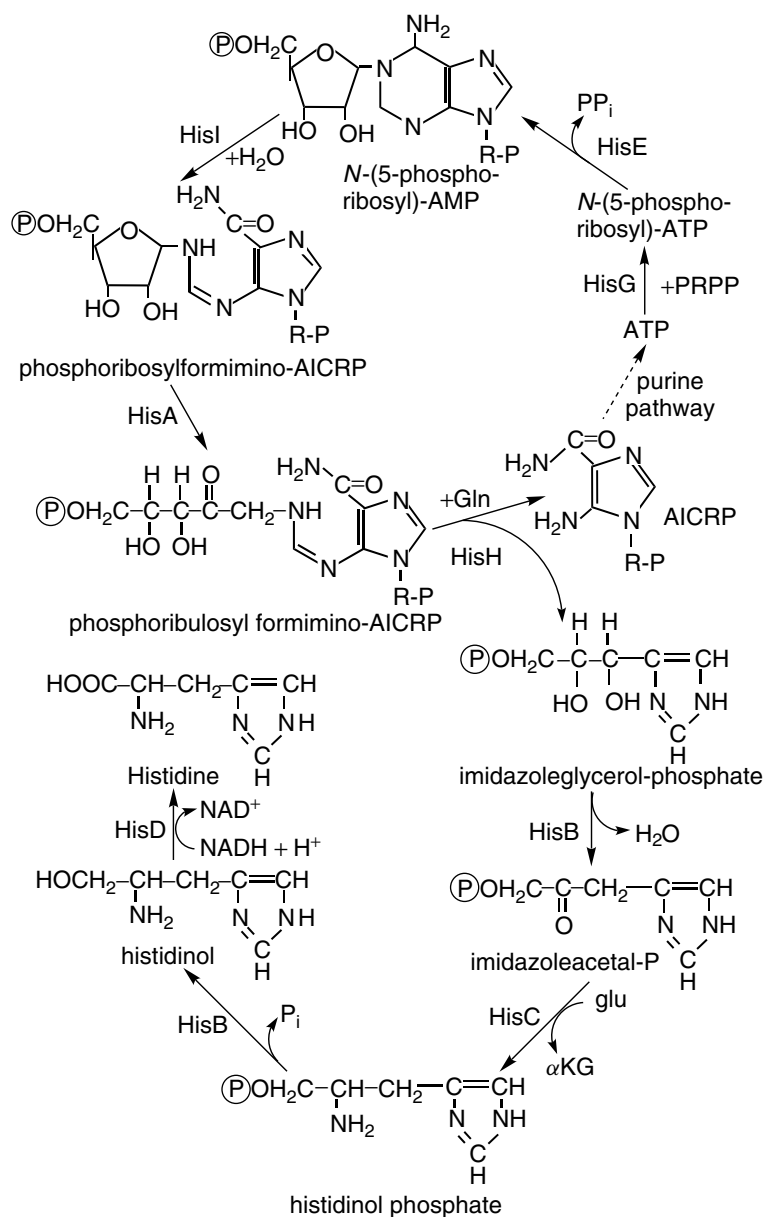
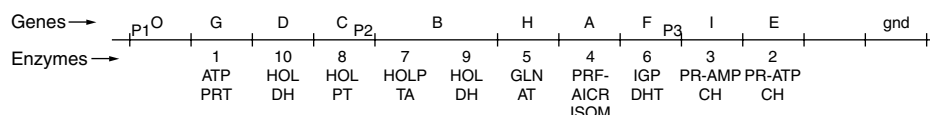
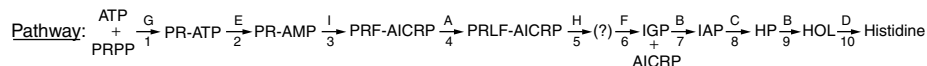


Fig. 15-25. Histidine biosynthesis. The pathway begins with the condensation of ATP and PRPP (5-phosphoribosyl 1-pyrophosphate) to form *N*-5-phosphoribosyl 1-pyrophosphate catalyzed by His G. Elimination of P_i and opening of the ring by HisI forms phosphoribosyl-forminino-5-aminoimidazole-4-carboxamide ribonucleotide. HisA mediates conversion to 5-phosphoribulosyl-forminino-5-aminoimidazole-4-carboxamide ribonucleotide. Cleavage of this intermediate by HisH results in the formation of imidazole glycerol-phosphate and AICRP (aminoimidazolecarboxamide ribonucleotide). AICRP is recycled through the purine pathway. HisB converts imidazole glycerol-phosphate to imidazole acetal-phosphate. Transamination by HisC yields histidinol phosphate. Removal of P_i by HisB yields histidinol. Via an enzyme-bound intermediate (histidinal), HisD converts histidinol to histidine.

The histidine operon of *Salmonella typhimurium*:



The his4 region of *Saccharomyces cerevisiae*:



Fig. 15-26. Arrangement of the genes in the histidine operon of *Salmonella enterica* and the *his4* region of *Saccharomyces cerevisiae*. The equivalent genes for the second, third, and tenth steps in *Neurospora crassa* are clustered in a manner similar to those in yeast. Abbreviations: (1) ATP-PRT, ATP-phosphoribosyltransferase; (10) HOL-DH, histidinol dehydrogenase; (8) HOL-PT, histidinol phosphatase; (7) HOLP-TA, histidinol phosphate transaminase; (9) histidinol dehydrogenase; (5) GLN-AT, glutamine amidotransferase; (4) PRF-AICR ISOM, phosphoribosyl-formimino-5-aminoimidazolecarboxamide ribotide isomerase; (6) IGP-DHT, imidazoleglycerolposphate dehydrogenase; (3) PR-AMP CH, phosphoribosyl-AMP cyclohydrolase; PR ATP CH, phosphoribosyl-ATP-cyclohydrolase. P1, P2, and P3 are promoters. In *S. enterica*, the genes are transcribed in a clockwise direction for O through E.

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CHAPTER 16

PURINES AND PYRIMIDINES

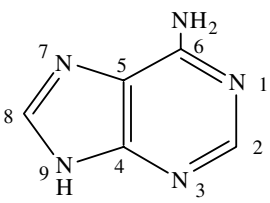
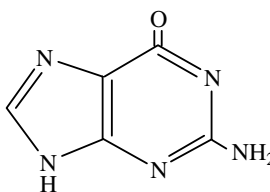
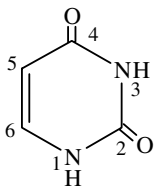
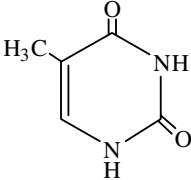
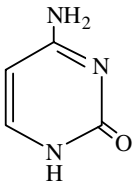
Our knowledge of the structure of nucleic acids and their subunits, the purines and pyrimidines, was gained by the isolation and chemical characterization of these structures from a variety of biological materials. The basic purine and pyrimidine structures and the numbering system recommended by *Chemical Abstracts* are shown in Table 16-1. The most commonly encountered purines are **adenine** (6-aminopurine) and **guanine** (2-amino-6-oxopurine). A number of methylated purines have been found in a variety of sources. Their significance is discussed in conjunction with their role in specialized functions.

Several pyrimidine derivatives have been characterized. **Uracil** (2,4-dioxypyrimidine), **cytosine** (4-amino-2-oxypyrimidine), and **thymine** (5-methyl-2,4-dioxypyrimidine or 5-methyluracil) are most common. Thymine is found only in DNA; uracil is present only, in RNA; 5-hydroxymethylcytosine is found in the DNA of T-even (T2, T4, T6, etc.) bacteriophages of *E. coli* and not in the DNA of the host cell (see Chapter 6). This most significant discovery by Wyatt and Cohen in 1953 proved to be of inestimable value in studying the synthesis of bacteriophage DNA, as it provided the distinguishing feature necessary to show that the synthesis of host cell DNA is terminated when bacteriophage DNA synthesis is initiated (see Chapter 6).

BIOSYNTHESIS OF PURINES

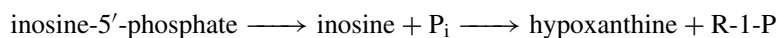
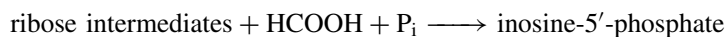
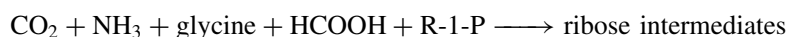
Evidence as to the origins of the purine molecule was obtained from studies with isotopically labeled precursors. Birds were used in the original experiments because of the high percentage of nitrogen excreted as uric acid or its derivatives (>80%). From these studies it was possible to determine that glycine was incorporated intact into the 4, 5, and 7 positions of the purine ring. Formate was consistently incorporated into the 2 and 8 positions and carbon dioxide into position 6 (Fig. 16-1). Pigeon liver homogenates were found to synthesize purines de novo from these same precursors,

TABLE 16-1. Basic Purine and Pyrimidine Structures and Numbering System ^a

Chemical Structure	Recommended IUPAC nName	Common Name Synonyms
	1 <i>H</i> -Purin-6-amine	Adenine 6-AminoPurine
	2-Amino-1,7-dihydro-6- <i>H</i> -purin-6-one	Guanine 2-Amino-6-oxopurine (2-amino-hypoxanthine)
	2,4(1 <i>H</i> ,3 <i>H</i>)-Pyrimidinedione	Uracil 2,4-Dioxypyrimidine
	5-Methyl-2-4-(1 <i>H</i> 3 <i>H</i>)-Pyrimidinedione	Thymine 5-Methyl-2,4-dioxypyrimidine (5-Methyl uracil)
	4-Amino-2(1 <i>H</i>)-pyrimidinedione	Cytosine 4-Amino-2-oxopyrimidine

^aThe chemical structures of these compounds are often shown as the tautomeric form with the maximum number of double bonds within the ring(s). However this is true only for adenine. The most commonly encountered tautomeric form is reflected in the name assigned by the IUPAC commission and give in the table. The 1977 *Chemical Abstracts* listing of chemical names gives the IPPAC recommendations for purine and pyrimidine nomenclature.

and the basic scheme for the formation of purines was considered to be



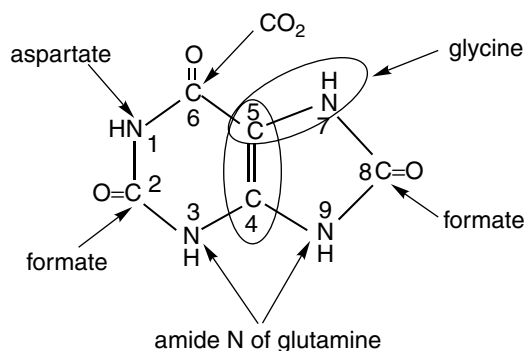


Fig. 16-1. Precursors of the purine molecule. In avian systems, the contribution of the amino nitrogen of aspartate to N-1 and the amide nitrogen of glutamine to N-3 and N-9 has been verified by ¹⁵N-labeling studies. Comparable isotope labeling studies have not been performed with microorganisms. However, other evidence indicates that these same donors provide the nitrogen for positions 1, 3, and 9 in the purines of microbial systems. The C-4, C-5, and N-7 are derived intact from glycine.

One of the first compounds suggested as a potential intermediate in the biosynthesis of purines by microorganisms was aminoimidazolecarboxamide (AICA). Somewhat ironically, AICA did not serve as an intermediate in the synthesis of IMP by the avian system. In microorganisms, considerable variation was observed in the ability of AICA to support the growth of purine-requiring auxotrophs. Eventually, it was demonstrated that the true intermediates in the pathway were the ribonucleotides rather than the free bases. Aminoimidazole and AICA were among the first compounds to be identified as degradative metabolites of purines by *C. cylindrosporum*. A purine-requiring mutant of *E. coli* accumulated an arylamine that differed from AICA in several respects and was converted to AICA by an AICA-accumulating mutant. It was found to be aminoimidazole or its ribonucleotide (AIR). Adenine auxotrophs of yeasts or yeasts grown under the condition of biotin deficiency accumulated aminoimidazole derivatives. Further work in a variety of systems eventually confirmed the participation of AIR, AICAR, and FAICAR as intermediates in the purine pathway (Fig. 16-2).

In the pigeon liver system, the formation of AIR from FGAM and the intermediates involved in the pathway to IMP and AMP were identified. Several microbial systems were also shown to follow the same pathway shown in Figure 16-2. The enzyme responsible for the conversion of AMP to AMP-S was shown to be the same enzyme that catalyzed the conversion of SAICAR to AICAR. The formation of GMP from IMP was also elucidated in microbial systems. Inosine-5-monophosphate (IMP) first undergoes a dehydrogenation to form XMP. Then, via an amido-transferase reaction, the amido-nitrogen of glutamine is transferred to the 2 position to form GMP.

The amino acid histidine and the essential B vitamins, folate, riboflavin, and thiamine, may be considered products of the purine biosynthetic pathway in that a portion of each of their structures is derived from purines. The regulatory systems required for coordinating the flow of metabolites through these multiple branches of the purine pathway are discussed later.

In histidine biosynthesis, as described in Chapter 15, ATP is the initial substrate. Coupling of PRPP at N-1 of the purine ring of ATP and ring cleavage results in

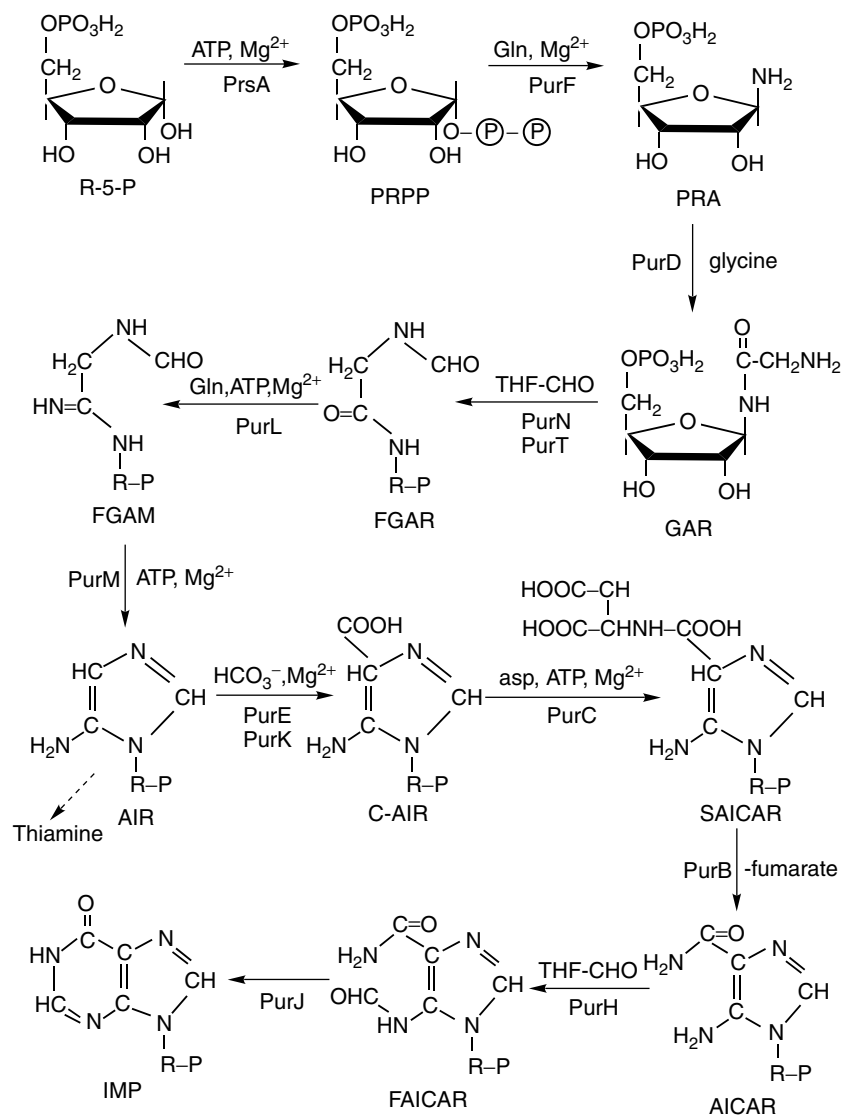


Fig. 16-2. Purine biosynthesis. R-5-P, ribose 5-phosphate; PRPP, 5-phosphoribosyl-1-pyrophosphate; PRA, 5-phosphoribosylamine; GAR, glycinamide ribonucleotide; FGAR, 1-*N*-formylglycinamide ribonucleotide; FGAM, α -*N*-formylglycinamidine ribonucleotide; AIR, aminoimidazole ribonucleotide; C-AIR, 5-amino-4-carboxyimidazole ribonucleotide; SAICAR, 5-amino-4-imidazole-(*N*-succinyl)-carboxamide ribonucleotide; AICAR, 5-aminoimidazole-4-carboxamide ribonucleotide; FAICAR, 5-formamidoimidazole-4-carboxamide ribonucleotide; IMP, inosine 5'-monophosphate. Enzyme designations for *E. coli* and *S. enterica* are: Prs, ribose phosphate pyrophosphokinase or PRPP synthetase; PurF, aminophosphoribosyltransferase; PurD, phosphoribosylglycinamide synthetase; PurI, PurL, phosphoribosylglycinamide formyltransferase; PurG, phosphoribosylformylglycinamidine synthetase; PurI, PurM, phosphoribosylaminoimidazole synthetase; PurE, phosphoribosylaminoimidazole carboxylase; PurC, phosphoribosylaminoimidazole succinocarboxamide synthetase; PurB, adenylosuccinate lyase; PurH, phosphoribosylaminoimidazolecarboxamide formyltransferase; PurJ, IMP cyclohydrolase.

formation of the histidine precursor imidazoleglycerol phosphate and AICAR. The AICAR is recycled through the purine biosynthetic pathway.

Synthesis of the pterin moiety of folate starts with GTP. Details of folate synthesis are discussed in Chapter 15 in conjunction with the derivation of PAB from chorismate as one of the several branches of the common aromatic amino acid biosynthetic pathway. The GTP cyclohydrolase I initiates the pathway, leading to the formation of a pterin derivative which, when coupled to PAB, forms 7,8-dihydropteroate, the precursor of 7,8-dihydrofolate (see Fig. 15-20).

The pyrimidine ring and the ribityl side chain of riboflavin are derived from GTP, as shown in Figure 16-3. The GTP cyclohydrolase II, encoded by *ribA*, cleaves the imidazole ring of GTP to form 2,5-diamino-6-ribosylamino-4(3*H*)-pyrimidinone

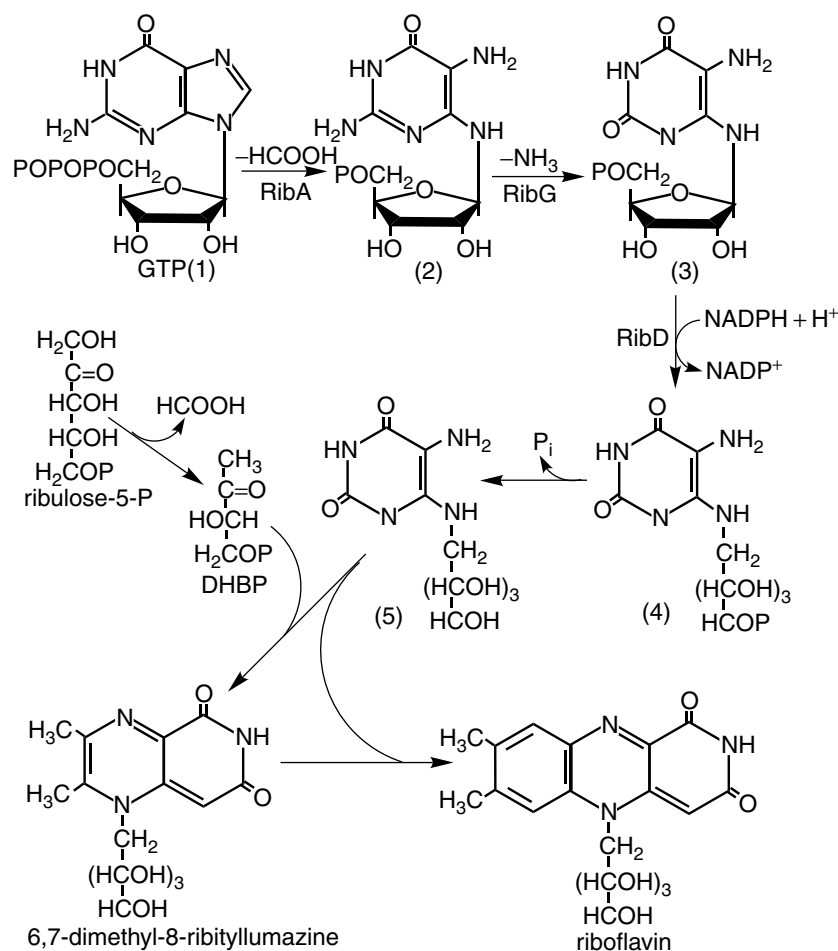


Fig. 16-3. Riboflavin biosynthesis. Enzyme designations are for *E. coli*. *RibA*, GTP cyclohydrolase II; *RibD*, pyrimidine deaminase; *RibG*, reductase; (2) 2-diamino-6-ribosylamino-4(3H)-pyrimidinedione 5'-phosphate; (3) 5'-amino-6-ribosylamino-2,4(1H,3H)-pyrimidinedione 5'-phosphate; (4) 5'-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione 5'-phosphate; (5) 5-amino-6-ribitylamino-2,4(1H,3H)-pyrimidinedione.

5'-phosphate (phosphoribosylamino-pyrimidine or PRP) with the release of formate and inorganic pyrophosphate. Ultimately, condensation of 5-amino-ribitylamino-2,4-(1*H*,3*H*)-pyrimidinedione with 3,4-dihydroxy-2-butanone 4-phosphate yields 6,7-dimethyl-8-ribityllumazine, the direct precursor of riboflavin (Fig. 16-3). Ribulose 5-phosphate is the precursor of 3,4-dihydroxy-2-butanone 4-phosphate, which forms the xylene ring of the riboflavin molecule.

The pyrimidine ring of thiamine (vitamin B₁) is synthesized from AIR as a branch of the purine biosynthetic pathway. Thiamine contains a pyrimidine ring and a thiazole ring. Two precursors, 4-amino-5-hydroxymethyl-2-methylpyrimidine pyrophosphate and 4-methyl-5-(β -hydroxyethyl)thiazole monophosphate, are synthesized separately and coupled to form thiamine monophosphate (TMP). This compound is then phosphorylated to form thiamine pyrophosphate (TPP), as shown in Figure 16-4. *Salmonella enterica* mutants blocked prior to the AIR step require both purine and thiamine for growth. Isotope labeling studies confirmed that AIR is the source of the carbon and nitrogen of the pyrimidine ring of thiamine. The genes coding for enzymes in the TPP pathway comprise a tightly linked cluster, *thiCEFGH*, at 90 minutes on the *E. coli* genetic map. The *thiC* gene product is required for the synthesis of the hydroxymethyl-pyrimidine precursor of TPP. The remaining enzymes (ThiEFGH) catalyze the synthesis of the thiazole precursor.

Under anaerobic conditions, *S. enterica* can form the pyrimidine moiety of thiamine independently of the *purF* locus. In the absence of oxygen, exogenous pantothenate satisfies the thiamine requirement of *purF* mutants. Only the PurF enzyme is bypassed, however, and PurD, PurG, and PurI are still required for the formation of AIR and its conversion to the pyrimidine ring of thiamine.

BIOSYNTHESIS OF PYRIMIDINES

Pyrimidine biosynthesis is initiated by the formation of carbamoyl phosphate (Fig. 16-5). In view of the importance of carbamoyl phosphate in both arginine and pyrimidine synthesis, coordinating the channeling of this intermediate into the two pathways is critical. In *Neurospora*, two carbamoyl phosphate gradients are maintained by two separate carbamoyl phosphate synthetases. Carbamoyl phosphate synthetase A provides a pool of carbamoyl phosphate specifically for the arginine pathway, as discussed in Chapter 15. Carbamoyl phosphate synthetase P provides a carbamoyl phosphate pool specifically for the pyrimidine pathway. Other organisms, for example, yeast, apparently achieve by regulation what *Neurospora* accomplishes by compartmentation.

Aspartate and carbamoyl phosphate are coupled to form carbamoyl aspartate (ureidosuccinate), which is cyclized by a separate enzyme to form dihydroorotate (Fig. 16-5). After oxidation to orotate, the nucleotide is formed by phosphoribosyl transferase. Note that, in comparison to purine biosynthesis, the nucleotide stage is established *after* completion of ring formation. Uridine-5'-phosphate (UMP) is converted to UTP. Then UTP is aminated to form cytidine triphosphate (CTP). There is no known enzyme that converts cytosine to CMP. For cytosine to serve as a nutritional source of pyrimidines, it must be deaminated to uracil.

The deoxynucleoside diphosphates are formed from the corresponding ribonucleoside diphosphate (NDP) by the action of thioredoxin, a sulfhydryl-containing protein

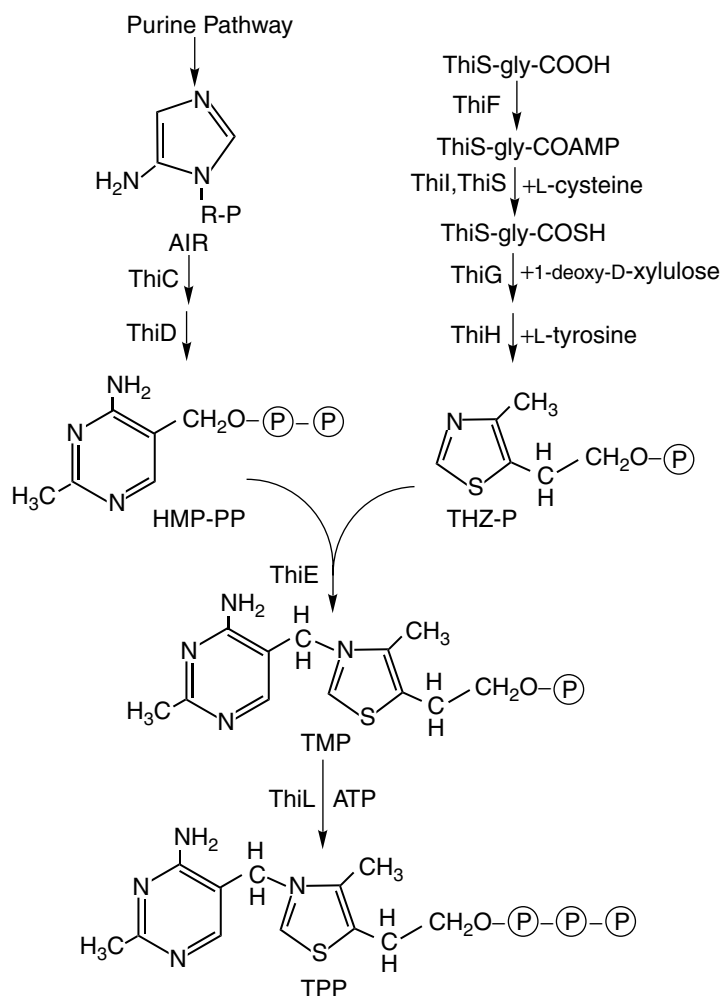
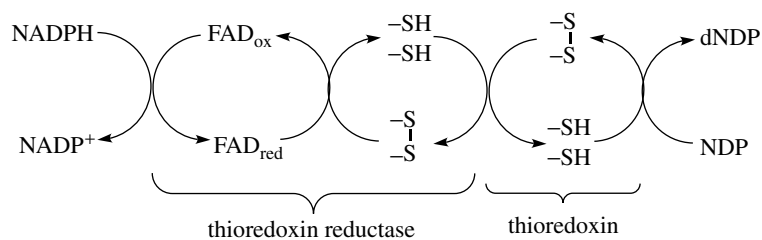


Fig. 16-4. Thiamine (vitamin B₁) biosynthesis. The enzyme designations are for *S. enterica*. TPP, thiamine pyrophosphate; TMP, thiamine monophosphate; HMP-PP, hydroxymethyl-pyrimidine pyrophosphate; THZ-P, 4-methyl-5- β -hydroxyethyl-thiazole monophosphate.

cofactor, and thioredoxin reductase, a system that undergoes cyclic reduction and reoxidation of FAD and NADPH:



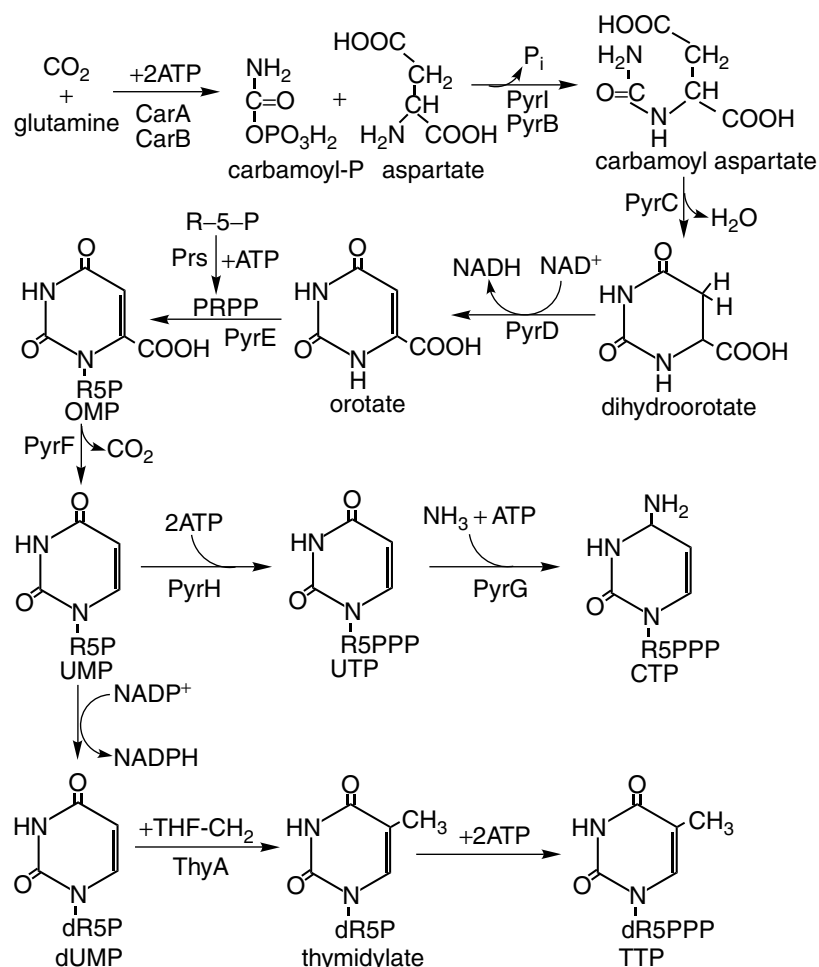


Fig. 16-5. Pyrimidine biosynthesis. Enzyme designations are for *E. coli*. CarA, CarB, carbamoylphosphate synthetases A and B; PyrI, PyrB, aspartate carbamoyltransferase; PyrC, dihydroorotase; PyrD, dihydroorotate dehydrogenase; Prs, PRPP synthetase; PyrE, orotate phosphoribosyltransferase; PyrF, orotidine-5'-phosphate (OMP) decarboxylase; PyrH, pyrimidine nucleotide phosphorylase; PyrG, cytidine triphosphate (CTP) synthetase; ThyA, thymidylate synthase. UMP, uridyate; UTP, uridine triphosphate; CTP, cytidine triphosphate; dUMP, deoxyuridine 5'-phosphate; TTP, thymidine triphosphate.

Thymidylate (TMP) is formed from dUMP by transfer of a methyl group from *N*₅,*N*₁₀-methylene tetrahydrofolate. In the formation of hydroxymethyl-deoxycytidylate, apparently no reduction accompanies the C-1 transfer from the *N*₅,*N*₁₀-methylene tetrahydrofolate, so the complete hydroxymethyl group is added. Both the purine and pyrimidine derivatives must be converted to the trinucleotide stage before they can be incorporated into nucleic acid.

In *E. coli* and *S. enterica*, dCTP deaminase forms most of the dUTP, the precursor of thymidylate. The dUTP is degraded by dUTPase (Dut), yielding PP_i and dUMP, the substrate for thymidylate synthase. The breakdown of dUTP prevents its incorporation

into DNA by DNA polymerase in place of dTTP. Accumulation of dUTP or incorporation of uracil into DNA is probably not lethal until at least 10% of DNA thymine is replaced by uracil. Above this level, excess uracil in DNA may result in degradation by repair enzymes, causing lethal double-strand breaks. Uracil-containing DNA may not be recognized by DNA-binding enzymes or may turn off protein synthesis by a regulatory system. Note the existence of a DNA repair system that deals with the problem (see Chapters 2 and 3).

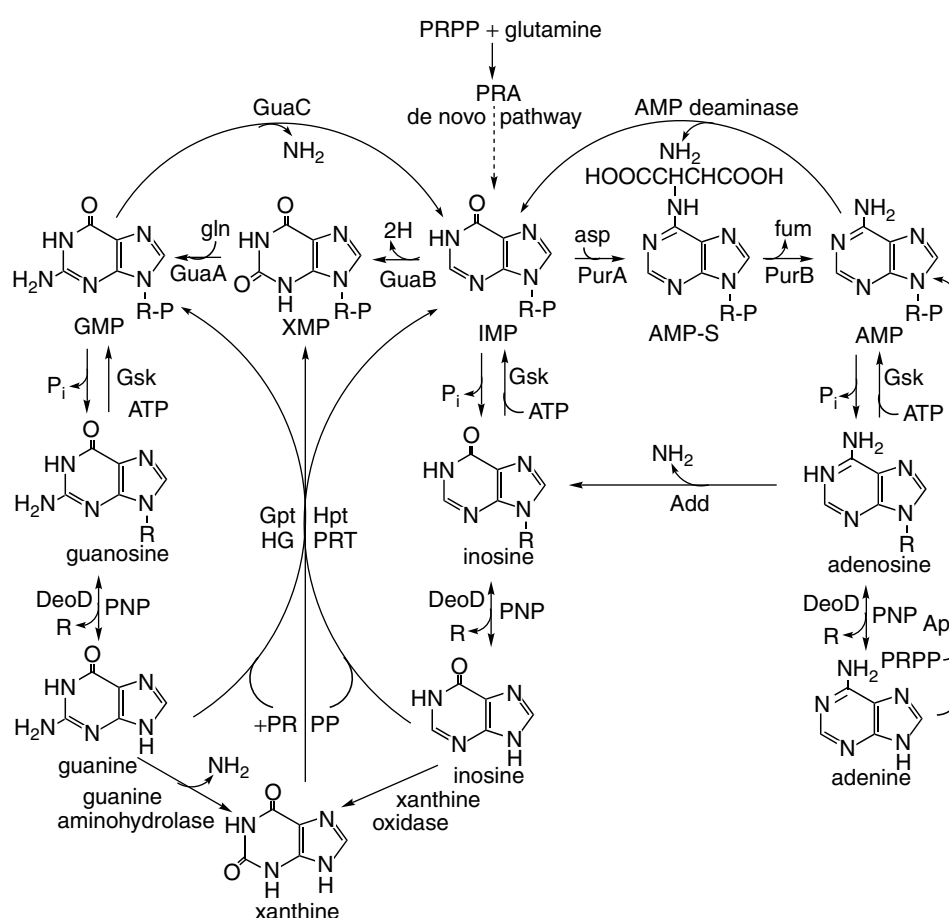


Fig. 16-6. Interconversion of purines and their derivatives. Enzyme designations are for *E. coli*. GuaA, GMP synthase; GuaB, IMP dehydrogenase; GuaC, GM reductase; PurA, adenylosuccinate synthetase; PurB, adenylosuccinate lyase; Hpt, hypoxanthine phosphoribosyltransferase (H-PRT); Gpt, guanosine/xanthine phosphoribosyltransferase (G-PRT); A-PRT, adenine phosphoribosyltransferase (Apt); Add, adenosine deaminase; gln, glutamine; asp, aspartate; fum, fumarate; PNP, purine nucleotide phosphorylase (DeoG); Ask, adenosine kinase; Gsk, guanosine-inosine kinase; PRPP, 5-phosphoribosylpyrophosphate; PRA, phosphoribosylamine; GMP, guanosine monophosphate; XMP, xanthine monophosphate; IMP, inosine monophosphate; AMP-S, succinyladenosine monophosphate; AMP, adenosine monophosphate.

INTERCONVERSION OF NUCLEOTIDES, NUCLEOSIDES, AND FREE BASES: SALVAGE PATHWAYS

It is often assumed that adenine and guanine or their ribonucleotides are freely interconverted by microorganisms. This is not the case. There is a limited ability to interconvert one base with another and an even more limited ability to interconvert nucleotides and nucleosides as shown in Figures 16-6 and 16-7. Adenine and guanine cannot be directly converted from one to the other. The free purine bases are converted directly to their nucleotides by pyrophosphorylases (phosphoribosyltransferases) by reaction with PRPP. Two distinct purine-utilizing enzymes, one specific for the conversion of adenine to AMP and the other specific for the conversion of hypoxanthine and guanine to IMP and GMP, respectively, are present in a variety of bacteria. These enzymes are subject to feedback inhibition and may play a role in the regulation of purine transport and incorporation into nucleic acids. Nucleotidases and nucleosidases are involved in the sequential degradation of the nucleosides to the free bases and are primarily catabolic enzymes. Glycohydrolases may degrade the nucleotides to the free bases and ribose phosphate.

As shown in Figure 16-6, the interconversion of the purine bases may be quite complex and often occurs only through a virtual maze of enzyme reactions. As a specific example, consider the pathway by which *S. enterica* converts exogenous adenine to guanine nucleotide (GMP). Adenine is converted to adenosine, followed by deamination to inosine and subsequent phosphorolysis to hypoxanthine. Hypoxanthine

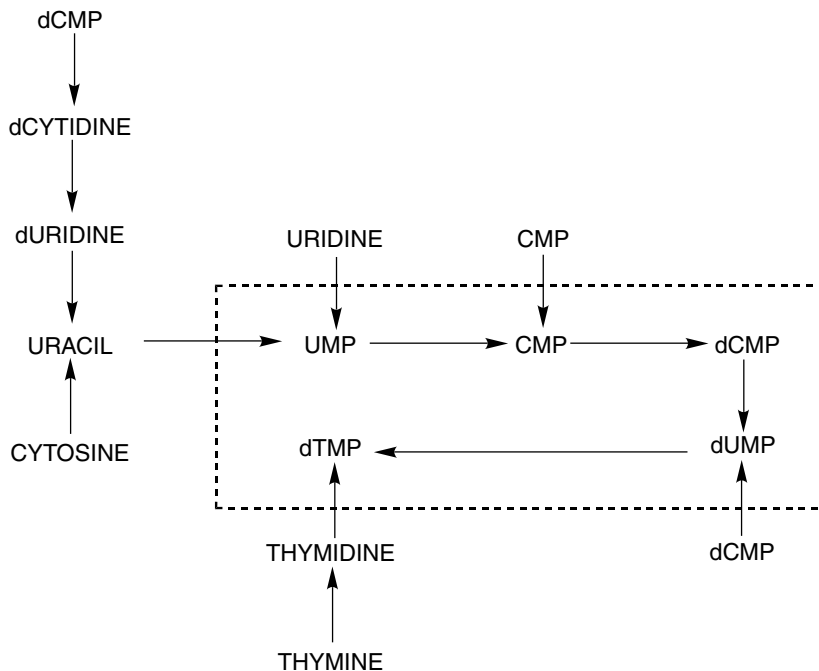
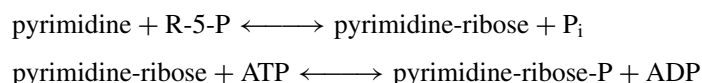


Fig. 16-7. Interconversion of pyrimidines and their derivatives. dCMP, deoxycytidine monophosphate; CMP, cytosine monophosphate; UMP, uridine monophosphate; dTMP, thymidine monophosphate.

is then converted to inosine monophosphate (IMP) by inosine monophosphate pyrophosphorylase. The IMP is then converted sequentially to XMP and GMP.

At the nutritional level, the ability of a given organism to interconvert adenine and guanine is reflected in the ability of the individual purines to promote the growth of purine-requiring strains. Adenine serves quite readily as the sole purine nutrient for many purine-requiring organisms while guanine shows a limited capability in this regard. The capacity of organisms to interconvert purines and their nucleotides also influences the action of a variety of purine antagonists. Regulation of the *gua* operon (which controls the conversion of IMP to XMP and GMP) is effected by adenine and guanine nucleotides rather than the free bases.

Some microorganisms, such as the lactobacilli, possess nucleoside-*N*-glycosyl transferases, which catalyze the transfer of the ribosyl or deoxyribosyl moieties from nucleosides to free bases. However, a wide variety of bacteria, including many of the enteric bacteria, appear to interconvert nucleosides and free bases through the coupling of nucleoside phosphorylases and do not possess nucleoside-*N*-glycosyl transferases. Both purine and pyrimidine bases may be converted to the nucleoside and nucleotide stages by the combined action of nucleoside phosphorylases and nucleoside kinases:



REGULATION OF PURINE AND PYRIMIDINE BIOSYNTHESIS

Both feedback inhibition and repression-depression control systems regulate the activities of the purine and pyrimidine pathways. As shown in Figure 16-8, the steps in the purine pathway subject to regulation are the first step, encoded by *purF*, and the enzymes immediately after the branch point at IMP (encoded by *purA*, *purB*, *guaB*, and *guaA*). The activity of PurF is inhibited by either AMP or GMP. Purine-requiring mutants that accumulate intermediates in the culture medium no longer do so when the purines or the nucleotides are added. The enzymes at the IMP branch point are affected by the specific nucleotide being synthesized. The IMP dehydrogenase is inhibited by feedback and its synthesis is repressed by GMP. The two enzymes that convert IMP to GMP are encoded by the *guaBA* operon and are coordinately controlled. The two genes encoding the enzymes responsible for the conversion of IMP to AMP, *purA* and *purB*, are subject to end product repression. Synthesis of PRA is inhibited by AMP through a feedback mechanism. Regulation of the enzymatic steps leading from IMP to AMP and GMP forms a metabolic "figure 8" insofar as the end products of the two branches loop back to inhibit the formation of the XMP or AMP-S intermediates (Fig. 16-8).

The purine pathway genes are distributed throughout the *E. coli* chromosome both as single genes and small operons. They are negatively regulated at the transcriptional level by the PurR protein and its corepressors. The *purR* gene encodes an aporepressor that combines with the purine corepressors hypoxanthine and guanine, resulting in an increased affinity for a 16 bp palindromic operator in each of the eight operons of the purine regulon. Other genes that supply intermediates for purine nucleotide synthesis or are involved in the synthesis or salvage of pyrimidine nucleotides are also part of

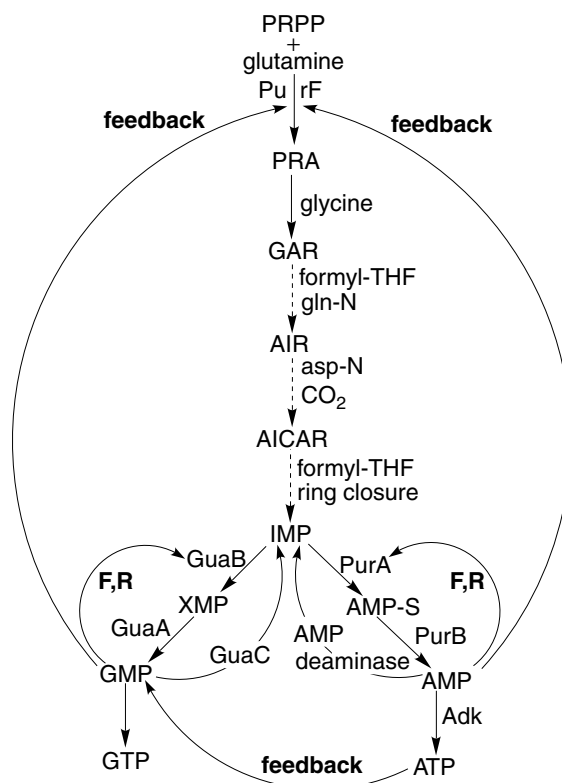
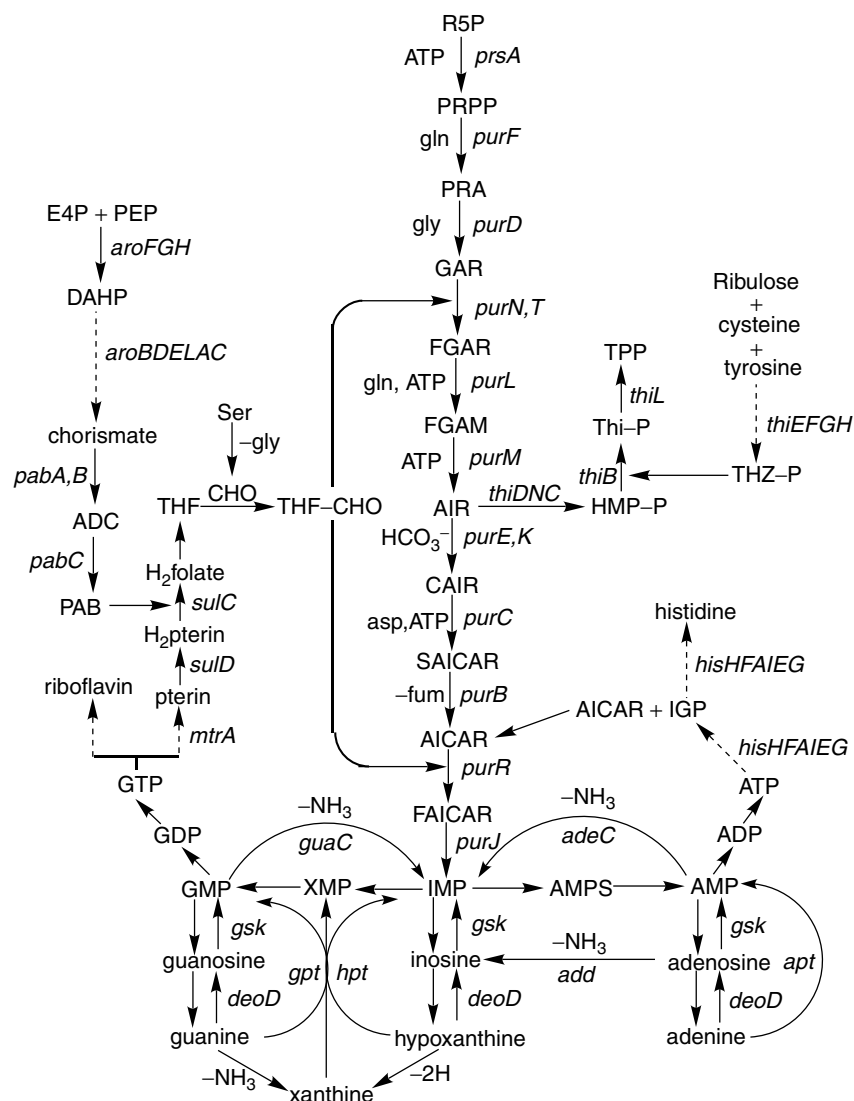


Fig. 16-8. Regulatory mechanisms in purine biosynthesis. PRPP, phosphoribosylpyrophosphate; PRA, phosphoribosylamine; GAR, glycinamide ribonucleotide; AIR, aminoimidazole ribonucleotide; AICAR, 5-aminoimidazole-4-carboxamide ribonucleotide; IMP, inosine monophosphate; XMP, xanthosine monophosphate; GMP, guanosine monophosphate; GTP, guanosine triphosphate; AMP-S, succinyladenosine monophosphate; AMP, adenosine monophosphate. Enzyme designations for *E. coli* are: PurF, aminophosphoribosyl transferase; GuaB, IMP dehydrogenase; GuaA, GMP synthase; PurA, adenylosuccinate synthase; PurB, adenylosuccinate lyase; Adk, adenylate kinase. AMP represses each of the individual steps.

the PurR-regulated regulon. This cross-pathway regulation is necessary to assure the proper supply of precursors for each of the pathways involved. These genes include *glyA* (encoding serine hydroxymethyltransferase, a major contributor of C-1 units), *codA* (encoding cytosine deaminase, a pyrimidine salvage enzyme), *pyrC* (encoding dihydroorotase), *pyrD* (encoding dihydroorotate dehydrogenase), *prsA* (encoding PRPP synthase), *speA* (encoding arginine decarboxylase), and *glnB* (encoding PII protein in nitrogen regulation). Each of these enzymes contains a site to which PurR binds in vitro and is coregulated in vivo by *purR*.

The expression of *E. coli purR* is autoregulated. Autoregulation at the level of transcription requires two operator sites, designated *purR*_{O1} (O₁) and *purR*_{O2} (O₂). Operator O₁ is in the region of DNA between the transcription start site and the site for translation initiation, and O₂ is in the protein-coding region. Operator site O₂, located within the *purR* coding sequence, binds the repressor with 6-fold lower affinity than O₁ but still appears to make an important contribution to the in vivo autoregulation.



R5P, ribose 5-phosphate; PRPP, 5-phosphoribosylpyrophosphate; PRA, 5-phosphoribosylamine; GAR, glycineamide ribonucleotide; FGAR, 1-*N*-formylglycineamide ribonucleotide; FGAM, α -*N*-formylglycineamidine ribonucleotide; SAICAR, 5-amino-4-imidazole(-*N*-succinyl)-carboxamide ribonucleotide; AICAR, 5-aminoimidazole-4-carboxamide ribonucleotide; FAICAR, 5-formamidoimidazole-4-carboxamide ribonucleotide; IMP, inosine 5'-monophosphate; AMP-S, *N*-succinyl-AMP; AMP, adenosine 5'-monophosphate; XMP, xanthosine 5'-monophosphate; GMP, guanosine 5'-monophosphate; GTP, guanosine triphosphate; ATP, adenosine triphosphate; THF, tetrahydrofolate; THF-CHO, *N*₅, *N*₁₀-tetrahydrofolate; E4P, erythrose 4-phosphate; PEP, phosphoenolpyruvate; DAHP, 3-hydroxy-D-arabino-heptulosonate 7-phosphate; ADC, 4-amino-4-deoxychorismate; PAB, *p*-aminobenzoate; H₂pterin, 6-CH₂OH-7,8-dihydroxy-pterate; H₂folate, 7,8-dihydrofolate; IGP, imidazoglycerol phosphate; HMP-P, 4-amino-5-hydroxymethylpyrimidine phosphate; THZ-P, 4-methyl-5-(β -hydroxyethyl) thiazole phosphate; Thi-P, thiamine monophosphate; TPP, thiamine pyrophosphate.

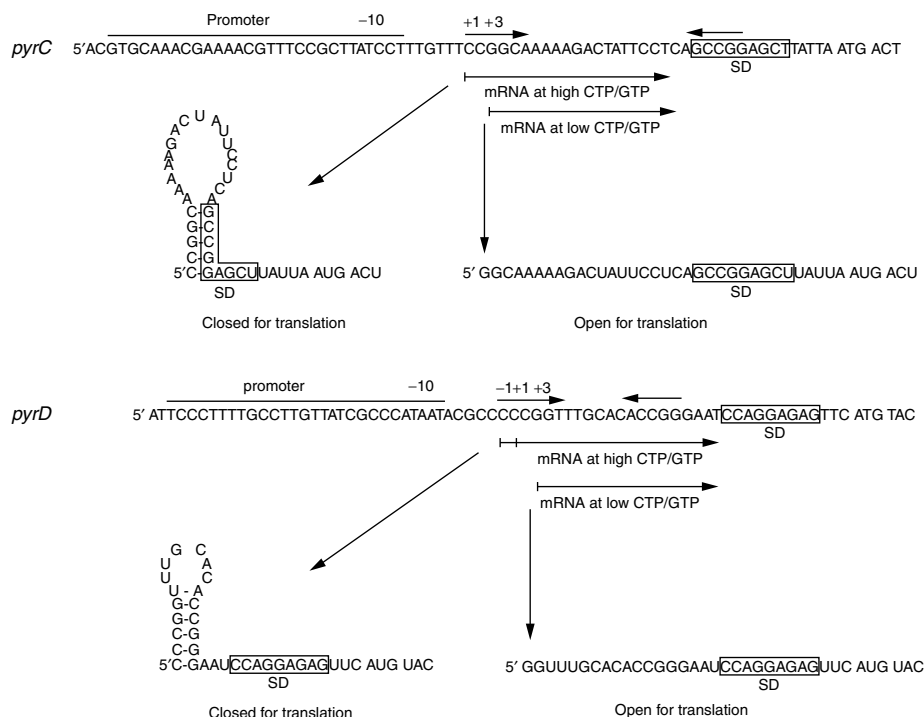


Fig. 16-10. Structure of the *S. enterica* serovar Typhimurium *pyrC* and *pyrD* promoter-leader regions and the proposed regulatory mechanism. Nucleotides are numbered relative to the promoter -10 region (overlined) where +1 at the consensus spacing of 7 bp downstream of the -10 element defines the *in vivo* transcriptional initiation point in repressing conditions. The Shine-Dalgarno (SD) regions are boxed, and the regions of hyphenated dyad symmetry are indicated by arrows above the sequence. Arrows below the sequence represent the transcripts arising in conditions of different CTP/GTP pool ratios. The putative secondary structures formed at the 5' ends of the transcripts are shown. (Source: from Sørensen, K. I., et al. *J. Bacteriol.* **175**:4137–4144, 1993.)

In *B. subtilis* the genes encoding enzymes for the 10 steps to IMP and the 4 genes encoding the conversion of IMP to AMP and GMP form a 12-gene *pur* operon. The *pur* operon is subject to dual control by adenine and guanine compounds. An adenine compound represses transcription initiation and a guanine compound regulates transcription termination-antitermination in a mRNA leader region preceding the first structural gene.

In considering the regulation of purine biosynthesis, complexities arise as a result of the branch points leading to the pyrimidine ring of thiamine, the pterin moiety of folate, the ribitylamino pyrimidine of riboflavin, and a portion of the ring structure of histidine (Fig. 16-9). The enzyme PRPP synthetase, encoded by *prsA*, participates in several biosynthetic pathways. It is involved in the biosynthesis of purine and pyrimidine nucleotides, tryptophan, histidine, and pyridine nucleotides [NAD(P)]. The PRPP synthetase enzyme is subject to repression by pyrimidine nucleotides and is also under the regulatory control of PurR. In view of the increasing number of genes that are subject to cross-pathway regulation, it seems likely that an even greater number of

genes containing sequences related to PurR-binding sites will be revealed through further research in this area.

Aspartate transcarbamylase (ATCase), the first enzyme specific to the pyrimidine pathway, represents one major site of regulatory control. In *E. coli*, the *pyrBI* operon encodes the catalytic (*pyrB*) and regulatory (*pyrI*) subunits of ATCase. The activity of ATCase is subject to allosteric regulation by the activator ATP and the inhibitors CTP and UTP, which bind to the regulatory subunit and alter substrate binding at the catalytic site. This dual control by purines and pyrimidines provides the cell with an efficient mechanism for maintaining the proper ratio of these compounds. The system serves as a prototype for allosteric regulatory systems. The level of *pyrBI* expression, which is negatively regulated by pyrimidine availability, also controls ATCase activity. This regulation occurs primarily through UTP-sensitive attenuation control, with additional pyrimidine-mediated regulation occurring independently at the level of transcription initiation (see Chapter 5). Of two promoters, P₁ and P₂, located upstream of *pyrB*, greater than 95% of the *pyrBI* transcripts are initiated at promoter P₂, with only a small portion of the level of ATCase influenced by the P₁ promoter.

In *S. enterica* and *E. coli*, expression of *pyrC* (encoding dihydroorotase) and *pyrD* (encoding dihydroorotase dehydrogenase) is regulated in response to fluctuations in the intracellular CTP/GTP pool ratio. High CTP/GTP pool ratios repress expression by production of an mRNA initiated with a CTP downstream of the leader region. As shown in Figure 16-10, this transcript is inefficiently translated because of its capacity to form a stable secondary structure (hairpin) at the 5' ends of the transcripts, thus sequestering sequences required for ribosomal binding. The potential for hairpin formation is controlled through CTP/GTP-modulated selection of the transcriptional start site. See Chapter 5 for an additional discussion of regulatory mechanisms.

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CHAPTER 17

BACTERIAL CELL DIVISION

Previous chapters discuss the processes of energy production and biosynthesis of amino acids, purines, pyrimidines, and other building blocks for macromolecular synthesis including chromosome replication, cell wall biosynthesis, and the formation of other vital cellular components such as the cell membrane. The flow of these metabolites and timing of the individual activities must be intricately coordinated so that they lead to the orderly process of cell duplication. In this chapter, emphasis is placed on the processes that take place during the cell cycle: the coordination of chromosome replication with formation of a division septum, the production and activity of various proteins essential to the cell division process, and the regulation of these events. Because more has come to be known about the genetics and regulation of cell division in *Escherichia coli*, this organism has become the paradigm for studies on bacterial cell division. Therefore, we consider cell division in gram-negative rods first.

CELL DIVISION IN GRAM-NEGATIVE RODS

As discussed in Chapter 7, the cell wall peptidoglycan of *E. coli* and other gram-negative bacteria is extended by a process of intercalation of new wall material orchestrated by a large number of enzymes. At some point, a series of events results in the assembly of a large group of proteins into a ring where a constriction occurs between two daughter cells. This ring structure and its component parts are called a **divisome**. Table 17-1 lists many of the genes that encode proteins contributing to peptidoglycan synthesis and cell division.

Under a variety of circumstances, cell division can be inhibited without disrupting the continued linear elongation of the cell. The action of certain β -lactam antibiotics, a variety of chemical and physical agents, alterations in the nutritional environment, and mutations that directly affect the process of cell division can result in the formation of filaments lacking septation. One very fruitful approach has been the use of mutants

TABLE 17-1. Function of *E. coli* Genes Involved in Cell Wall Synthesis and Cell Division

Gene	Map Location	Reported Function
<i>P_{mra}</i>	2	Promoter for first 9 genes (<i>mraA</i> to <i>ftsW</i>) in 2 min <i>mra</i> cluster ^a
<i>mraA</i>	2	D-Alanine carboxypeptidase
<i>mraB</i>	2	Peptidoglycan-synthesizing enzyme
<i>mrhA</i>	2	UDP- <i>N</i> -acetylglucosaminyl-3-enolpyruvate reductase
<i>mrhB</i>	2	Peptidoglycan (PG)-synthesizing enzyme
<i>mrhC</i>	2	PG-synthesizing enzyme
<i>mrcA</i>	2	PBP1A (<i>ponA</i>); PG-synthesizing enzyme
<i>mrcB</i>	2	PBP1B (<i>ponB</i>); PG-synthesizing enzyme
<i>mraY</i>	2	UDP- <i>N</i> -acetylmuramoylpentapeptide:undecaprenol-PO ₄ phosphatase
<i>ftsL</i>	2	Essential cytoplasmic membrane protein involved in cell division; depletion of FtsL results in Y shapes, etc.
<i>fts36</i>	2	(<i>fts33</i>), filamentous; maps upstream of <i>ftsI</i>
<i>ftsI</i>	2	PBP3 (<i>pbpB</i> ; <i>sep</i>), membrane protein, septum PG formation
<i>murE</i>	2	<i>meso</i> -DAP-adding enzyme
<i>murF</i>	2	D-Alanyl-D-alanine adding enzyme
<i>murD</i>	2	D-Glutamic acid adding enzyme
<i>ftsW</i>	2	Septum PG formation; may pair with PBP-3 like RodA/PBP-2
<i>murG</i>	2	<i>N</i> -acetylglucosaminyl transferase; final step in lipid cycle of PG synthesis
<i>murC</i>	2	L-Alanine adding enzyme
<i>ddlB</i>	2	D-Alanyl-D-alanine ligase
<i>dacA</i>	2	PBP5; D-alanine carboxypeptidase
<i>dacB</i>	2	PBP4; D-alanine carboxypeptidase
<i>dacC</i>	2	PBP6; D-alanine carboxypeptidase
<i>ftsA</i>	2	Forms filaments with constrictions at septal sites
<i>ftsZ</i>	2	Mg ²⁺ -dependent GTPase; FtsZ forms ring structure at cell division site; facilitates integration of PBP3 into CM; FtsZ overproduction induces minicell formation; forms filaments with no sign of septa; reduced D-alanine carboxypeptidase activity; interacts with SOS-response-associated cell division; inhibits SulB (SfiB) action
<i>sulB</i>	2	(Suppressor of <i>lon</i> , or <i>sfiB</i> , suppressor of filamentation); allelic to <i>ftsZ</i> ; functions in septation process
<i>envA</i>	2	(<i>divC</i>); forms chains of unseparated cells; reduced levels of NAc muramyl L-alanine amidase
<i>ftsM</i>	2	(<i>supU</i> ; <i>serU</i>); encodes tRNA ^{Ser} ₂ ; regulation of cell division
<i>secA</i>	2	(<i>azi</i> ; <i>pea</i>); secretion of envelope proteins; terminal end of 2 min region
<i>mrcB</i>	3.6	(<i>pbpF</i> , <i>ponB</i>); peptidoglycan synthetase; PBP1Bs
<i>rodA</i>	14.3	(<i>mrdB</i>); affects cell shape, mecillinam sensitivity
<i>rodB</i>	14.3	(<i>mrdA</i> , <i>pbpA</i>), PBP2; mecillinam resistance
<i>mukB</i>	21.0	Required for chromosome partitioning; DNA binding; <i>mukB</i> mutants are anucleate
<i>minB</i>	26.4	<i>min</i> operon; formation of minicells containing no DNA, positioning division septum
<i>minC</i>	26.4	Inhibition of FtsZ ring at division site
<i>minD</i>	26.4	Affects cell division and growth; membrane ATPase that activates MinC
<i>minE</i>	26.4	Reverses inhibition by MinC of FtsZ ring

(continued overleaf)

TABLE 17-1. (continued)

Gene	Map Location	Reported Function
<i>dicB</i>	35.5	Control of cell division
<i>xerD</i>	41.7	<i>cer</i> -specific recombination
<i>pbpG</i>	47.9	PBP7
<i>zipA</i>	54.5	FtsZ interacting protein; essential gene affects cell division and growth; septal ring structural protein
<i>parC</i>	68.2	Cell partitioning; topoisomerase IV subunit A
<i>parE</i>	68.4	nfxD; cell partitioning; topoisomerase IV subunit B
<i>murA</i>	71.8	<i>mrbA</i> , <i>murZ</i> ; UDP- <i>N</i> -enoylpyruvoyl transferase; phosphomycin resistance
<i>xerC</i>	86.1	<i>cer</i> -specific recombination
<i>murI</i>	89.7	<i>mbrC</i> , <i>dga</i> , <i>glr</i> ; glutamate racemase; D-glutamate synthesis; essential for peptidoglycan synthesis
<i>murB</i>	89.9	UDP- <i>N</i> -acetylglucosaminyl-3-enolpyruvate reductase
<i>murH</i>	99.3	Terminal stage in peptidoglycan synthesis, incorporating disaccharide peptide units into wall

^aIndividual promoters are distributed throughout the 2 min region as well.

defective in cell division. Temperature-sensitive mutants that continue to elongate at 42 °C and form long aseptate filaments containing nucleoids at regular intervals but divide normally at the permissive temperature (28 °C) are designated *fts* for filamentation temperature sensitive. The *fts* genes can be designated as functioning early or late in the division process on the basis of their morphology.

Mutants showing no signs of septation, such as *ftsZ*, apparently act early in the sequence, whereas mutants in *ftsA* and *ftsI* give rise to filaments with indications of an attempt to form septa. By combining a *ftsZ* mutant with morphology mutants such as *rodA* or *pbpA* that govern the cylindrical shape of the cell, it was possible to discern that FtsZ acts at an early stage. Combination with *ftsA*, *ftsQ*, and *ftsI* mutations resulted in elongated morphology with indications of initiation of septation, suggesting that these genes act at a later stage. Mutation in a cell separation gene (*envA*) causes *E. coli* cells to form chains of cells during fast growth in rich medium. The mutation is associated with low levels of *N*-acetylglucosaminyl-L-alanine amidase, an enzyme involved in splitting peptidoglycan molecules between the *N*-acetylmuramic acid residue and the pentapeptide side chain.

The proposed steps in the cell division process of *E. coli* are shown in Table 17-2. Three main groups of enzymes are involved. The first group contains enzymes that form the peptidoglycan sacculus around the growing cell. In the absence of modifying activities, this set of enzymes produces the minimum amount of peptidoglycan needed to completely surround the cell—that is, it produces spherical cells that enlarge without division. The second groups modifying reactions—contains for example, the RodA-PBP2 system—that results in the development of a cylindrical shape to the peptidoglycan sacculus. These functions must be active over the entire surface (except at the poles) because preexisting cylindrical surfaces are converted to spherical surfaces after inactivation of either of these two gene products. Also, a spherical surface formed

TABLE 17-2. Genes Involved in Morphogenesis and Cell Division in *Escherichia coli*

Genes for Peptidoglycan Synthesis	Genes for Rod Cell Shape	Early Acting Cell Division Genes		Late Acting Cell Division Genes	Cell Division Genes Present at Constriction
<i>mraA</i>	<i>rodA</i>	<i>ftsZ</i>	<i>zipA</i>	<i>ftsZ</i>	<i>ftsZ</i>
<i>mraB</i>	<i>pbpA</i>	<i>minE</i>	<i>ftsA</i>	<i>zipA</i>	<i>minB</i>
<i>mrhA</i>			<i>ftsW</i>	<i>ftsA</i>	<i>ftsA</i>
<i>mrhB</i>				<i>ftsW</i>	<i>ftsW</i>
<i>mrhC</i>				<i>ftsI</i>	<i>ftsI</i>
<i>mrcA</i>				<i>ftsL</i>	<i>ftsL</i>
<i>mrcB</i>				<i>ftsN</i>	<i>ftsN</i>
<i>mraY</i>				<i>ftsQ</i>	<i>ftsQ</i>
<i>murE</i>				<i>ftsK</i>	<i>ftsK</i>
<i>murF</i>				<i>envA</i>	
<i>murG</i>					
<i>murC</i>					
<i>ddl</i>					
<i>dacA</i>					
<i>dacB</i>					
<i>dacC</i>					

in the absence of these products is replaced by or converted to a cylindrical surface after reactivation of this system.

The third group, responsible for cell division through the formation of new septation sites, acts periodically and is confined to the site of septation at the cell surface. A switch to formation of the type of peptidoglycan (murein) required for septation occurs late in the *E. coli* cell cycle. Inhibition by β -lactam agents gives rise to long, multinucleate nonseptate filaments. Benzylpenicillin and cephalixin selectively bind to PBP3, a high-molecular-weight protein specifically involved in the synthesis of peptidoglycan at the septum. The PBP3 protein is encoded by *ftsI* (*pbpB*) and displays both transpeptidation and transglycosylation activity. The FtsI protein interacts with the other gene products (FtsA, FtsQ, FtsW, and RodA) to maintain the rod shape. FtsW pairs with PBP3 in the same manner that RodA and PBP2 interact to regulate peptidoglycan synthesis. The PBP2–RodA and PBP3–FtsW pairs of membrane proteins carry out alternating morphogenetic modifications of the growing murein sacculus.

Switching from cell elongation to septum formation may result from the change in the relative availability of the different peptide chains required by the two competing systems. Using the technique of fusing a green fluorescent protein (GFP) derived from *Aequorea victoria*, a marine jellyfish, to the FtsI protein, it has been possible to show that this protein forms a septal ring. This penicillin-binding protein (FtsI) is a late recruit to the division site, and its septal location is dependent on interaction with FtsZ, FtsA, FtsQ, and FtsL as well as an intact membrane anchor.

The FtsZ protein acts at an early stage of the septation process (Table 17-2 and Fig. 17-1). FtsZ is a Mg-dependent GTPase dispersed throughout the cell. Just prior to the onset of septal invagination, FtsZ is recruited from the cytoplasm to the division site,